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DERO HE · DEROD · DERO-MINER · DERO-WALLET-CLI · HARNESS

DERO ENHANCEMENT LEDGER

Every change made to this DERO HE source tree from Round 1 (2026-07-08) to Round 29 (2026-11-08). The tree runs on one workstation: the node `derod`, the miner `dero-miner` (CPU threads plus a Metal GPU worker), the wallet CLI, and a Python evidence harness that checks the source and the running node. Each round lists what changed, the problem that triggered it, a code excerpt from git, and the measured result with its source. Refuted ideas are kept as evidence.

29
rounds (plus 4b–4d, 6b, 9b, 23b, 23c, 28d)

~8.1k → 88–92k
live H/s run means, before R14 to R23 (miner PAUSED since R24)

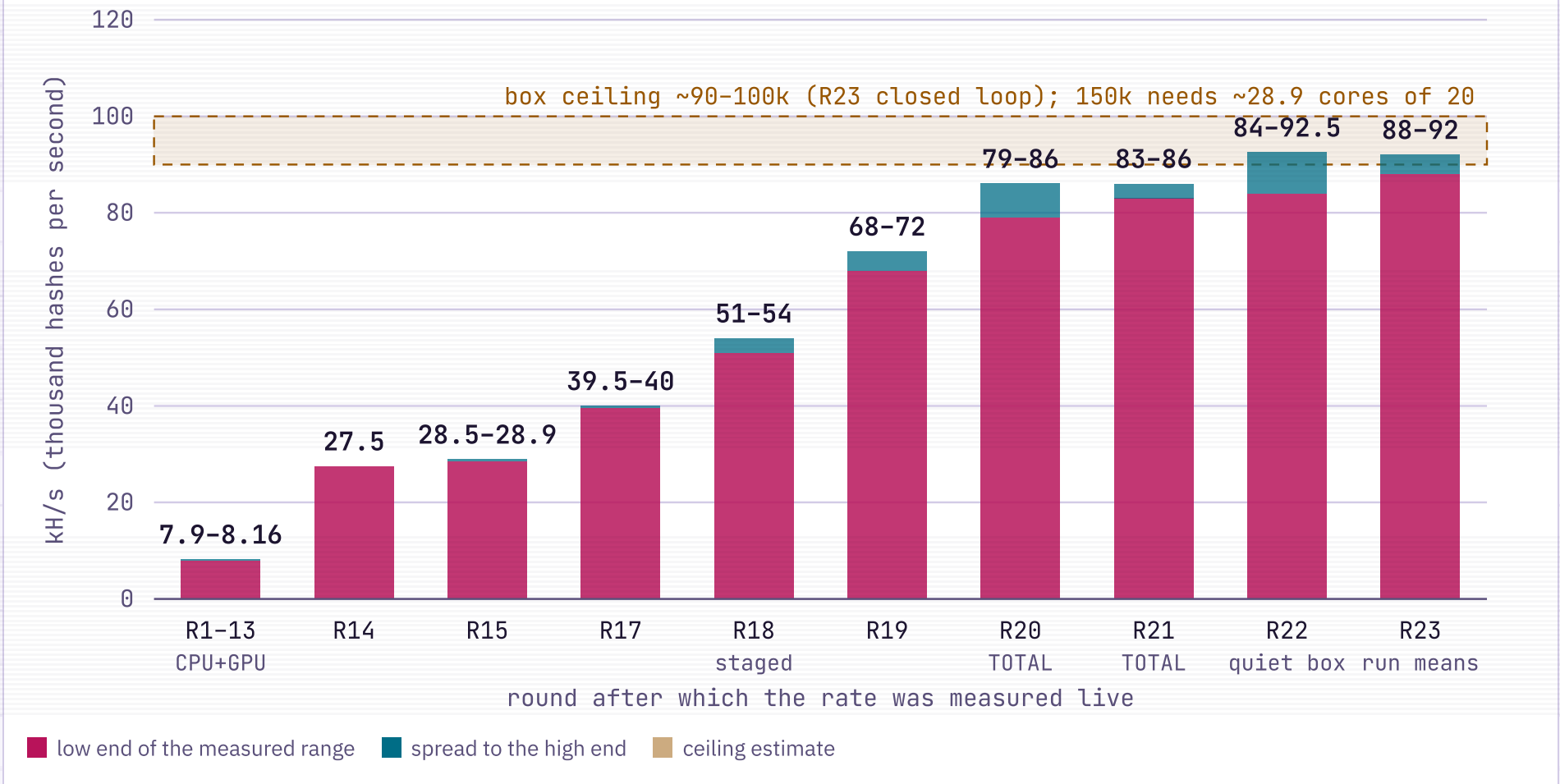
1619/1619
Go mutations caught (R29 full run)

30/33 (+3 PAUSED, miner)
project gates (R29 closing project-only run)

35/35
harness KATs (R29 closing project-only run)

0
shadow mismatches in every passing pipeline, R11 to R29

v2
forensic dataset of the HF3 window, by defect class: replay G28, audit G29, every figure traced G30, bounds G31, classes G32 (Monetary forensics)

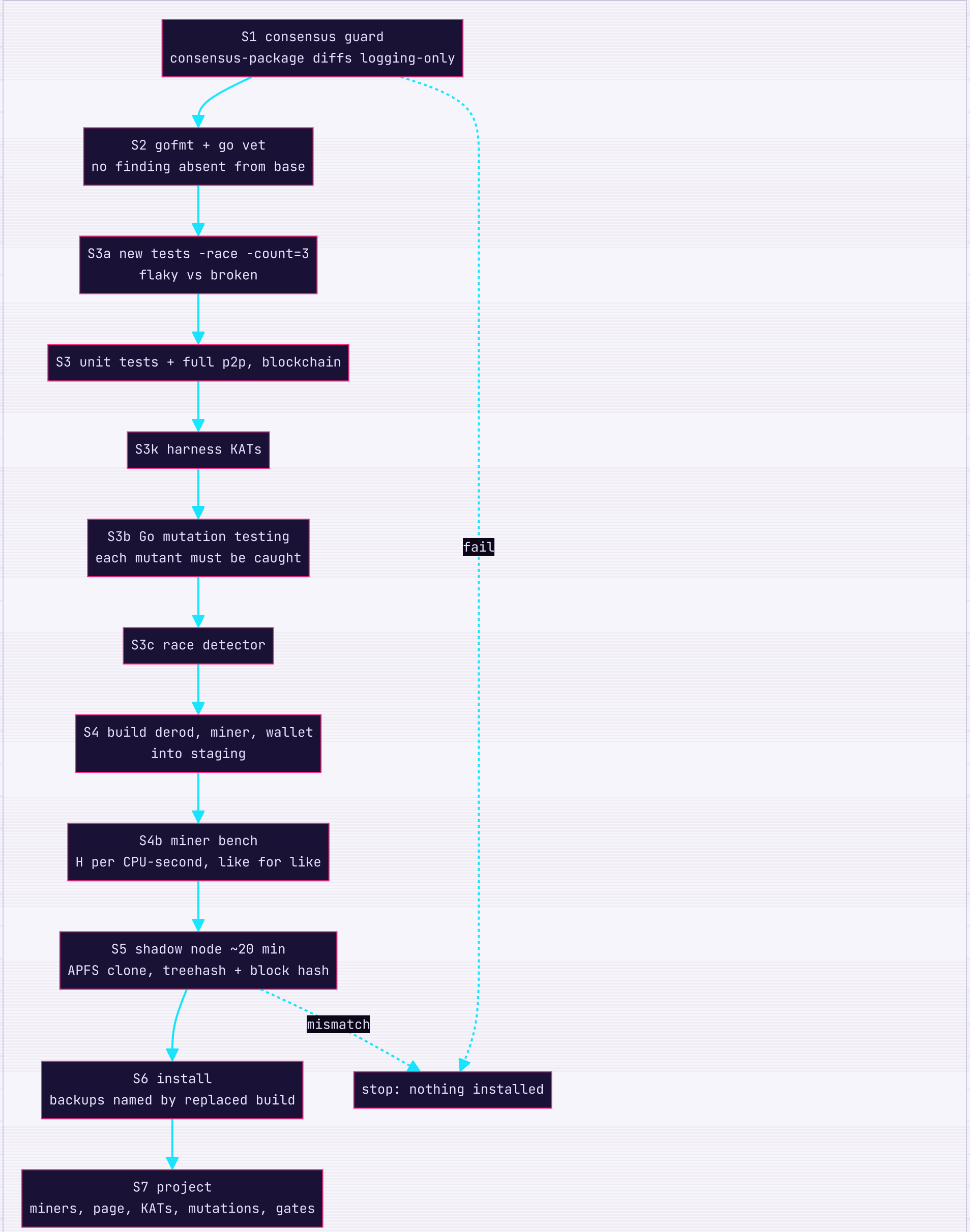


Sources: R1–13 from RUN_REPORT Round 1 (7.9–8.16 KH/s incl. GPU ~1.25k; CPU-only ~6.9k by `--bench`, ARCHITECTURE §13.1); R14 ARCHITECTURE §31.1; R15 §32.1 (TOTAL 28.5–28.9k); R17 §34.1 and RUN_REPORT Round 18 (~39.5–40k); R18 §35.1 (split mode staged rate, 51–54k); R19 RUN_REPORT Round 20 (68–72k); R20 RUN_REPORT Round 21 (TOTAL 79–86k; mean 72.5k over a window that includes an agent-load dip); R21 RUN_REPORT Round 22 (TOTAL 83–86k, staged ~79k). R22 RUN_REPORT Round 23 (84–86k, then 92.5k on the quiet box; the 43–67k contended dip is not drawn); R23 run ledger means of the r23 runs (87,668 at 07:07, 92,194 at 18:03). R16 has no live figure in the sources and is not drawn; the band is the R23 box-ceiling estimate (ARCHITECTURE §39.1). The R23 soundness review is summarised in Round 23. Round 23b has no live H/s figure yet and is not drawn; its tiles read the second R23b pipeline run.

METHOD: OPEN LOOP TO CLOSED LOOP

Each round opens with discovery: multi-agent workflows read the code and the live node, and every finding goes to an adversarial verifier before anything is written. The closed loop is `harness/pipeline/fix_build_verify.py`. A stage that fails stops the run before install, so `bin/` only receives binaries that passed every earlier stage. Every stage appends a JSON event to `harness/loop/runs/pipeline_<ts>.jsonl`.

Monetary forensics (R29)



WHAT EACH CHECK PROVES

- **Consensus guard (G13, DK26).** Changed lines in consensus packages must be logging-only, checked against a baseline copy of bf274e5. From R14 it also proves `astrobwtv3.SuffixSort` is assigned only in `cmd/dero-miner`.
- **Mutation testing.** Each new test must catch a deliberate break of the code it covers. Counts grew from 15/15 (R12) to 743/743 (R23), 853/853 in the first R23b run (stopped by its restore check) and 853/853 in the second, 955/955 in R23c; from R22 a mutation that does not compile is INVALID, not caught.
- **Miner bench.** From R18 the gate uses hashes per CPU-second from getrusage, because wall-clock ratios on a shared box were misleading (the R17 ×1.59). From R21 staged and installed use the same sorter.
- **Shadow.** The new derod runs on a clone of the live datadir with the live node as its only peer. Wherever both report the same topoheight, the balance-tree root and block hash must match. Typical run: 120 blocks, 240 treehashes.
- **Install.** The running derod keeps its old inode; the operator restarts it.

R1	R2	R3	R4	R5	R6	R6b	R7	R8	R9	R11
2026-07-08	2026-07-08	2026-08-09/10	2026-08-19	2026-08-21	2026-08-24	2026-08-25	2026-10-20	2026-10-20	2026-10-20	2026-10

R1 → R29: WHAT CHANGED, MEASURED

Every feature area this tree changed, from its first measured value to its latest. Each value is copied from an Evidence row of the round it names; the factor is the ratio of the two cited numbers (worst end first), drawn as a two-bar strip: grey is the base, coloured is the latest. Where Round 1 had no comparable number, the base is the round that first measured it ("new in R*k*"). The R23c column reads PIPE23C: the r23c miner was measured live on 2026-11-02 01:08–01:28 (harness/runtime/r23c_live_window_20261102.txt); a field still unmeasured shows ◌ pending measurement.

improved: 16 · fixed: 9 · new capability: 6 · open: 3 · held: 2 · improved (estimate): 1

Mining hashrate & sort · GPU · Payouts & uptime · Submission path & orphans · P2P & sync · Security, DoS & agent bans · RPC & websockets · Crypto & soundness · Emission audit · Harness, gates, KATs & mutations · Ops & tools

MINING HASHRATE & SORT

Live miner hashrate, TOTAL — improved

R1	→ R23
7.8–8.16 KH/s (CPU+GPU)	88–92k H/s (run means)

×10.8–11.8

R23c: 94.3k H/s TOTAL, window mean 2026-11-02 01:08-01:28

R1: RUN_REPORT 2026-07-08 18:25-18:31; R23: run ledger bin/dero-miner.runs.jsonl (87,668 and 92,194)

Suffix sort, S4b like for like, 2 threads — improved

R15	→ R15
SA-IS 880 H/s (the R1–R13 sorter, same A/B)	descriptor 3,866 H/s

×4.4

RUN_REPORT Round 15 (idle box)

CPU sort time per text (lower is better) — improved

before R19	→ R19
253 µs	175 µs (bit-exact, 2,048 hashes)

×1.45

R23c: 169.22 µs (r23b 177.38 µs in the same harness; thread CPU, 1 thread, median of 12 pairs)

ARCHITECTURE §35.1

Hashrate ceiling of the box — improved (estimate)

R1	→ R23b
"M1 hardware ceiling" ~7,680 H/s (refuted in R14: it was SA-IS)	~100k H/s (estimate from the per-core sort rate)

ARCHITECTURE §13.1, §30.5; R23b workflow HR23b-6

GPU

Hashes the GPU works on per second (not like for like: R1 full hash, R21 split) — improved

R1	→ R21
~1,230–1,257 H/s (full hash on Metal)	staged ~79k H/s (GPU stage 1 + final SHA, CPU sorts)

R23c: 87.9k H/s staged, same window

RUN_REPORT 2026-07-08; RUN_REPORT Round 22 (r21 staged)

--gpu-max cap at 90 — fixed

R21	→ R23b
GPU busy 94.6–96.0 %: not enforced (FOR22-1)	duty bucket since R22; busy 90–92 % of a 96 % cap

R23c: 95.6 % of the 96 % cap

RUN_REPORT Round 22; R23b workflow FOR-8

GPU failure handling — fixed

R18	→ R19
GPU server exit left the worker dead (07:40 incident)	auto-respawn + bit-exact self-test + audit-strike policy

ARCHITECTURE §34; RUN_REPORT Round 19

PAYOUTS & UPTIME

Paid fraction of submitted shares — new capability

R13	→ R23b
new in R13 (DERO.GetMiningOutcome); not measurable in R1	0.957 (46 submitted, 44 paid)

RUN_REPORT Round 13; DERO.GetMiningOutcome 19:28 (R23b)

Miner uptime — open

R23b	→ R23b
0.312 of 72 h; 9 of 9 runs ended by a typed console exit	--supervised + tools/launchd shipped; 24 h idle-ledger read open

R23c: ◌ pending measurement

R23b workflow PAY-1; ARCHITECTURE §40.2

Shares found vs expectation — improved

R2	→ R23b
2 found, 2.3 expected (71 min)	46 found, 43.47 expected (z = +0.38)

ARCHITECTURE §19.1; R23b workflow FOR-2

SUBMISSION PATH & ORPHANS

Dead normal work per block at the normal → final flip (lower is better) — fixed

before R17	→ R17
82.5 ms	17.7 ms (refresh_due, shadow A/B)

×4.7

ARCHITECTURE §33.2

Final share → next-height job (own blocks) (lower is better) — improved

R22	→ R23
252 ms	32 ms (one getwork dispatcher)

×7.9

RUN_REPORT Round 23; ARCHITECTURE §39.5

KCP packets per RPC message (lower is better) — improved

before R17	→ R17
4	1 (identical bytes)

×4.0

ARCHITECTURE §33.3

Orphaned shares (lower is better)
improved

R17
3 of 23 r16 shares (9th-slot races)

→ R23b
2.7 % of regular shares

R17 forensics (23 shares followed); R23b workflow PAY-4 (all shares vs regular shares: not like for like)

P2P & SYNC

Outgoing peers improved

R1
2 peers, ~5,100 blocks behind

→ R23b
14 = every usable known peer

×7.0

RUN_REPORT Round 1; DERO.GetSyncInfo 19:13 (R23b)

Inbound peers open

R2
0 (KCP_PassiveOpens = 0)

→ R23b
0: forward UDP 11011 (user action)

ARCHITECTURE §19.4; R23b workflow SP23b-1

Block fetch, emulated 170 ms RTT
improved

before R18
4.5 blocks/s

→ R18
41.8 blocks/s (pipelined, treehash 31/31)

×9.3

ARCHITECTURE §34.4

Retransmits (modelled) (lower is better)
improved

before R17
18.6 %

→ R17
5.4 % (one packet per message)

×3.4

ARCHITECTURE §33.3

SECURITY, DOS & AGENT BANS

Distributed probing, 1000 keys × 8 probes
new capability

before R23b
0 banned

→ R23b
900 banned (global anomaly budget)

R23b workflow SEC23b-2

Unknown tx type on SendRawTransaction (lower is better) fixed

before R23b
1 recovered panic per call

→ R23b
0 (refused from the header)

R23b workflow SEC23b-4

Remote crashes found and fixed fixed

R13
remote-triggerable crash caught in pre-build review

→ R21
getwork blob-length panic SEC21-01 fixed

RUN_REPORT Round 13; RUN_REPORT Round 21

RPC & WEBSOCKETS

DERO.GetTxStatus, 1 txid p50 (lower is better) improved

before R23b
23.7 ms

→ R23b
1.0 ms (settled cache, verify on hit)

×24

R23b workflow RPCWS23B-A (mainnet clone)

Is the node at the tip? new capability

R7
GetInfo said OK while the node was 40.4 days behind

→ R12
DERO.GetSyncInfo (lag, target height)

RUN_REPORT Round 7.2; RUN_REPORT Round 12

rpc.serverInfo to remote callers fixed

R23
method list, start time, loopback counters leaked

→ R23b
−32601 for remote and cross-site callers

commit 3224c07; R23b workflow SEC23b-1

CRYPTO & SOUNDNESS

Range-proof soundness defect (C1) open

R8
CONFIRMED by the 199-agent audit

→ R23b
still in the code; needs a coordinated fork

RUN_REPORT Round 8; dero-soundness-r23.html §3

Wallet ring-order nonce reuse (C3) fixed

R22
two payloads shared a blinding scalar

→ R22
distinct ring order per payload

RUN_REPORT Round 22; ARCHITECTURE §38.5

HF3 table Parity fixed

R23b
set on the sync path only

→ R23b
every block-add path, under hf3_exec_mu

commit 3224c07 p2p/hf3_parity.go; R23b workflow CR23b-1

EMISSION AUDIT

Mainnet blocks audited, 0 alerts
new capability

R23
new in R23: 13,100+ (offline copy)

→ R23b
+19,183 more (archive mode)

dero-soundness-r23.html §1; R23b workflow SND-1

Auditor CPU per block (lower is better)
held

R23
4.9–5.2 ms (offline)

→ R23b
4.48–5.66 ms (archive)

×0.87–1.16

RUN_REPORT Round 23; R23b workflow SND-1

HARNESS, GATES, KATS & MUTATIONS

Project gates improved

R7 → R23b
12/12 (G1–G12; R1 had 6 ad-hoc gates) → 22/22

×1.83

R23c 22/22

RUN_REPORT Round 7.1; pipeline record pipeline_20261030T215745

Harness KATs improved

R7 → R23b
20/20 (DK01–DK20) → 34/34

×1.70

R23c 35/35

RUN_REPORT Round 7.1; pipeline record pipeline_20261030T215745

Go mutations caught improved

R12 → R23b
15/15 → 853/853

×57

R23c 955/955

RUN_REPORT Round 12; pipeline record pipeline_20261030T215745

Race-detector packages, 0 races improved

R13 → R23b
6 packages → 26 packages 0 races

×4.3

R23c 26 packages 0 races

RUN_REPORT Round 13; pipeline record pipeline_20261030T215745

Shadow mismatches vs the live node held

R11 → R23b
0 (240/240 treehash) → 0 (120/120 blocks)

R23c 0

RUN_REPORT Round 11; pipeline record pipeline_20261030T215745

OPS & TOOLS

Miner log across restarts fixed

R19 → R20
truncated on every start (r19 run record lost) → append, rotate at 32 MiB, 0600

RUN_REPORT Round 20

Typed console exit new capability

R23b → R23b
stopped the miner (9 of 9 runs) → --supervised ignores it; launchd KeepAlive

R23b workflow PAY-1

Sync ETA formula (tip_age/18)/(rate – 1/18) new capability

R9b → R9b
new in R9b → within 4 min over 11.98 h

RUN_REPORT Round 9b

ROUND 24 NODE FIGURES (LIVE R24 NODE, BENCH AND SHADOW)

The comparison above and the curve below stay on R23c, the latest round with a MEASURED miner: R24 changed only the node and the miner track is paused (rft §0.3). The r24 derod was read live on 2026-11-02 (restart 15:31:03, reads 15:41:24 and 16:32:01); the first table puts the node-side figures that both binaries report the same way next to each other. The hour had no tx-carrying block and no sync, so the compact-block and prewarm counters are ○ pending measurement.

MEASURED	R23C LIVE	R24 LIVE	KIND	SOURCE
GetMonetaryIntegrity agrees with GetEmissionAudit (disclosure check S3)	FAIL (17:56 UTC)	PASS ×2 (23:41, 00:32 UTC)	live	harness/runtime/disclosure_live_20261102T175638Z.txt → disclosure_live_20261103T003206Z.txt
JSON-RPC POST without Content-Type	empty 415	415 + JSON-RPC -32600 body	live	r24_dimension_results RPCWS24-1 → harness/runtime/r24_live_first_20261102.txt
Emission audit, alerts / payees explained	0 / 3,331 of 3,331 (489 audited, 01:28)	0 / 1,447 of 1,447 (219 audited, 16:32)	live	harness/runtime/r23c_live_extract_20261102.txt → r24_live_hour_20261102.txt
Emission audit CPU, mean µs per block	3,525	3,389	live	harness/runtime/r23c_live_extract_20261102.txt → r24_live_hour_20261102.txt (no auditor change in R24: like for like, within run noise)
Proof scalars ≥ Order, last 100,000 blocks	0 of 15,866	0 of 15,782	live census	harness/runtime/disclosure_census_20261102T175740Z.json → disclosure_census_20261103T004204Z.json

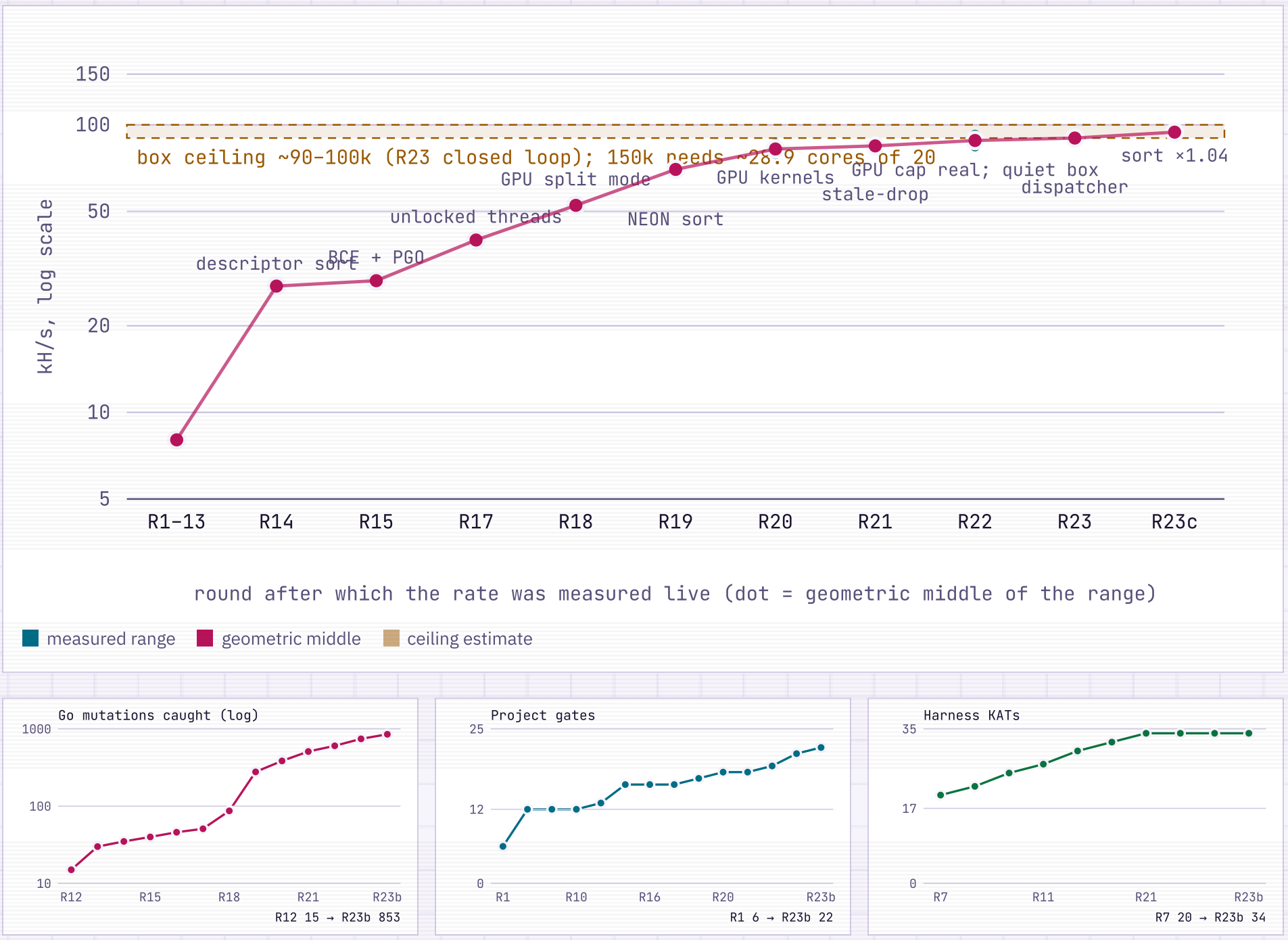
The rows below are bench or shadow A/Bs of the R24 dimensions, verifier figures where a verifier re-ran them (source: harness/loop/runs/r24_dimension_results_20261102.json, by finding id); none is a live reading.

MEASURED	BEFORE (BASE)	R24	KIND	SOURCE
Remote RPC probes through the aggregate after pre-warming (scenario S1)	4,800	573	bench	security SEC24-1
Penalty-table peak records with every ban running	85,536	65,536	bench	security SEC24-3
CPU of 6 concurrent identical tx submissions	466–477 ms	87–97 ms	bench	txpool TXP-3
Replayed proof-invalid tx via Admit	44–46 ms	2.5–3.3 ms	bench	txpool TXP-4
CPU per synced block (prewarm difficulty single-flight)	base = 1	×0.947 (6/6 pairs)	shadow	chainperf CP24-1
Catch-up blocks/s from a live source (cross-response overlap)	base = 1	×1.114 (6/6 pairs)	shadow	syncp2p SP24-3
Prewarm PoW per synced block (B34 counter)	17.48–17.50	9.99–10.00	shadow	syncp2p SP24-2

MEASURED	BEFORE (BASE)	R24	KIND	SOURCE
Loopback RPC CPU (regression check)	base = 1	×0.987	bench	rpcws RPCWS24-PERF

THE ENHANCEMENT CURVE, R1 → R29

Speed and verification grew together. The first chart is the live hashrate of each round on a log axis, so equal steps are equal factors; each label names the lever of that step. The three small panels below are the closed loop growing: Go mutations caught, project gates and harness KATs per round, each on its own axis.



Sources: hashrate points as in the overview note (R1–13 to R23); R23c: harness/runtime/r23c_live_window_20261102.txt (bin/dero-miner.log 01:08–01:28, n=81 TOTAL lines); verification counts: R1 RUN_REPORT Round 1 gate ledger (6 ad-hoc gates); R7 Round 7.1; R8 Round 8; R10 Round 10; R11 commit 4e489d1; R12–R18 the rounds' pipeline rows; R19 16/17 gates (G17 correctly failed); R20–R23 RUN_REPORT; R23b pipeline record pipeline_20261030T215745. R16 has no live hashrate figure in the sources and is not drawn.

▶ Data tables

R1 · 2026-07-08 · Release142 local build

NETWORK LATENCY: --LOWLATENCY KCP, LATENCY-ORDERED BROADCAST, MINER SUBMIT PATH

live 7.8–8.16 KH/s gates 6/6

WHAT CHANGED

- `--lowlatency` derod flag selects the KCP profile `SetNoDelay(1,10,2,1)`: 10 ms flush, fast retransmit, congestion control off. The old default logged "turbo mode" while it ran the 40 ms congestion-controlled profile.
- `broadcast_MiniBlock` sends to peers sorted by measured latency, lowest first. Same peers, same bytes.
- Miner: its own websocket dialer with explicit `TCP_NODELAY`, reconnect back-off 1 s to 8 s (was a flat 10 s), and `submitShare()` serialises JSON outside `connection_mutex` and records submit latency.
- `net_metrics.sh` scrapes the miniblock propagation histogram; `run_derod_lowlatency.sh` launches derod with `--min-peers=40 --lowlatency`.

WHY

> A miniblock is paid only if it is among the first 9 normal miniblocks to reach the node that builds the block (arrival order, no freshness gate). Propagation latency from this node is therefore the

network quantity that affects reward (ARCHITECTURE §18).

CODE

The Release142 working copy was not under git; these hunks entered the git tree verbatim in the Round 4 port commit.

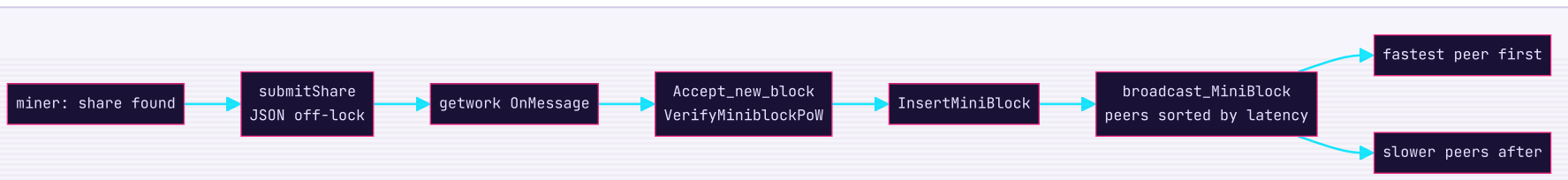
p2p/connection_pool.go · lines 557–574 · commit d43e08e

```
    // Dispatch to the LOWEST-LATENCY peers first. The miniblock is the reward-critical object:
    // it is orphaned (no reward) if it reaches the integrator that finds the final block AFTER
    // that block's first-9 normal-miniblock slots fill (arrival-order cap; no freshness gate).
    // Front-loading the fastest propagation paths minimises that orphan window. This mirrors
    // Broadcast_Block (already latency-sorted, line ~407). Order-only change: same peers, same
    // bytes, no consensus impact. Latency is written via atomic.StoreInt64 (common.go), read atomically.
    connections := make([]*Connection, 0, len(unique_map))
    for _, v := range unique_map {
        connections = append(connections, v)
    }
    sort.SliceStable(connections, func(i, j int) bool {
        return atomic.LoadInt64(&connections[i].Latency) < atomic.LoadInt64(&connections[j].Latency)
    })

    //connection.logger.V(4).Info("Sending mini block to peer ")

    for _, v := range connections {
        select {
```

FLOW



EVIDENCE

MEASURED	RESULT	SOURCE
Gates G-B, G-A, G-C, G-tree, G-E, G-live	6/6 PASS	RUN_REPORT, Round 1 gate ledger
Live miner throughput, 23 samples	7.8–8.16 KH/s, GPU ~1,230–1,257 H/s	RUN_REPORT 2026-07-08 18:25–18:31
Rejected shares	0	RUN_REPORT live pipeline
Node state during the test	~5,100 blocks behind, 2 peers, 0 inbound	RUN_REPORT dominant finding
Live propagation p50/p90 under --lowlatency	deferred (node not synced)	RUN_REPORT unverified ledger

Finding that re-ranked the levers: mining while behind produces stale miniblocks that peers reject, so staying synced and keeping peers outranks transport micro-latency.

R2 · 2026-07-08 · Release142 local build

REWARD DEEP DIVE: GROUND-TRUTH AUDIT TOOLS

finds 2 vs 2.3 expected

WHAT CHANGED

- `reward_audit.py` : scans `DERO.GetBlock` `block_header.miners[]` for the operator address and counts paid miniblocks, orphan and side blocks. It is the only on-chain source for the paid fraction.
- `mining_dashboard.sh` : one-view monitor (sync, peers, KCP retransmits, propagation p50/p90, paid share). Four bugs in the agent draft were fixed before use.
- `run_derod_lowlatency.sh` pins `--p2p-bind=0.0.0.0:11011` so a router forward can target a fixed port.

WHY

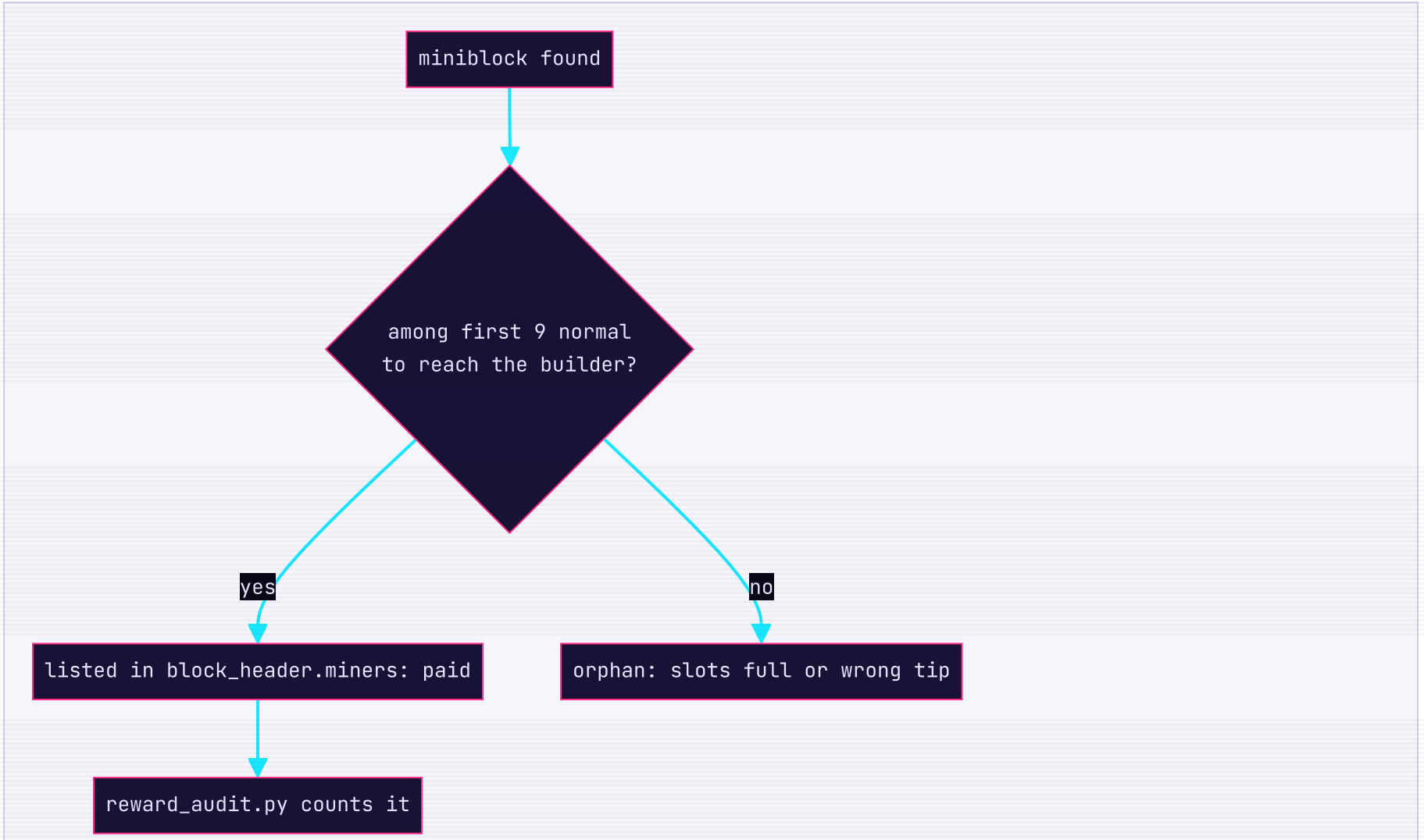
> The node counter and the miner counter both report local acceptance, not inclusion in a canonical block. A paid-share number needed a chain scan. Open-loop run: 11 agents (5 analysers, adversarial verification, synthesis), workflow wf_42fdf10b.

CODE

reward_audit.py · lines 62-81 · commit d43e08e

```
    scanned = len(blocks)
    our_minis = net_minis = our_atomic = 0
    orphan_blocks = side_blocks = 0
    blocks_with_us = 0
    per_slotcount = collections.Counter()   # distribution of slots/block
    our_by_height = {}                       # height → our count (for found→paid join)
    for h in sorted(blocks):
        bh = blocks[h]
        if bh.get("orphan_status"):
            orphan_blocks += 1; continue
        if bh.get("sideblock"):
            side_blocks += 1  # side blocks still count minis toward payout in DERO DAG; keep but flag
        ms = bh.get("miners") or []
        slots = len(ms)
        per_slotcount[slots] += 1
        net_minis += slots
        ours = sum(1 for m in ms if m == a.addr)
        if ours:
            blocks_with_us += 1
            our_by_height[h] = ours
```

FLOW



EVIDENCE

MEASURED	RESULT	SOURCE
Miniblocks per block	10 (399/399 and 2500/2500 blocks)	ARCHITECTURE §19.1
Finds vs expectation, 71 min at ~8 KH/s	`2 found, 2.3 expected`	ARCHITECTURE §19.1
KCP retransmit ratio, steady state	21–54 %, 94 % RTO-driven (path loss)	ARCHITECTURE §19.2
Re-enable KCP congestion control	`REFUTED · refuted: cannot reduce RTO loss`	ARCHITECTURE §19.2
--min-peers=40	`REFUTED · no-op: outbound peer supply exhausted`	ARCHITECTURE §19.4
Inbound peers	`KCP_PassiveOpens = 0`	ARCHITECTURE §19.4
Integrator address left on the dev default	`+11 % lost; operator node already correct`	ARCHITECTURE §7A
Tip-adoption lag	~300 ms, same with or without the miner	ARCHITECTURE §7A
Network hashrate formula used here (D×10/18)	`superseded in R15: NW ≈ D`	ARCHITECTURE §31.2

R3 · 2026-08-09/10 · Release142 (no code change)

SYNC STALL AT TOPO 7,412,674: RECOVERY PLAYBOOK

sync to tip `0 verification errors`

WHAT CHANGED

- No code change. The round produced a recovery procedure: salvage the reward record, archive the datadir (move, never delete), fresh fastsync bootstrap, sync to tip, restart the miner.
- `pop` was checked against the source and refuted as a fix: it only zeroes topo records and cannot reach below the prune point 7,280,004.

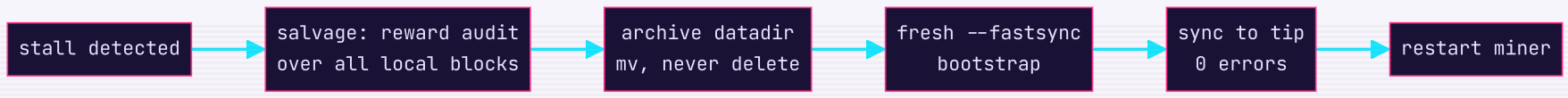
WHY

After a month offline derod replayed 108,686 blocks at 5.4 blocks/s, then rejected the canonical block at topo 7,412,675 with "transaction statement 0 verification failed": 206 identical failures in 3 h 53 min. The first verdict was a corrupt balance-tree leaf. Round 4 revised it: a second, independent datadir failed the same way at 7,467,516, so the cause is a pre-HF3 era of transactions that fail fresh replay verification.

CODE

No code snippet: this round changed no source.

FLOW



EVIDENCE

MEASURED	RESULT	SOURCE
Salvage audit	`132,669 local blocks scanned, 16 paid miniblocks, 0 fetch misses`	RUN_REPORT Round 3 S1
Fresh fastsync bootstrap	1 h 43 min	RUN_REPORT Round 3 S2
Sync to tip	`topo = height = 7,453,737, 0 verification errors, 0 panics`	RUN_REPORT Round 3 S3
Miner after recovery	TOTAL 7,950–8,213 H/s, GPU ~1,250 H/s	RUN_REPORT Round 3 S4
pop as a fix	REFUTED · refuted (cannot pass prune point 7,280,004)	RUN_REPORT Round 3 root cause
"Corrupt local leaf" root cause	REFUTED · revised in R4	RUN_REPORT Round 4

R4 · 2026-08-19 → 08-21 · 3.6.0-152.DEROHE.STARGATE+14082026

HARD-FORK MIGRATION TO RELEASE152, TWO BOOTSTRAP PANICS FIXED, HF3 CROSSED (4, 4B, 4C, 4D)

fill_common panics `47 → 0` HF3 `crossed, 0 panics`

WHAT CHANGED

- **4.** Built from tag Release152 (Release151 plus an HF3 verification hotfix), new git root on branch local-enhancements. All Round 1–3 enhancements ported with 0 patch fuzz (commit d43e08e). Go 1.20.3 to 1.26.6 because the vendored kcp-go needs math/rand/v2.
- **4b.** `fill_common` no longer panics when the top block is unreadable mid-bootstrap; it reports a genesis-empty view (da2fd6a). `LocatePruneTopo` in the handshake is wrapped in a recover at the p2p call site (3c5ffbf).
- **4c.** Console errors at tip diagnosed as benign by census. No code change, no restart, 13 h before the fork.
- **4d.** HF3 crossed at block 7,504,640 with header major_version 3.

WHY

> HF3 at 7,504,640 required an upgrade. During the operator's bootstrap 47 "Recovered while handling connection" panics killed peer connections (topo and state records land before block data), and fixing that exposed 10 "invalid prune detected" panics in the handshake.

CODE

Two snippets: the 4b fixes as committed.

p2p/common.go · lines 16-44 · commit da2fd6a

```
    version, err := chain.ReadBlockSnapshotVersion(chain.Get_Top_ID())
    if err ≠ nil {
        fill_common_empty(common)
        return
    }

    if common.StateHash, err = chain.Load_Merkle_Hash(version); err ≠ nil {
        fill_common_empty(common)
        return
    }

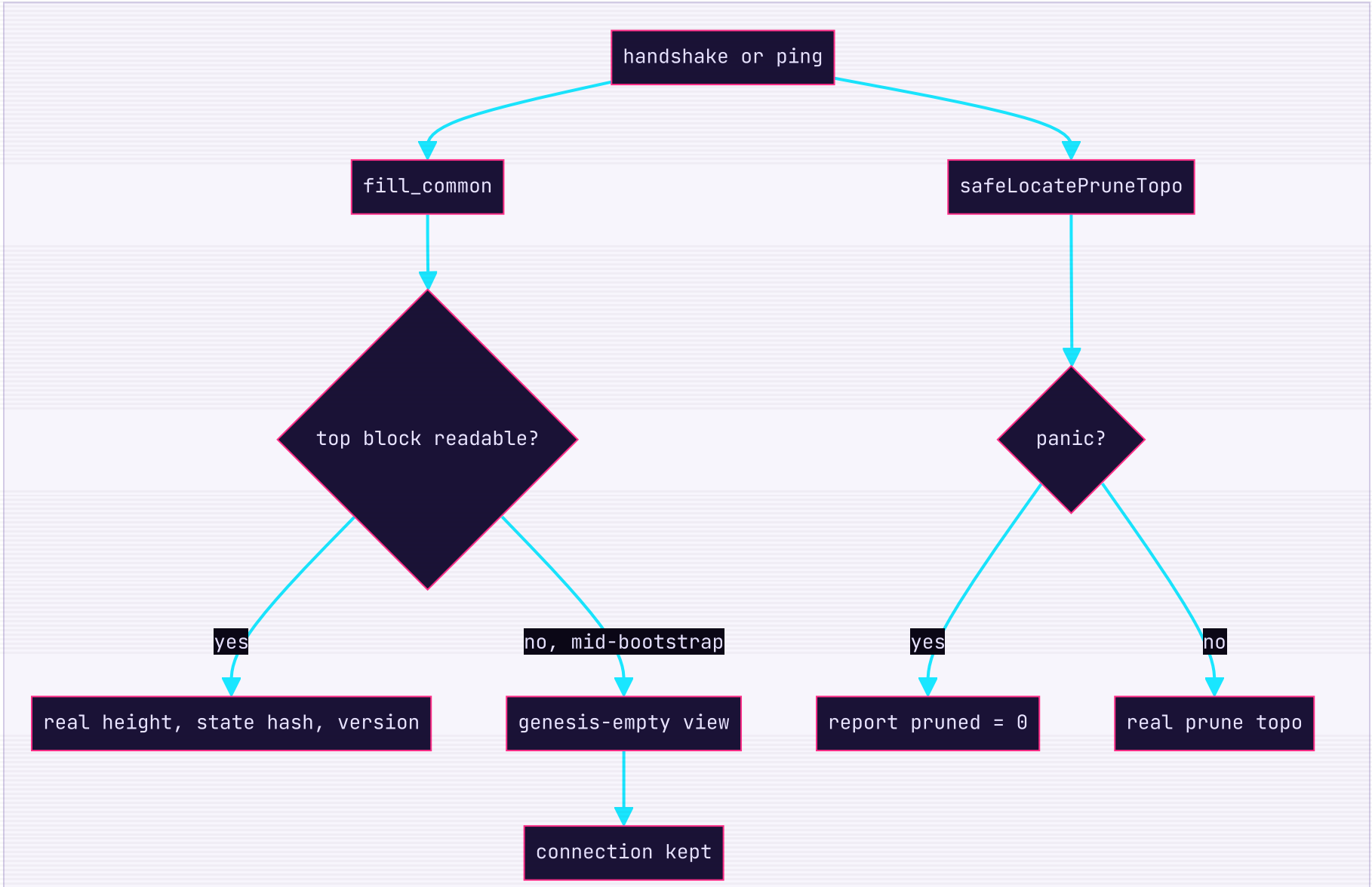
    common.Top_Version = uint64(chain.Get_Current_Version_at_Height(int64(common.Height))) // this must be taken fro
    common.T0 = globals.TimeSkipP2P().UTC().UnixMicro()
}

// During a fastsync bootstrap, topo/state records are committed before the corresponding
// block data reaches bltx_store, so reading the top block can fail transiently. Panicking
// here (the old behavior) killed every connection whose handshake/ping fired in that window,
// leaving the node with only zombie peers when the bootstrap finished. Report a genesis-empty
// view instead — the same story a brand-new node tells — and let the next exchange after
// bootstrap completion carry the real values.
func fill_common_empty(common *Common_Struct) {
    common.Height = 0
    common.TopoHeight = 0
    common.StateHash = crypto.Hash{}
    common.Top_Version = 0
    common.T0 = globals.TimeSkipP2P().UTC().UnixMicro()
}
```

p2p/common.go · lines 37-49 · commit 3c5ffbf

```
// Mid-bootstrap the topo store is written non-contiguously and LocatePruneTopo can
// panic("invalid prune detected") (storetopo.go:178). A handshake must never lose the
// connection over that transient — report unpruned (0) and let later exchanges carry
// the real value. Guarded here at the p2p call site so blockchain semantics are untouched.
func safeLocatePruneTopo() (pruned int64) {
    defer func() {
        if r := recover(); r ≠ nil {
            pruned = 0
        }
    }()
    return chain.LocatePruneTopo()
}
```

FLOW



EVIDENCE

MEASURED	RESULT	SOURCE
Port: patch fuzz	`0`	RUN_REPORT Round 4 gate ledger
Offline 20-thread CPU bench after port	~6,800 H/s (parity)	RUN_REPORT Round 4 gate ledger
Replay wall on a second datadir	`topo 7,467,516, same signature`	RUN_REPORT Round 4 finding
fill_common panics after da2fd6a	`47 before, 0 through a full bootstrap`	RUN_REPORT Round 4b

MEASURED	RESULT	SOURCE
Node at tip after fresh bootstrap	`7,499,969, 34 peers, 0 verification errors`	RUN_REPORT Round 4b
Error census at tip (4c)	`88 peer churn, 12 remote disk errors, 10 closed pipe, 0 verification failures`	RUN_REPORT Round 4c
HF3 crossing (4d)	`major_version 3, 0 verification errors, 0 panics`	RUN_REPORT Round 4d
Local grandfathering of failing txs	`REFUTED · refuted as unsafe (needs per-tx parity forensics)`	RUN_REPORT Round 4b

R5 · 2026-08-21 · 3.6.0-152 + local commits

CHAIN SPLIT FOUND; V2-REMNANT PEER FILTER; UPSTREAM GRANDFATHER TABLE

peers serving v3 `29 of 33`

WHAT CHANGED

- 4712b7b adopts the upstream HF3 grandfather table: 75 entries, covering all three replay walls this node hit. Entries affect only replay of pre-HF3 heights. Ping log demoted from Error to Info.
- 3195f3b: `Connection.update()` drops a peer whose height is at least MAJOR_HF3_HEIGHT + STABLE_LIMIT while it reports Top_Version below 3.
- 0de1854: such peers are also banned for 1 h, which stops redial churn.
- Restart procedure hardened: match the exact daemon pid, wait for process exit and port release, verify a single instance.

WHY

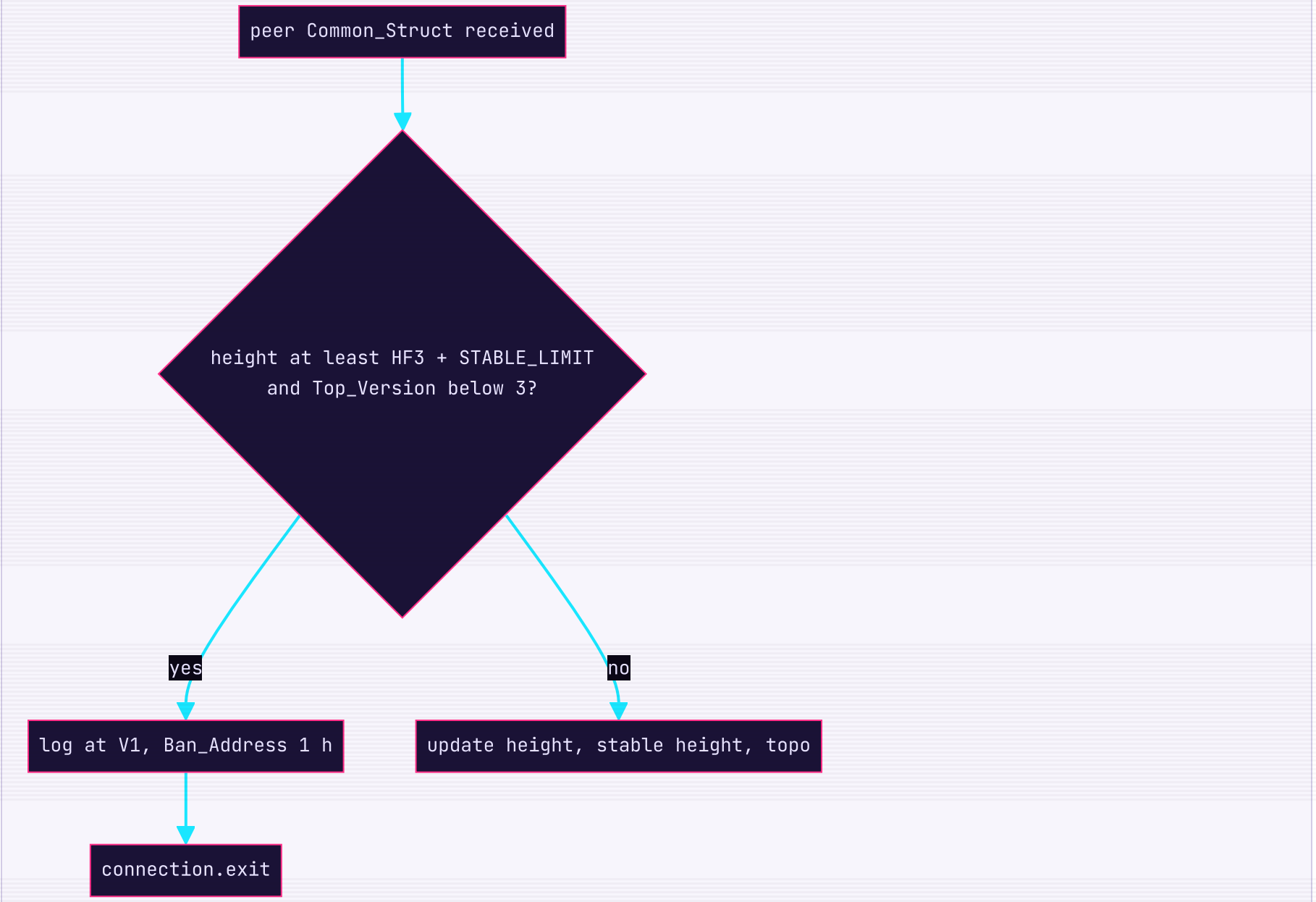
> After the operator popped ~2,900 blocks to below the fork, four peers offered chains sharing ancestry only to 7,504,602. A non-upgraded v2 chain was still running. Its peers caused "deviated too much" churn, and a v2 peer chosen with --sync-node would rewind ~1,000 blocks onto the wrong branch.

CODE

p2p/common.go · lines 83–95 · commit 0de1854

```
// HF3 chain-split guard: a peer whose chain is past the HF3 activation height but
// still reports Top_Version < 3 is following the non-upgraded v2 remnant chain.
// Its blocks can never be accepted here ("invalid fork version") and its sync
// offers only produce "deviated too much" churn — and would trigger a deep
// Rewind_Chain if --sync-node were ever set. Drop it. The STABLE_LIMIT margin
// avoids flapping on peers that are legitimately mid-sync near the boundary.
if common.Height ≥ globals.Config.MAJOR_HF3_HEIGHT+config.STABLE_LIMIT && common.Top_Version < 3 {
    connection.logger.V(1).Info("dropping peer on pre-HF3 (v2) chain", "peer_height", common.Height, "peer_top_v
    Ban_Address(Address(connection), 3600) // stop the redial churn; v2 peers that upgrade are welcome back in a
    connection.exit()
    return
}
```

FLOW



EVIDENCE

MEASURED	RESULT	SOURCE
Resync after the pop	`7,502,573 → tip, 0 errors`	RUN_REPORT Round 5
Chain fingerprint at 7,504,640	v3 e454090e… (ours) vs v2 54219c50…	RUN_REPORT Round 5
Difficulty, v3 vs v2 remnant	4.1–4.5 M vs ~2.1 M	RUN_REPORT Round 5
Peers serving v3	29 of 33	RUN_REPORT Round 5
v2 drop events, first 15 min before the 1 h ban	`141`	RUN_REPORT Round 5 addendum
Node health when popped	`healthy: ~198 blocks/h, 0 real errors`	RUN_REPORT Round 5 forensics

R6 · 2026-08-24 · 3.6.0-152 + d28c5b7, 9612278

64-HOUR HEALTH AUDIT; TWO LATENT P2P DEFECTS FIXED

blocks `12,453 at 18.52 s/block`

WHAT CHANGED

- d28c5b7: `rpc_changeset.go` was the only `Load_BL_FROM_ID` call site of 23 that discarded the error. On a pruned node a changeset request below the prune point returned a nil block, and `bl.Serialize()` panicked and dropped a good peer. Now the error is logged at V(1) and returned.
- 9612278: `peers.json` and `ban_list.json` were written only on clean shutdown. A 5-minute ticker in `maintain_connection_to_peers()` now snapshots both.
- Found, not code: the p2p listener is KCP over UDP 11011. There is no TCP listener on 11011, so only a UDP forward can bring inbound peers.

WHY

> Asked to check node health. The audit found the defects by census, not by symptom: one recovered panic in 64 hours, and peer files last written three minutes before the run began.

CODE

p2p/rpc_changeset.go · lines 63–73 · commit d28c5b7

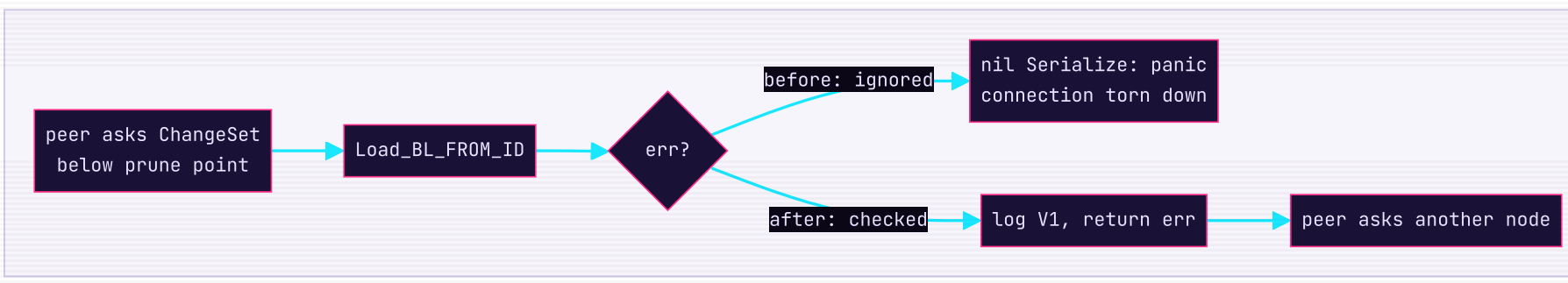
```
        // Load_BL_FROM_ID returns (nil, err) when the block body is not on disk — routine on a
        // pruned node (a peer may ask for a changeset below our prune point) and possible in a
        // race right after a reorg. The upstream code discarded this error and called
        // bl.Serialize() on the nil block, panicking inside handle_connection_panic and killing
        // an otherwise good peer connection. Report the error instead; the peer just asks elsewhere.
        bl, err := chain.Load_BL_FROM_ID(blid)
        if err ≠ nil {
            c.logger.V(1).Info("cannot serve changeset, block body unavailable", "topo", topo, "blid", blid, "err",
            return err
        }
        cbl.Block = bl.Serialize()
```

p2p/controller.go · lines 490–505 · commit 9612278

```
    // Upstream writes peers.json / ban_list.json only from P2P_Shutdown(), so any exit that is
    // not a clean SIGINT — SIGKILL, OOM, power loss, a duplicate instance losing its port binds —
    // discards every peer the node learned since startup and it cold-starts from the seed list.
    // Snapshot both lists periodically so a hard kill costs at most one interval of peer knowledge.
    // Two small JSON writes; the mutexes taken here are the same ones shutdown already uses.
    persist := time.NewTicker(5 * time.Minute)

    for {
        select {
        case ←Exit_Event:
            return
        case ←persist.C:
            save_peer_list()
            save_ban_list()
            continue
        case ←delay.C:
```

FLOW



EVIDENCE

MEASURED	RESULT	SOURCE
Blocks extended over 230,657 s	`12,453 at 18.52 s/block`	RUN_REPORT Round 6
Blocks per hour	`186–199, no hour below 186`	RUN_REPORT Round 6
Reorgs / resyncs	`0 / 3`	RUN_REPORT Round 6
ERROR lines	316 (4.9/h): 315 remote or churn, 1 real defect	RUN_REPORT Round 6 census
Persistence ticker, isolated testnet	`both files written at T+311 s while running`	RUN_REPORT Round 6 finding 2
Peer pool	`40 addresses, 17 v2-remnant, 24 outgoing held`	RUN_REPORT Round 6 finding 4
Tests	`go vet clean; p2p, blockchain, block tests ok`	RUN_REPORT Round 6 gates

The node was not restarted: it was extending at 18.5 s/block, the state the Round 5 rule says not to disturb. Both fixes took effect on the next operator restart.

R6b · 2026-08-25 · 3.6.0-153.DEROHE.STARGATE+21082026

WALLET CLI BUILT; TREE CONFIRMED EQUAL TO RELEASE153

tree vs Release153 `diff empty`

WHAT CHANGED

- `bin/dero-wallet-cli` built for the first time (13.3 MB). Earlier builds targeted only derod and dero-miner.
- Release142 → Release151/152 wallet changes catalogued: balance decode from the last known value, two sender_idx off-by-one fixes, SyncHistory rollback rewrite, TokenAdd nil-map panic, read locks on read paths, Connected=false after a failed connect, a TransferPayload0 early return.
- Release153 differs from Release152 in three files (grandfather table, ping log level, version). The first two were already adopted in R5, so bf274e5 only bumps the version string.

WHY

> There was no wallet binary in bin/, and the version string lagged the code.

CODE

config/version.go · lines 21–23 · commit bf274e5

```
// right now it has to be manually changed
// do we need to include git commitsha??
var Version = semver.MustParse("3.6.0-153.DEROHE.STARGATE+21082026")
```

EVIDENCE

MEASURED	RESULT	SOURCE
Release153 vs tree, grandfather table	`diff empty, 75 entries on both`	RUN_REPORT Round 6b
Changes under walletapi/ and cmd/dero-wallet-cli/	`0 (diff vs Release153 empty)`	RUN_REPORT Round 6b
Upstream walletapi tests	`red on pristine Release153 too (throwaway worktree)`	RUN_REPORT Round 6b
Miner flags after rebuild	`--cpu-max, --gpu, --gpu-max intact`	RUN_REPORT Round 6b

R7 · 2026-10-20 · harness only

EVIDENCE HARNESS; THE NODE THAT ONLY LOOKED HEALTHY

gates `12/12` node lag found `40.4 days`

WHAT CHANGED

- `harness/`: corpus preparation, 7 miners, page generator, index, gate runner, KAT suite, hooks. Every row on a generated page needs a resolved source anchor or it is not drawn.
- Gate G9 compares the page with the live daemon and reads the tip block's own age. KAT DK20 covers it.
- The `Recover while executing SC` path was documented as a deterministic upstream defect that must not be patched locally (a change would alter fees and fork the node).

WHY

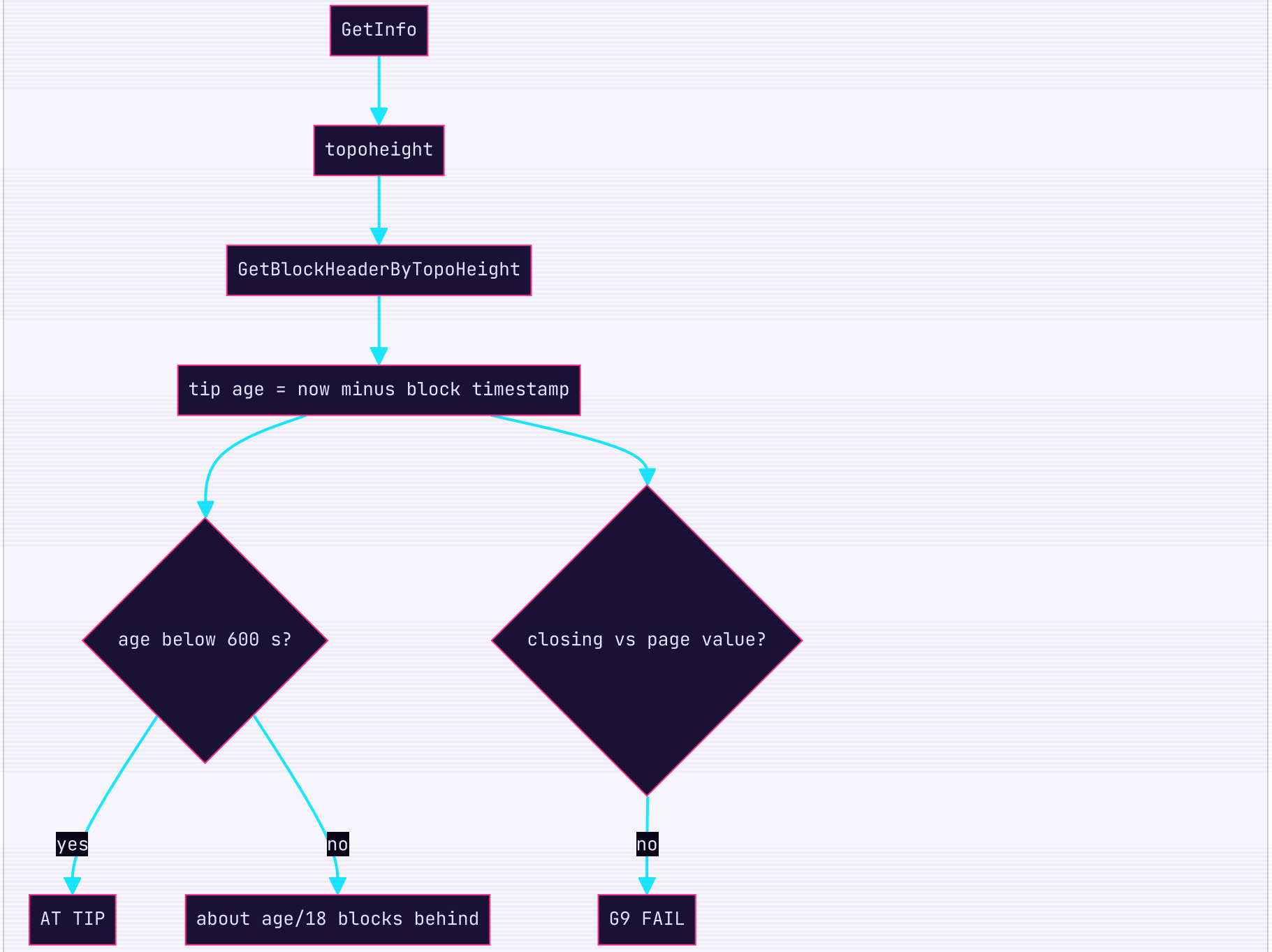
> The harness computed an impossible cadence: 7,345 blocks in 1,667 s. Following it showed the node was 40.4 days behind. GetInfo looked healthy: averageblocktime50 is derived from block timestamps and target_height was commented out.

CODE

harness/run_gates.py · lines 183–198 · commit 36cfd62

```
        # The decisive reading: the tip block's own age. GetInfo alone cannot distinguish a node at the tip from
        # one replaying history (finding find-attip), so the gate asks the chain, not the summary.
        hreq = urllib.request.Request('http://127.0.0.1:10102/json_rpc',
                                      data=json.dumps({'jsonrpc': '2.0', 'id': '1',
                                                       'method': 'DERO.GetBlockHeaderByTopoHeight',
                                                       'params': {'topoheight': gi['topoheight']}}).encode(),
                                      headers={'Content-Type': 'application/json'})
        bh = json.loads(urllib.request.urlopen(hreq, timeout=8).read())['result']['block_header']
        import time as _t
        age = _t.time() - (bh['timestamp'] / 1000)
        was = float(shown.get('tip_age') or age)
        closing = age ⩽ was + 60          # catching up, or at worst holding station
        at_tip = age < 600
        ok9 = same_ver and grew and gi['status'] == 'OK' and closing
        d9 = (f"live height {gi['height']} (page {shown.get('height')}, "
              f"{gi['height'] - int(shown.get('height', 0))} newer) · version "
```

FLOW



EVIDENCE

MEASURED	RESULT	SOURCE
Gates G1–G12	12/12 PASS	RUN_REPORT Round 7.1
KATs DK01–DK20	20/20	RUN_REPORT Round 7.1
Mutation proofs	13/13 (each flips its own KAT)	RUN_REPORT Round 7.1
Anchors	372/372 resolved, 278 rows, 0 refused	RUN_REPORT Round 7.1
Node lag found	40.4 days, ≈194,000 blocks, ingest 3.5–3.6 blocks/s	RUN_REPORT Round 7.2
"Fee lost" on SC revert	REFUTED · corrected in R11: the fee is kept	RUN_REPORT Round 11

R8 · 2026-10-20 · harness only, no consensus change

AUDIT OF THE FOUNDATION'S DISCLOSED RANGE-PROOF SOUNDNESS DEFECT

C1 CONFIRMED

WHAT CHANGED

- Audited the Foundation's disclosed range-proof soundness defect against this tree. Claims were confirmed or refuted one by one; no consensus change was made.
- Harness: `harness/external/` added as an EXT root, a press-claim miner and an anchor-re-resolving forensics ingest. KAT DK21: a quoted claim is classed EXT and attributed, never asserted. KAT DK22: no verdict is proved only by the audited document.
- Operator note: `total_supply` is computed from the emission formula, so it prints the same number whatever the balances hold.

WHY

> The Foundation published the defect and withdrew support for the chain. The request was independent forensics folded into the harness.

CODE

Harness code only. The audit details stay in ARCHITECTURE §25.

harness/kat/verify_deroarch.py · lines 355–369 · commit dbe37c3

```
def dk21(d, V, H, st):
    """A quoted claim is never an established fact. Every row anchored in the external document must carry the
    EXT class and must attribute what it says, and the page must never present one as the tree's own finding."""
    bad = []
    for r in V:
        ext = [e for e in r['evidence'] if '/harness/external/' in e['abs']]
        if not ext:
            continue
        if r['class'] ≠ 'EXT':
            bad.append(f"{r['id']}: anchored in the external document but classed {r['class']}")
        c = str(r.get('claim') or '')
        if not re.search(r'(states|said|asserts|claims|according to|Foundation)', c):
            bad.append(f"{r['id']}: quotes the document without attributing it")
    return not bad, (f'{sum(1 for r in V if any("/harness/external/" in e["abs"] for e in r["evidence"]))} rows cite
                     f'the external document; all EXT and attributed' if not bad else f'{len(bad)}: {bad[:3]}')
```

FLOW



EVIDENCE

MEASURED	RESULT	SOURCE
Workflow size	199 agents: 6 investigations × 3 lenses + synthesis	RUN_REPORT Round 8
Findings surviving review / re-resolution	64 → 39 → 17	RUN_REPORT Round 8
C1: a soundness defect exists	CONFIRMED	RUN_REPORT Round 8
C2: mechanism as stated	REFUTED · REFUTED (consequence CONFIRMED)	RUN_REPORT Round 8
C3–C7 / C8	PARTLY-CONFIRMED / CANNOT-DETERMINE-LOCALLY	RUN_REPORT Round 8
Gates / KATs / mutations	12/12, 22/22, 14/14	RUN_REPORT Round 8

R9 · R9b · R10 · 2026-10-20 → 10-21 · harness only

COHERENCE PASS, TIP REACHED, EVERY CONSOLE ERROR TRIAGED

tip reached topo 7,789,273

WHAT CHANGED

- **R9** (54d9476): section-by-section coherence pass and the missing diagrams.
- **R9b**: the node reached the tip; three harness defects fixed (a one-way DK20 mutation that became a no-op at tip, the press file classed as an experiment, a gate label counted twice).
- **R10**: `mine_triage.py` triaged the operator's 20-hour console capture. The zero-signer keyhash is checked by computing blake2s-256 of 33 zero bytes, so no trust in the log text is needed.

WHY

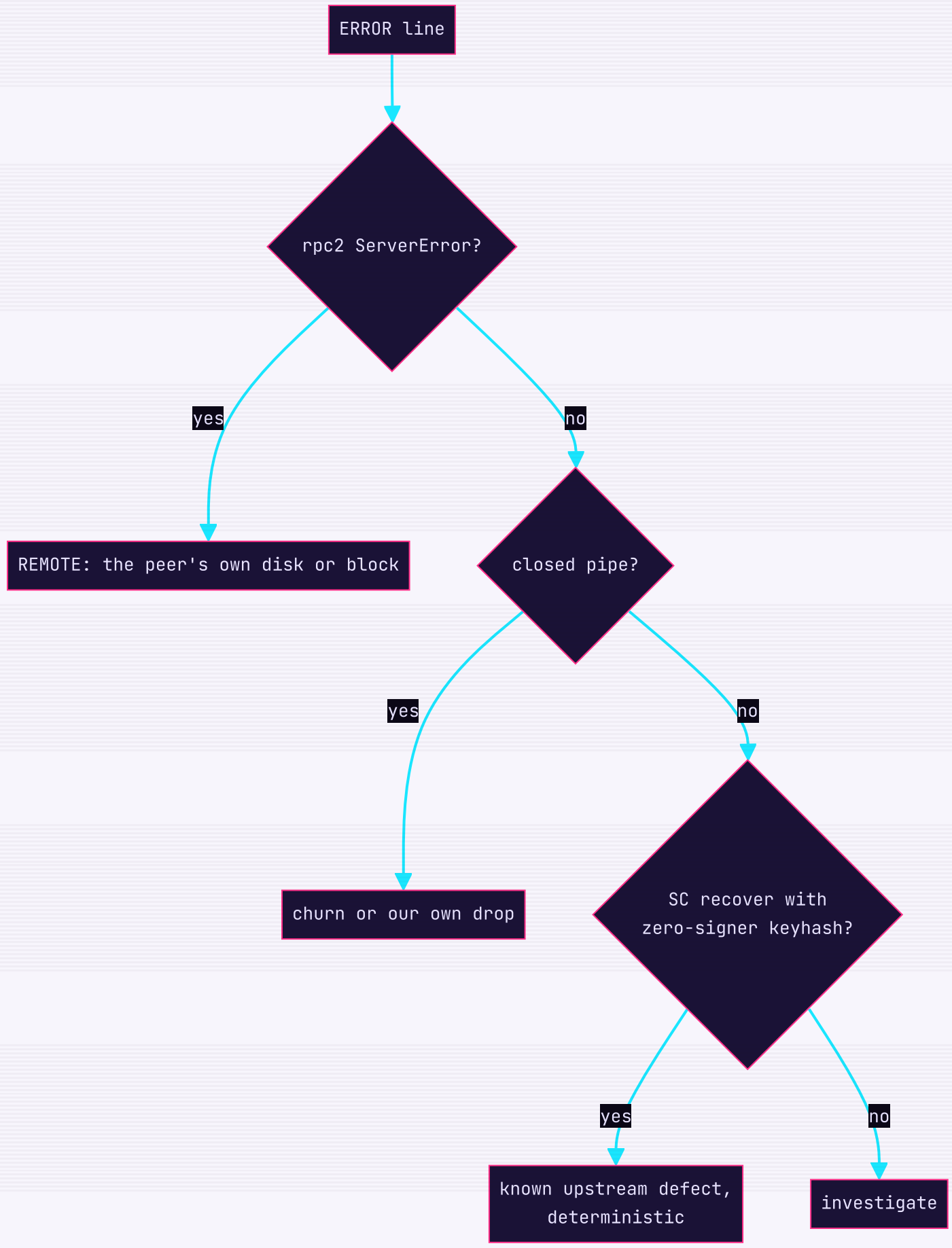
> 15 ERROR lines on the console needed an origin rule: which are this node's faults and which are a peer's, echoed back over rpc2.

CODE

harness/miners/mine_triage.py · lines 130–136 · commit 7488df4

```
    # ---- 1. anonymous SC tx → ErrorRevert on the zero signer → recovered panic
    sc = of(lambda e: e['msg'].startswith('Recover while executing SC'))
    if sc:
        keys = sorted({k for e in sc for k in re.findall(r'keyhash ([0-9a-f]{64}) not found', e['kv'])})
        zero = hashlib.blake2s(bytes(33), digest_size=32).hexdigest()
        txs = [re.search(r'"txid": "([0-9a-f]{64})"', e['kv']).group(1) for e in sc]
        nosig = [t for t in txs if any(f.startswith(f'tx {t} ') and 'signer=NONE' in f for f in facts())]
```

FLOW



EVIDENCE

MEASURED	RESULT	SOURCE
Tip reached (R9b)	`04:13:35, topo 7,789,273`	RUN_REPORT Round 9b
Catch-up over 11.98 h	182,821 blocks at 4.24 blocks/s	RUN_REPORT Round 9b
ETA formula (tip_age/18)/(rate − 1/18)	`within 4 min`	RUN_REPORT Round 9b
Console ERROR lines triaged (R10)	`15 lines, 6 causes, 0 need action`	RUN_REPORT Round 10
Adversarial verifiers on the 6 rows	`all PARTLY: causes upheld, wording corrected`	RUN_REPORT Round 10
Gates / KATs / mutations	`12/12, 25/25, 17/17`	RUN_REPORT Round 10

R11 · 2026-10-21 · 3.6.0-153.DEROHE.STARGATE+21102026

CONSOLE ERRORS FIXED IN CODE; CONSENSUS GUARD AND SHADOW VERIFICATION

shadow `240/240 treehash` gates `13/13`

WHAT CHANGED

- The zero-signer SC revert logs one V(1) line instead of an ERROR with a 3 KB stack. The SC parse failure is a V(1) INFO.
- `p2p/object_errors.go` classifies GetObject failures as remote, closed or other; remote and closed log at V(1).
- P5: a tx-only fetch the peer fails no longer drops the connection. The peer is dropped on the 8th consecutive failure; only a delivered tx resets the count.
- Harness: `consensus_guard.py` (G13, DK26), a baseline copy of bf274e5 (DK27), and `pipeline/fix_build_verify.py`.

WHY

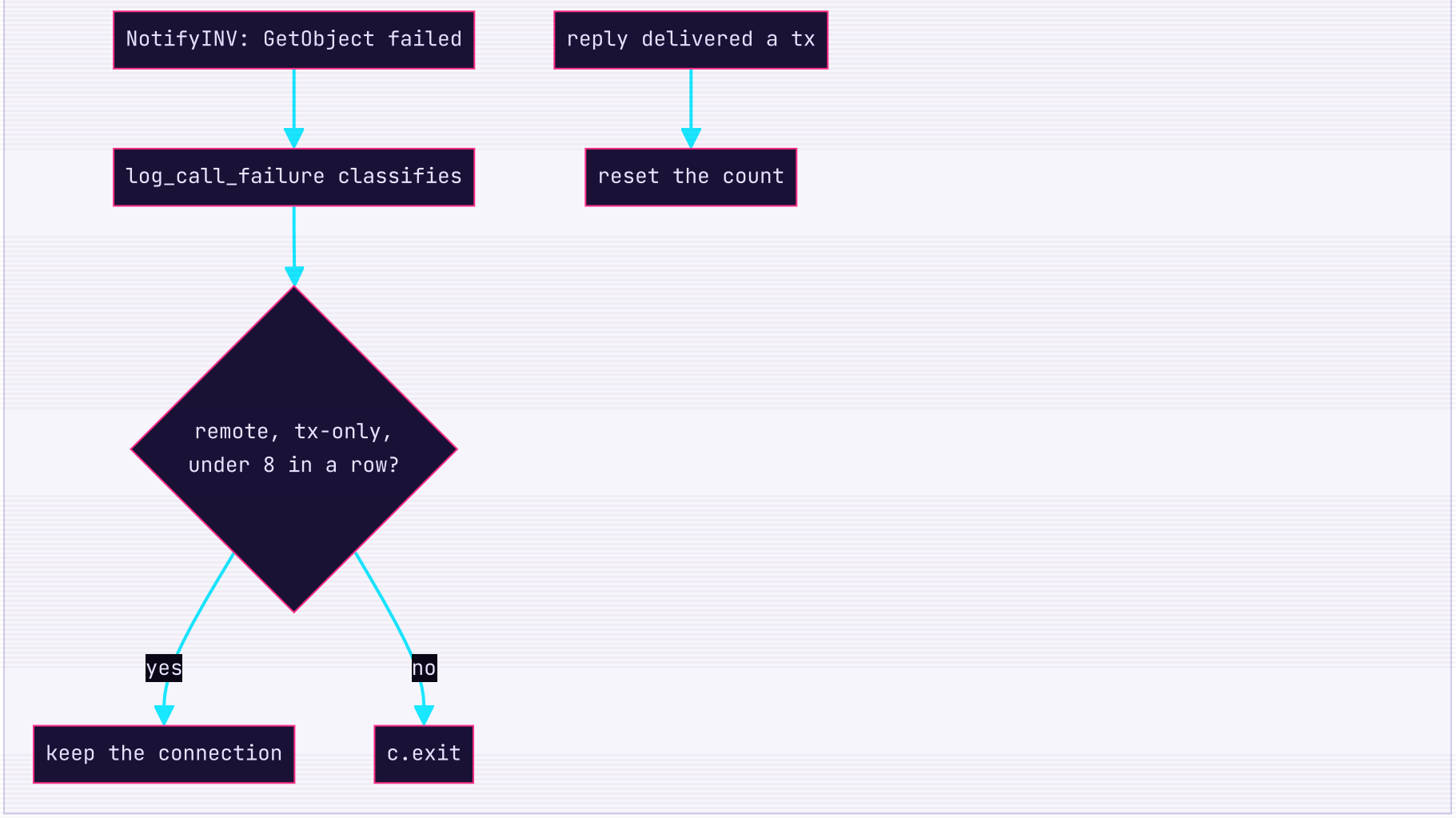
> NotifyINV called c.exit() on any failure, so one peer that lost one mempool entry cost the whole connection and every call pending on it.

CODE

p2p/rpc_notifications.go · lines 94–110 · commit 4e489d1

```
        if err = c.Client.Call("Peer.GetObject", need, &oresponse); err ≠ nil {
            kind := log_call_failure(c.logger, err, "Call failed GetObject", "need_objects", need)
            // a peer that announced a transaction and then could not serve it (typically mined or
            // evicted in between) keeps the connection; blocks, chunks and other failures do not
            if kind == call_failure_remote && is_tx_only(need) && c.keep_after_tx_fetch_failure() {
                c.update(&request.Common)
                fill_common(&response.Common)
                return nil
            }
            c.exit()
            return
        } else { // process the response
            c.note_tx_fetch_reply(need, len(oresponse.Txs))
            if err = c.process_object_response(oresponse, request.Sent, false); err ≠ nil {
                return
            }
        }
```

FLOW



EVIDENCE

MEASURED	RESULT	SOURCE
Consensus guard vs bf274e5	PASS	RUN_REPORT Round 11
New unit tests under -race	10	RUN_REPORT Round 11
Adversarial code reviews	2 reviews, 3 real issues found and fixed	RUN_REPORT Round 11
Shadow vs live node	240/240 treehash, 120/120 block hash, 0 regressions	RUN_REPORT Round 11
Gates / KATs / mutations	13/13, 27/27, 19/19	commit 4e489d1 message
Pipeline iterations before PASS	5, every failure in the verification machinery	RUN_REPORT Round 11
After operator restart	0 ERROR lines in the first 1.6 h	RUN_REPORT Round 11

R12 · 2026-10-22 · 3.6.0-153.DEROHE.STARGATE+22102026

INTELLIGENCE HARNESS; 11 NON-CONSENSUS FIXES

mutations 15/15

WHAT CHANGED

- New RPC `DERO.GetSyncInfo` classifies the node as synced, syncing, stalled or isolated from the tip age and the peers' median height. GetInfo `target_height` is filled.
- Fixed: a registration-tx GetTransaction panic, stack traces in RPC errors, atomic peers.json and ban_list.json writes, handshake, TreeSection, accept-loop and avg-offset fixes, the logr DPANIC call; 5 new counters.
- Harness: `log_intel`, `datadir_intel`, `go_mutate`; gates G14 (datadir) and G15 (log health); KATs DK28 and DK29; pipeline stages S3b mutations, S3c race, S5 feature probes.

WHY

R7 showed GetInfo cannot tell whether the node is at the tip. The node needed to answer that itself.

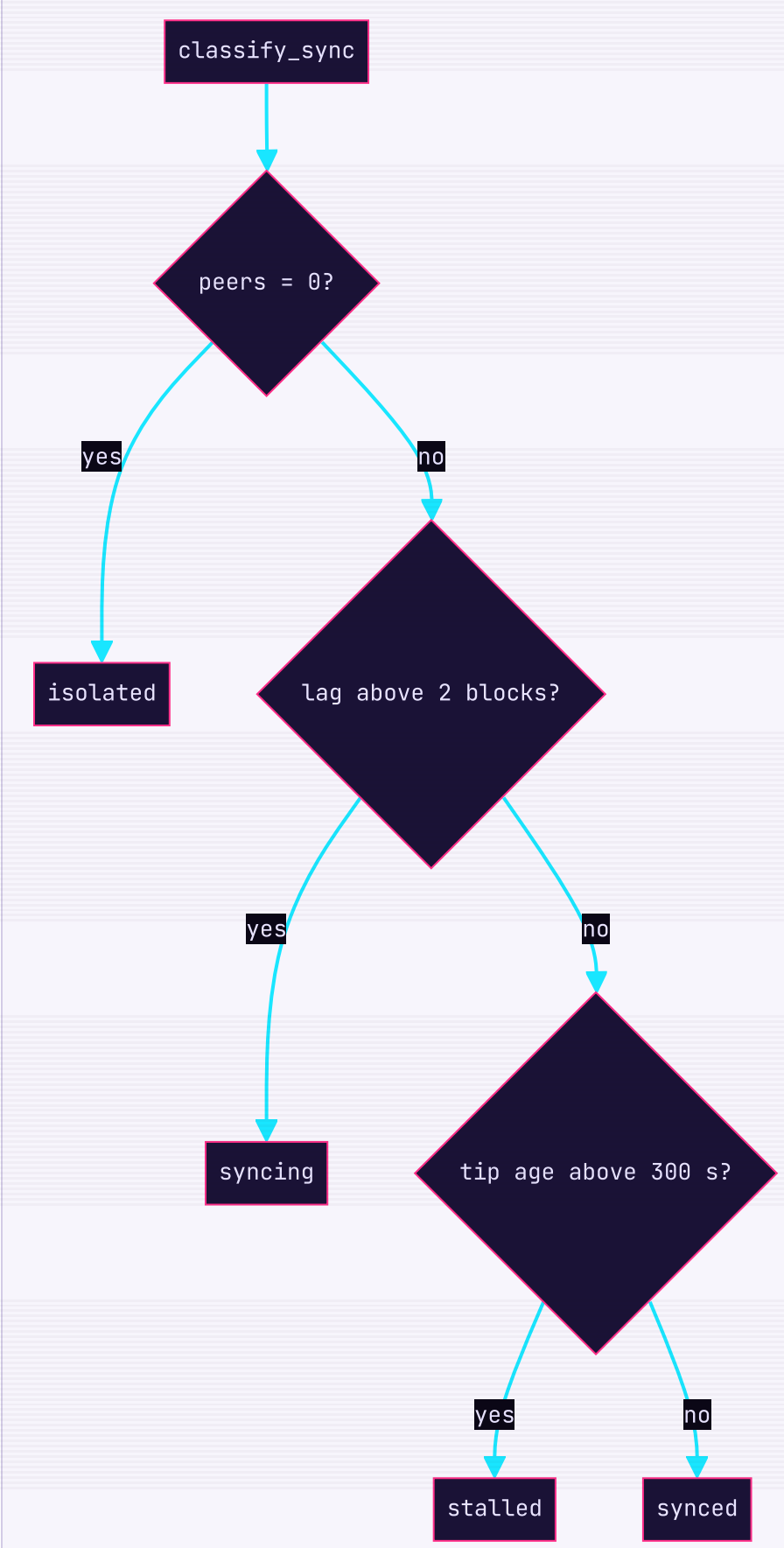
CODE

cmd/derod/rpc/rpc_dero_getsyncinfo.go · lines 64–79 · commit 86958c4

```
const SYNC_LAG_BLOCKS = 2
const STALL_TIP_AGE_S = 300

// classify_sync is a pure function so the thresholds are unit-tested (rpc_dero_getsyncinfo_test.go).
func classify_sync(peers uint64, lag int64, tip_age_s float64) (status, reason string) {
    switch {
    case peers == 0:
        return "isolated", "no handshaken peers; height cannot be compared with the network"
    case lag > SYNC_LAG_BLOCKS:
        return "syncing", fmt.Sprintf("%d block(s) behind the peers' median height", lag)
    case tip_age_s > STALL_TIP_AGE_S:
        return "stalled", fmt.Sprintf("tip block is %.0f s old and no peer is ahead", tip_age_s)
    }
    return "synced", fmt.Sprintf("within %d block(s) of the peers' median and the tip is %.0f s old", SYNC_LAG_BLOCK
}
```

FLOW



EVIDENCE

MEASURED	RESULT	SOURCE
Discovery	8 investigations, 124 findings: 99 confirmed, 25 partly, 0 refuted	RUN_REPORT Round 12
Go mutations / races	`15/15, 0 races`	RUN_REPORT Round 12
Shadow	`240/240 treehash + 120/120 block hash`	RUN_REPORT Round 12
Feature probes	`candidate shows every fix, control every defect`	RUN_REPORT Round 12
Mean block interval at tip	18.84 s over 303 blocks	RUN_REPORT Round 12
Stale statements corrected in docs	23	RUN_REPORT Round 12

R13 · 2026-10-22 · 3.6.0-153.DEROHE.STARGATE+22102026.r13

P2P HARDENING, CONCURRENCY FIXES, MINING OUTCOME RPC, COVERAGE RATCHET

mutations `30/30` gates `16/16`

WHAT CHANGED

- Frame reader bounded by received bytes; per-method outgoing deadlines; caps on heavy handlers.
- Misbehaviour scoring per IP with a 10-minute half-life and escalating temporary bans (`p2p/penalty.go`). Seed, priority and exclusive nodes are never banned by score.
- `update()` race fixes; `exit()` leaves the pool at once.
- `DERO.GetMiningOutcome`, a stale-work signal, inbound reachability. Coverage miner, risk register, G16 ratchet, DK30.

WHY

> Upstream logs "should be banned" at a dozen sites and at most disconnects, so a misbehaving peer could reconnect at once.

CODE

p2p/penalty.go · lines 37–50 · commit a027e29

```
const PENALTY_BAN_SCORE = 100.0            // a score reaching this bans the IP
const PENALTY_HALF_LIFE = 10 * time.Minute // scores halve every 10 minutes: occasional faults never add up to a ban
const PENALTY_BAN_BASE = 600               // first ban 10 minutes; each further ban doubles
const PENALTY_BAN_MAX = 86400              // up to 24 hours

// points per kind of fault; the reasons are a fixed set so the metric stays bounded
var PENALTY_POINTS = map[string]float64{
    "invalid_pow":           50, // a block whose proof of work fails: two in a short time ban
    "invalid_pow_miniblock": 25, // a miniblock judged against our view of its tips: weaker evidence while we catch
    "undecodable":           50, // a block or tx that does not deserialize
    "malformed_request":     20, // a request outside the protocol's bounds
    "timeout":               10, // a call the peer never answered
    "busy":                  2,  // a heavy request beyond the concurrency cap
}
```

p2p/penalty.go · lines 111–123 · commit a027e29

```
    m.score = m.score*math.Pow(0.5, now.Sub(m.at).Seconds()/PENALTY_HALF_LIFE.Seconds()) + points
    m.at = now
    if m.score < PENALTY_BAN_SCORE || never_ban(ip) {
        return 0
    }
    m.offences++
    m.score = 0
    ban_seconds = uint64(PENALTY_BAN_BASE) << uint(m.offences-1)
    if ban_seconds > PENALTY_BAN_MAX || m.offences > 8 {
        ban_seconds = PENALTY_BAN_MAX
    }
    return ban_seconds
}
```

FLOW



EVIDENCE

MEASURED	RESULT	SOURCE
Pre-build adversarial review	`4 reviewers, ~20 issues, incl. a remote-triggerable crash`	RUN_REPORT Round 13
Go mutations / races	`30/30, 0 races over 6 packages`	RUN_REPORT Round 13
Shadow	`240/240 treehash + 120/120 block hash`	RUN_REPORT Round 13
Probe	`candidate answers GetMiningOutcome; control says "no such method"`	RUN_REPORT Round 13
Gates / KATs / mutation proofs	`16/16, 30/30, 22/22`	RUN_REPORT Round 13
Pipeline runs	`3 (stale mutation spec, round-frozen probe, PASS)`	RUN_REPORT Round 13

R14 · 2026-10-22 · +22102026.r14

DESCRIPTOR SUFFIX SORT IN DERO-MINER

S4b ×3.95 live ~27.5 KH/s

WHAT CHANGED

- The descriptor suffix sort (third-party MIT code; licence in THIRD_PARTY_NOTICES.md) is ported to `cmd/dero-miner/dsa`. It runs behind the nil hook `astrobwtv3.SuffixSort`; `pow.go` falls back to the built-in SA-IS when the hook is nil or declines.
- Consensus guard gained an "assigned" check: a non-test assignment of SuffixSort outside cmd/dero-miner/ is a violation, so derod verifies every PoW with the reference SA-IS.
- Three guards: a 51-input startup self-test, a sampled dsa.Verify (1 in 256), and a watcher that disables the fast path on any new recovered panic.
- Engine: GPU wins re-hashed on the CPU before submit; getwork no longer leaks one fd per reconnect; atomic `job_counter`; `--sort` and `--cpu-qos`.

WHY

> SA-IS was 77 % of a hash. A suffix array is unique, so any correct sorter yields the identical hash.
> The sort exploits stage 1's ~270 near-copy 256-byte runs, which a generic SA-IS cannot.

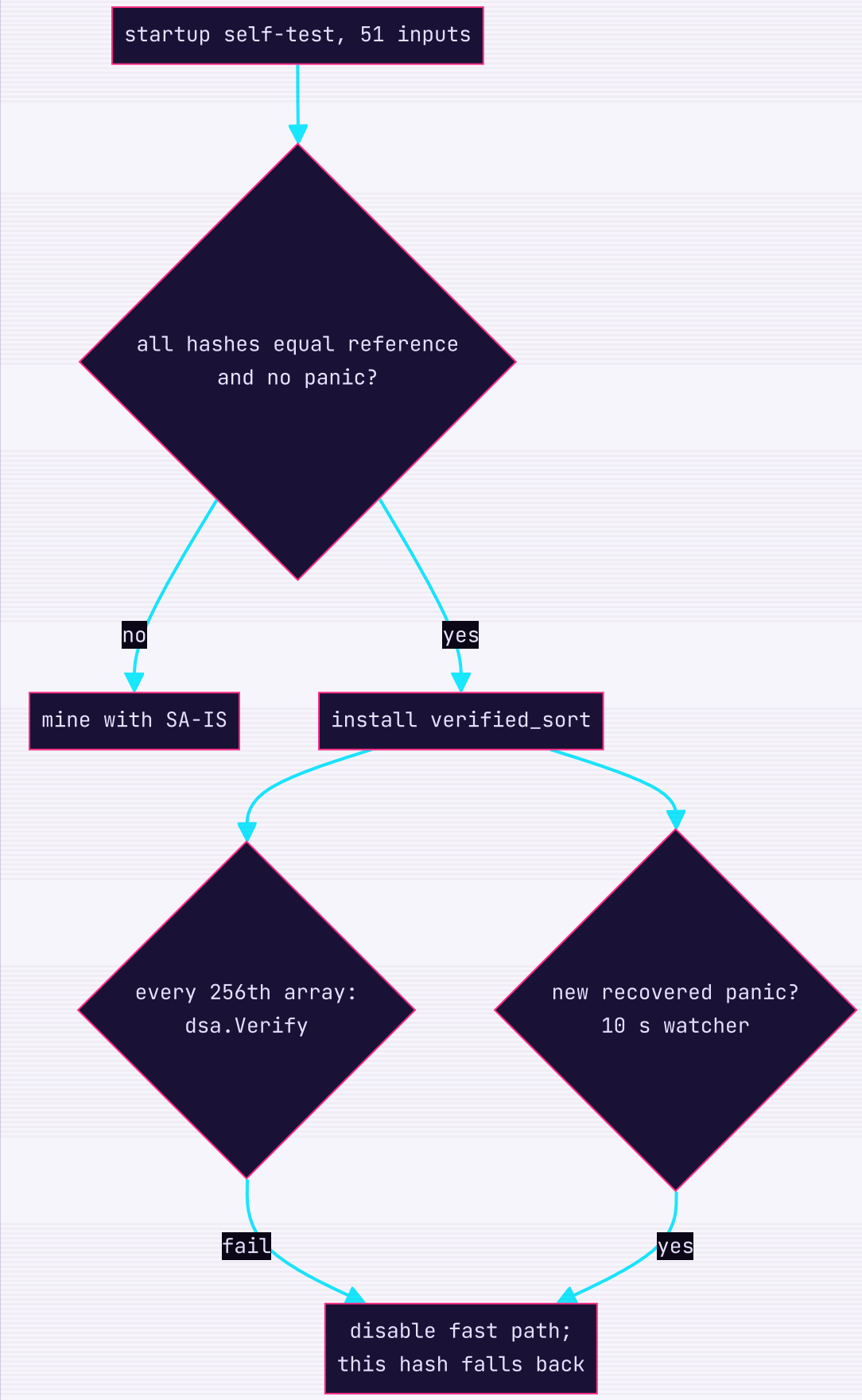
CODE

cmd/dero-miner/fastsort.go · lines 32-55 · commit 52c02df

```
func verified_sort(text []byte, sa []int32) bool {
    if atomic.LoadInt32(&fast_sort_disabled) == 1 {
        return false
    }
    if !fast_sorter(text, sa) {
        atomic.AddUint64(&fast_sort_fallbacks, 1)
        return false
    }
    if atomic.AddUint64(&fast_sort_calls, 1)%FAST_SORT_CHECK_EVERY == 0 {
        atomic.AddUint64(&fast_sort_checked, 1)
        if !dsa.Verify(text, sa) {
            disable_fast_sort("a sampled suffix array failed verification")
            return false // this hash falls back to SA-IS, so it is still correct
        }
    }
    return true
}

func disable_fast_sort(why string) {
    if atomic.CompareAndSwapInt32(&fast_sort_disabled, 0, 1) {
        fast_sort_reason.Store(why)
        logger.Error(fmt.Errorf("%s", why), "descriptor sort DISABLED, falling back to the built-in SA-IS")
    }
}
```

FLOW



EVIDENCE

MEASURED	RESULT	SOURCE
Pipeline pipeline_20261022T154341	ok	RUN_REPORT Round 14
S4b at 2 threads, median	587.9 → 2,324.1 H/s (×3.95)	RUN_REPORT Round 14
Hash equivalence	652/652 identical, 0 declines	ARCHITECTURE §30.5
Go mutations / races	35/35 (5 new), 8 packages 0 races	RUN_REPORT Round 14
Shadow	120 blocks / 240 treehashes, 0 mismatches	RUN_REPORT Round 14
Live r14 run, ~63 min	~27.5 KH/s TOTAL vs ~7.9 before (×3.5)	ARCHITECTURE §31.1
"M1 hardware ceiling" (§13–14)	REFUTED · refuted: it was the ceiling of SA-IS	ARCHITECTURE §30.5
Missed mutant (duplicate bitmap in dsa.Verify)	equivalent mutant; code simplified, rerun 2/2	RUN_REPORT Round 14

R15 · 2026-10-22 · +22102026.r15

HONEST OPERATOR NUMBERS; DSA BOUNDS-CHECK ELIMINATION AND PGO

staged +3.3 % NW ≈ D

WHAT CHANGED

- Miner: `est_earned` counts final miniblocks; counters fold across reconnects; "NW" divides a HighDiff job's 9·D back to D (`nw_from_job`); MINING @ uses a 15 s window.
- derod: hashrate estimates count finals and clip to uptime; GetBlockTemplate returns 9·D for a final template; "Block Minted" no longer subtracts rejected from accepted.
- p2p: the v2-remnant ban escalates 1 h → 24 h; `feed_chunk` skips PoW verification of a chunk whose tip is unknown (`unknown_tip`), which removed a recovered panic.
- Miner speed: dsa loops resliced to drop bounds checks; Verify sampling 1 in 1024; PGO profile for dero-miner only.
- Pipeline: NETWORK-mode shadows when the live node is down.

WHY

> A 12-agent dive over the live r14 run found the operator numbers were wrong: finals were dropped from estimates and the miner showed a 9× network jump on final jobs. Network hashrate derived from consensus code: 9 normal miniblocks at D plus 1 final at 9·D per 18 s block is 18·D hashes per 18 s, so NW ≈ D.

CODE

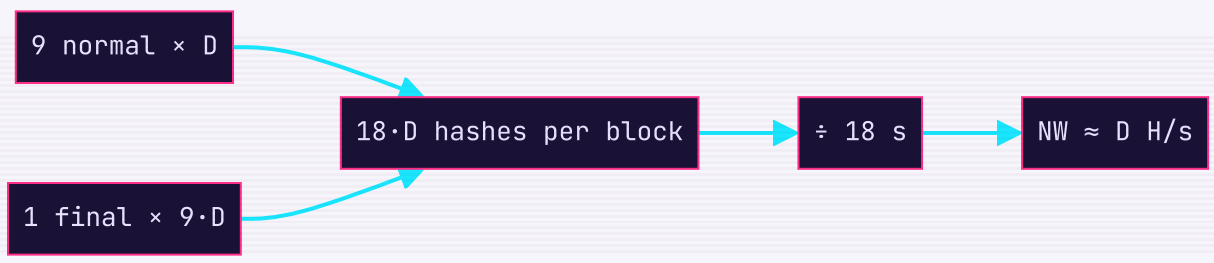
`cmd/dero-miner/stats.go` · lines 30–42 · commit `c84d0aa`

```
// nw_from_job estimates the network hashrate from a job. Per block 9 normal miniblocks at difficulty D and one
// final at 9D (config.MINIBLOCK_HIGHDIFF) are found in BLOCK_TIME = 18 s: 18D hashes per 18 s, i.e. about D H/s.
// A final job carries 9D, which the prompt used to show as a 9x network jump. Byte 0 bit 4 of the hashing blob is
// the HighDiff flag (block/miniblock.go Serialize).
func nw_from_job(blob string, diff uint64) uint64 {
    var b [1]byte
    if len(blob) ⩾ 2 {
        if _, err := hex.Decode(b[:], []byte(blob[:2])); err == nil && b[0]&0x10 ≠ 0 {
            return diff / config.MINIBLOCK_HIGHDIFF
        }
    }
    return diff
}
```

`p2p/chunk_server.go` · lines 402–410 · commit `c84d0aa`

```
// unknown_tip reports whether any tip has no stored height (Load_Block_Height returns -1 for an unknown block)
func unknown_tip(height func(crypto.Hash) int64, tips []crypto.Hash) bool {
    for _, t := range tips {
        if height(t) < 0 {
            return true
        }
    }
    return false
}
```

FLOW



EVIDENCE

MEASURED	RESULT	SOURCE
Live r14 run	`0 ERROR/WARN, 0 panics, 6/6 paid, P(X≤6) = 0.115`	RUN_REPORT Round 15
First pipeline run	`shadow FAIL: live node stopped, shadows lost their only peer`	RUN_REPORT Round 15
Rerun pipeline_20261022T224618	`guard, format, tests PASS; 40/40 mutations; 0 races`	RUN_REPORT Round 15
S4b idle box, 2 threads	`SA-IS 880 → descriptor 3,866 H/s (×4.39)`	RUN_REPORT Round 15
Staged r15 vs installed r14	`+3.3 % (3,851–3,885 vs 3,711–3,762, ranges disjoint)`	RUN_REPORT Round 15
Verify at 1/256	cost 0.52 % of hashrate → moved to 1/1024	ARCHITECTURE §31.5
PGO	median −4.5 % time per hash (noisy)	ARCHITECTURE §31.5
Old ×10/18 share formula	`REFUTED · understated share ~1.8×`	ARCHITECTURE §31.2

`R16 · 2026-10-23 · +22102026.r16`

NATIVE C DESCRIPTOR, PAIRED SHA-256, DSA TAIL BUG

C+pair `×1.349`

WHAT CHANGED

- `cmd/dero-miner/dsac`: the C descriptor sort via cgo (-O3 -mcpu=apple-m1, darwin/arm64 only), five bounds guards added, fallback chain C → Go dsa → SA-IS. `--sort=auto|native|go|descriptor|sais`.
- `cmd/dero-miner/astropair`: each thread hashes two nonces so the two final SHA-256 passes interleave. Stage 1 is a generated copy of pow.go guarded by `TestStageIsCurrent`; the SHA block function is GOROOT's BSD-3 asm.
- dsa tail fix: the tail walk could add up to 255 descriptors without a room check. Grow and retry; regression test `TestFreshScratchTail`.
- `--bench` logs to `<exe>.bench.log` (it had truncated the live log); GPU stale-share accounting.

WHY

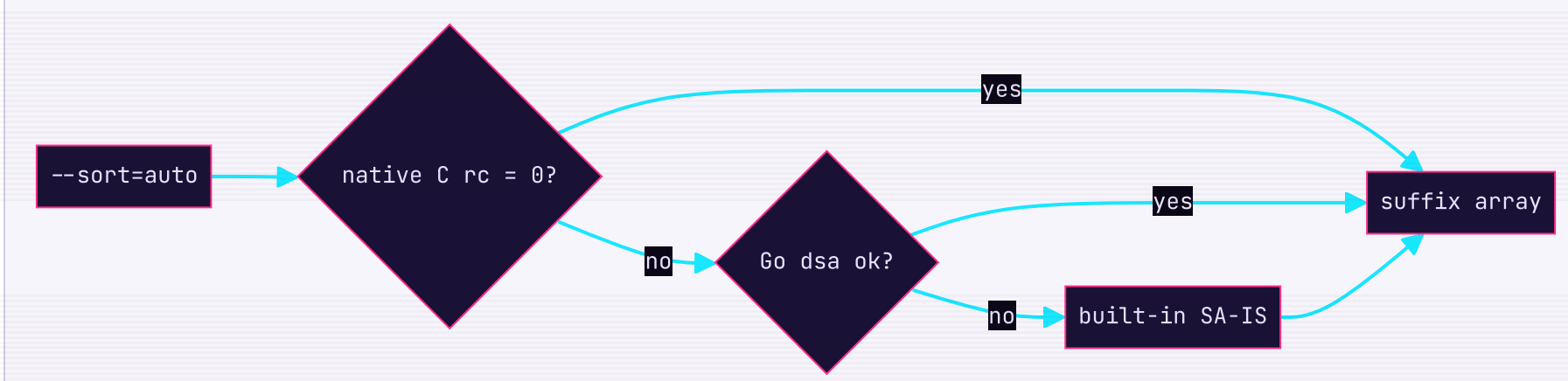
> The faster sort made SHA a larger share of each hash. A reported 31.1 → 27.9 KH/s "regression" needed checking first.

CODE

cmd/dero-miner/dsa/dsa.go · lines 569–583 · commit 8456fcf

```
    tail := fullBlocks * blockSize
    // [comment line trimmed]
    if tail < n && descLen+(n-tail) > s.dcap {
        full := n + blockSize + 8
        if s.dcap ⩾ full {
            return false
        }
        s.dcap = full
        s.desc = make([]desc, full)
        s.desc2 = make([]desc, full)
        goto retry
    }
```

FLOW



EVIDENCE

MEASURED	RESULT	SOURCE
31.091 KH/s "regression"	REFUTED · refuted: 1 s window aliasing 4,096-hash GPU batches (Monte Carlo max 31,091)	ARCHITECTURE §32.1
r15/r14 paired median	1.037 over 30 pairs	ARCHITECTURE §32.1
C sort bit-exactness	652/652; ASan, UBSan, race clean	RUN_REPORT Round 16
Paired SHA bit-exactness	605 inputs; final SHA stage 27 % faster	RUN_REPORT Round 16
4-way A/B vs r15, 8 rounds, 2 threads	C+pair 1.349 · C 1.217 · Go 1.029	RUN_REPORT Round 16
First pipeline run	test FAIL: index 764/764 on a fresh scratch under -race	RUN_REPORT Round 16
Rerun pipeline_20261023T061124	46/46 mutations, 10 packages 0 races, gates 16/16	RUN_REPORT Round 16
S4b staged vs installed r15	×1.40 (3,280–3,834 vs 2,434–2,908 H/s)	RUN_REPORT Round 16
Smaller GPU batches for fewer orphans	REFUTED · refuted: 2.7–3.7 s per dispatch at any size	ARCHITECTURE §32.5

R17 · 2026-10-23 · +22102026.r17

SHARE SUBMISSION PATH: TEMPLATE-CACHE FIX, PROPAGATION, UNLOCKED THREADS

dead work 82.5 → 17.7 ms/block paid 30/30

WHAT CHANGED

- derod: `refresh_due` schedules a job resend when the 100 ms template cache expires. A 9th normal miniblock arriving inside that window had left miners on dead normal work until the 500 ms timer.
- p2p: each RPC message coalesced into one write (one KCP packet instead of four, identical bytes); first-relay scoring for dial selection; whitelisted peers kept 24 h; inbound reachability counts public IPs only; v2-ban offences persist; GetInfo peerlist sizes assigned; slots_full orphan reason.
- Miner: on darwin with no QoS the mining goroutines are no longer `LockOSThread`ed; paid vs accepted read from `DERO.GetMiningOutcome`; expected DERO/day and a luck ratio.

WHY

> Two workflows (10 + 8 agents) followed 23 r16 shares from find to block. All 3 orphans were CPU normal shares that lost the race for the 9th slot by 0.13–0.74 s. Find → submit takes ~55 µs; the

race is decided by p2p hops to the builder.

CODE

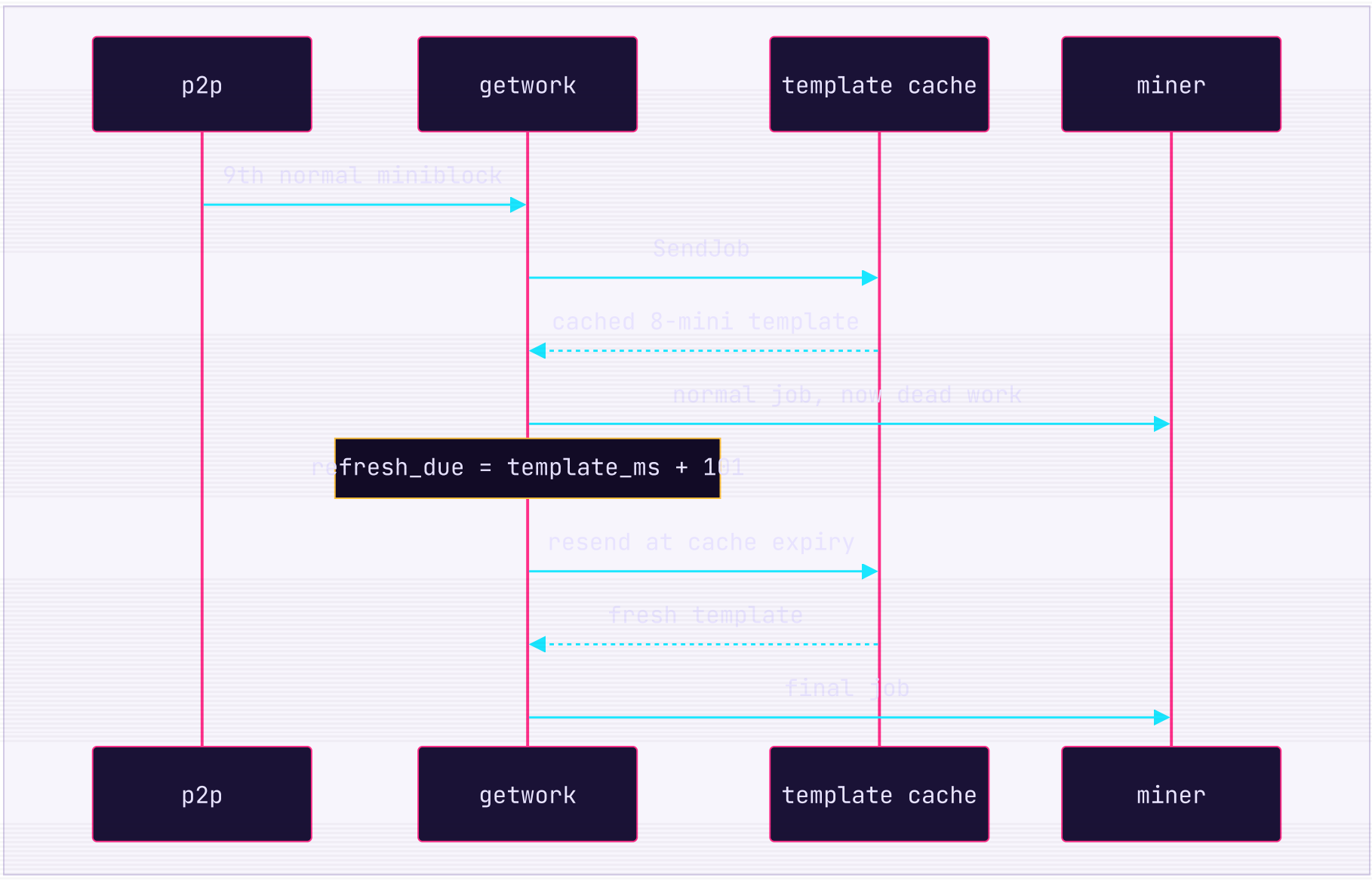
cmd/derod/rpc/websocket_getwork_server.go · lines 161–170 · commit 24cfc30

```
// refresh_due returns when to resend for a template made at template_ms by a send started at trigger_ms,
// or 0 when the template is newer than the send. The cache expires once now > template_ms+100. A template stamped i
// the trigger's own millisecond may still be the cached one from before the miniblock landed; one spare resend is
// cheaper than mining dead work until the 500 ms timer (Round 17 verifier).
func refresh_due(template_ms, trigger_ms int64) int64 {
    if template_ms ⩽ trigger_ms {
        return template_ms + 101
    }
    return 0
}
```

cmd/dero-miner/miner.go · lines 504–508 · commit 24cfc30

```
// should_lock_mining_thread: only an OS-level per-thread setting (affinity on linux/windows, a QoS class on darwin)
// is worth a locked goroutine
func should_lock_mining_thread(goos string, qos int) bool {
    return !(goos == "darwin" && qos == 0)
}
```

FLOW



EVIDENCE

MEASURED	RESULT	SOURCE
Template cache hit on the flip	17 % of blocks, 86 ms dead work per block	RUN_REPORT Round 17
Shadow A/B after refresh_due	82.5 → 17.7 ms per block, 0 timer flips	ARCHITECTURE §33.2
KCP packets per message	4 → 1; modelled retransmits 18.6 % → 5.4 %	ARCHITECTURE §33.3
Unlocked threads, 2 threads	×1.16–1.23 per thread	ARCHITECTURE §33.4
Pipeline pipeline_20261023T094538	51/51 mutations, 10 packages 0 races, shadow 0 mismatches	RUN_REPORT Round 17
S4b staged vs r16 ×1.59	REFUTED · artefact of a 2-thread wall bench; CPU efficiency ~×1.03–1.04 (R18)	ARCHITECTURE §34.1
Re-send own miniblock, latency send order, -O2/-flto, PGO refresh	REFUTED · refuted (no effect or not significant)	ARCHITECTURE §33.5
Live r17 run	30/30 shares paid, 0 orphans, ~39.5–40k H/s	RUN_REPORT Round 18; ARCHITECTURE §34.1

R18 · 2026-10-24 · +22102026.r18

HASHRATE CEILING, GPU SPLIT MODE, GPU AUTO-RESPAWN, PIPELINED SYNC, PEER-LAYER SECURITY

pipelined sync ×9.2 live staged 51–54k

WHAT CHANGED

- `--gpu-mode=split` (opt-in): the GPU computes stage 1 and the final SHA-256 into shared memory; CPU threads only sort. Every win is re-hashed on the CPU before submit.
- GPU auto-respawn: a supervisor kills and reaps the server, backs off 1 s doubling to 60 s (reset after 5 min healthy), relaunches, re-runs the bit-exact self-test. Only a self-test mismatch disables the GPU.
- Sync: batched, pipelined block fetch (`sync_pipeline.go`) with fallback to batch 1 after any block over 64 KiB; PoW prewarm only for blocks with exactly 10 miniblocks; KCP window 32 → 256 segments.
- Security: wallet RPC Origin/Host guard; at most 2 outbound connections per IPv4 /16; gossip accepts only literal public IP:port; NotifyINV bounded and scored; derod HTTP body limits; DERO.GetSupplyAudit.
- Go modernisation: zero-alloc CBOR append_frame, 17 Connection fields as typed atomics (9 races fixed), derod exit double-close fixed with sync.Once.

WHY

> The live hashrate dropped from 40k to 21–34k at 07:40: an agent's long Metal benchmark tripped the GPU watchdog, the GPU server exited with EOF, and the worker never restarted. Separately, the question was whether 2× was reachable.

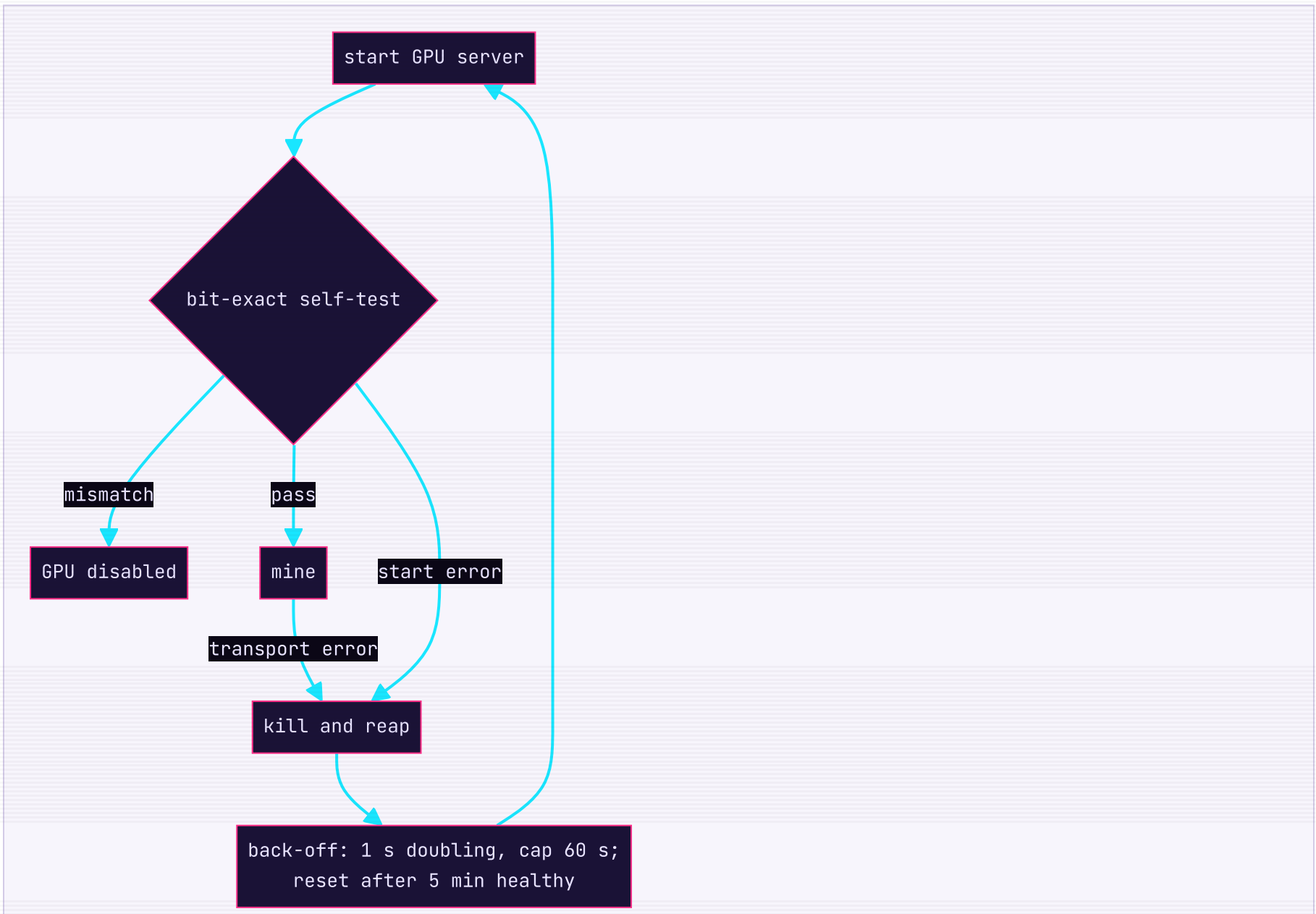
CODE

cmd/dero-miner/gpu_worker.go · lines 163–179 · commit 9703e42

```
func (b *gpuBackoff) next(now time.Time) time.Duration {
    if b.cur < b.min {
        b.cur = b.min
    }
    if !b.lastStart.IsZero() && now.Sub(b.lastStart) ⩾ b.healthy {
        b.cur = b.min
    }
    b.lastStart = time.Time{} // further attempts of this outage keep doubling
    d := b.cur
    b.cur *= 2
    if b.cur > b.max {
        b.cur = b.max
    }
    return d
}

// gpuWorker supervises the full-hash server and mines with it; start and sleep are replaceable for tests.
```

FLOW



EVIDENCE

MEASURED	RESULT	SOURCE
CPU efficiency at 20 threads	~2.33k H per CPU-second	ARCHITECTURE §34.1
Full-hash ceiling	`16 P × 2.36k + 4 E × 1.04k + GPU 1.24k ≈ 43k H/s`	ARCHITECTURE §34.1

MEASURED	RESULT	SOURCE
Split vs full, T=20, contended	×1.22–1.34 (verifier)	ARCHITECTURE §34.2
Pipelined sync, emulated 170 ms RTT	4.5 → 41.8 blocks/s (×9.2), treehash 31/31	ARCHITECTURE §34.4
Unbounded prewarm, 16,383-miniblock block	2 GB and 4.4 s CPU before the bound	ARCHITECTURE §34.4
CBOR append_frame	49.4 → 6.9 µs per block frame	ARCHITECTURE §34.6
Metal suffix-sort spike	REFUTED · 652/652 bit-exact, ~24.9k SA/s; not merged (failed consensus guard)	RUN_REPORT Round 18
Pipeline pipeline_20261024T101720	87/87 mutations, 15 packages 0 races, 2,335.6 staged H/cpu-s	RUN_REPORT Round 18
Project gates after G14 retention fix	16/16	RUN_REPORT Round 18
Live r18 split, staged	51–54k H/s (GPU near binding)	ARCHITECTURE §35.1

R19 · 2026-10-24 → 10-26 · +22102026.r19

NEON SORT, SPLIT AUDIT-STRIKE POLICY, PLATFORM ROUND

sort 253 → 175 µs live 68–72k

WHAT CHANGED

- NEON ports of the C sort's AVX2-only fast paths (column mask, uniform-span emitter, constant-column slide, singleton scatter).
- Split audit-strike policy: an audit difference kills the live GPU server and the respawn re-runs the bit-exact self-test. Each server generation takes at most one strike; 3 strikes in 1 h or a self-test mismatch disables the engine.
- Sync: fastsync TreeSection fetch pipelined (depth 4, max 6) with fallback to depth 1 after any error; duplicate concurrent tx admissions collapse into one.
- Security: inbound handshakes on the default KCP profile with a 6 s deadline (UDP reflection); bans checked before the accept token; IPv6 bans per /64; wallet log redaction (0600); XSWD on loopback by default; RPC server timeouts.
- RPC: DERO.Subscribe, /health, GetBlockHeadersRange, GetMiniBlocks, GetDifficultyHistory, GetTxPoolDetail, GetPeerInfo, Prometheus /metrics and /dashboard.
- Wallet: atomic saves (temp + fsync + rename, 0600, .bak); --password honoured on restore; sync_loop exits on Close; GetTransfersPaged.

WHY

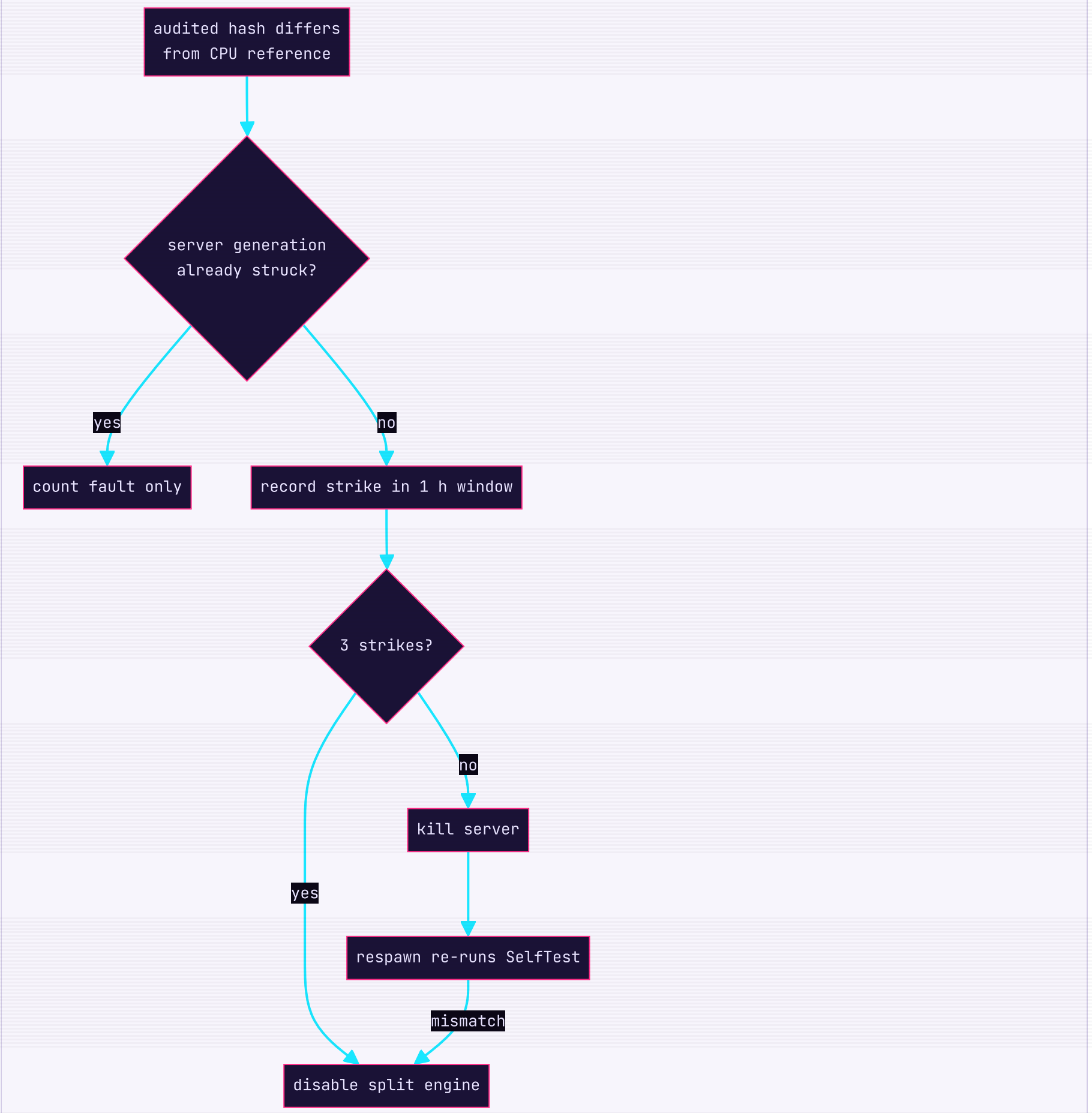
> On 2026-10-26 18:23:41 the live r18 miner disabled split mode for good after 1 audit difference in 15,830 audits. The likely cause was a stale re-staging race, fixed in the same round. Winning shares were always CPU re-hashed, so no wrong share could be sent.

CODE

cmd/dero-miner/gsplit/engine.go · lines 859–886 · commit e1bae2c

```
func (e *Engine) auditFault(why string, now time.Time, gen uint64) {
    e.sAuditFaults.Add(1)
    e.strikeMu.Lock()
    if gen ≠ 0 && gen ⩽ e.struckGen {
        e.strikeMu.Unlock()
        return
    }
    if gen > e.struckGen {
        e.struckGen = gen
    }
    keep := e.strikes[:0]
    for _, t := range e.strikes {
        if now.Sub(t) < e.cfg.AuditStrikeWindow {
            keep = append(keep, t)
        }
    }
    e.strikes = append(keep, now)
    n := len(e.strikes)
    e.strikeMu.Unlock()
    if n ⩾ e.cfg.AuditStrikes {
        e.Disable(fmt.Sprintf("%s (%d audit differences within %s)", why, n, e.cfg.AuditStrikeWindow))
        return
    }
    e.cfg.Logf("GPU split %s: restarting the GPU server and re-running the bit-exact self-test (strike %d of %d)", w
    if tr, ok := e.live.Load().(transport); ok {
        tr.kill() // the reply loop sees the server exit and respawns it; the respawn re-runs the SelfTest
    }
}
```

FLOW



EVIDENCE

MEASURED	RESULT	SOURCE
Sort time per text	253 → 175 µs, bit-exact on 2,048 hashes	ARCHITECTURE §35.1
S4b H/cpu-s, 4 threads, vs r18	×1.604; split CPU side 5,028 H/cpu-s	RUN_REPORT Round 19
Hybrid GPU-side sort	REFUTED · dropped: upper bound ~+1.2 % at 989 SA/s	ARCHITECTURE §35.1

MEASURED	RESULT	SOURCE
TreeSection pipelining, loopback	1.8–2.2×	ARCHITECTURE §35.3
Wallet history sync	7.6–8.7× faster	ARCHITECTURE §35.5
Pipeline attempts	4 failures (one unseen ~41 h by a silent watcher), then PASS	RUN_REPORT Round 19
Final pipeline_20261026T185835	278/278 mutations, 19 packages 0 races, shadow 0 mismatches	RUN_REPORT Round 19
Project gates	16/17: G17 correctly failed on the still-running r18 miner	RUN_REPORT Round 19
Live r19 miner while mining	~68–72k H/s (derod estimator 74.8k ±13 %)	RUN_REPORT Round 20

R20 · 2026-10-28 · +22102026.r20

UPTIME FORENSICS, GPU KERNELS, MINER LOG RETENTION, RPC HOST/ORIGIN GUARD

live TOTAL 79–86k

WHAT CHANGED

- GPU kernels in `gpu_split_server.swift`: SHA rounds unrolled, 8-wide stage-1 threadgroups, overlapped buffers. CPU: 16-lane NEON singleton scatter; `feeder_cpu_backlog_ms` separates CPU-bound from GPU-bound waits.
- Miner log: append, never truncate; rotate at 32 MiB into .1 to .3; mode 0600 (`minerlog.go`). Exit reasons logged; SIGHUP honoured only when not ignored, so nohup keeps working; getwork read deadline.
- derod: getwork connect and disconnect presence lines and counters; configurable log rotation; darwin process metrics; free_disk_space_bytes of the datadir.
- p2p: per-peer tx-ask hold; compact block reconstruction from local pools (no protocol change); TLS 1.2 floor; inbound per-netgroup cap and accept bucket.
- RPC: Host/Origin guard on loopback binds (`DERO_RPC_ALLOWED_HOSTS`, `DERO_RPC_ALLOWED_ORIGINS`); `DERO.GetTxStatus` for up to 64 txids.
- Wallet: parallel history discovery plus prefetch; daemon calls bounded at 2 min; HTTP server timeouts. Harness: S3a fails closed on failing or racy new tests, S3k runs KATs before install, G18 reads live /metrics.

WHY

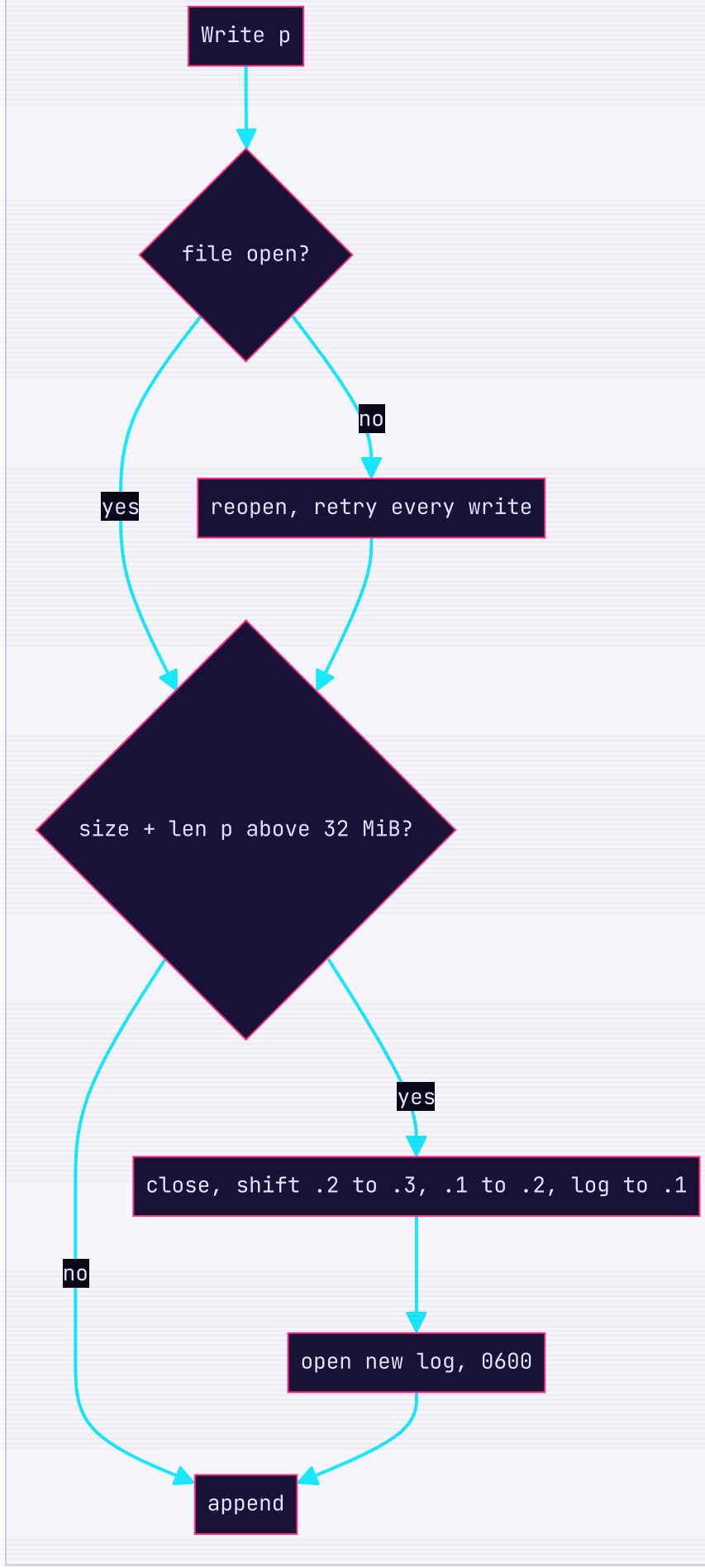
> The r19 run had 40 shares in 26.7 h. The ledger showed this was miner-off time, not a hashing problem; the miner had been connected for roughly 35–40 minutes of the run. The miner log had been truncated on every start, so its own record of that run was gone. DNS rebinding and cross-site websocket hijacking were both confirmed on the live node.

CODE

cmd/dero-miner/minerlog.go · lines 70–92 · commit 3ed6e8f

```
func (r *rotating_log) Write(p []byte) (int, error) {
    r.mu.Lock()
    defer r.mu.Unlock()
    if r.closed {
        return 0, os.ErrClosed
    }
    if r.f == nil { // the reopen after a rotation failed (r20 review): retry on every write, never stay silent
        if err := r.open(); err ≠ nil {
            return 0, err
        }
    }
    if r.max > 0 && r.size > 0 && r.size+int64(len(p)) > r.max {
        r.f.Close() // close before renaming: Windows cannot rename an open file
        r.f = nil
        r.shift()
        if err := r.open(); err ≠ nil {
            return 0, err
        }
    }
    n, err := r.f.Write(p)
    r.size += int64(n)
    return n, err
}
```

FLOW



EVIDENCE

MEASURED	RESULT	SOURCE
Workflow wf_12dfa98b-8f5	17 agents, 8 dimensions, 0 findings refuted	RUN_REPORT Round 20
GPU SHA unroll / stage-1 / overlap	`×1.8–2.0 / ×1.8 proxy / ×1.50–1.63 proxy`	RUN_REPORT Round 20
GPU kernel bit-exactness	`14,354 real-GPU hashes, 0 mismatches`	RUN_REPORT Round 20
NEON singleton scatter	`sort ×1.06, own H/cpu-s ×1.03`	RUN_REPORT Round 20
tx-ask hold, live replay	`23 failed fetches → 10 asks`	RUN_REPORT Round 20
Wallet parallel history	`×2.3–5.9`	RUN_REPORT Round 20
Pipeline pipeline_20261028T115635	`first attempt PASS: 385/385 mutations, 21 packages 0 races, 70 new tests ×3 0 flaky`	RUN_REPORT Round 20
S4b ×1.654 vs r19	`REFUTED · corrected in R21: SA-IS vs native C; like for like ×1.04`	RUN_REPORT Round 21
Project	`KATs 32/32, mutations 27/27, gates 18/18`	RUN_REPORT Round 20
Live r20 run	`TOTAL 79–86k H/s, staged 73–80k (r19 ~57k)`	RUN_REPORT Round 21

R21 · 2026-10-28 · +22102026.r21

CPU-SORT-BOUND SPLIT MINER; STALE-DROP ON THE FINAL FLIP; PLATFORM ROUND

S4b `×1.041` mutations `510/510`

WHAT CHANGED

- Miner: split stale-drop on the same-height normal → final flip (`split_stale`); cgo yield every 8 sorts; `--gpu-max` debt charged on the wall-clock busy union; zero-run sort tie-break keys; opt-in GPU sort with an ABBA probe (default off); getwork blob-length panic fixed (SEC21-01, remote crash); unpredictable shm names; autoreleasepool leak fix; run ledger.
- p2p: grey-peer quarantine; known-tx inventory; GetObject unrequested-tx amplification cut; node-wide inbound frame byte budget; lagging-peer gauge.
- RPC/ws: per-socket and node-wide budgets for foreign-origin websockets; `DERO.WatchTx` and TxPool pushes; getwork job frame encoded once per job; `ALLOWED_ORIGINS=null` bypass closed.
- Ops: `--prune-history` preflight (it could destroy the balance tree); stray tracked root binary untracked. Harness: S4b compares like for like, regime cards, run ledger.

WHY

> In the live r20 run the bottleneck moved to the CPU sort: feeder_cpu_backlog_ms ~39–44 s/min and ~186–192 µs of CPU per hash, with the GPU ~92 % busy. r20 dropped staged texts only on a new height, so after each normal → final flip the CPU still sorted texts whose shares could only be orphaned.

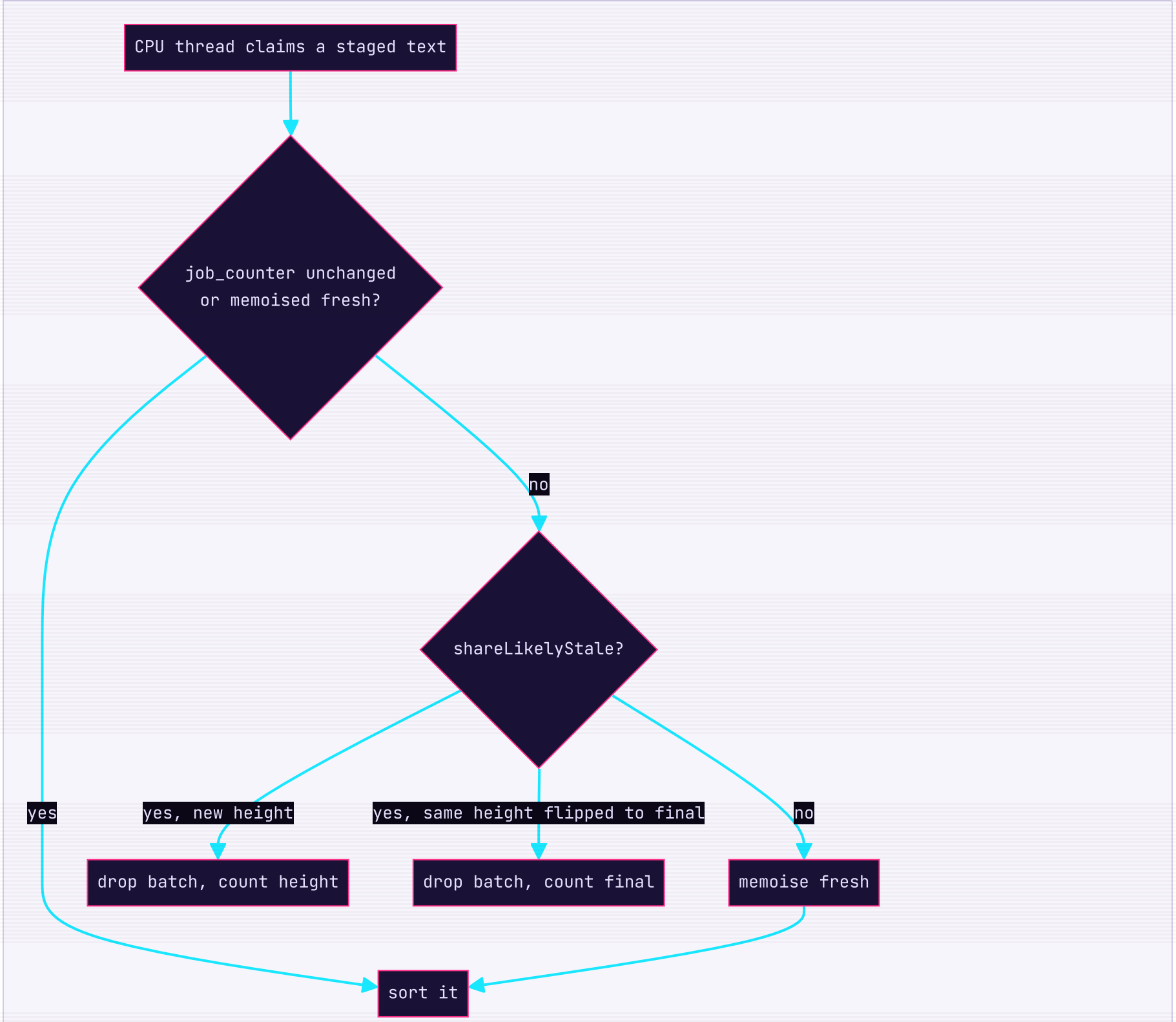
CODE

cmd/dero-miner/gpu_split.go · lines 172-195 · commit a421554

```
func split_stale(tag interface{}) bool {
    sj := tag.(*splitJob)
    if jc := atomic.LoadInt64(&job_counter); jc == sj.counter || jc == atomic.LoadInt64(&sj.fresh) {
        return false
    }
    mutex.RLock()
    cur := job
    jc := atomic.LoadInt64(&job_counter) // job and job_counter change together under mutex
    mutex.RUnlock()
    if !shareLikelyStale(sj.job, cur) {
        atomic.StoreInt64(&sj.fresh, jc)
        return false
    }
    if cur.Height ≠ sj.job.Height {
        atomic.AddUint64(&split_stale_height, 1)
    } else {
        atomic.AddUint64(&split_stale_final, 1)
    }
    return true
}

// split_found checks one split hash against its job's difficulty; a win is re-hashed on the CPU, then submitted.
func split_found(tag interface{}, blob []byte, h [32]byte) {
    sj := tag.(*splitJob)
```

FLOW



EVIDENCE

MEASURED	RESULT	SOURCE
Workflow wf_3a03ca93-a3e	19 agents, 9 dimensions; 2 findings refuted	RUN_REPORT Round 21
Likely-stale split shares before the fix	`5 of 5 orphaned (slots_full)`	RUN_REPORT Round 21; commit a421554
cgo yield	`split bench ×1.056 (author-measured)`	RUN_REPORT Round 21
Zero-run tie-break keys	`sort ×1.015–1.022`	RUN_REPORT Round 21
Getwork job encode	`×1.7–2.3`	RUN_REPORT Round 21
GPU sort (op 3)	`REFUTED · net loss ~9 % live, ~+2 % at best: off by default`	RUN_REPORT Round 21
Grey-peer quarantine	`~10 failed handshakes/min stopped`	RUN_REPORT Round 21
Pipeline runs	`4 failures (mutation, flaky, install gate, race tail), then PASS`	RUN_REPORT Round 21
pipeline_20261028T185146	`510/510 mutations, 102 tests ×3 0 flaky, 21 packages 0 races`	RUN_REPORT Round 21
S4b like for like vs r20	`×1.041; split CPU side 5,792 H/cpu-s (r20 5,457)`	RUN_REPORT Round 21
Shadow	`120/120 blocks, 0 mismatches`	RUN_REPORT Round 21
GPU server install	`real-GPU bit-exact on 4,102 hashes; stamps match`	RUN_REPORT Round 21
Project	`KATs 34/34, mutations 31/31, gates 18/18`	RUN_REPORT Round 21

R22 · 2026-10-29 · +22102026.r22

THE GPU CAP MADE REAL; R22 STAGE-1 KERNELS; CRYPTO-LAYER AUDIT; PLATFORM ROUND

GPU cap `made real` mutations `605/605`

WHAT CHANGED

- Miner: `--gpu-max` is now a duty bucket (`gsplit/duty.go`): each GPU busy span is charged at busy×100/cap, wall time is credited, at most 100 ms is banked, and the feeder stages nothing while in debt; its waits are counted as `feeder_throttle_ms` .
- Split pipeline: exact precomputed PoW-target compare (`powtarget.go`); GPU SHA skipped for batches whose job went stale mid-sort (`StaleQuiet` keeps one verdict per drop); `collect_ms` ;

opt-in ABBA duty tuner `--cpu-max=auto[:floor]` / `--gpu-max=auto[:floor]`.

- GPU kernels r22: table-driven stage-1 ops (tables built once on the GPU by `k_lut` from the op's own statements) replace a 256-way switch; per-kernel spans `gpu_s1_ms_per_cb` / `gpu_sha_ms_per_cb`.
- p2p: stuck-peer eviction (r21 F4); `--p2p-nat=probe` (read-only); known-tx full-set sweep fix; per-peer INV txid bucket (64/s); short GetObject replies counted as failures; TLS 1.2 AEAD/curve floor. F6 mutex profiles of miner and derod: no lock limits throughput.
- RPC: opt-in `--rpc-tx-cache` (GetTransaction verify-on-hit cache, off by default: hit/miss timing is a cross-client oracle on a public RPC).
- Crypto: node-local `glue/txcanon` round-trip admission; canonical keys in GetRandomAddress; wallet ring-order nonce-reuse fix (C3); signature input checks (C5); EncryptWithKey no longer seals in place (C6).
- Miner hardening: mainnet submit token bucket, LastError log gate, BUILD_STAMPS hex check, prune preflight Lstat.
- Harness: binding classifier and gate G19; `miner_runs.py` validator with derod and chain joins; go_mutate INVALID class; atomic snapshots; S4c GPU like-for-like bench; `race_split` (1fdfa51, KAT IK38).

WHY

> The live r21 run made 83–86k H/s TOTAL (staged ~79k) with the GPU busy 94.6–96.0 % under `--gpu-max=90`: the cap was not enforced, because the feeder slept off debt while the ring was full anyway (FOR22-1). The CPU sort backlog was 74 %, so the miner was co-bound. A dip to 49k came from other sessions' CPU jobs at nice 0; the measured law is H/s = 4,817 × miner cores + 8,000 (r = 0.991). On the node, 15 outgoing peers were the whole reachable set and 2 of them were frozen far below the tip; the router answers no NAT mapping protocol. The crypto-layer audit looked for decoder, wallet and proof bugs that could cost a mined block or leak on chain without a fork.

CODE

Excerpts in order: the duty bucket (charge and settle), the r22 stage-1 table step (Metal source generated by the Swift server), the exact PoW-target compare, stuck-peer selection, the txcanon admission check, the wallet ring-order pick (C3), and S3a's race_split from the closed-loop fix.

`cmd/dero-miner/gsplit/duty.go · lines 22–49 · commit dd15e82`

```
// dutyBurstNs: the most GPU idle time the bucket banks (a long pause cannot buy an unbounded full-duty burst).
const dutyBurstNs = int64(100 * time.Millisecond)

// dutyMaxSleep bounds one feeder sleep so a stop request is seen quickly.
const dutyMaxSleep = 20 * time.Millisecond

// dutyCharge is the wall time a busy span of busyNs costs under a cap% duty cycle.
func dutyCharge(busyNs int64, cap int) int64 {
    if cap < 1 || cap ⩾ 100 {
        return 0
    }
    return busyNs * 100 / int64(cap)
}

// settleDuty credits the wall time since the previous settle (feeder goroutine only: dutyLast is not shared) and
// returns the ns still owed (⩽ 0: the feeder may stage). The credit is clamped at dutyBurstNs.
func (e *Engine) settleDuty(now int64) int64 {
    el := now - e.dutyLast
    if el < 0 {
        el = 0
    }
    e.dutyLast = now
    d := e.debt.Add(-el)
    if d < -dutyBurstNs {
        d = e.debt.Add(-dutyBurstNs - d) // charges racing in from the reply loop are kept
    }
    return d
}
```

`gpu-miner-poc/gpu_split_server.swift · lines 280–292 · commit dd15e82`

```
let lutStep = """
            { uint dd=OPD[op];
              if(dd & 0x10000u){ switch(op){
\(lutSpecial)              default: break; } }
              else { if(dd & 0x20000u) rc4_ksa(&r, s); uchar y=s[p2];
                uchar am1=(dd&2u)?y:uchar(0xff), xm1=(dd&1u)?y:uchar(0), am2=(dd&8u)?y:uchar(0xff), xm2=(dd&4u)?y:uc
                device const uchar* T0=LUT+op*768u;
                for(uint w=p1&~3u;w<p2;w+=4){ uint v=*(threadgroup const uint*)(s+w); uint rr=0;
                  for(uint q=0;q<4;q++){ uchar x=T0[(v>>(8u*q))&0xffu]; x=(x&am1)^xm1; x=T0[256u+x]; x=(x&am2)^xm2;
                  uint lo=(w<p1)?(p1-w):0u; uint hi=min(4u,p2-w); uint m=(hi⩾4u?0xffffffffu:((1u<<(8u*hi))-1u)) & ~
                  *(threadgroup uint*)(s+w)=(v&~m)|(rr&m); } } }

"""
```

cmd/dero-miner/powtarget.go · lines 46-61 · commit dd15e82

```
// meets reports CheckPowHashBig(h, d) for the difficulty the target was made from.
func (t *powTarget) meets(h crypto.Hash) bool {
    if t.all {
        return true
    }
    if t.slow ≠ nil {
        return CheckPowHashBig(h, t.slow)
    }
    for i := 3; i ⩾ 0; i-- { // the hash is a little-endian 256-bit integer (HashToBig reverses it)
        hw := binary.LittleEndian.Uint64(h[8*i:])
        if hw ≠ t.w[i] {
            return hw < t.w[i]
        }
    }
    return true // equal: CheckPowHashBig accepts hash == target
}
```

p2p/stuck_peer.go · lines 73-103 · commit dd15e82

```
// stuck_select updates track with views (dropping connections that are gone) and returns at most one view to evict.
func stuck_select(views []stuck_view, our_height int64, now time.Time, track map[*Connection]*stuck_track) []stuck_v
    present := make(map[*Connection]bool, len(views))
    for _, v := range views {
        present[v.c] = true
        if t := track[v.c]; t == nil || t.height ≠ v.height {
            track[v.c] = &stuck_track{height: v.height, since: now}
        }
    }
    for c := range track {
        if !present[c] {
            delete(track, c)
        }
    }
    healthy := 0
    var cands []stuck_view
    for _, v := range views {
        if our_height-v.height ⩽ STUCK_PEER_BLOCKS {
            healthy++
            continue
        }
        if v.outgoing && !v.exempt && now.Sub(track[v.c].since) ⩾ STUCK_PEER_WINDOW {
            cands = append(cands, v)
        }
    }
    if len(cands) == 0 || healthy < STUCK_MIN_HEALTHY || len(cands) > healthy {
        return nil
    }
    sort.SliceStable(cands, func(i, j int) bool { return cands[i].height < cands[j].height })
    return cands[:1]
}
```

glue/txcanon/txcanon.go · lines 70-95 · commit dd15e82

```
// Check reports whether tx, decoded from raw, may enter the mempool. canonical is true when raw is exactly the
// node's own encoding of tx. A non-canonical encoding is still accepted when its canonical form decodes back to a
// transaction with the same txid and the same canonical bytes (e.g. an undefined point flag byte); otherwise the
// error is ErrNotRoundTrip.
func Check(raw []byte, tx *transaction.Transaction) (canonical bool, err error) {
    ser, err := serialize(tx)
    if err ≠ nil {
        rejected.Inc()
        return false, fmt.Errorf("%w: %v", ErrNotRoundTrip, err)
    }
    if bytes.Equal(raw, ser) {
        return true, nil
    }
    var back transaction.Transaction
    if err = decode(&back, ser); err ≠ nil {
        rejected.Inc()
        return false, fmt.Errorf("%w: %v", ErrNotRoundTrip, err)
    }
    ser2, err := serialize(&back)
    if err ≠ nil || back.GetHash() ≠ tx.GetHash() || !bytes.Equal(ser2, ser) {
        rejected.Inc()
        return false, ErrNotRoundTrip
    }
    nonCanonical.Inc()
    return false, nil
}
```

walletapi/transaction_build.go · lines 355-372 · commit dd15e82

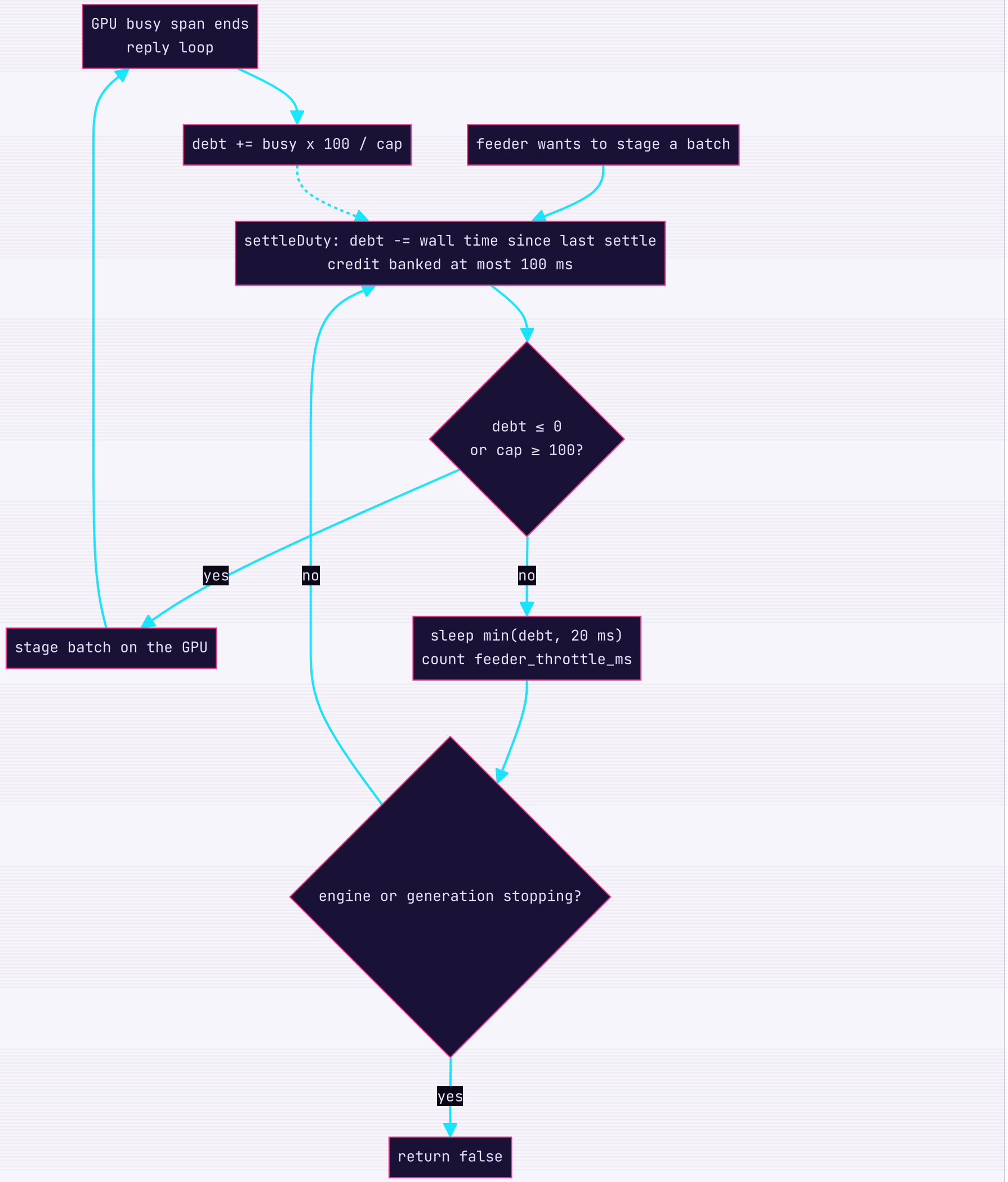
```
func pick_ring_order(crand *mathrand.Rand, witness_index []int, used map[string]bool, key func([]int) string) bool {
    for attempt := 0; attempt < RING_ORDER_ATTEMPTS; attempt++ {
        crand.Shuffle(len(witness_index), func(i, j int) {
            witness_index[i], witness_index[j] = witness_index[j], witness_index[i]
        })
        // make sure sender and receiver are not both odd or both even
        if witness_index[0]%2 == witness_index[1]%2 {
            continue
        }
        k := key(witness_index)
        if used[k] {
            continue
        }
        used[k] = true
        return true
    }
    return false
}
```

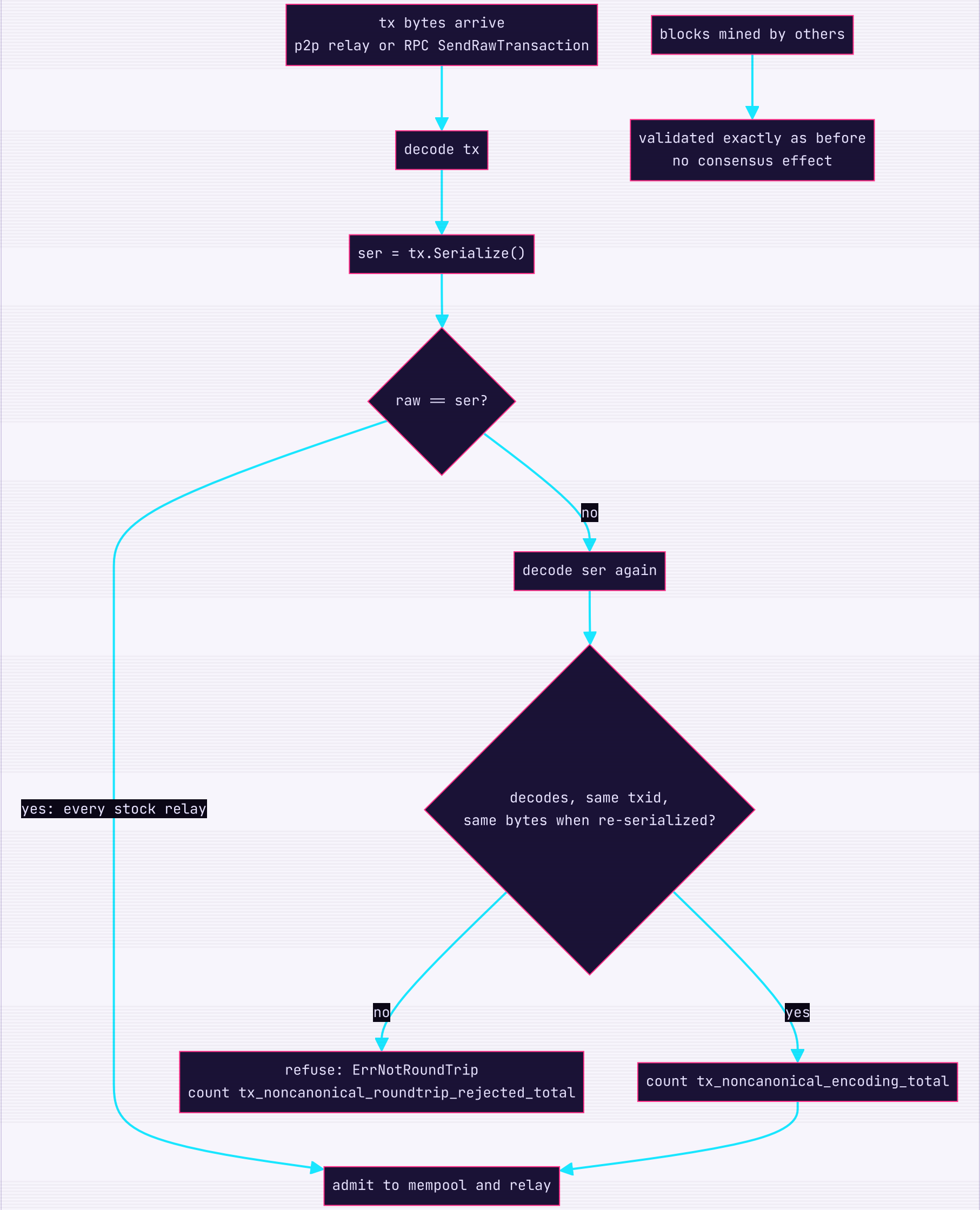
harness/pipeline/fix_build_verify.py · lines 673-686 · commit 1fdfa51

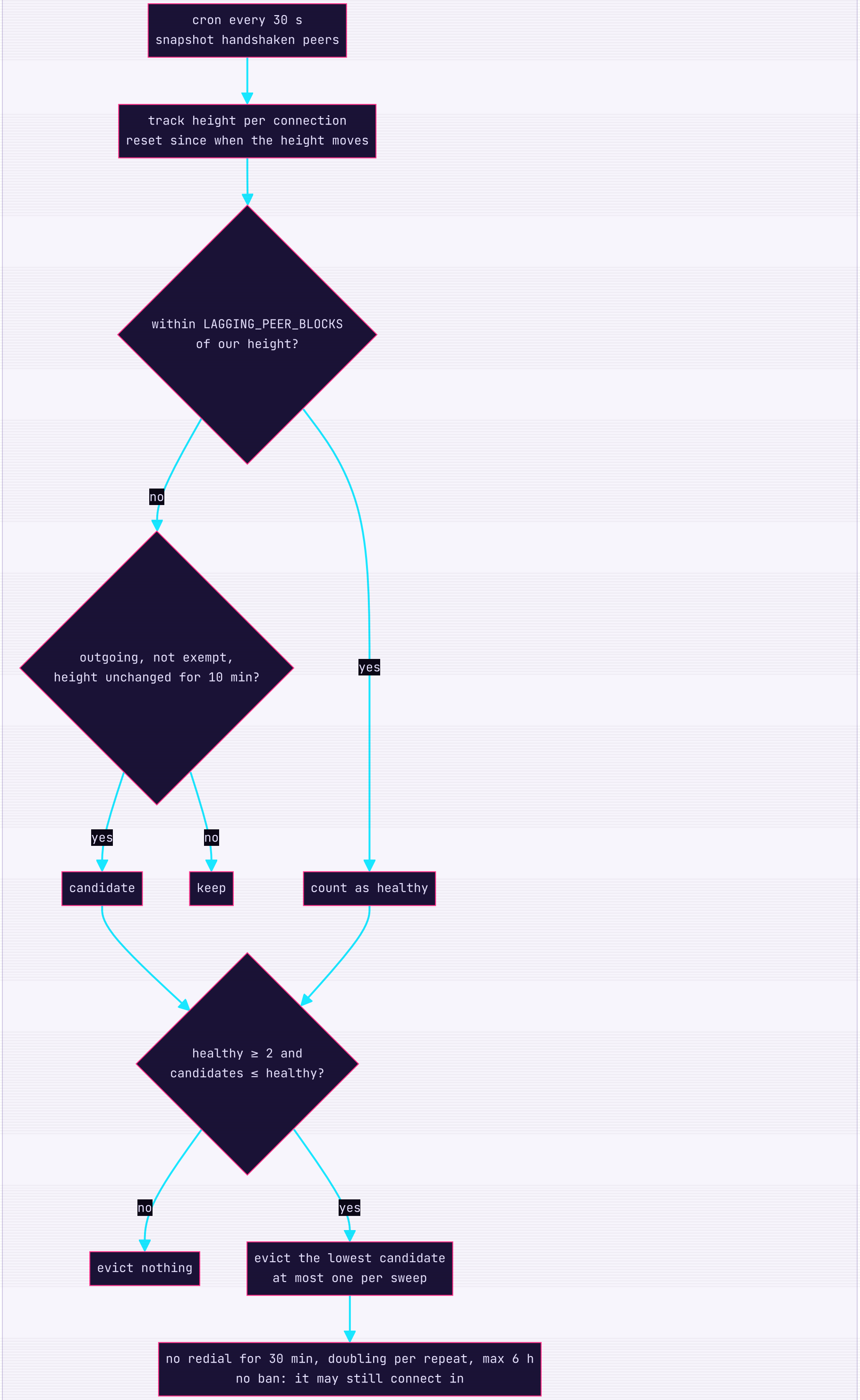
```
NORACE_TAG = re.compile(r'^//go:build\s+.*!race\b', re.M)


def race_split(names, sources):
    """R22: (race, norace) — a test defined in a file built only without -race (//go:build !race) never compiles und
    -race, so S3a must run it without -race; under -race it 'never ran' and S3a called it broken (the walletapi
    Test_R22_MultiPayload_Distinct_Blinders, whose package harness races upstream)"""
    norace = set()
    for text in sources:
        if NORACE_TAG.search(text):
            norace |= {n for n in names if re.search(r'^func ' + re.escape(n) + r'\(', text, re.M)}
    return [n for n in names if n not in norace], [n for n in names if n in norace]
```

FLOW


GPU busy span ends
reply loop
debt += busy x 100 / cap
feeder wants to stage a batch
settleDuty: debt -= wall time since last settle
credit banked at most 100 ms
debt ≤ 0
or cap ≥ 100?
yes
no
no
stage batch on the GPU
sleep min(debt, 20 ms)
count feeder_throttle_ms
engine or generation stopping?
yes
return false


tx bytes arrive
p2p relay or RPC SendRawTransaction
blocks mined by others
decode tx
validated exactly as before
no consensus effect
ser = tx.Serialize()
raw == ser?
no
decode ser again
yes: every stock relay
decodes, same txid,
same bytes when re-serialized?
no
yes
refuse: ErrNotRoundTrip
count tx_noncanonical_roundtrip_rejected_total
count tx_noncanonical_encoding_total
admit to mempool and relay



EVIDENCE

MEASURED	RESULT	SOURCE
Workflow wf_ff537f61-150	19 agents, 9 dimensions; patch 151 files; consensus guard PASS (only config/version.go)	RUN_REPORT Round 22
Live r21 TOTAL / staged	`83–86k / ~79k H/s`	RUN_REPORT Round 22
GPU duty with --gpu-max=90 (r21)	`94.6–96.0 %: cap not enforced (FOR22-1)`	RUN_REPORT Round 22; ARCHITECTURE §38.5
CPU sort backlog (r21)	`74 %: co-bound`	RUN_REPORT Round 22

MEASURED	RESULT	SOURCE
Contention law, 24 windows of 30 s	H/s = 4,817 × cores + 8,000, r = 0.991; 13.0 of 18 cores gave 70.6k	ARCHITECTURE §38.1
CPU-own hashes per idle try	exactly 2.0; 1.4–4.0k H/s, 2–5 % of all hashes	ARCHITECTURE §38.1
Duty bucket at --gpu-max=90	~4k H/s (-5 % staged), estimate; restart with --gpu-max=96	ARCHITECTURE §38.5, §38.6
r22 stage-1 kernels	GPU time ×0.742, bit-exact; end to end ≈ 0 at 4096,4,2	ARCHITECTURE §38.5
Geometry DERO_GSPLIT=8192,5,3 proxy	×1.55–2.1, contended, unverified live	ARCHITECTURE §38.5
Exact PoW-target compare	×92 per check, 0 allocs; live 0–2 % (estimate)	ARCHITECTURE §38.5
CPU sort phase profile	REFUTED · merge 43 %, radix ≤ 30 %, runs+walk 24 %; six variants refuted	ARCHITECTURE §38.5
Known-tx full-set sweep fix	×150–500	RUN_REPORT Round 22
--rpc-tx-cache (off by default)	×5.4–7.2 CPU/call upper bound at 100 % hits; 0 on the live node	ARCHITECTURE §38.5
Frozen outgoing peers (r21)	2 of 15 slots	ARCHITECTURE §38.5; commit dd15e82 p2p/stuck_peer.go
go_mutate before R22	22 of 492 mutations never compiled, counted as caught	ARCHITECTURE §38.3
Chain join, live run	22 paid of 24 found; r21 20:20 run 2 of 3 paid (exit record said 0)	ARCHITECTURE §38.2
derod over 721 s	goroutines 122 (max 157), RSS 196 MB flat, fds 59–61, 0.075 cores	ARCHITECTURE §38.4
Graviton balances	7.80 GB, ~91 MB/day; disk not a constraint	ARCHITECTURE §38.4
Run 1 pipeline_20261029T094334	FAIL at S3a (!race test run with -race); fixed 1fdfa51, IK38	ARCHITECTURE §38.6
pipeline_20261029T094741	PASS: 93 new tests ×3 0 flaky; mutations 605/605; race 22 packages 0 races	ARCHITECTURE §38.6
S4b like for like vs r21	×0.983 H/cpu-s (noise, sort unchanged)	ARCHITECTURE §38.6
Shadow	120/120 blocks, 240 treehashes, 0 mismatches	ARCHITECTURE §38.6
GPU server install	kernels=r22, real-GPU bit-exact on 4,102 hashes; stamps match	ARCHITECTURE §38.6
Project	KATs 34/34; harness KATs R19 10/10, R20 8/8, R21 8/8, R22 10/10 (47/47 mutations); gates 19/19	ARCHITECTURE §38.6

CRYPTO-LAYER AUDIT: C1–C9 CLASSIFICATION

None of these is the disclosed range-proof soundness defect (R8). Nothing here changes which blocks or transactions are valid: consensus-visible items are pinned by tests and documented for an upstream fix.

ID	FINDING	CLASS	DISPOSITION	SOURCE
C1	Compressed-point decoder accepts x ≥ P and returns the point at infinity; the identity has no decodable encoding	consensus-visible: decoder pinned by tests, not changed	node-local mitigation shipped: txcanon admission (p2p relay and SendRawTransaction); upstream fix needs coordinated activation	RUN_REPORT Round 22; ARCHITECTURE §38.5; R22 workflow crypto C1
C2	Flag bytes 0x02–0xff accepted (decode to P or −P)	consensus-visible	RPC-local mitigation shipped: canonical keys in GetRandomAddress; mainnet 0 of 2,141,950 keys affected	RUN_REPORT Round 22; ARCHITECTURE §38.5; R22 workflow crypto C2
C3	Two payloads with the same ring order shared the blinding scalar and payload keystream	wallet bug	fixed: a distinct ring order per payload; a third ringsize-2 payload to one destination is refused (ErrRingOrderExhausted)	RUN_REPORT Round 22; ARCHITECTURE §38.5; R22 workflow crypto C3
C4	Payment-proof amount bounds do not authenticate the amount	format change needed (proof/)	documented; fix is a new proof format or a receiver-side wallet check	RUN_REPORT Round 22; ARCHITECTURE §38.5; R22 workflow crypto C4
C5	Signature check accepted out-of-range scalars and non-canonical signer addresses	wallet bug	fixed: signature input checks	RUN_REPORT Round 22; ARCHITECTURE §38.5; R22 workflow crypto C5
C6	EncryptWithKey sealed into the caller's plaintext buffer	wallet bug (latent)	fixed: no longer seals in place	RUN_REPORT Round 22; ARCHITECTURE §38.5; R22 workflow crypto C6

ID	FINDING	CLASS	DISPOSITION	SOURCE
C7	Proof decoders treat a short read as success; Serialize then panics	consensus-code robustness	documented upstream; relay side covered by txcanon (no panic)	RUN_REPORT Round 22; ARCHITECTURE §38.5; R22 workflow crypto C7
C8	Hash/Key UnmarshalText accept trailing garbage after 64 hex digits	consensus dir, RPC-input leniency	documented	RUN_REPORT Round 22; ARCHITECTURE §38.5; R22 workflow crypto C8
C9	Fuzz and differential runs of the decoders	verification	no new panics; no third non-canonical class found	RUN_REPORT Round 22; ARCHITECTURE §38.5; R22 workflow crypto C9

R23 · 2026-10-30 · +22102026.r23

CONTENDED-BOX FORENSICS; PAYOUT PATH; DOS, AGENT AND INFLATION PROTECTIONS; SOUNDNESS REVIEW

final share → job `252 → 32 ms` mutations `743/743`

WHAT CHANGED

- Payouts: one getwork dispatcher (`job_dispatch.go`): every chain source only kicks it, it sends at most one job per observed state change (height, miniblock count, block-added event), retries a half-added block 2, 4, 8 ms apart, fires a template refresh exactly when due and keeps an unchanged job alive every 2 s; templates are numbered under `sendjob_mu` and `deliver_job` never sends a miner an older job after a newer one.
- Inflation: per-block emission auditor (`glue/emission` , `DERO.GetEmissionAudit` , offline `tools/emission-audit`): the balance-tree diff must equal the published transfer deltas plus the formula coinbase plus SC payouts; alert `dero_emission_audit_alerts_total` ; transient errors retried ×3 and excused.
- Security: remote RPC weighted bucket, behaviour scoring and escalating bans (loopback exempt); unknown-method, fuzz and hammer bans for automated clients (IPv6 keyed per /64); getwork per-IP cap 32, ban after 8 bad shares across every session of the key, one-slot gate for local shares; `glue/txprecheck` refuses structurally invalid txs before state reads and scores the relaying peer; getwork TLS floor.
- Crypto: C4 receiver payment-proof check, burn-aware (CR23-1); txcanon observes only txs of accepted blocks (CR23-2); integrated-address port kept and size-checked (CR23-5); zero seed refused; random TLS serial.
- RPC/ws and p2p: finality fields (`DERO.GetTxStatus`), `DERO.GetFeeEstimate` , per-miner job numbering newest-last, walletapi reconnect fixes, Connected=false on drop; fast stuck-peer re-eviction.
- Miner: run-ledger Hashes/Height restored; contention fields; noise-gated duty tuner. Harness: contention census and gate G20, S4c GPU self-A/B path, payout ledger and gate G21, S3 runs race-skipping new tests without -race (4ecb0f2), `mine_localdiff` splits every hunk at `+func` (aab15f3).
- Soundness: page `artifacts/dero-soundness-r23.html` (d5b7097) and the HF3 exemption replay tests `walletapi/r23_hf3_replay_test.go` and `blockchain/r23_hf3_table_test.go` (21cd5b7), wired into the pipeline.

WHY

> The live r22 run fell from 84–86k to 43–67k H/s. It was not r22: six renderer processes of another desktop app ran at ~100 % CPU each at nice 0 (load ~42). With them closed (05:38) the same r22 miner sampled 92.5k H/s on 17.31 of 18 budget cores, where the contention law predicts 91,382. So ~90–100k is the box ceiling and the remaining payout levers are uptime and the paid fraction, not H/s. Two of those levers were open: after a final share the miner waited 180–252 ms for the next-height job, and the r22 run ledger read 0 hashes in every row.

CODE

Excerpts in order: the dispatcher decision, newest-last delivery, the emission auditor's checks, the txprecheck ring rules, the RPC ban escalation, the getwork bad-share run, the burn-aware receiver proof binding (CR23-1), the run-ledger fix, the localdiff per-func split, and cases 1–3 of the HF3 exemption replay test.

cmd/derod/rpc/job_dispatch.go · lines 116-135 · commit 1f78612

```
// decide says whether to send a job now. trigger is the time handed to send_job for refresh_due: the moment the sta
// changed (a template stamped at or before it may predate the change), or 0 for a keepalive (no change to miss).
func (d *dispatch_core) decide(st dispatch_state, now int64) (reason string, trigger int64) {
    if !d.primed || st ≠ d.last {
        d.primed, d.last, d.changed_ms, d.retries = true, st, now, 0
        return "change", now
    }
    if d.failed && d.retries < JOB_RETRY_MAX && now-d.changed_ms < JOB_RETRY_WINDOW {
        d.retries++
        return "retry", d.changed_ms
    }
    if due := atomic.LoadInt64(d.due); due ≠ 0 && now ⩾ due && atomic.CompareAndSwapInt64(d.due, due, 0) {
        return "refresh", d.changed_ms
    }
    if now-d.last_send_ms ⩾ d.keepalive_ms {
        return "keepalive", 0
    }
    return "", 0
}
```

cmd/derod/rpc/getwork_dispatch.go · lines 70-84 · commit 1f78612

```
// deliver_job writes msg to w unless this session already got job seq or a newer one. sent=false, err=nil: supersed
func deliver_job(sess *user_session, w job_writer, seq uint64, msg []byte) (sent bool, err error) {
    sess.send_mu.Lock()
    defer sess.send_mu.Unlock()
    if seq ⩽ sess.sent_seq {
        getwork_jobs_superseded.Inc()
        return false, nil
    }
    sess.sent_seq = seq
    w.SetWriteDeadline(time.Now().Add(JOB_WRITE_DEADLINE))
    if err = w.WriteMessage(websocket.TextMessage, msg); err ≠ nil {
        return false, err
    }
    return true, nil
}
```

glue/emission/emission.go · lines 469-500 · commit 1f78612

```
    // 6. the checks
    if !samePoint(dR, sumD) {
        r.alert("Right components moved by something other than the published D values")
    }
    expected := new(bn256.G1).Add(sumC, mulBase(new(big.Int).Add(new(big.Int).SetUint64(sp.Paid), new(big.Int).SetUi
    residual := new(bn256.G1).Add(dL, new(bn256.G1).Neg(expected)) // must be P*G
    r.Verdict = VerdictOK
    if r.SCCalls == 0 { // no contract ran: every plaintext movement is pinned exactly
        if scDelta.Sign() ≠ 0 {
            r.alert("SC-held plaintext DERO changed by %s in a block without an SC call", scDelta)
        }
        if !samePoint(residual, zeroPoint()) {
            r.alert("Left components moved by plaintext value not explained by transfers, coinbase (%d) or registrat
        }
    } else {
        max := p.MaxSCSubset
        if max ⩽ 0 {
            max = 12
        }
        if len(deposits) > max {
            r.Verdict = VerdictSCUnexplained
            r.note("%d SC deposits: payout candidates not enumerated", len(deposits))
        } else if P, ok := matchPayout(residual, deposits, scDelta); ok {
            r.SCPayout = P
            r.Verdict = VerdictOKSC
            if P > r.MaxPlainFlow {
                r.MaxPlainFlow = P
            }
        } else {
            r.Verdict = VerdictSCUnexplained
            r.note("Left residual matches no SC payout candidate (an SC payout that panicked part-way burns the rest
        }
```

glue/txprecheck/txprecheck.go · lines 91–119 · commit 1f78612

```
    for t := range tx.Payloads {
        st := &tx.Payloads[t].Statement
        bpp := int(st.Bytes_per_publickey)
        if bpp == 0 {
            return refuse("ring", "payload %d: zero bytes per key pointer", t)
        }
        n := len(st.Publickeylist_pointers) / bpp
        if st.RingSize != uint64(n) {
            return refuse("ring", "payload %d: ring size %d but %d pointers", t, st.RingSize, n)
        }
        if st.RingSize < 2 || st.RingSize > 128 {
            return refuse("ring", "payload %d: ring size %d outside 2..128", t, st.RingSize)
        }
        if !crypto.IsPowerOf2(n) {
            return refuse("ring", "payload %d: ring size %d not a power of two", t, n)
        }
        seen := make(map[string]bool, n)
        for i := 0; i < n; i++ {
            seen[string(st.Publickeylist_pointers[i*bpp:(i+1)*bpp])] = true
        }
        if len(seen) != n {
            return refuse("ring_duplicate", "payload %d: duplicate ring member", t)
        }
        if uint64(len(st.C)) < st.RingSize || st.D == nil || tx.Payloads[t].Proof == nil {
            return refuse("statement", "payload %d: incomplete statement", t)
        }
    }
    return nil
}
```

cmd/derod/rpc/rpc_abuse.go · lines 226–253 · commit 1f78612

```
// add_locked adds behaviour points and bans when the score is reached; returns the ban length (0 = not banned now).
func (t *abuse_table) add_locked(e *abuse_entry, pts float64, now time.Time) time.Duration {
    if e.overflow { // shared by many keys: scoring it would let attackers ban every new honest client (r23 merge)
        return 0
    }
    e.score = decayed(e.score, e.score_at, now) + pts
    e.score_at = now
    if e.score < ABUSE_SCORE_BAN {
        return 0
    }
    if e.strikes > 0 && now.Sub(e.last_ban) > ABUSE_STRIKE_FORGET {
        e.strikes = 0
    }
    e.strikes++
    d := ABUSE_BAN_BASE
    for i := uint(1); i < e.strikes && d < ABUSE_BAN_MAX; i++ {
        d *= 2
    }
    if d > ABUSE_BAN_MAX {
        d = ABUSE_BAN_MAX
    }
    e.score = 0
    e.banned_until = now.Add(d)
    e.last_ban = now
    abuse_bans.Inc()
    return d
}
```

cmd/derod/rpc/getwork_guard.go · lines 153–172 · commit 1f78612

```
// note returns true when the session must be closed and its key banned (once).
func (g *share_guard) note(bad bool) bool {
    g.mu.Lock()
    defer g.mu.Unlock()
    if g.closed {
        return false
    }
    if !bad {
        g.bad_run = 0
        return false
    }
    getwork_bad_shares.Inc()
    g.bad_run++
    if g.bad_run >= GETWORK_BAD_SHARE_LIMIT {
        g.closed = true
        return true
    }
    return false
}
```

walletapi/payment_proof_verify.go · lines 103–115 · commit 1f78612

```
            // the binding: B + burn*G must be sk*D for the D of this very statement
            burn := tx.Payloads[t].BurnValue
            if burn > amount {
                continue
            }
            bound := blinder
            if burn > 0 {
                bound = new(bn256.G1).Add(blinder, new(bn256.G1).ScalarMult(crypto.G, new(big.Int).SetUint64(burn)))
            }
            if bytes.Equal(new(bn256.G1).ScalarMult(st.D, secret).EncodeCompressed(), bound.EncodeCompressed()) {
                return amount - burn, t, nil
            }
        }
```

cmd/dero-miner/runledger.go · lines 79–84 · commit 1f78612

```
func run_record_from(m miner_metrics, start, now time.Time, exp float64) run_record {
    r := run_record{Kind: "run", RunID: fmt.Sprintf("%x", start.UnixNano()), Version: config.Version.String(),
        Start: start.Format(time.RFC3339), Last: now.Format(time.RFC3339), End: "running", UptimeS: math.Round(m.upt
        Threads: m.threads, CPUMax: cpu_duty_pct(), GPUMax: gpu_duty_pct(), // r22: the caps in force (the tuner may
        Hashes: m.hashes, Height: m.status.height, // r23 forensics: the r22 merge had pushed these two into the com
        Blocks: m.status.accepted_blocks, Minis: m.status.accepted_minis, Rejected: m.status.rejected,
```

harness/miners/mine_localdiff.py · lines 49–60 · commit aab15f3

```
        # R20: a NEW file is one hunk, so its sole anchor was its longest line, and r19 lengthening a line in
        # supply_audit_for moved the only citation out of audit_sc_balances (G16 lost it). R23: a modified file has
        # the same problem when new funcs are appended right after an earlier added func (object_errors.go: the r23
        # decode_block_tx/observe_block_txs joined the r19 deserialize_tx hunk, and a longer comment took its only
        # citation). So every hunk is split at each added top-level func.
        hunks = [h for hk in hunks for h in re.split(r'(?m)^(?=\+func )', hk)]
        for hunk in hunks:
            hadd = [l[1:].strip() for l in hunk.splitlines() if l.startswith('+') and not l.startswith('+++')]
            # a segment that starts at an added func cites that declaration first: its longest line may be the doc
            # comment of the NEXT func, which lies outside this one (R23 object_errors.go)
            first = [hadd[0]] if hadd and hunk.startswith('+func ') else []
            for cand in first + sorted(hadd, key=len, reverse=True):
```

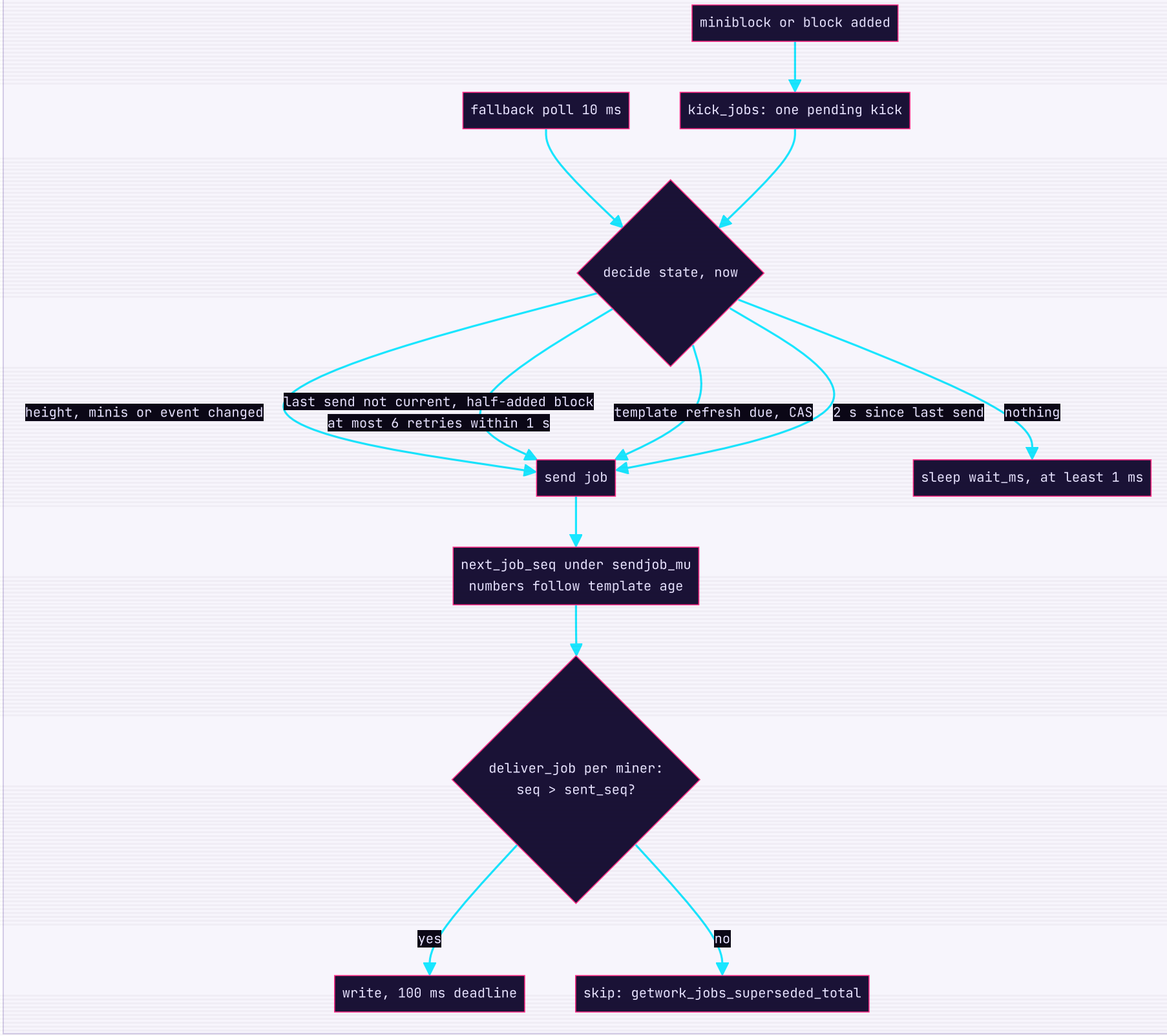
walletapi/r23_hf3_replay_test.go · lines 146–162 · commit 21cd5b7

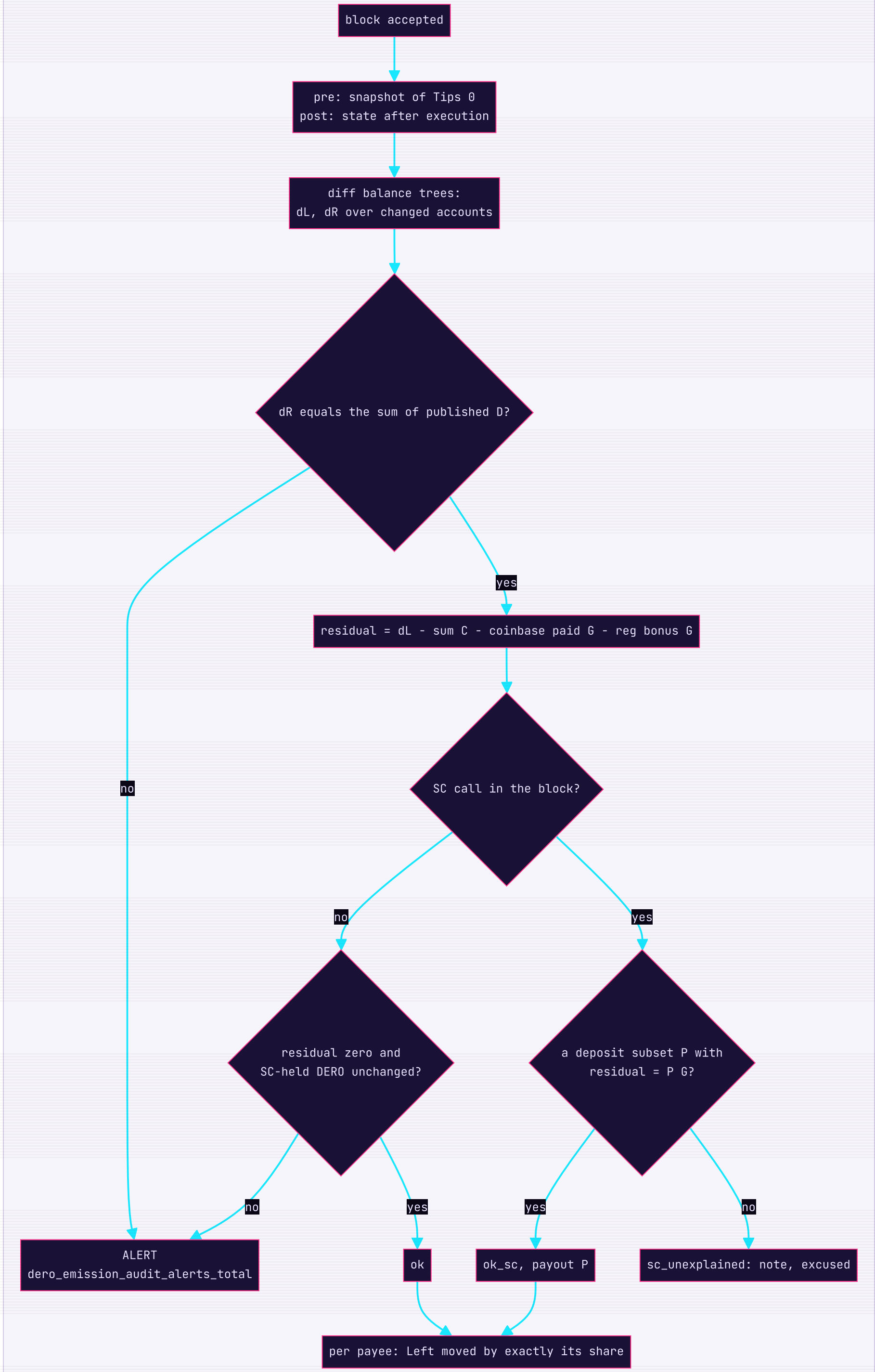
```
    // 1. post-HF3: the exemption does not reach the proof
    if err := chain.Verify_Transaction_NonCoinbase(fresh(forged)); err == nil {
        t.Fatal("post-HF3: an HF3-table TXID let a forged proof verify (exemption is not height-bound)")
    }

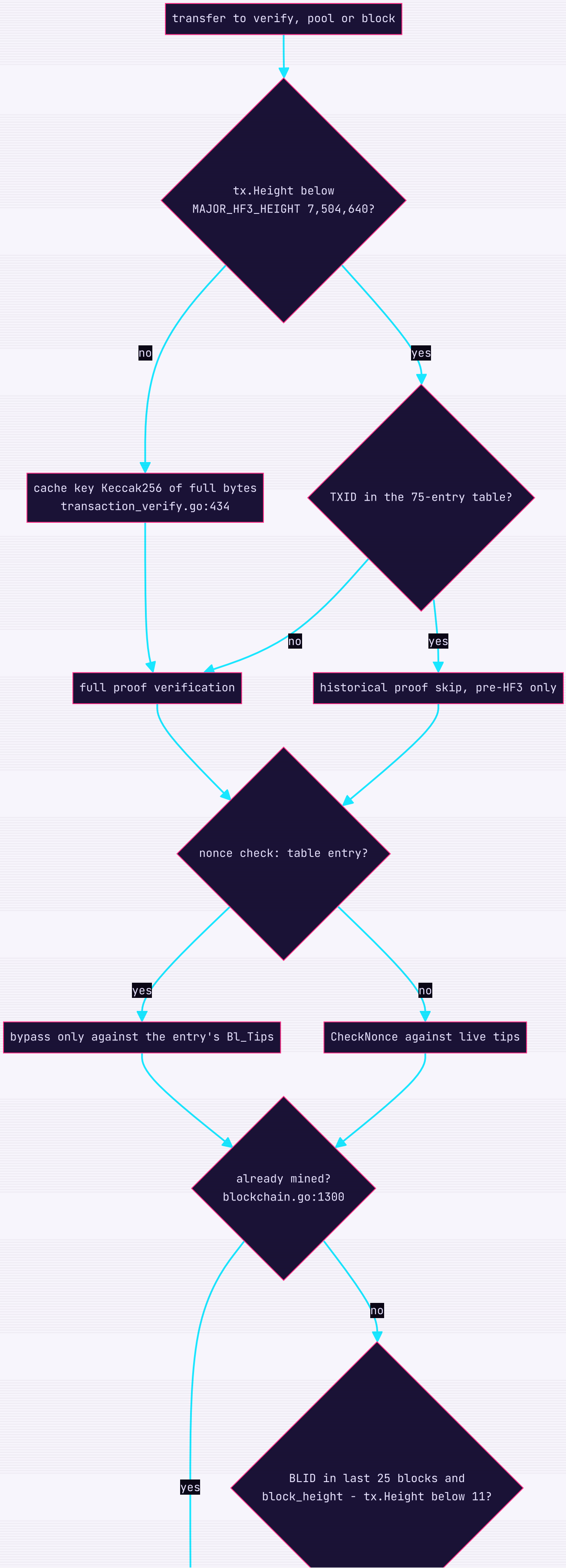
    // 2. pre-HF3: the same entry does skip the proof (the historical bypass exists and this test reaches it)
    globals.Config.MAJOR_HF3_HEIGHT = int64(built.Height) + 1
    err = chain.Verify_Transaction_NonCoinbase(fresh(forged))
    globals.Config.MAJOR_HF3_HEIGHT = hf3
    if err != nil {
        t.Fatalf("pre-HF3: the table entry should skip the proof (historical behaviour), got %v", err)
    }

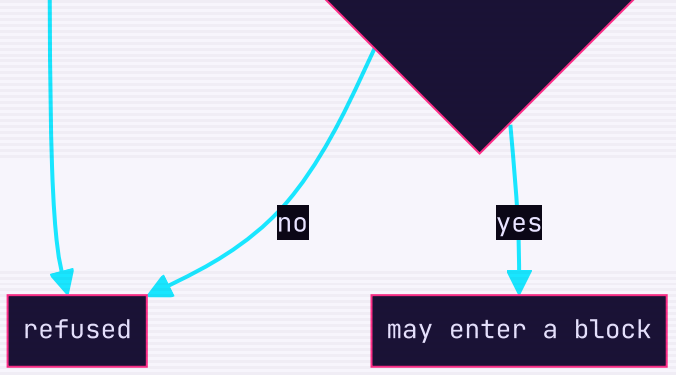
    // 3. back at post-HF3 the TXID-keyed cache entry the exemption left behind is unreachable
    if err := chain.Verify_Transaction_NonCoinbase(fresh(forged)); err == nil {
        t.Fatal("post-HF3: the TXID-keyed cache entry from the exemption let a forged proof verify")
    }
```

FLOW


miniblock or block added
fallback poll 10 ms
kick_jobs: one pending kick
decide state, now
height, minis or event changed
last send not current, half-added block
at most 6 retries within 1 s
template refresh due, CAS
2 s since last send
nothing
send job
sleep wait_ms, at least 1 ms
next_job_seq under sendjob_mu
numbers follow template age
deliver_job per miner:
seq > sent_seq?
yes
no
write, 100 ms deadline
skip: getwork_jobs_superseded_total


block accepted
pre: snapshot of Tips 0
post: state after execution
diff balance trees:
dL, dR over changed accounts
dR equals the sum of published D?
yes
residual = dL - sum C - coinbase paid G - reg bonus G
SC call in the block?
no
no
yes
residual zero and
SC-held DERO unchanged?
a deposit subset P with
residual = P G?
no
yes
yes
no
ALERT
dero_emission_audit_alerts_total
ok
ok_sc, payout P
sc_unexplained: note, excused
per payee: Left moved by exactly its share


transfer to verify, pool or block
tx.Height below MAJOR_HF3_HEIGHT 7,504,640?
no
yes
cache key Keccak256 of full bytes transaction_verify.go:434
TXID in the 75-entry table?
no
yes
full proof verification
historical proof skip, pre-HF3 only
nonce check: table entry?
yes
no
bypass only against the entry's Bl_Tips
CheckNonce against live tips
already mined? blockchain.go:1300
yes
no
BLID in last 25 blocks and block_height - tx.Height below 11?



EVIDENCE

MEASURED	RESULT	SOURCE
Workflow wf_f6115a56-623	23 agents; patch 108 files, merged as 1f78612	RUN_REPORT Round 23
Live r22 dip	84–86k → 43–67k H/s: six renderer processes ~100 % CPU each, nice 0, load ~42; not r22	RUN_REPORT Round 23
Same r22 miner, renderers closed 05:38	92.5k H/s at 17.31/18 cores; law 91,382 (G20 census quiet)	RUN_REPORT Round 23; ARCHITECTURE §39.1
Contended census 01:53, 15 s	miner 11.32 of 18 cores, foreign 6.59 (renderers ×17: 4.73); loss estimate 32k H/s	ARCHITECTURE §39.1
Law vs sample	64,552 predicted vs 63,687 sampled (-1.3 %); ps 11.74 vs process_cpu 11.78 cores	ARCHITECTURE §39.1
Final share → next-height job	252 → 32 ms (own blocks), 180 → 30 ms (P2P blocks)	RUN_REPORT Round 23; ARCHITECTURE §39.5
First hash after restart / first job after connect	5.5 → 0.5 s / 0.84–1.0 s → 6–37 ms (loopback)	ARCHITECTURE §39.5
Payout gain of the dispatcher	+0.3–2 % (≈1.5 %), estimate until 24 h of getwork_stale_height_jobs_total / blocks	RUN_REPORT Round 23
r22 run ledger	hashes 0, height 0 in every row (fields inside a comment); fixed, TestRunRecordFrom, mutation gm-r23-runledger-hashes-height CAUGHT	ARCHITECTURE §39.4; commit 1f78612
r23 run ledger, 07:07:52 run	120,515,560 hashes over 1,375 s (87,648 H/s) vs avg_hs 87,668; height 7,831,820	run ledger bin/dero-miner.runs.jsonl
HR23-1 "+17 % at --cpu-max=90"	REFUTED · 90 vs 100 within noise	RUN_REPORT Round 23
Payout ledger 01:51, seven runs	506.7 shares expected, 479 found (luck 0.945, [437.06, 523.88]), 466 paid; 14.33 of 15.58 DERO	ARCHITECTURE §39.3
Payout ledger 06:09 (G21)	paid 680 of 758.57 expected (20.91 of 23.33 DERO); found 699 in [648.13, 752.8]: below expectation beyond luck, contended hours included	ARCHITECTURE §39.3
Reward per miniblock at height 7,830,715	123000 >> 2 = 30,750 atomic per block ÷ 10 = 3,075	ARCHITECTURE §39.3
Emission auditor CPU	4.9–5.2 ms/block offline; live mean 6.7 ms	RUN_REPORT Round 23; dero-soundness-r23.html §1
Emission auditor, offline mainnet copy	13,100+ blocks between 7.50M and 7.83M clean	dero-soundness-r23.html §1
Emission auditor, live after the 06:52 restart	2,196 audited, 0 alerts; observed 66,850,500 = formula; payees 13,223 / 13,229 verified	dero-soundness-r23.html §1
Emission auditor, live read 19:16	2,456 audited, 0 alerts; observed 74,722,500 = formula; mean 6.70 ms, max 22.6 ms; 26 reorgs	DERO.GetEmissionAudit, 2026-10-30 19:16
Mining outcome since 06:52, read 19:16	37 submitted, 33 paid, 2 orphaned (slots_full), 2 pending; paid fraction 0.943	DERO.GetMiningOutcome, 2026-10-30 19:16
Node, read 19:16	synced, lag 0; 13 outgoing, 0 inbound; 8 tx-fetch failures, 0 bans	DERO.GetSyncInfo, 2026-10-30 19:16
Automated-client bans	unknown method 20/min, fuzz ~34, hammer ~100; IPv6 /64	ARCHITECTURE §39.6
Remote RPC bucket	200 weighted calls/s, burst 2000; ban at score 100 (half-life 60 s), 30 s doubling to 1 h	commit 1f78612 cmd/derod/rpc/rpc_abuse.go
Pipeline runs 1 and 2	run 1 lacked --contended-ok; run 2 FAIL S3 (test skips under -race) → 4ecb0f2	RUN_REPORT Round 23
pipeline_20261030T043422	new tests 116 ×3 0 flaky; mutations 743/743; race 25 packages 0 races; S4b ×0.999; S4c SKIP (server unchanged)	RUN_REPORT Round 23; ARCHITECTURE §39.10
Shadow	120/120 blocks, 240 treehashes, 0 mismatches; install r23	RUN_REPORT Round 23
S7 failures	G16 (localdiff hunk merge lost deserialize_tx → aab15f3), then DK18 (test comment reworded, 9ae3e7d)	RUN_REPORT Round 23
--project-only pipeline_20261030T062228	PASS: KATs 34/34, mutations 31/31, gates 21/21	RUN_REPORT Round 23
HF3 soundness tests (21cd5b7)	Test_R23_HF3_Exemption_Is_Height_Bound PASS (8.87 s, no -race); TestR23HF3TableIsHistorical, TestR23TxidCommitsToHeightNotProof PASS	go test at 21cd5b7, 2026-10-30

SOUNDNESS REVIEW: FOUR CONCERNS AND THE 23.9M CLAIM

Summarised from the project's soundness review page, dero-soundness-r23.html (same folder), which holds the evidence for each row. No consensus code was changed and the chain is not forked. The page describes the defect class only; no coin-creating proof was built or described.

CONCERN	STATUS	WHAT THE REVIEW STATES	SOURCE
1 · Alleged inflation	Checked every block	Visible issuance (block rewards, fees, smart-contract payouts) is checked on every block: the r23 emission auditor has 0 alerts. Inflation hidden inside encrypted amounts is still invisible: total_supply is CalcSupply(height), a pure formula, and would print the same number on an inflated chain; balances are ElGamal ciphertexts that no one can add up.	dero-soundness-r23.html §1
2 · Double-spend chaos	Fixed by HF3	Real in code and closed by the HF3 hard fork at height 7,504,640 (21 August 2026). A transaction's ID covers only its header, not its proof bytes, so two bodies could share one ID; before HF3 the store kept whichever arrived last. HF3 rejects stored bytes that differ from the incoming body, keys the validity cache on Keccak256 of the full transaction, and rejects duplicate registrations. R22 adds byte-exact re-serialisation at admission; R23 refuses structurally invalid transactions before any database read. Still open: the ID still excludes the proof (a consensus rule); blockchain/storefs.go writes non-atomically (reported only); 75 historical transactions (67 "duplicates", 8 "false parity", heights 7,415,466–7,497,942) are exempt from proof verification, but only below HF3. The cache key is the bare ID only when the transaction's reference height is below 7,504,640 (transaction_verify.go:434), and no block after HF3 can carry them, because of the 11-block validity window (corrected in R23b).	dero-soundness-r23.html §2
3 · Range-proof soundness	Open, needs a fork	Disclosed by the Foundation on 2026-10-20 and still in the code. The audit confirms a soundness defect exists (C1) and refutes the stated mechanism (C2): no generator is adjustable; the free parameter is the scalar μ, which no challenge hash binds, with the same consequence. Sites: cryptography/crypto/proof_verify.go:430 and :452, proof_innerproduct.go:307. Changing the verifier changes which proofs are valid and forks this node off the network (gate G13 blocks that on purpose); the Foundation ended support with no corrective fork, so no release carries a fix.	dero-soundness-r23.html §3
4 · Soundness of Release 151	Consensus-exact	Content-identical to upstream Release153 plus node and miner enhancements that do not change consensus: it agrees exactly with the network and carries the network's open range-proof defect. G13: every consensus-package change since upstream is logging-only (7 files); R23 shadow 120/120 blocks and 240 tree hashes, 0 mismatches.	dero-soundness-r23.html §4
The 23,886,517.3 DERO claim	Not checkable	"Spendable coin created (floor): 23,886,517.3 DERO" is the Foundation's 2026-10-20 press text, recorded as an external claim. It concerns the HF3 double-spend, not the range-proof defect; audit verdict C6 is unchanged: PARTLY-CONFIRMED. Confirmed in code: MAJOR_HF3_HEIGHT = 7504640, the duplicate-execution mechanism and the 75-entry grandfather table. The table holds no amount field. Correction (R23b): this node's store begins at 7,498,540, its fastsync bootstrap, which is above all 75 affected heights (the highest is 7,497,942). The archived pre-HF3 datadir holds exactly one of the 75 bodies. Even an unpruned archive holds only one body per ID, because the store overwrites: it shows the body it kept, never the one it lost, and the amounts stay encrypted. Only the Foundation's off-chain method can produce the figure: treat it as their claim, unverified.	dero-soundness-r23.html, "The 23,886,517.3 DERO claim"

HF3 EXEMPTION: WHAT THE R23 REPLAY TEST PINS

- Proof skip is bound to pre-HF3 heights by the cache key: `blockchain/transaction_verify.go:434` uses the TXID only when `tx.Height < MAJOR_HF3_HEIGHT`, else `Keccak256(tx.Serialize())`. A same-TXID body with one changed proof byte is refused at a post-HF3 height and, after a pre-HF3 skip, the TXID-keyed cache entry it left behind is unreachable.
- The nonce bypass answers only for the entry's own `Bl_Tips`; against the live tips a replay of a mined transfer fails the nonce check and the pool refuses it.
- Execution swaps in the entry's `Parity` (`blockchain/transaction_execute.go:244`); the 75 literals set only `Tx_Parity` and `p2p/chain_sync.go:397` sets `Parity` only for blocks below HF3 during sync. A mismatched Parity leaves the real nonce slot unadvanced, yet the replay is still refused by the already-mined guard (`blockchain/blockchain.go:1300`) and by the block rules: BLID in the last 25 blocks and `block_height - tx.Height < TX_VALIDITY_HEIGHT` = 11 (`blockchain/blockchain.go:760-790`).
- The TXID (`Keccak(SerializeHeader)`) commits to Height, BLID and the payload statements and excludes the proofs (`TestR23TxidCommitsToHeightNotProof`); all 75 table entries lie below 7,504,640 at heights 7,415,466–7,497,942 with one 32-byte tip each (`TestR23HF3TableIsHistorical`).

R23b · 2026-10-30 · +22102026.r23b

UPTIME LEVER; HF3 PARITY ON EVERY PATH; ADMISSION WINDOW; EMISSION ARCHIVE

uptime 0.312 measured gates 22/22

WHAT CHANGED

- Payouts: `--supervised` makes the miner ignore a typed console exit/quit/bye (it stops only on a signal, which the run ledger records); `tools/launchd` writes a launchd plist (KeepAlive, Nice 0, absolute miner path, always `--supervised`); derod exports `getwork_miner_absent_seconds`

and logs a rate-limited WARNING when no miner has been connected for 10 min, then hourly; opt-in `--phase-duty` (off by default) with per-phase metrics.

- Hashrate: `--cpu-max` is charged on the CPU the mining threads actually got (`cmd/dero-miner/cpushare.go`): on a quiet box the sleep is exactly the r22 sleep; `DERO_CPU_DUTY_CLOCK=wall` restores the wall-clock rule. Clang PGO of the C sort was tried and refuted.
- Double-spend and crypto: the HF3 table Parity is applied on every way a block reaches the chain (sync, chunked, compact, rewind, console import), atomically under `hf3_exec_mu` (CR23b-1); relay refuses non-canonical registration keys and unreduced signatures (CR23b-2, policy only); random getwork TLS certificate serial (CR23b-3).
- Security: `rpc.serverInfo` and the other jrpc2 built-ins are closed to remote and cross-site callers (SEC23b-1); a global anomaly budget across all remote keys scores anomalies ×4 during a probing surge (SEC23b-2); `SendRawTransaction` reads the height from the raw header and refuses out-of-window transactions before `Deserialize` (SEC23b-3), so an unknown tx type no longer panics there (SEC23b-4); relay and RPC refuse the 75 grandfathered HF3 TXIDs by ID before verification (SEC23b-5).
- Inflation monitoring: the emission auditor books fees, SC flows, payees and burns per replaced block under reorgs (FOR-5, SND-4); archive mode and an HF3 forensic report in `tools/emission-audit` (SND-1, SND-2); a crash-safe series file and `issued_emission` in `DERO.GetEmissionAudit` (SND-3); `DERO.GetMonetaryIntegrity` with a table digest anyone can recompute.
- RPC/ws and p2p: settled-transaction cache for `DERO.GetTxStatus` (verify on hit); a `Reorg` websocket event and a live finality monitor (a local pop or `Rewind_Chain` is `local_rewind`, never a violation); lazy sync height gate (SP23b-4); `tx_precheck_checked_total` (FOR-6).
- Harness: gate G22 and stage S3s back every claim unit of the soundness page; the idle ledger prices the gaps between miner runs; G17 shows the binding label next to the regime label; G20 fails with no A/B record; S3a re-runs tests that skip under -race; `mine_localdiff` cites a repeated line at its own copy; DK18 catches unlabelled seed phrases. Test helper fix bfe3e54: `with_clients` cleanup restores `getwork_session_count`.
- Soundness page `artifacts/dero-soundness-r23.html` corrected (SND-7); the two R23 rows that quoted the old wording are corrected on this page too (Round 23 soundness review).

WHY

> The r23 miner ran well (88–92k H/s run means), but it ran only 31 % of the 72 h before 19:10: all nine runs in the run ledger ended with a typed console exit, and the longest gap was 10.5 h. The idle ledger prices that at about 50.5 DERO forgone against 23.1 DERO paid (estimate), so uptime, not H/s, is the largest payout lever. A final-phase hash earns 1/9 of a regular-phase hash while the final phase takes 51 % of the wall time. The platform dimensions found four gaps: jrpc2 answered rpc.serverInfo to remote callers (method list, start time, counters that include loopback traffic); per-key scoring let many addresses probe just under their ban lines; SendRawTransaction admitted valid but stale transfers that can never be mined, and panicked on an unknown tx type; and the HF3 table Parity was set only on the sync path, so a historical block relayed as chunks could execute a table transaction with a stale value and stall a node replaying the pre-HF3 range. The soundness review had stated that this node was pruned above all but the last few affected heights; the archive audit showed its store begins above all 75.

CODE

Excerpts in order: the HF3 helper every block-add path now calls (CR23b-1), the SendRawTransaction header peek and admission call (SEC23b-3), the window rule, rpc.serverInfo closed to remote callers (SEC23b-1), the global anomaly budget (SEC23b-2), the verify-on-hit lookup of the settled GetTxStatus cache, the CPU-share duty sleep, and the --supervised console exit.

p2p/hf3_parity.go · lines 84-100 · commit 3224c07

```
// hf3_add runs add for cbl, preparing the HF3 table first (and atomically with add) when cbl carries a table tx.
func hf3_add(cbl *block.Complete_Block, verify func(*transaction.Transaction), add func(*block.Complete_Block) (erro
    if cbl == nil || cbl.Bl == nil || !hf3_carries(cbl.Bl.Height, cbl.Txs) {
        return add(cbl)
    }
    hf3_exec_mu.Lock()
    defer hf3_exec_mu.Unlock()
    metrics.Set.GetOrCreateCounter("p2p_hf3_table_blocks_total").Inc()
    apply_hf3_parity(cbl.Bl.Height, cbl.Txs, verify)
    return add(cbl)
}

// Add_Complete_Block_HF3 is c.Add_Complete_Block with the HF3 table prepared exactly as the sync path prepares it.
// Every caller outside the blockchain package that offers a block to the chain uses it (TestR23bAddCompleteBlockCen
func Add_Complete_Block_HF3(c *blockchain.Blockchain, cbl *block.Complete_Block) (error, bool) {
    return hf3_add(cbl, func(tx *transaction.Transaction) { c.Verify_Transaction_NonCoinbase(tx) }, c.Add_Complete_B
}
```

cmd/derod/rpc/rpc_dero_sendrawtransaction.go · lines 56-81 · commit 3224c07

```
    // r23b: a height no block on top of our tip can hold, or a header Deserialize refuses (an unknown type panics
    // there, and the recover below would log a full stack at ERROR per call), is refused from the header alone,
    // before any curve point is decoded (glue/txprecheck/window.go)
    tip_height := send_tx_tip()
    if err = txprecheck.PeekWindow(tx_bytes, tip_height); err ≠ nil {
        err = fmt.Errorf("Transaction rejected by daemon err '%s'", err)
        return
    }

    // fmt.Printf("txbytes length %d data %s\n", len(p.Tx_as_hex), p.Tx_as_hex)
    // lets add tx to pool, if we can do it, so  can every one else
    err = tx.Deserialize(tx_bytes)
    if err ≠ nil {
        return
    }

    // lets try to add it to pool
    // r22: refuse a tx whose re-serialization (what we relay, store and mine) no node could decode (glue/txcanon)
    // r23: and one whose structure consensus refuses, before any state is read for it (glue/txprecheck)
    // r23b: and a grandfathered HF3 TXID or an out-of-window height (AdmitAt), exactly as the p2p relay path does;
    // the txid is computed once here and reused (it was computed for the reply anyway)
    txid := tx.GetHash()
    if err = txprecheck.AdmitAt(&tx, txid, tip_height, func(tx *transaction.Transaction) error { return txcanon.Admi
        p2p.Broadcast_Tx(&tx, 0) // broadcast tx
        result.Status = "OK"
        result.TXID = fmt.Sprintf("%s", txid)
```

glue/txprecheck/window.go · lines 111-125 · commit 3224c07

```
// Window refuses a transaction height that no block extending our tip (at tip_height) can hold.
func Window(tx_height uint64, tip_height int64) error {
    if tip_height < 0 {
        return nil
    }
    if uint64(tip_height) > tx_height+blockchain.TX_VALIDITY_HEIGHT {
        window_too_old.Inc()
        return ErrWindow
    }
    if tx_height > uint64(tip_height) {
        window_too_new.Inc()
        return ErrWindow
    }
    return nil
}
```

cmd/derod/rpc/rpc_abuse.go · lines 403-426 · commit 3224c07

```
// jrpc2 serves the reserved "rpc." names itself, before any assigner sees them, unless DisableBuiltin is set. Its
// rpc.serverInfo returned to ANY caller, remote included: the full method list, the node's start time (restart
// detection) and the server's request and byte counters, which on the shared /json_rpc bridge count loopback calls
// too, so a remote poller could watch the local wallet's and miner's RPC activity (measured: a remote request read
// rpc.requests 4 after 3 loopback pings). On a remote websocket the call also bypassed remote_assigner, so it was
// neither charged nor scored. Remote callers now get the method-not-found answer of any unknown name (and its score
// loopback keeps the built-ins.
var remote_options = remote_server_options(options)

func remote_server_options(o *jrpc2.ServerOptions) *jrpc2.ServerOptions {
    c := *o
    c.DisableBuiltin = true
    return &c
}

// ws_server_options: a remote socket, and a cross-site page's socket over loopback (a web page must not watch the
// local wallet's RPC activity either), get no built-ins; the user's own loopback tools keep them.
func ws_server_options(c *ws_client) *jrpc2.ServerOptions {
    if c.abuse_key ≠ "" || c.foreign {
        return remote_options
    }
    return options
}
```

cmd/derod/rpc/rpc_abuse.go · lines 81-90 · commit 3224c07

```
// r23b: global (all remote keys together) anomaly budget. Per-key scoring alone lets a fleet of N addresses each
// stay just under its ban line: at ABUSE_HALF_LIFE 60 s a key settles at score = rate*pts*86.6 s, so a key sending
// fewer than 0.23 unknown names/s (13.9/min) is never banned, and N keys probe 13.9*N names/min. Anomalies (unknown
// names, invalid params/requests, parse errors) from every remote key draw on one bucket; while it is empty the nod
// is in a probing surge and every anomaly scores ABUSE_SURGE_FACTOR times its points, so the per-key line drops to
// 3.5/min. An honest client is unaffected unless it ITSELF sends several invalid calls a minute during a surge;
// loopback and the foreign key never reach this code; successes and ordinary handler errors never draw on it.
const ABUSE_GLOBAL_ANOMALY_RATE = 10.0   // anomalies/s across all remote keys before a surge
const ABUSE_GLOBAL_ANOMALY_BURST = 600.0 // a minute of the rate
const ABUSE_SURGE_FACTOR = 4.0
```

cmd/derod/rpc/tx_status_final_cache.go · lines 74-104 · commit 3224c07

```
// lookup returns the cached block of txid if blid_at(topo) still names it; a mismatch drops the entry.
func (c *final_tx_cache) lookup(txid crypto.Hash, blid_at func(int64) (crypto.Hash, error)) (e final_tx_entry, ok bo
    c.Lock()
    if !c.on {
        c.Unlock()
        return e, false
    }
    e, ok = c.by_tx[txid]
    if !ok {
        if e, ok = c.old[txid]; ok {
            c.by_tx[txid] = e // promote
            delete(c.old, txid)
        }
    }
    c.Unlock()
    if !ok {
        final_tx_misses.Inc()
        return e, false
    }
    if b, err := blid_at(e.topo); err ≠ nil || b ≠ e.blid { // verify on hit
        c.Lock()
        delete(c.by_tx, txid)
        delete(c.old, txid)
        delete(c.by_bl, e.blid)
        c.Unlock()
        final_tx_stale.Inc()
        return final_tx_entry{}, false
    }
    final_tx_hits.Inc()
    return e, true
}
```

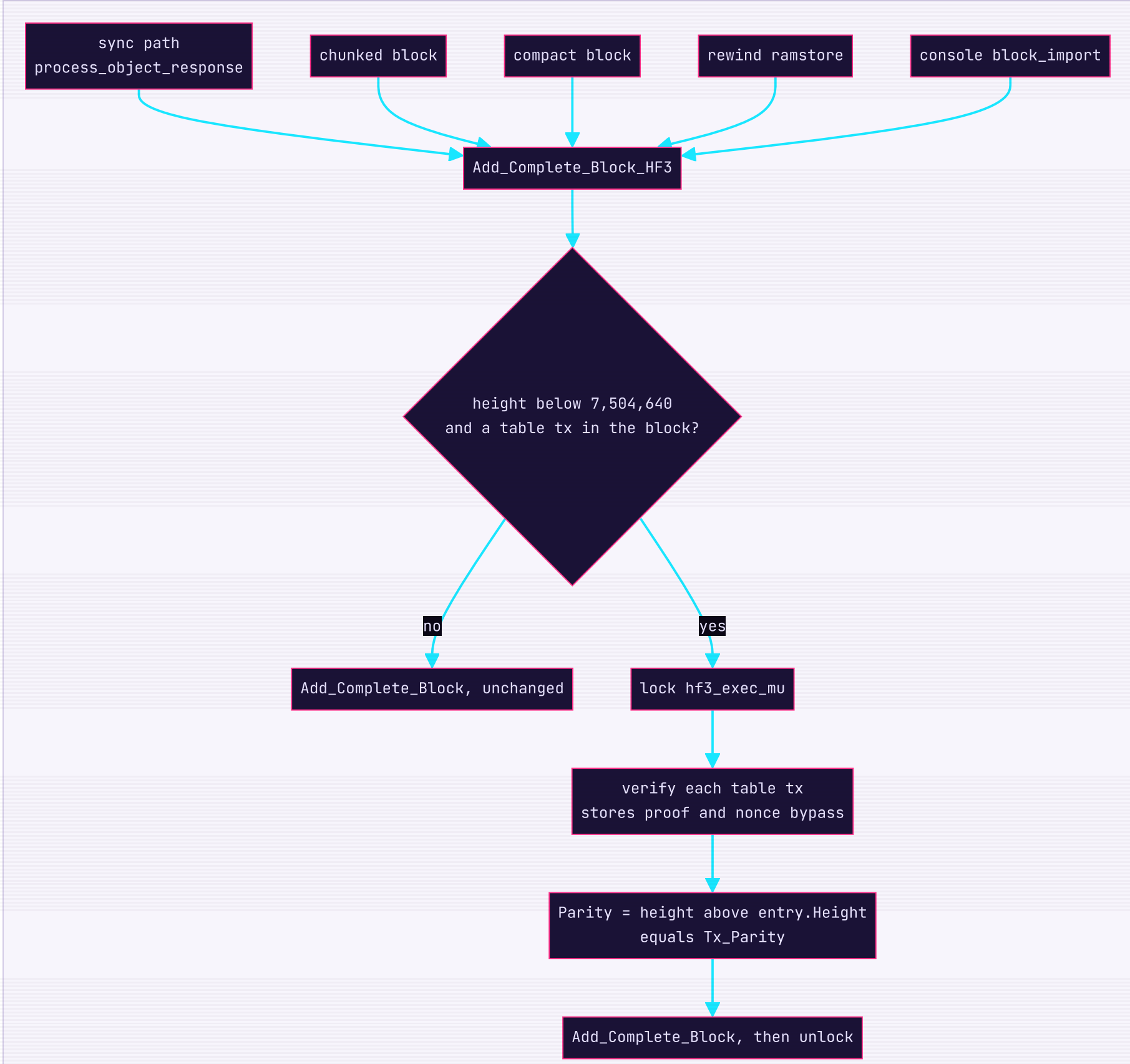
cmd/dero-miner/cpushare.go · lines 73-87 · commit 3224c07

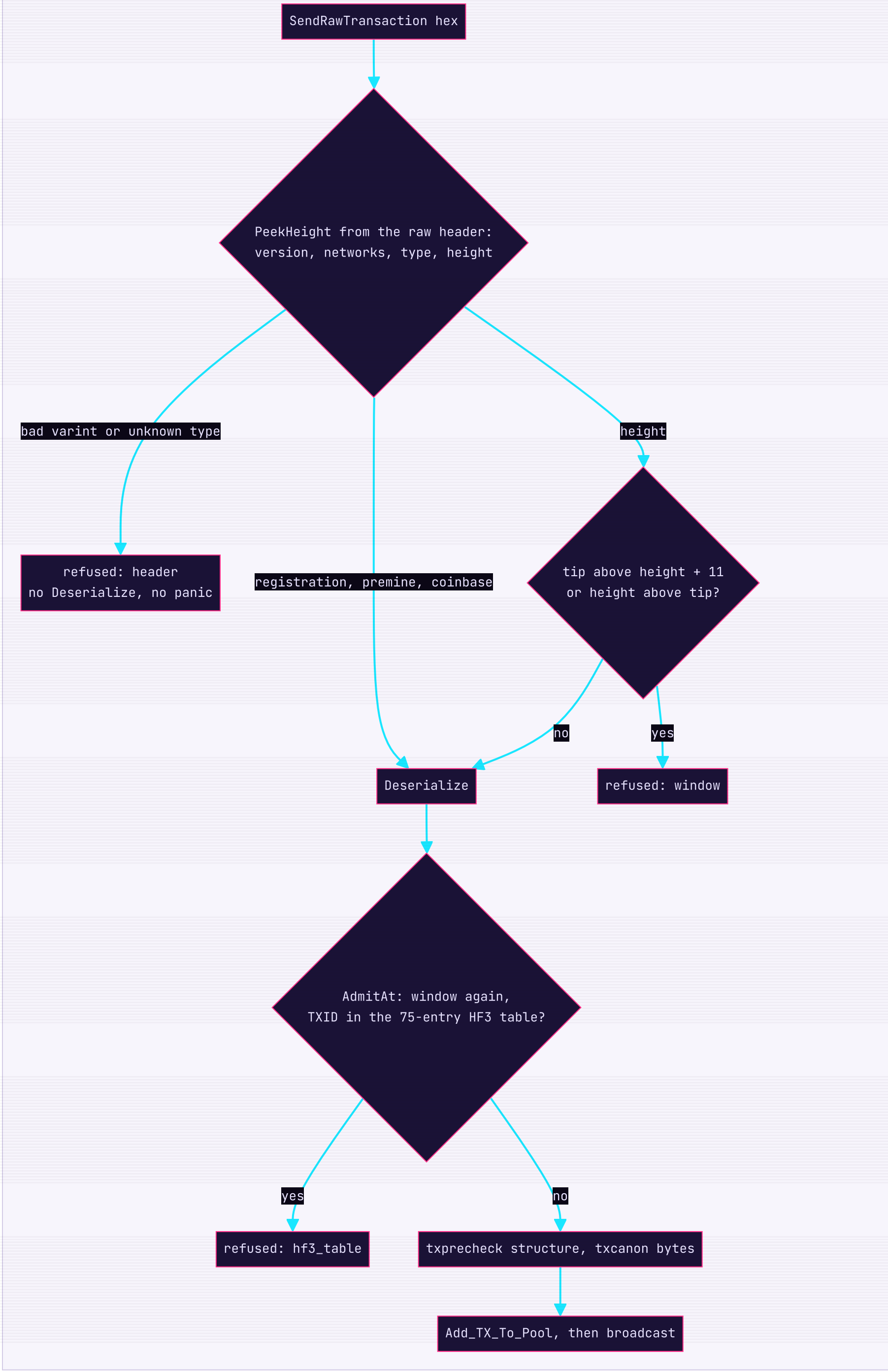
```
// throttle_sleep: the duty sleep after busy wall time at cap duty (percent) for a thread that was on a CPU for
// share/1000 of it. share 1000 gives busy*(100-duty)/duty exactly (the r22 sleep).
func throttle_sleep(busy time.Duration, duty int, share int64) time.Duration {
    if duty ≥ 100 || duty < 1 || busy ≤ 0 {
        return 0
    }
    if share > 1000 {
        share = 1000
    }
    num := share*100 - 1000*int64(duty)
    if num ≤ 0 {
        return 0
    }
    return time.Duration(int64(busy) * num / (1000 * int64(duty)))
}
```

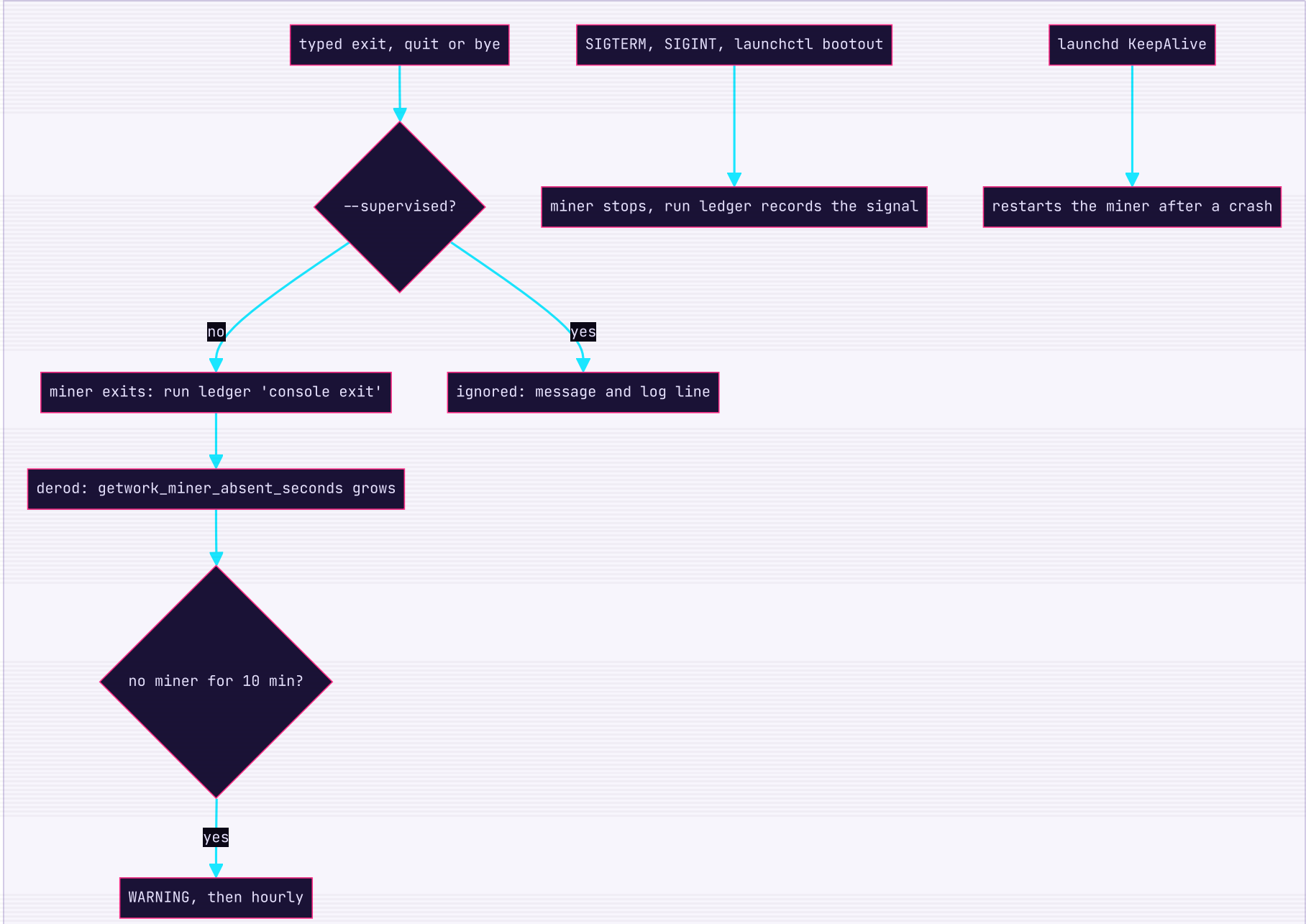
cmd/dero-miner/miner.go · lines 146-159 · commit 3224c07

```
// console_exit handles a typed exit word: ignored (message, log) when supervised, else exit("console <word>").
// It reports whether exit was called. The console loop calls it for every exit word (TestConsoleLoopWiresExit).
func console_exit(line string, supervised bool, out io.Writer, exit func(reason string)) bool {
    w := strings.ToLower(strings.TrimSpace(line))
    if !console_exit_allowed(supervised) {
        fmt.Fprintln(out, supervised_exit_message)
        logger.Info("console exit ignored", "reason", "--supervised", "line", w)
        return false
    }
    logger.Info("miner exiting", "reason", "console "+w)
    exit("console " + w)
    return true
}
```

FLOW


sync path
process_object_response
chunked block
compact block
rewind ramstore
console block_import
Add_Complete_Block_HF3
height below 7,504,640
and a table tx in the block?
no
yes
Add_Complete_Block, unchanged
lock hf3_exec_mu
verify each table tx
stores proof and nonce bypass
Parity = height above entry.Height
equals Tx_Parity
Add_Complete_Block, then unlock


SendRawTransaction hex
PeekHeight from the raw header:
version, networks, type, height
bad varint or unknown type
refused: header
no Deserialize, no panic
registration, premine, coinbase
height
tip above height + 11
or height above tip?
no
yes
Deserialize
refused: window
AdmitAt: window again,
TXID in the 75-entry HF3 table?
yes
no
refused: hf3_table
txprecheck structure, txcanon bytes
Add_TX_To_Pool, then broadcast



EVIDENCE

MEASURED	RESULT	SOURCE
Workflow (R23b)	23 agents over 11 dimensions; merged as 3224c07 (83 files), test-helper fix bfe3e54	R23b workflow summary; commits 3224c07, bfe3e54
Miner uptime, 72 h to 19:10	connected 22.44 of 72 h (0.312); 9 of 9 runs ended "console exit"; longest gap 10.5 h	R23b workflow PAY-1; commit 3224c07 cmd/dero-miner/miner.go
Idle ledger since 2026-10-28 20:20	idle 31.4 of 47.2 h (uptime 33 %); ≈50.5 DERO forgone (≈45.6 at realised luck) against 23.1 DERO paid, estimate	R23b workflow H23b-2; merge report
--supervised shadow test	a typed exit left the --supervised miner running and stopped the plain one	R23b workflow PAY-1
Uptime payout estimate	at ~99 % uptime under launchd: ≈ +27 DERO/day by the merge formula, +31.8 by the payouts dimension (estimate, assumes the typed exits were unintended)	merge report; R23b workflow PAY-1
Phase economics, 400 mainnet blocks	final-phase hash earns 1/9 of a regular one; final phase 0.511 of wall time; mean block 18.46 s	R23b workflow PAY-3
--phase-duty=balanced	-1.7 % to +8.9 % DERO/day, estimate; off by default; GPU stays at --gpu-max	merge report
Mining outcome since 06:52, read 19:28	46 submitted, 44 paid, 2 orphaned (slots_full); paid fraction 0.957; found 46 vs 43.47 expected (z = +0.38)	DERO.GetMiningOutcome 19:28; R23b workflow FOR-2
The two orphans	9th-slot arrival races, not software faults: 2 × 0.03075 = 0.0615 DERO; orphans 2.7 % of regular shares	R23b workflow FOR-3, PAY-4
Payout-path counter getwork_stale_height_jobs_total	225 over 152 miner-connected blocks = 1.48 per block; dead time ≤ ~5–10 ms per 18.5 s block, under 0.1 % (estimate)	derod /metrics 19:28; R23b workflow FOR-4
Regime of the 19:10 run	CPU-bound: H/s tracks cores (r = 0.993) at ~5.45k H/cpu-s; GPU busy 90–92 % of its 96 % cap	R23b workflow FOR-8
Agent work at nice 15	miner ~85k → 49k H/s at load 60–115; a 6-thread nice-15 job takes ~2.6 cores (~15k H/s, estimate); utility QoS ×0.12 of that	R23b workflow HR23b-2
--cpu-max on CPU share, contended proxy	wall 90 vs no cap ×0.861; CPU-share 90 vs wall 90 ×1.037 and ×1.123; CPU-share 90 vs no cap ×0.995. Live +4–12 % while contended is an estimate	R23b workflow HR23b-3; commit 3224c07 cmd/dero-miner/cpushare.go
GPU geometry 8192 (proxy)	staged capacity ×1.34–1.47 (8192×2), ×1.52–1.74 (8192×3); live ≈ 0 while CPU-bound, +5–8 % on a quiet box (estimate)	R23b workflow HR23b-4
Clang PGO of the C descriptor sort	REFUTED · ×0.99–1.01, bit-exact	R23b workflow HR23b-5
Ceiling recomputed	~100k H/s; 101.8–104.2k on all 20 cores with nothing else running; 150k needs ×1.44 per-core sort speed	R23b workflow HR23b-6
Outgoing peers	14 = every usable known peer (17 known − 2 stuck − 1 v2-chain); 0 inbound	DERO.GetSyncInfo 19:13; R23b workflow SP23b-1
Loopback catch-up profile	miniblock PoW 43 %, Add_Complete_Block 24 %, p2p's own code 4.2 %	R23b workflow SP23b-3

MEASURED	RESULT	SOURCE
Emission auditor during sync (A/B)	`off 19.23, 22.18 blocks/s; on 23.83, 25.7, 16.82 blocks/s (load 58–90): no measurable cost`	R23b workflow SP23b-3
Lazy sync height gate	Get_Height reads 1,020 → 267 in the stream test; CPU 41.28 → 41.28 ms/block (n = 6 + 6)	R23b workflow SP23b-4
Settled GetTxStatus cache, mainnet clone p50	`1 txid 23.7 / 18.1 → 1.0 / 0.9 ms (~20×); 8 txids 105 / 84 → 1.2 / 1.4 ms (60–70×)`	R23b workflow RPCWS23B-A
Reorg event	`push order Reorg, Block, Height; observer 0.15–0.34 ms per block; live auditor had seen 26 reorgs with no client signal`	R23b workflow RPCWS23B-B, FOR-5
GetMonetaryIntegrity	~1 ms p50 per call, against 1.1–1.9 ms for GetEmissionAudit	R23b workflow RPCWS23B-C
rpc.serverInfo	`before: a remote request read rpc.requests 4 after 3 loopback pings; now remote HTTP, remote ws and foreign ws get -32601 with no start time or counters; loopback keeps it`	commit 3224c07 cmd/derod/rpc/rpc_abuse.go; R23b workflow SEC23b-1
Global anomaly budget, simulated attack	`1000 keys × 8 probes over 70 s: 0 banned before, 900 after; unbanned probe rate per key 13.9 → 3.5 per min`	R23b workflow SEC23b-2
Stale transfer on SendRawTransaction	`Deserialize ~2.5 ms → header refusal 1.6–2.0 µs`	R23b workflow SEC23b-3
Unknown tx type on SendRawTransaction	`recovered panics per call 1 → 0`	R23b workflow SEC23b-4
HF3 TXID refusal	`65–77 ns/op, 0 allocs`	R23b workflow SEC23b-5
HF3 Parity outside the sync path	`chunked, compact, rewind and console-import paths left it stale; table entries with Tx_Parity false need Parity = true in their own block (58 of 75)`	commit 3224c07 p2p/hf3_parity.go; R23b workflow CR23b-1
Same-TXID second body after HF3 (simulator)	`refused by the byte-mismatch rule and never executes; mutation CAUGHT`	R23b workflow SND-6; dero-soundness-r23.html §2
Emission auditor, archive mode	`19,183 more mainnet blocks, 0 alerts: 14,222 pre-HF3, 960 at the start of this store, 4,001 recent; issued = formula in every range; 4.48–5.66 ms CPU per block`	R23b workflow SND-1; dero-soundness-r23.html §1
HF3 forensic report	`75 entries in 0.065 s CPU; 1 of 75 bodies in any local store; its block audits clean`	R23b workflow SND-2
Burns in the plaintext-flow watch	`never seen before (a BURN_TX value is not serialised); now counted per block`	R23b workflow SND-4
G22 soundness gate	`86 page units; 32/32 on-page and 6/6 off-page code claims backed; gates 21 → 22`	ARCHITECTURE §40.1, §40.5; merge report
mine_localdiff citations	`39 of 2,924 outside their own hunk → 0 of 2,924`	R23b workflow H23b-6; ARCHITECTURE §40.4
Merge checks (3224c07)	`consensus guard PASS (only the config/version.go string); go test -race ok on 10 touched packages; 851 Go mutations in go_mutations.json; R23b harness KATs 5/5, 32/32 mutations; verify_deroarch 34/34 after a page rebuild`	merge report; ARCHITECTURE §40.5
pipeline_20261030T205204 (run 1)	`FAIL at mutate: 853/853 caught, but the restore check of ./cmd/derod/rpc/ was BROKEN: TestAbsenceClockFollowsRegisterUnregister left a ghost getwork session, so TestR23GetworkFanoutOrderedOverSockets failed in drain_miners when run with the package's mutation tests → fixed in bfe3e54`	commit bfe3e54; pipeline record pipeline_20261030T205204
pipeline_20261030T215745 (run 2)	`PASS: new tests 82 ×3, 0 flaky; mutations 853/853; race 26 packages 0 races; S4b ×1.011; shadow 120/120 blocks, 240 treehashes, 0 mismatches; gates 22/22; KATs 34/34`	pipeline record pipeline_20261030T215745

R23c · 2026-11-01 · +22102026.r23c

FASTER DESCRIPTOR SORT, SYNC PREWARM OVERLAP, AGGREGATE DOS SCORING, LOSSLESS WEBSOCKET PUSHES, PER-ACCOUNT EMISSION AUDIT

sort CPU `×1.035–1.053` gates `22/22` live `94.3k H/s` H/cpu-s `5,435`

WHAT CHANGED

- Suffix sort: in `cmd/dero-miner/dsac/descriptor.c`, a NEON pair write for two consecutive singleton groups (`DSA_PAIR`) and a branch-free group split in the column walk (`DSA_GSCAN`); `DSA_NO_PAIR` / `DSA_NO_GSCAN` restore the r22 code. Thread CPU per sort ×1.035–1.053, bit-exact on 120,000 fresh stage-1 texts. The lever asked for, fusing radix pass 2 with the singleton write-out, was measured and refuted.
- Miner: phase-exact expected shares (`expectedSharesPhased`, `final_hash_fraction` in REWARD STATS). The CPU path, the CPU→GPU handoff and the GPU flow were measured and need no engine change; tests and mutations pin what was measured, and the `gpu_split.go` comment now names `DERO_GSPLIT=8192,3,2` as the quiet-box A/B.
- P2P and sync: the next synced block's PoW prewarm starts once every tip has its topo record, during the previous block's topo fsync (`DERO_SYNC_PREWARM_OVERLAP=0` disables it); ban-list

lookups use an index, the penalty table is pruned and bounded; 4 score bans in one /24 or /48 within 1 h refuse new INBOUND sessions from it (outbound and existing peers untouched).

- Security: trusted remote keys (10 min old, 64 admitted calls, no recent ban) score ×1 during a surge (RPC-1); untrusted keys share a /24 or /48 record with its own bucket and ban line (RPC-2); the per-key ban is checked before the global budget is drawn (SEC23b-2).
- RPC and websockets: per-client ordered queues for Block, MiniBlock and TxPool pushes and one ordered push worker; at most 256 txids or SC keys per `DERO.GetTransaction` / `DERO.GetSC` call for remote and cross-site callers (loopback unbounded), refused bodies scored; `DERO.GetMonetaryIntegrity` reads a summary instead of copying the report lists and now carries issued emission, burns and payees.
- Wallet: `walletapi` history resync after a reorg plans the cut in one call and keeps the valid entries (RPCWS23B-E).
- Inflation monitoring: the emission auditor checks every account, not only block sums (a misattributed miniblock share or swapped commitments passed as ok before); a block above height 399,000 must have 10 miniblocks and 1 final; a rewind of 26 or more topoheights restarts the segment instead of double counting; auditor CPU ×0.64. Read-only `tools/storecheck` scans the block store.
- Payouts: derod counts dead normal jobs after the 9th miniblock and the normal→final flip delay (`getwork_dead_normal_jobs_total`, `getwork_final_flip_delay_seconds`); `harness/intel/payout_census.py` classifies every found share by loss mechanism.
- Harness: `harness/round.json` round registry (checked at BEGIN, KAT RR1–RR4); `harness/pipeline/benchlock.py` in the repo; `mine_evolution.py` and DK35 for the src-arch evolution section; this page's R1 → R29 comparison and enhancement curve; `artifacts/dero-rft.md`. After the full run: the install backup tag keeps its letter suffix (4f0dda2) and the evolution snapshot keeps line 1 stable (8a086cd).
- Live measurement of the r23c miner, 2026-11-02 01:08–01:28 (n=81): `94.3k H/s TOTAL mean` `5,435 H/cpu-s (+5.2 %)` `GPU busy 95.6 % of 96: GPU co-binds`; uptime under launchd, the 8192,3,2 ABBA and the --phase-duty A/B: `◌ pending measurement`.

WHY

> R23b left the miner CPU-sort-bound on a contended box and near the GPU cap on a quiet one (H/s tracks cores at ~5.45k H/cpu-s; GPU busy 90–92 % of its 96 % cap, R23b workflow FOR-8), so the request was a faster CPU sort to feed the GPU, made part of Round 23. Before install, the r23b miner made 90.5k H/s on 17.53 cores (5,164 H/cpu-s, GPU busy 93.1–95.3 %) in the quiet window 09:27–09:37, and about 57.5k on 10.38 cores while other jobs loaded the box (10:05). The miner workflow found that the CPU path outside the sort is 1.4–1.5 % of a worker, that the handoff loses nothing, and that only a bigger GPU batch helps the GPU; the gain from a faster sort depends on the regime. The node workflow covered the open R23b items: the surge trigger, the missing /24 key, SEC23b-2, RPCWS23B-C and -E, and CR23b-4, which was refuted. It also found new gaps: ban lookups that scanned every ban, unbounded GetTransaction lists, lost websocket pushes and an auditor that checked only block sums.

CODE

Excerpts in order: the NEON pair write (DSA_PAIR), the prewarm overlap, the aggregate inbound-ban rule, the remote item bound, the one-call history resync plan, and the per-account emission check.

cmd/dero-miner/dsac/descriptor.c · lines 2015–2046 · commit 091b35d

```
        /* derohe r23c (sortfuse): the NEON form of the AVX2 pair path above. Two consecutive singleton groups of at
         * most sixteen positions each go out in one iteration: both arena loads are issued before either store, and
         * the second group's sixteen lanes start at out + la, so they overwrite exactly the first group's overhang
         * (stores land in program order). `out + la + 16 ⩽ n` keeps both windows inside sa, and each arena read en
         * at offset + 15 ⩽ arena_len + 15, inside the arena's sixteen words of slack (offset + len ⩽ arena_len is
         * checked for both, as on the single path). */
        if (i + 2 < desc_len &&
            !DSA_KEY_EQ(ds[i].key, ds[i + 1].key) &&
            !DSA_KEY_EQ(ds[i + 1].key, ds[i + 2].key)) {
            const Desc da = ds[i], db = ds[i + 1];
            const uint32_t la = dsa_len(da), lb = dsa_len(db);
            if (la ⩽ 16 && lb ⩽ 16 && out + la + 16 ⩽ n &&
                (size_t)dsa_off(da) + la ⩽ arena_len && (size_t)dsa_off(db) + lb ⩽ arena_len) {
                const Arena* sa_ = s→arena + dsa_off(da);
                const Arena* sb_ = s→arena + dsa_off(db);
                const uint16x8_t a0 = vld1q_u16(sa_), a1 = vld1q_u16(sa_ + 8);
                const uint16x8_t b0 = vld1q_u16(sb_), b1 = vld1q_u16(sb_ + 8);
                const uint32x4_t ra = vdupq_n_u32(DSA_REL_OF(da.key));
                const uint32x4_t rb = vdupq_n_u32(DSA_REL_OF(db.key));
                uint32_t* const oa = (uint32_t*)(sa + out);
                uint32_t* const ob = oa + la;
                vst1q_u32(oa, vorrq_u32(vshll_n_u16(vget_low_u16(a0), 8), ra));
                vst1q_u32(oa + 4, vorrq_u32(vshll_high_n_u16(a0, 8), ra));
                vst1q_u32(oa + 8, vorrq_u32(vshll_n_u16(vget_low_u16(a1), 8), ra));
                vst1q_u32(oa + 12, vorrq_u32(vshll_high_n_u16(a1, 8), ra));
                vst1q_u32(ob, vorrq_u32(vshll_n_u16(vget_low_u16(b0), 8), rb));
                vst1q_u32(ob + 4, vorrq_u32(vshll_high_n_u16(b0, 8), rb));
                vst1q_u32(ob + 8, vorrq_u32(vshll_n_u16(vget_low_u16(b1), 8), rb));
                vst1q_u32(ob + 12, vorrq_u32(vshll_high_n_u16(b1, 8), rb));
                out += la + lb;
                i += 2;
                PROF_STAT(ST_KEYGRP, 2);
```

p2p/sync_prewarm_overlap.go · lines 56–90 · commit 091b35d

```
func start_prewarm_overlap(ready func() bool, prewarm func() int) *prewarm_overlap {
    o := &prewarm_overlap{stop: make(chan struct{}), done: make(chan struct{})}
    go func() {
        defer close(o.done)
        defer func() { recover() }() // a failure only means no cache entry; the chain re-verifies
        t := time.NewTicker(PREWARM_OVERLAP_POLL)
        defer t.Stop()
        for {
            stopped := false
            select {
            case ←o.stop:
                stopped = true
            default:
            }
            if ready() {
                if stopped {
                    prewarm_overlap_late.Inc()
                } else {
                    prewarm_overlap_early.Inc()
                }
                prewarm()
                return
            }
            if stopped {
                prewarm_overlap_missed.Inc()
                return
            }
            select {
            case ←o.stop:
            case ←t.C:
            }
        }
    }()
    return o
}
```

p2p/penalty.go · lines 264–276 · commit 091b35d

```
// ---- r23c (p2psec): aggregate inbound bans for rotating identities ----------------------------------------------
//
// A score ban covers one IPv4 address or one IPv6 /64 (penalty_scope). An operator holding a /24 (256 addresses) or
// /48 (65,536 /64s) gets a fresh score for every address it moves to, so misbehaviour was bounded per address, neve
// per operator. When PENALTY_AGG_MEMBERS distinct scopes of one aggregate (IPv4 /24, IPv6 /48) are score-banned
// within PENALTY_AGG_WINDOW, new INBOUND sessions from the whole aggregate are refused for PENALTY_AGG_BAN_BASE,
// doubling per repeat up to PENALTY_BAN_MAX.
// It is inbound-only on purpose: the aggregate may also hold honest nodes, and an attacker must not be able to make
// us drop or stop dialling OUR chosen outbound peers (that would help an eclipse); connections already established
// are kept, and the ban list (ban_list.json, outbound dialling) is untouched. Seed/priority/exclusive nodes inside
// aggregate exempt it. Nothing here changes which blocks or transactions are valid.
const PENALTY_AGG_MEMBERS = 4
const PENALTY_AGG_WINDOW = time.Hour
```

cmd/derod/rpc/param_bounds.go · lines 67–73 · commit 091b35d

```
func check_tx_hashes(ctx context.Context, n int) error {
    if n > MAX_REMOTE_TX_HASHES && caller_is_limited(ctx) {
        return tx_hashes_bound_error(n)
    }
    return nil
}
```

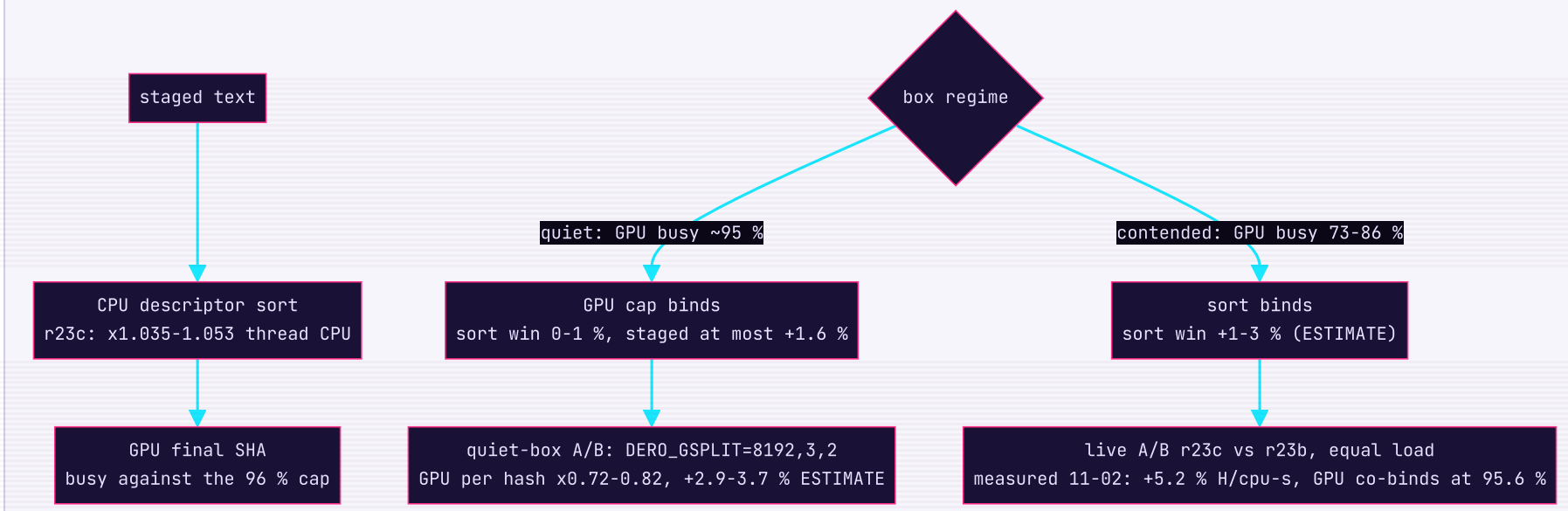
walletapi/history_sync_plan.go · lines 44–71 · commit 091b35d

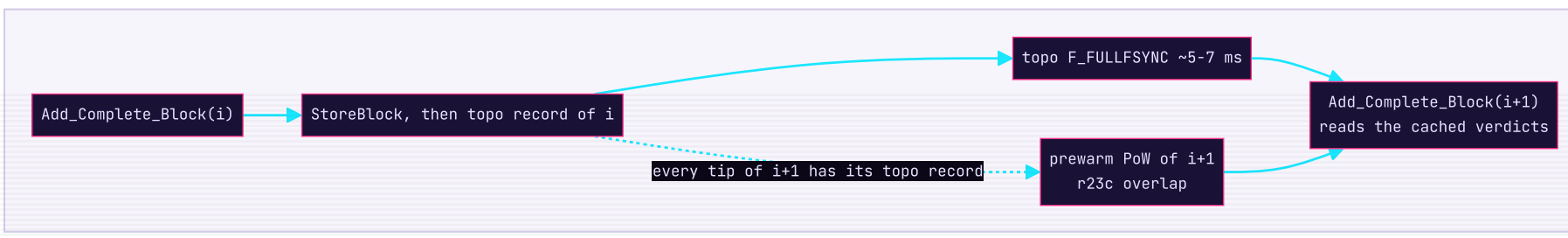
```
func plan_history_sync(entries []rpc.Entry, registration, end_topo int64, header func(topo int64) (string, error)) (
    i := len(entries) - 1
    for i ⩾ 0 {
        topo := entries[i].TopoHeight
        if registration ⩾ topo { // chain pruned on the server: keep old history, rescan from registration (as befo
            return history_sync_plan{keep: i + 1, start: registration, calls: p.calls}, nil
        }
        hash, herr := header(topo)
        p.calls++
        stale := false
        switch {
        case herr ≠ nil && topo > end_topo: // the daemon's chain no longer reaches this entry
            stale = true
        case herr ≠ nil:
            return history_sync_plan{keep: i + 1, start: registration, calls: p.calls}, herr
        case hash ≠ entries[i].BlockHash:
            stale = true
        }
        if !stale { // a matching block: everything up to here stands, resume right after it
            return history_sync_plan{keep: i + 1, start: topo + 1, calls: p.calls}, nil
        }
        for i ⩾ 0 && entries[i].TopoHeight == topo { // the whole group of that block
            i--
        }
    }
    return history_sync_plan{keep: 0, start: registration, calls: p.calls}, nil
}
```

glue/emission/emission.go · lines 634–650 · commit 091b35d

```
            wantL := e.L
            if v := payees[k]; v > 0 {
                wantL = new(bn256.G1).Add(e.L, mul(v))
            }
            okR := samePoint(a.R, e.R)
            okL := samePoint(a.L, wantL)
            if !okR {
                r.alert("account %x: Right component moved by something other than the D values of its own rings", k
            }
            if r.SCCalls == 0 && !okL {
                r.alert("account %x: Left component moved by something other than its own ring commitments, coinbase
            }
            if okR && okL {
                exact[k] = true
                r.AccountsExact++
            }
        }
```

FLOW





EVIDENCE

MEASURED	RESULT	SOURCE
Workflows (R23c)	3 workflows, 12 dimensions, 24 agents (every builder checked by an adversarial verifier or reviewer); merged as 091b35d (69 files); consensus guard PASS	workflow journals wf_2624ea65-0c5, wf_5c7e92c4-c90, wf_d2cb875d-e03; commit 091b35d
r23b baseline, quiet window 09:27–09:37	90.5k H/s TOTAL mean of 43 lines at 17.53 cores, 5,164 H/cpu-s; GPU busy 93.1–95.3 % of 96 %	bin/dero-miner.log 09:27:14-09:37:45 (read-only)
r23b baseline, contended sample 10:05	57,471 H/s at 10.38 cores, GPU busy 75.8 %, load 61	R23c hashrate watcher log, line 10:05:12
Descriptor sort DSA_PAIR + DSA_GSCAN, thread CPU per sort	×1.047 at 1 thread (12/12 pairs, 177.38 → 169.22 µs), ×1.037 at 6 threads (6/6); verifier ×1.0355-1.0534 at 6 threads, A/A 0.996-1.001	R23c workflow sortfuse; verify-sortfuse
Same change, whole hash and live	Go-level HashC child CPU 0.988 (5/6, verifier; builder 0.973). Live ESTIMATE: +1-3 % H/cpu-s CPU-bound, 0-1 % on a quiet box	verify-sortfuse (required change)
Bit-exactness of the sort change	120,000 fresh stage-1 texts: 0 mismatches, 0 declined (builder, and verifier with its own permutation checker and the SA-IS hash); 480 + 272 adversarial texts: 0 mismatches	R23c workflow sortfuse; verify-sortfuse; dsac/r23c_fresh_test.go
Fused radix pass 2 + singleton write-out	REFUTED · ×1.137 slower at 1 thread (0/8), ×1.151 at 6 threads (0/6); exact-store form ×1.307: ~19.8k random 64-byte read-modify-write windows over a 275 KB sa	R23c workflow sortfuse
CPU time outside the sort	REFUTED · 1.4–1.5 % of a split worker in process (engine/sort 1.007), ~0.6 % more live; live P-core split rate 5,963 H per P-core-second at P-IPC 4.40; 16–17 % of CPU-seconds on E-cores at 0.38× P	R23c workflow cpupath; verify-cpupath
Overhead A/B test fix (test only)	engine/sort at matched T = 4: 1.030 → 1.006 (7 pairs); both base legs lacked the yield; miner binary sha256 unchanged	verify-cpupath; cmd/dero-miner/gsplit/overhead_test.go
CPU → GPU handoff	REFUTED · backlog and idle tries alternate ~21×/s; an idle-try own pair costs 1.673–1.696× a staged sort; wakeup lever ≤ 0.4 %; chunked SHA -24 to -64 % (model); hashes_total - split_hashed = 2 × idle tries exactly	R23c workflow handoff; verify-handoff
Does a faster sort feed the GPU?	regime-dependent: quiet box (GPU ~95 % of 96) at most +1.6 % staged; contended windows GPU busy 73.3–85.7 % with backlog 54–56 s/min, where the sort binds	verify-handoff (live split series 09:19–10:21)
GPU geometry DERO_GSPLIT=8192,3,2 (proxy next to the live miner)	GPU busy per hash ×0.72–0.82 (28/28 light-regime pairs below 1); +2.9–3.7 % TOTAL on a quiet box, ESTIMATE (CPU capacity 93.15–93.9k); 8.8 GB shm; 8192,5,3 no better than 8192,4,2	R23c workflow gpuflow; verify-gpuflow
GPU levers	REFUTED · warp-specialised final SHA (pipeline ×0.96–0.99), inflight 3 (live miner ×0.92), depth alone, cross-batch SHA coalescing (×0.963 at depth 4)	R23c workflow gpuflow
Sync PoW prewarm overlap (loopback shadow)	blocks/s median ×1.255 (7/7 pairs, 37.2 → 49.7), verifier ×1.331 (6/6); CPU per block 0.991–0.993; treehash identical in every run; overlap off = base	R23c workflow chainperf; verify-chainperf
Verification cost, real mainnet txs	full verify 148–159 ms/tx median; 5.8–6.0 → 25.5–27.6 tx/s at 20 workers; demand 0.000139–0.000185 TPS	chainperf CP23c-3, CP23c-4; verify-chainperf
Ban-list lookup	20,000 bans 8,489,862 → 42.4 ns/op; 1 ban 261.8 → 59.6 ns (CPU/op 0.230); 0 mismatches against the old scan over ~72k lookups	p2psec P2P-1; verify-p2psec
Penalty table	with 5,000 banned records 45,646 → 234 ns per fault; the 65,536 bound is soft (68,537 records with every ban running)	p2psec P2P-2; verify-p2psec
Inbound /24 aggregate ban	after 4 score bans in one /24: 100/100 new addresses admitted → 0/100; the verifier fixed a stale fast-path flag after a ban ended (×1.889 admission CPU)	p2psec P2P-3; verify-p2psec (agg_soonest)
Remote RPC surge, established key	a wallet at 12 invalid calls/min during a surge: banned after its 11th call → never banned; a fresh key is still banned	p2psec RPC-1
Remote RPC /24 aggregate	16 rotating keys over 30 min: 4,800/4,800 probes through → 572 (-88 %); keys pre-warmed to trusted escape it (4,800/4,800)	p2psec RPC-2; verify-p2psec

MEASURED	RESULT	SOURCE
SEC23b-2 order	a banned key with 5,000 invalid-params replies drew 580 global tokens and a surge → 0 tokens, no surge	p2psec RPC-3; verify-p2psec
Normal-path cost of the security changes	CPU/op ratios 0.978–1.052 (builder), 0.988–1.042 (verifier); worst +3 to +6 ns per call or session	p2psec COST; verify-p2psec
Websocket pushes to a busy client	300 back-to-back pairs: ~50 % of bundles lost → 600/600; fan-out CPU ×1.085–1.122 per 100 clients, once per block	rpcws RPCWS23C-2; verify-rpcws
GetTransaction amplification	100,000 unknown txids = 934–965 ms CPU, 421 MB; now ≤ 256 items per remote call, refused bodies scored; body precheck 37 → 0.7 µs	rpcws RPCWS23C-4; verify-rpcws
Wallet history resync after a reorg	2-group reorg 3 calls → 1; a shallow reorg kept 0 entries → keeps the valid ones; an entry above the tip never synced → treated as stale	rpcws RPCWS23C-1; verify-rpcws
GetMonetaryIntegrity emission view	5.4–7.8 µs, 9,984 B, 4 allocs → 0.34–0.43 µs, 256 B, 2 allocs	rpcws RPCWS23C-5; verify-rpcws
Emission auditor, per-account check	15,000 + 13,000 mainnet blocks: 0 alerts, every account exact (102,611/102,611 in the first set), payees explained 101,882/101,882 and 92,092/92,092; base had passed a misattributed share and swapped commitments as ok	cryptoinfl CI-1; verify-cryptoinfl
Auditor deep rewind	a rewind of 26 or more topoheights double counted (verified 117 over a span of 116, consistent false) → segment restart; the verifier found IssuedEmission was not reset (issued_matches_formula false on an honest chain) → fixed	cryptoinfl CI-3; verify-cryptoinfl
Auditor CPU per block	×0.643 (7/7 pairs), verifier ×0.638	cryptoinfl CI-6; verify-cryptoinfl
Receiver payment proof (CI-9)	the disclosed Keccak(sk·E) is the V2 payload key and E is unbound: a crafted proof can decrypt another V2 payload (payload privacy only)	verify-cryptoinfl CI-9 (builder refutation not upheld)
Payout census, 1,791 found shares	paid 97.04 % (0.9708 on re-run); stale-flagged 1.06 %, slot race 1.40 %, other 0.22 %, lost finals 0.28 %	payouts PAY23c-2; verify-payouts
The 5 lost finals	3 reached our chain first and lost the lower-BLID tie-break; 2 were inbound (by 54 and 2 ms)	verify-payouts PAY23c-8 (evidence corrected)
Expected shares and luck	the 5/(9D) model over-stated expectation ×1.034–1.072 (live 1.038 on re-read) → phase-exact pricing	payouts PAY23c-3; verify-payouts
Miner uptime	0.2667 of 86.46 h (≈112 DERO forgone) / 0.2735 of 87.26 h (≈102 DERO), ESTIMATE; 34 % of the idle hours overlapped derod downtime; the live miner was not under launchd	payouts PAY23c-1; verify-payouts
Flip-delay hook cost	619–1,916 ns per send against a 4.13–4.2 ms send_job (niced, not under benchlock: the lock never freed)	payouts PAY23c-4; verify-payouts
CR23b-4 rand.Read error check	REFUTED · go1.26.6 crypto/rand.Read never returns an error and calls fatal() on failure: a check would be dead code	cryptoinfl CI-7; rpcws RPCWS23C-6; payouts
Install backup naming	the tag regex dropped the letter suffix, so r23b read as r23 (whose backup existed) and was replaced with no copy → regex r\d+[a-z]?; r23b rebuilt from cd1be86 into bin/*.22102026.r23b-backup (vcs.revision cd1be86, modified=false)	commit 4f0dda2 harness/pipeline/fix_build_verify.py
pipeline_20261101T120821 (full run)	guard PASS; format PASS (3 pre-existing vet unreachable-code findings); flaky 82 ×3, 0 flaky; tests PASS; soundness 29 tests; KATs PASS; mutations 955/955 (upstream failures recorded 3); race 26 packages 0 races; build r23c; S4b REFUSED (contended: foreign 4.88 cores; descriptor/SA-IS 4.58, not evidence); S4c SKIP (server sha 5aff39c8f529 unchanged); shadow 120/120 blocks, 240 treehashes, 0 mismatches, 0 regressions; install PASS; S7 FAIL gates 21/22 (G22: the SOUNDNESS_CLAIMS auditor-metric anchor emission_audit.go:243 had moved to :265) → fixed in 4f0dda2	pipeline record pipeline_20261101T120821

MEASURED	RESULT	SOURCE
pipeline_20261101T140320 and pipeline_20261101T141454 (project-only 1, 2)	PASS: KATs 35/35, harness mutations 34/34, gates 22/22, but 1 refused row each: line 1 of evolution_git.tsv carried HEAD, so mine_localdiff's first-line anchor went stale → mine_evolution.py keeps line 1 stable (8a086cd)	pipeline records pipeline_20261101T140320, pipeline_20261101T141454
pipeline_20261101T142234 (project-only 3)	PASS: KATs 35/35, harness mutations 34/34, gates 22/22, 0 refused rows (REJECTED.json [])	pipeline record pipeline_20261101T142234
r23c live window 2026-11-02 01:08–01:28 (n=81 TOTAL lines; load avg 50–57: a niced scratch backup ~71 % of a core and macOS system services)	TOTAL 94,273 H/s mean (median 96,297, min 86,209, max 97,651) at 17.37 cores; 5,435 H/cpu-s; staged 87,904; GPU busy 95.6 % of 96; 28 found miniblocks	harness/runtime/r23c_live_window_20261102.txt
Live A/B, r23c against r23b, H per CPU-second	+5.2 % against the r23b quiet baseline 5,164 (09:27–09:37), +3.2 % against same-day r23b 5,265 (11-01 15:14); TOTAL +4.2 % against 90.5k. Above the 0–1 % quiet-box estimate: sort win CONFIRMED, magnitude PLAUSIBLE (part may be load or regime variance)	harness/runtime/r23c_live_window_20261102.txt; R23c hashrate watcher log 15:14
Binding regime, r23c live	GPU co-binds: busy 95.6 % of the 96 % cap, duty bucket sleeping 15.2 s/min, CPU backlog 36.9 s/min (23.9–53.7), ring full 44.7 s/min; so the next H/s needs GPU-side work (8192,3,2 ABBA, user-run)	harness/runtime/r23c_live_extract_20261102.txt (21 GPU split lines)
GPU split health, r23c live	0 restarts, 0 audit faults, 0 GPU/CPU mismatches, descriptor fallbacks 0 (verified 83,996 → 194,464); dropped_stale +956,878 in 20 min (~0.9 % of staged); 0 WARN/ERROR lines	harness/runtime/r23c_live_extract_20261102.txt
Payouts, r23c live	GetMiningOutcome 01:28: submitted 45, paid 39, orphaned 2 (slots_full), pending 4, paid fraction 0.9512; REWARD STATS luck 0.98 on 44.7 expected shares, final_hash_fraction 0.573	harness/runtime/r23c_live_window_20261102.txt; r23c_live_extract_20261102.txt
Emission auditor, r23c live (partial)	OK: 489 audited, 0 alerts, segment 7,844,298–7,844,779, emission consistent, payees 3,331/3,331 explained, 3,525 µs/block mean (max 27,276); the 24 h read stays pending	harness/runtime/r23c_live_window_20261102.txt (DERO.GetEmissionAudit 01:28)
Getwork flip counters, r23c derod	over 7,163 s: 16 dead normal jobs, 128 final flips; flip delay count 14, mean 52.7 ms (about 5× the simulated 10.36 ms, small sample)	harness/runtime/r23c_live_extract_20261102.txt (derod /metrics 01:29)
derod health, r23c	GetSyncInfo 01:29: synced, lag 0, tip age 8 s, 13 outgoing / 0 incoming peers, 0 recovered panics; derod.log since 23:00: 1 ERROR (a peer's short GetObject reply at start), 0 WARN. JSON-RPC now needs Content-Type: application/json (empty body without it)	harness/runtime/r23c_live_extract_20261102.txt
Whole r23c run 00:53:03 → 02:04:08 (typed console exit)	4,265 s at avg 94,009 H/s; 83 miniblocks + 11 blocks accepted, luck 1.06 on 88.4 expected; GetMiningOutcome 02:08: 95 submitted, 92 paid, 3 orphaned (slots_full), paid fraction 0.9684; emission audit 622 blocks, 0 alerts; flip delay n=30 mean 61.2 ms	bin/dero-miner.runs.jsonl; harness/runtime/r23c_live_extract_20261102.txt
Miner uptime under launchd	◌ pending measurement: every r23c run (5 of 5, 11-01 16:02 to 11-02 02:04) ran from a terminal and ended by a typed console exit	bin/dero-miner.runs.jsonl; harness/runtime/r23c_live_extract_20261102.txt

R24 · 2026-11-02 · +22102026.r24

NODE-ONLY ROUND: RPC AND P2P ABUSE BOUNDS, MEMPOOL ADMISSION POLICY, SYNC PREWARM FIXES, INTEGRITY RPC FIX, AREA INTELLIGENCE AND THE DISCLOSURE TRACK

miner ◌ PAUSED (miner) mutations 1108/1108 gates 22/25 (+3 PAUSED, miner) live r24 node S3 PASS, 0 ERROR

WHAT CHANGED

- Miner track ◌ PAUSED (miner): the user paused miner enhancements (rft §0.3, §2.7), so no miner code changed, no live miner ran, and the miner gates G17, G18, G21 and the benches S4b / S4c report PAUSED while `harness/round.json` says `tracks.miner = paused`. The miner still builds and passes its tests.
- Security: trust is earned by honest calls and lost by probing; a trusted key's unknown names still score into its /24 or /48 aggregate, and probes never earn trust (B28). Aggregate-only refusals are rate-limited instead of scored. The penalty table bound is hard (`penalty_evict_locked`). A

returning-peer accept lane (`p2p/proven_lane.go`) keeps peers with a verified handshake reachable when spoofed datagrams drain the shared accept budget (B30, node-local part).

- RPC and websockets: a remote `/json_rpc` body may name at most 1,024 txids + SC keys and pays item tokens; the adversarial verifier found two bypasses (an ignored params field, a non-object batch element) and closed both before the merge (B29). JSON-RPC without the exact `Content-Type: application/json` now gets a JSON-RPC −32600 body instead of an empty 415 (B38).
- Tx pool: `glue/txpolicy` verifies identical bytes once for RPC and relay together, remembers proof-invalid bytes, and caps the pool with fee-rate eviction that never evicts a tx a mining template carries; mined txs no longer count as pending. B19 (integer fee ranking) is verified; its fix would touch consensus code and is not shipped.
- Sync: compact-block txs are observed once the block is accepted (B18); prewarm counters count each block once (B34); a cross-response prewarm overlap peeks the next pipelined response; the difficulty read in the prewarm is single-flight; the chain locator no longer spins on a read error.
- Crypto and supply: `DERO.GetMonetaryIntegrity` published wrong totals on the live r23c node (issued_matches_formula false, payees 0, while the auditor said true); it now shares the auditor's field mapping (CI24-1). Decoys from `DERO.GetRandomAddress` are corrected for tree depth (CI24-3). The wallet withholds V2 receiver proofs whose ephemeral key repeats (CI24-2).
- Getwork and ops: the r23c "5× flip delay" was refuted as a defect (a conditional mean compared with an unconditional one); new L9 and slow-flip counters; `tools/launchd-derod` (derod supervision, not installed), `tools/nodehealth` and `tools/getworkprobe`.
- Harness: area intelligence (`harness/areas.json` , 14 areas, gate G23) and the Foundation-disclosure track (`harness/loop/disclosure_track.json` , gate G24), both standing per-round steps; S3 never excuses a first-run data race of a listed flaky test.
- Live r24 readings (measured 2026-11-02, derod restarted onto r24 at 15:31:03): `S3 PASS` the integrity RPC agrees with the auditor (BD7 closed), `B38 confirmed` a wrong Content-Type gets a JSON-RPC 415 body, the node is synced with 0 ERROR / WARN; the compact-block and prewarm counters wait for a tx-carrying block and a sync (`◌ pending measurement`).

WHY

> The user paused miner enhancements on 2026-11-02 and will not run the miner until they resume, so Round 24 covers the node: the "Open after Round 23c" items that were node-side (B18, B19, B28–B30, B32, B34, B35, B38, B39) and what the live r23c node showed. Two live readings drove the round: GetMonetaryIntegrity disagreed with GetEmissionAudit on the running node (CI24-1), and the compact-block path, which carries every live tx block, was the one block path the round-trip census never saw (B18). Every dimension's builder was checked by an adversarial verifier; the rpcws verifier did not uphold the B29 patch as built (two bypasses) and its fixed patch is the one merged. The user also asked (2026-11-02) that area intelligence and the Foundation-disclosure track run every round; both shipped in R24 itself.

CODE

Excerpts in order: trust lost by probing (B28), the hard penalty bound, per-element batch item counting (B29), shared verification of identical bytes, the integrity-RPC field mapping (CI24-1), and the compact-block observation (B18).

`cmd/derod/rpc/rpc_abuse.go · lines 112–123 · commit 59eee1d`

```
// r24 (security, B28): trust is earned by honest calls and LOST by probing.
//
// r23c trust was permanent while a key was not banned: 16 keys pre-warmed with 70 cheap GetInfo calls each over
// 10.5 min were then exempt from the /24 aggregate and from the surge, and probed 4,800 unknown names in 30 min wit
// none refused (fresh keys: 572), and 64 pre-warmed keys in 64 /24s probed 19,200/19,200 through a held surge
// (fresh: 1,920). Two rules close that, both on the unknown-method branch only (the known-method path is unchanged)
//   - every unknown name of a trusted key still scores into its aggregate (trusted keys stay exempt from the
//     aggregate's ban and bucket), so a pre-warmed fleet in one /24 bans the /24 like a fresh one;
//   - a trusted key loses its trust (and must earn it again: ABUSE_TRUST_AGE and ABUSE_TRUST_CALLS from now) when i
//     own decayed unknown-name score reaches ABUSE_TRUST_PROBE_MAX. An unknown name it sends while its aggregate is
//     banned is refused (r24 verify: rate_limited, trust kept until the probe line; see admit).
//
```

p2p/penalty.go · lines 89–112 · commit 59eee1d

```
func penalty_evict_locked(now time.Time) {
    type cand struct {
        k    string
        rank int64 // 0 = no running ban (dropped first), else the ban's end
    }
    c := make([]cand, 0, len(penalty_map))
    for k, v := range penalty_map {
        r := int64(0)
        if now.Before(v.banned_until) {
            r = v.banned_until.UnixNano()
        }
        c = append(c, cand{k, r})
    }
    sort.Slice(c, func(i, j int) bool { return c[i].rank < c[j].rank })
    n := len(c) - PENALTY_EVICT_TO
    for i := 0; i < n; i++ {
        delete(penalty_map, c[i].k)
    }
    if n > 0 {
        atomic.AddUint64(&penalty_evictions, uint64(n))
    }
}

var penalty_last_prune time.Time
```

cmd/derod/rpc/param_bounds.go · lines 131–152 · commit 59eee1d

```
// r24 verify: a batch is split per element as jrpc2 splits it (decode_requests), so one non-object element no longe
// hides the rest of the batch from the bound.
func remote_body_items(body []byte) ([]int, error) {
    if !body_may_name_bounded(body) { // r23c verify: most remote bodies (SendRawTransaction hex, GetInfo) skip the
        return nil, nil
    }
    var reqs []bounded_request
    if decode_requests(body, func(n int) { reqs = make([]bounded_request, n) }, func(i int, el []byte) error {
        return json.Unmarshal(el, &reqs[i])
    }) ≠ nil {
        return nil, nil // not JSON: the bridge refuses the whole body without running any of it
    }
    var items []int
    total := 0
    for i, r := range reqs {
        n, err := request_items(r.Method, r.Params)
        if err ≠ nil {
            return nil, err
        }
        if n == 0 {
            continue
        }
```

glue/txpolicy/txpolicy.go · lines 19–26 · commit 59eee1d

```
//  1. Shared verification of identical bytes (flight). Add_TX_To_Pool caches a positive verdict only AFTER the proo
//     verify (transaction_verify.go:466), so N concurrent submissions of the same bytes verified N times: measured
//     6 copies = 454 ms CPU for one 75 ms ring-16 tx (r24 evidence txflow_base_ring16.txt, dup_concurrent). p2p
//     already collapsed concurrent RELAYS (p2p/tx_flow.go, r19); SendRawTransaction did not, and an RPC copy never
//     shared with a relayed copy. Now every admission of the same Keccak256(raw bytes) shares one Add_TX_To_Pool
//     call. A follower gets the leader's error, or ErrDuplicate when the leader's admission succeeded (before r24 a
//     follower that verified second got "rejected by pool by mempool": an error, never a second relay).
//  2. Known-invalid bytes (negative verdict cache). A transaction whose PROOF failed (Add_TX_To_Pool: "could not be
```

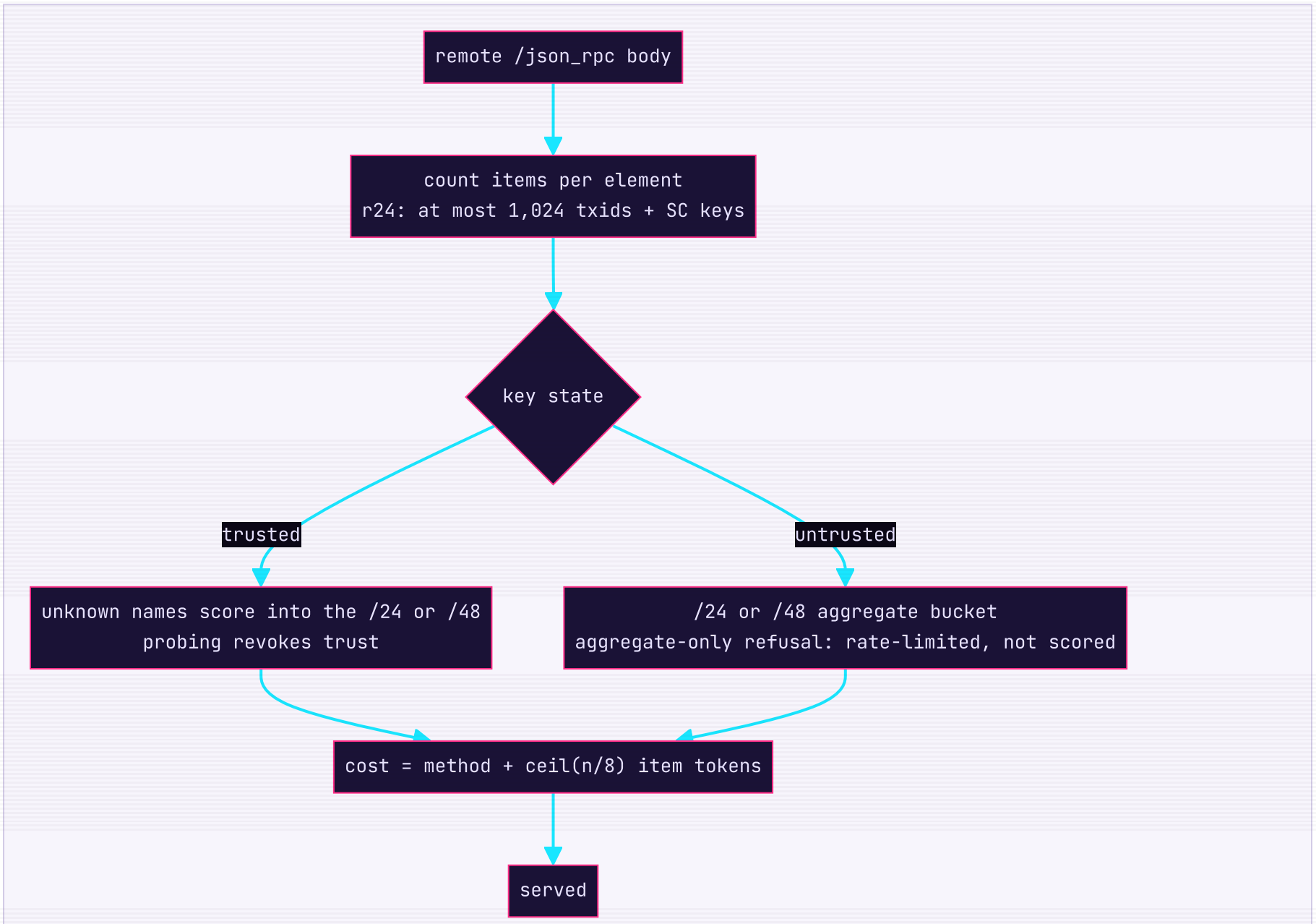
cmd/derod/rpc/emission_audit.go · lines 596–614 · commit 59eee1d

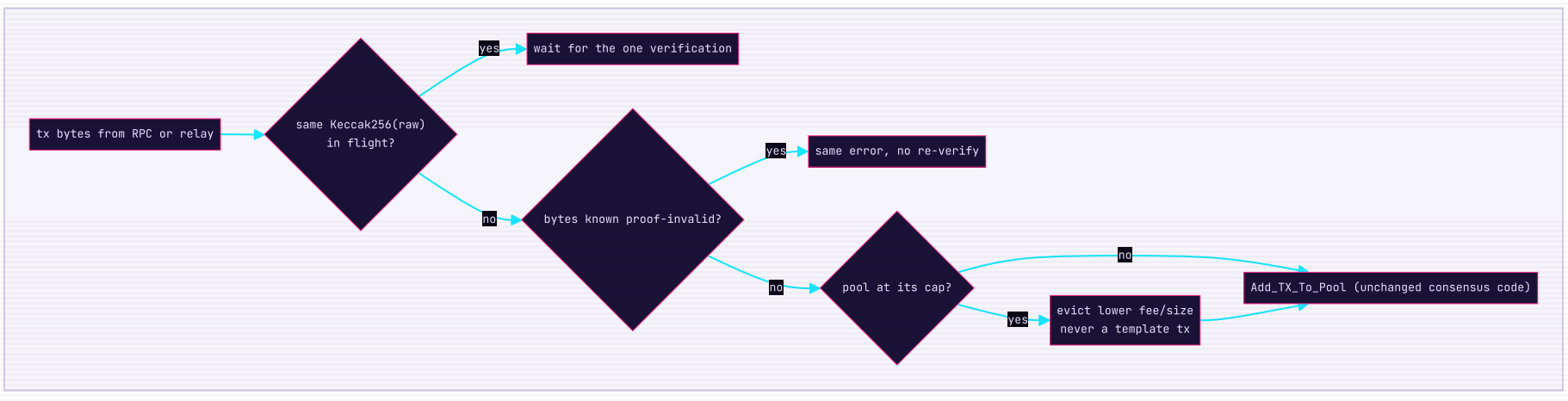
```
// r24 (cryptoinfl CI24-1): the r23c version listed the fields by hand and missed the seven that r23c CI-4 added to
// MonetaryIntegrity_Emission (issued_emission, issued_matches_formula, burns, payees, payees_verified,
// payees_explained, deep_rewinds), so the live r23c node published issued_matches_formula=false and payees=0 next t
// a GetEmissionAudit that said true and 7,117. It now copies the result struct (slice headers only, no list copy) a
// goes through emission_summary, the one field mapping both paths share.
func (a *emission_auditor) summary() MonetaryIntegrity_Emission {
    a.Lock()
    r := a.res
    v := make(map[string]uint64, len(a.res.Verdicts))
    for k, n := range a.res.Verdicts {
        v[k] = n
    }
    a.Unlock()
    r.Verdicts = v
    r.LastAlerts, r.Notable, r.Recent, r.Series.Last = nil, nil, nil, nil
    r.Status = emission_status(r.Enabled, r.Alerts, r.Audited)
    return emission_summary(r)
}
```

p2p/compact_block.go · lines 102–121 · commit 59eee1d

```
// r24 (B18): raws are the pool bytes txs were decoded from; the txs of an ACCEPTED compact block are observed
// (txcanon.Observe, a counter only) exactly as the sync and chunk paths observe theirs. Before r24 this path, which
// carried every live tx block (p2p_compact_block_hits_total 3, block_tx_roundtrip_observed_total 0 on r23c), was th
// one block path the round-trip census never saw. A block already in the chain is not observed again.
func (connection *Connection) add_compact_block(bl *block.Block, txs []*transaction.Transaction, raws [][]byte) bool
    cp := *bl // the chain gets its own copy of the header too
    cbl := block.Complete_Block{Bl: &cp, Txs: txs}
    blid := cp.GetHash()
    if chain.Is_Block_Topological_order(blid) || chain.Is_Block_Tip(blid) {
        return true
    }
    if _, ok := Add_Complete_Block_HF3(chain, &cbl); !ok {
        return false
    }
    connection.logger.V(2).Info("block assembled from first chunk and local pools", "blid", blid, "txcount", len(txs
    Broadcast_Block(&cbl, connection.Peer_ID)
    observe_block_txs(raws, txs) // r24 (B18): only txs of an accepted block are observed
    return true
}
```

FLOW





EVIDENCE

MEASURED	RESULT	SOURCE
Workflows (R24)	2 workflows: 8 node dimensions (harness, security, cryptoinfl, getworkops, rpcws, txpool, chainperf, syncp2p) and 2 more (areaintel, disclosure), each builder checked by an adversarial verifier; combined patch md5 d19d3f9a38b1cece1ffd4cff5087167f, 79 files, +9,431/−303, merged as 59eee1d; area intelligence and the disclosure track integrated as 76fd69b; consensus guard PASS	`workflow journals wf_d8346d53-96e, wf_04f8d64d-c27; harness/loop/runs/r24_dimension_results_20261102.json; git show --stat 59eee1d`
Tests added by the dimensions	`108 (harness 4, security 15, cryptoinfl 16, getworkops 19, rpcws 9, txpool 28, chainperf 6, syncp2p 11); the S3a stage ran 120 new tests ×3, 0 flaky`	`harness/loop/runs/r24_dimension_results_20261102.json tests_added; pipeline_20261102T110200 stage flaky`

MEASURED	RESULT	SOURCE
Pre-warm escape of aggregate RPC scoring (bench)	scenario S1 (pre-warmed keys): 4,800 → 573 probes through (fresh keys 572); scenario S3: 19,200 → 2,752	harness/loop/runs/r24_dimension_results_20261102.json security SEC24-1 (verifier, scenarios_3way)
Verifier defects fixed before the merge (security)	an aggregate-only refusal did not charge the key's own bucket (18,401 of 20,000 calls in 1 s, 0 banned); trust revoked on the first unknown name let a neighbour deny an honest trusted wallet (1,767/1,800 refused in 60 min) → both fixed	harness/loop/runs/r24_dimension_results_20261102.json security SEC24-1, SEC24-2 (verifier)
Penalty table bound (bench)	peak 85,536 → 65,536 records; one eviction pause at most 12.9 ms, once per 4,096 new records	harness/loop/runs/r24_dimension_results_20261102.json security SEC24-3 (verifier)
Normal-path cost of the security changes (bench A/B)	normal paths ×0.990–1.013 (6 ABBA pairs, verifier); AddPenaltyNewBanned ×0.907	harness/loop/runs/r24_dimension_results_20261102.json security SEC24-AB (verifier)
B29 cost of a remote 256 × 256 GetTransaction batch on base (bench)	5.8–6.0 s child CPU under the bench lock (verifier; builder 5.3 s CPU, 470 MB): CPU-bound, not the ~0.6 s of R23c	harness/loop/runs/r24_dimension_results_20261102.json rpcws RPCWS24-3
B29 bypasses found by the verifier	an ignored params field skipped every item bound; one non-object batch element skipped all remote admission (5,100 of 5,100 Pings served, rpc.serverInfo answered a remote caller) → both closed in the merged patch	harness/loop/runs/r24_dimension_results_20261102.json rpcws RPCWS24-2 (verifier, not upheld as built)
RPC regression check (bench A/B)	loopback ×0.987 (verifier; builder ×0.984); wallet 32-txid ×1.017 (builder ×0.995)	harness/loop/runs/r24_dimension_results_20261102.json rpcws RPCWS24-PERF
B38 Content-Type (live read-only probe)	no Content-Type, text/plain, charset variants: empty 415; GET: empty 405; exact application/json: 200 → now a JSON-RPC -32600 body (upstream jrpc2 bridge exact compare, not a regression)	harness/loop/runs/r24_dimension_results_20261102.json rpcws RPCWS24-1
Six concurrent identical submissions (bench)	CPU 466–477 ms → 87–97 ms, one accepted in each arm (4 runs)	harness/loop/runs/r24_dimension_results_20261102.json txpool TXP-3 (verifier)
Replayed proof-invalid tx (bench)	44–46 ms → 2.5–3.3 ms via Admit; 43.8–52.7 ms → 0.016–0.018 ms via RPC	harness/loop/runs/r24_dimension_results_20261102.json txpool TXP-4 (verifier)
Pool-cap eviction of a template tx (verifier, simulator)	the node rejected its own found final miniblock ("tx … not found") → protected template-tx set; cost 0 below the cap, 21–25 ns per new 0-tx template	harness/loop/runs/r24_dimension_results_20261102.json txpool TXP-5 (verifier)
B19 integer fee ranking (mainnet scan)	58/58 non-registration txs in the last 20,000 blocks rank 0 (verifier 16/16 in 5,000); the optional fix makes the consensus guard FAIL	harness/loop/runs/r24_dimension_results_20261102.json txpool TXP-1
Sync prewarm difficulty herd (shadow A/B)	CPU per synced block geomean ×0.947, 6/6 below 1 (verifier; builder ×0.916, 9/10); blocks/s gain not established (×1.008 verifier, run position dominates)	harness/loop/runs/r24_dimension_results_20261102.json chainperf CP24-1, CP24-3
Cross-response prewarm overlap (shadow A/B)	catch-up ×1.114 from a live source, 6/6 faster (verifier; builder ×1.132 on 3 pairs), CPU per block ×1.009; synced chain matched the source 24/24	harness/loop/runs/r24_dimension_results_20261102.json syncp2p SP24-3 (verifier)
B34 prewarm counters (live r23c and shadow)	live 1,040 = (88 + 16) × 10; shadow base 17.48–17.50 PoW per block → 9.99–10.00 (24/24 runs)	harness/loop/runs/r24_dimension_results_20261102.json syncp2p SP24-2 (verifier)
B18 compact-block observation (live r23c)	6 compact hits, 0 observed on the running node → observed after acceptance (proved by unit tests; the shadows had 0 compact hits)	harness/loop/runs/r24_dimension_results_20261102.json syncp2p SP24-1 (verifier)
B39 flip delay (live counters re-read)	REFUTED · 238 flips, 34 dead jobs, 30 dead windows, 61.2 ms conditional mean against the model's 51.46 ms: z = 1.60, not significant; the 5× compared a conditional with an unconditional mean	harness/loop/runs/r24_dimension_results_20261102.json getworkops GWO24-1 (verifier)
CI24-1 integrity RPC on the live r23c node	GetEmissionAudit issued_matches_formula true, payees 7,117; GetMonetaryIntegrity false, payees 0 (an r23c merge listed the fields by hand) → shared field mapping	harness/loop/runs/r24_dimension_results_20261102.json cryptoinfl CI24-1; harness/runtime/disclosure_live_20261102T175638Z.txt (check S3 FAIL)

MEASURED	RESULT	SOURCE
CI24-3 decoy depth bias (mainnet tree clone)	leaf depth 18–25: the 1.9 % of accounts at depth 19 got 7.7 % of the draws; modelled best-guess attacker right in 17.4 % of 16-member rings vs the ideal 6.25 % (ESTIMATE) → depth-corrected sampler, modelled 7.0 % at R24_DECOY_SLACK=2	harness/loop/runs/r24_dimension_results_20261102.json cryptoinfl CI24-3
Disclosure track re-test	11 claims re-tested (3 CONFIRMED, 7 PARTLY-CONFIRMED, 1 CANNOT-DETERMINE-LOCALLY), anchors 149/149 at 04ceb55 and 150/150 after the merge, 0 drift, press rows 14/14 mapped, exploit-lint 0, 11 new facts, 13 fixes (4 fork-required designs; node-local: 4 implemented, 2 feasible but not implemented, 3 infeasible)	harness/runtime/disclosure_track_20261102T182602Z.json, disclosure_track_20261102T184720Z.json
Proof-scalar canonicality census (read-only RPC)	last 100,000 blocks (tip 7,846,426): 347 tx bodies, 1,308 payloads, 15,866 proof scalars, 0 ≥ Order, 0 parse errors	harness/runtime/disclosure_census_20261102T175740Z.json
Area intelligence	14 areas (13 active, 1 paused): OK 10, WATCH 3, ATTENTION 0, PAUSED 1; 22 proposals (0 high, 3 medium, 19 low), 0 refused values	harness/runtime/area_intel.json, area_proposals.json (run 20261102T123330)
pipeline_20261102T090301 (full run 1, install)	guard PASS; format PASS; flaky 120 ×3, 0 flaky; tests PASS; soundness 29; KATs PASS; mutations 1108/1108; race 26 packages 0 races; S4b / S4c PAUSED (miner); shadow 120/120 blocks, 240 treehashes, 0 mismatches; install PASS (3.6.0-153.DEROHE.STARGATE+22102026.r24, backups bin/*.22102026.r23c-backup); S7 FAIL at mine_evolution (KeyError 'R24') → fixed in bed3ef4	pipeline record pipeline_20261102T090301
pipeline_20261102T103606 (project-only 2)	PASS: KATs 35/35, harness mutations 34/34, gates 19/22 (+3 PAUSED, miner)	pipeline record pipeline_20261102T103606
pipeline_20261102T104901 (full run 3)	S3 FAIL: the opt-in simulator tests (R24_EVICT, R24_TXPOOL_LOAD) were registered in NEW_TESTS and SKIP without their env → kept out of NEW_TESTS, KAT NT1 refuses an env-gated skip (cc1c391)	pipeline record pipeline_20261102T104901
pipeline_20261102T110200 (full run 4)	guard PASS; format PASS; flaky 120 ×3, 0 flaky; tests PASS; soundness 29 tests; KATs 17 files PASS; mutations 1108/1108 (upstream failures recorded 3); race 26 packages 0 races; S4b / S4c PAUSED (miner); shadow 120/120 blocks, 240 treehashes, 0 mismatches, 0 regressions; install SKIP (--no-install); S7 PASS gates 21/24 (+3 PAUSED, miner)	pipeline record pipeline_20261102T110200
pipeline_20261102T123214 (project-only, after the registry commit)	PASS: KATs 35/35, harness mutations 34/34, gates 21/24 (+3 PAUSED, miner); G22, G23, G24 PASS	pipeline record pipeline_20261102T123214; harness/loop/GATES.md
pipeline_20261102T125906 (project-only, closing run after the R24 artifacts commit)	PASS: KATs 35/35, harness mutations 34/34, gates 22/25 (+3 PAUSED, miner); G22, G23, G24, G25 PASS; coherence 955 statements, 0 drifts; 0 refused rows	pipeline record pipeline_20261102T125906; harness/loop/GATES.md
Live r24 node: restart and reads	2026-11-02 15:31:03 (r23c shut down cleanly 15:30:46); reads 15:41:24 and 16:32:01	derod.log (its --log-dir) banner Version 3.6.0-153.DEROHE.STARGATE+22102026.r24 at 15:31:03; harness/runtime/r24_live_first_20261102.txt, r24_live_hour_20261102.txt
Live r24 node health	synced, lag 0, tip age 22.5–27.5 s, 12 outgoing / 0 incoming; 0 ERROR / 0 WARN since the restart; 0 recovered panics, 0 p2p bans	harness/runtime/r24_live_first_20261102.txt and r24_live_hour_20261102.txt (DERO.GetSyncInfo; errors since restart 0)
Live r24 integrity RPC = auditor (CI24-1, BD7)	S1–S4 PASS twice (23:41 and 00:32 UTC); S3: 18 shared fields equal	harness/runtime/disclosure_live_20261102T234142Z.txt, disclosure_live_20261103T003206Z.txt (check PASS S3)
Live r24 JSON-RPC without Content-Type (B38)	HTTP 415 with the JSON-RPC -32600 body	harness/runtime/r24_live_first_20261102.txt (== no content-type)
Live r24 emission audit after 1 h	219 audited, 0 alerts, payees 1,447 / 1,447, mean 3,389 µs/block, 2 reorgs; the 24 h read is pending	harness/runtime/r24_live_hour_20261102.txt (DERO.GetEmissionAudit 16:32)
Live r24 proof-scalar census	last 100,000 blocks: 15,782 proof scalars, 0 non-canonical	harness/runtime/disclosure_census_20261103T004204Z.json (tip 7,847,749)

MEASURED	RESULT	SOURCE
Live r24 new counters (B18, B34, security, flips)	◌ all 0 after 1 h (no tx block, no sync, RPC on loopback, 0 inbound, no miner): B18 and B34 stay pending measurement, the flip counters PAUSED (miner)	harness/runtime/r24_live_hour_20261102.txt (== metrics); r24_live_txcount_20261102.txt (blocks 217 tx_total 0)
Live miner	◌ PAUSED (miner): no live miner this round	artifacts/dero-rft.md §0.3; harness/round.json tracks.miner

R25 · 2026-11-03 · +22102026.r25

NODE-ONLY ROUND PLUS THE MONETARY-FORENSICS ENGINE: TYPED RPC ERRORS, CONSENSUS AND PEER COUNTERS, SYNC HOLD, DATASET V1 OF THE HF3 WINDOW WITH REPLAY, AUDIT AND ITS FRONTEND

miner ◌ PAUSED (miner) mutations 1271/1271 gates 24/27 (+3 PAUSED, miner) forensics dataset v1: replay + audit + G30

live r25 node ◌ pending (restart)

WHAT CHANGED

- Miner track ◌ PAUSED (miner): no miner code changed; G17, G18, G21 and S4b / S4c report PAUSED while `harness/round.json` says `tracks.miner = paused`.
- RPC (rpcrobust): a GetEncryptedBalance or GetSC below the prune point gets the typed error `-32031` before any read (B85; the R25 discovery probe had made the node log 4 recovered panics); all 31 `panic(` lines of `cmd/derod/rpc` were audited and every input-reachable one returns a typed error (B86); GetBlockHeaderByHash no longer shadows its error (B80); GetTransaction gains `txs_status`; GetInfo `total_supply_basis` says total_supply is formula-computed (BD6). A standing derod error census (`harness/intel/error_census.py`) and a prune guard for every harness probe (`harness/kit/pruneguard.py`) run each S7.
- Consensus observability: the seven CO24 node-side gaps are closed as logging, counters and RPC fields only (B55–B61), with the new read-only `DERO.GetConsensusStats`; consensus guard PASS.
- Peers: per-peer offence scores, ban reasons and sources, Chain-request, INV-clip, unserved-object and stuck-hold counters (B63–B69); a peer that answers a sync with nothing is held from sync instead of re-picked every 4 s (B47); no observed peer address is left in tracked Go sources (B70) or new runtime snapshots (B71).
- Security and pool: netgroup quota on the proven lane and a quarter burst for fresh remote keys (B48), rebroadcast skips mined txs (B42), the pool cap holds under concurrency (B44), relay/RPC admission refuses non-reduced proof scalars (BD5).
- Crypto and disclosure: BD9 settled by an executable honest-proof test (2 of 13 scalar classes refused, 11 accepted under one TXID); B22 refuted above height 399,000 (exactly one final miniblock; 20,000 of 20,000 live blocks); the disclosure track re-tested every claim (G24).
- Perf and ops: the sync-CPU herd fix cuts CPU per synced block ×0.923 but sync blocks/s is set by the one-at-a-time block add (B46, refuted as a lever); `tools/launchd-derod -preflight` (B35); B39 read live on the r24 binary for 75 min.
- Monetary forensics (new, rft §0.8): dataset v1 of the HF3 window with content-pinned replay (G28), an independent second-implementation audit whose one overturn (A4b) was corrected at its source and kept on record (G29), and this page's forensics sections, where every figure names its dataset file and SHA-256 (G30). Go to Monetary forensics.
- Soundness review, now one page per round (rft §0.6): dero-soundness-r29.html re-verifies the review on the r25 build (G22, forensics grades, audit) and the Round 23 page stays unchanged as history.
- Tooling track (new, rft §0.9): area `tooling` with `tooling_intel.py`, a scrubbed memory mirror, the meta review, the resume checkpoint (`round_checkpoint.py`, §0.10) and a KAT for every R24/R25 tooling defect class.
- Live r25 readings `measured 2026-11-04`: restart 06:57:28, 63 min read-only; 60/60 samples synced, lag 0, 14–15 outgoing / 0 incoming, 0 ERROR / 0 WARN, 0 recovered panics, 0 bans; GetConsensusStats, refusal counters (insert_failed 834, stale_height 11), alt_blocks_count 7 = 7, fees and first-seen fields, 2 reorgs logged and persisted, peer observatory in counter mode, ban_reasons.json, typed not-found −32032, txs_status, total_supply_basis, B18 observed 4; pending an event: B47 hold, B48 lane quota, B57, B60 across a restart, B61; B85 −32031 unit-proven only (never sent live).

WHY

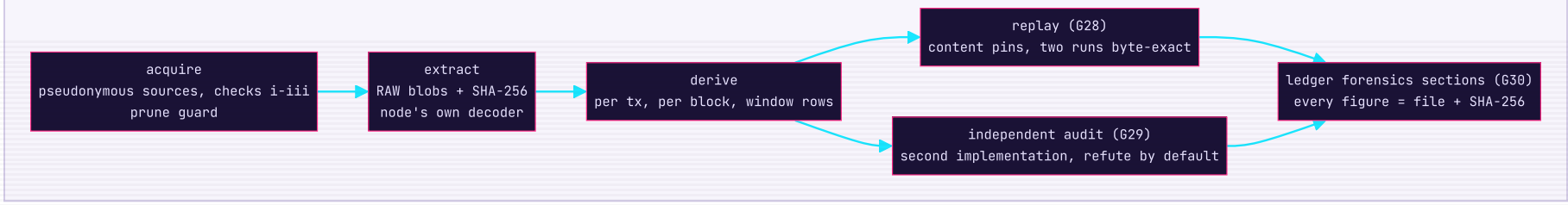
> The user asked (2026-11-03, rft §2.11) for a reproducible monetary-forensics engine for the Foundation's range-proof disclosure that separates what can be proved from what is speculation, and for the tooling around the source to be massively enhanced; the miner stays paused. The R25 discovery found that the live node can serve none of the HF3 window (everything lies below its prune point), that a state read below that point made the node log recovered panics (B85), and that seven consensus and seven peer facts the observatories needed were not readable from the

node (CO24). Every node dimension's builder was checked by an adversarial verifier, and every verifier upheld its dimension (with fixes merged: B41 completed, B47 wiring tested, the perfops preflight false PASS closed). The forensics stages ran as their own workflow (acquire, extract, replay, audit); the audit refuted by default and overturned one claim.

CODE

R25 adds no code excerpt here: the RPC and observability changes are counters and typed errors, and the forensics engine is described by its dataset below.

FLOW



EVIDENCE

MEASURED	RESULT	SOURCE
Workflows (R25)	node workflow wf_73b0a863-dde: 7 dimensions (rpcrobust, consensus, peers, secpool, cryptodisc, perfops, harnesstool), 14 agents, every verifier upheld; forensics workflow wf_f130a2af-f58: 4 stages; node merge 1b047f7 (171 files, +14,731 / -477); forensics integrated as 84a0103	workflow journals; harness/loop/runs/r25_dimension_results_20261103.json; git show --stat 1b047f7
Tests added by the dimensions	78 (rpcrobust 20, consensus 12, peers 19, secpool 5, cryptodisc 3, perfops 11, harnesstool 8); Go mutations unioned 1108 → 1271	harness/loop/runs/r25_dimension_results_20261103.json tests_added; merge report
B85 below the prune point (shadow on an APFS clone, base vs r25)	base: recovered panics and ERROR lines for state reads below prunedheight; r25: typed -32031, a rate-limited WARNING, no stack, a metric	harness/loop/runs/r25_dimension_results_20261103.json rpcrobust B85
B86 panic(audit	31 lines in 6 handler files: every input-reachable site returns a typed error, invariants kept with a written justification	harness/loop/runs/r25_dimension_results_20261103.json rpcrobust B86
B51 live CPU per GetRandomAddress (r24 binary)	25.1–40.9 ms CPU/call over 2 runs × 120 calls (verifier 18.3 and 27.0 ms; 1.0–2.3× the calibration); loopback only	harness/loop/runs/r25_dimension_results_20261103.json cryptodisc CD25-1 (live read booked to the r24 binary)
B22 final-miniblock count (live)	REFUTED · 20,000 of 20,000 blocks have one final and 10 miniblocks; consensus forces F = 1 from height 399,000	harness/loop/runs/r25_dimension_results_20261103.json cryptodisc CD25-3; harness/runtime/r25_cryptodisc_finals_20261103.txt
B46 sync CPU herd fix (36-run shadow campaign)	REFUTED · CPU per synced block ×0.923; sync blocks/s unchanged: the one-at-a-time block add (guarded) sets it	harness/loop/runs/r25_dimension_results_20261103.json perfops PO25-1
B39 flip delay (live, 75 min, r24 binary)	dead windows explained by the 100 ms template cache, 0 slow windows	harness/loop/runs/r25_dimension_results_20261103.json perfops PO25-2
Peers shadow	120/120 blocks equal, treehash equal at every matching sample; Go mutations 29/29, KAT mutations 23/23	harness/loop/runs/r25_dimension_results_20261103.json peers PEERS25-15
pipeline_20261103T125920 (full run 1)	S3a FAIL: TestR25ConsensusRPCOnSimulatorChain flaky with -count > 1 (a fixed ghost-block hash already settled) → fresh hash per run (99ef779)	pipeline record pipeline_20261103T125920
pipeline_20261103T130341 (full run 2)	S1–S5 PASS; S6 refused by the provenance guard because a commit landed mid-run (self-caused)	pipeline record pipeline_20261103T130341
pipeline_20261103T144257 (full run 3, install)	S1–S6 PASS, installed 3.6.0-153.DEROHE.STARGATE+22102026.r25 (backups bin/*.22102026.r24-backup, renamed from r24i/r24p after the tag-reader fix); S7 FAIL: a triage row missing from src-arch (verify_deroarch) → fixed in d0b5762	pipeline record pipeline_20261103T144257
pipeline_20261103T165745 (full run 7)	S5 FAIL: the shadow control was bin/derod, which after the install is r25 → the control is now the live node's build (354a8e0, KAT IT3)	pipeline record pipeline_20261103T165745

MEASURED	RESULT	SOURCE
pipeline_20261103T182557 (full run 8)	guard PASS; format PASS; flaky 88 ×3, 0 flaky; tests PASS; soundness 29 tests; KATs 35 result files PASS; mutations 1271/1271 (upstream failures recorded 3); race 26 packages 0 races; S4b / S4c PAUSED (miner); shadow 120/120 blocks, 240 treehashes, 0 mismatches, 0 regressions (control bin/derod.22102026.r24-backup (the live node's build) against the r25 candidate; DERO.GetConsensusStats: control no such method, candidate ok); install SKIP (--no-install); S7 PASS gates 24/27 (+3 PAUSED, miner)	pipeline record pipeline_20261103T182557
pipeline_20261103T164836 (project-only, run 6)	PASS: KATs 35/35, harness mutations 34/34, gates 24/27 (+3 PAUSED, miner)	pipeline record pipeline_20261103T164836
pipeline_20261104T053904 (project-only, closing run after the R25 artifacts commit)	PASS: KATs 35/35, harness mutations 34/34, gates 27/30 (+3 PAUSED, miner); G23-G30 PASS; coherence 0 drifts (run 9, pipeline_20261104T052308, failed only G7: the src-arch forensic line overflowed at 480 px, fixed at its source)	pipeline record pipeline_20261104T053904; harness/loop/GATES.md
Forensic dataset v1	see Monetary forensics (every figure there traces to the dataset, G30)	forensics/r25/manifest.json; forensics/ledger.json
Live r25 node: restart and window	3.6.0-153.DEROHE.STARGATE+22102026.r25 running since 2026-11-04 06:57:28 (pid 34438), read to 08:00:18	harness/runtime/r25_live_window_20261104.txt (ps lstart, GetInfo version)
Live r25 node health	60/60 samples synced, lag 0, 14–15 outgoing / 0 incoming, 0 ERROR / 0 WARN, 0 recovered panics, 0 bans	harness/runtime/r25_live_window_20261104.txt (series lines; harness/runtime/error_census_20261104T080034.json)
Live r25 features confirmed	GetConsensusStats, refusal counters (insert_failed 834, stale_height 11), alt_blocks_count 7 = 7, fees and first-seen fields, 2 reorgs logged and persisted, peer observatory in counter mode, ban_reasons.json, typed not-found -32032, txs_status, total_supply_basis, B18 observed 4	harness/runtime/r25_live_window_20261104.txt (GetConsensusStats, metrics, sidecars, own calls); harness/runtime/peer_intel.json evidence_mode
Live r25 features without an event in the window	◌ B47 hold, B48 lane quota, B57, B60 across a restart, B61; B85 -32031 unit-proven only (never sent live)	harness/runtime/r25_live_window_20261104.txt (metrics: p2p_sync_peer_holds_total 0, proven lane 0, block_rejected 0; rpc_state_unavailable_total absent)
Live r25 emission audit	227 audited, 0 alerts, issued = formula 6,918,750, payees 1,451 / 1,451, mean 4,473 µs/block, 2 reorgs; the 24 h read is pending (from 2026-11-05 06:57)	harness/runtime/r25_live_window_20261104.txt (GetEmissionAudit 08:00:18)
Live r25 disclosure track	S1–S4 PASS (15:11 UTC); census 15,440 proof scalars, 0 non-canonical	harness/runtime/disclosure_live_20261104T151132Z.txt; disclosure_census_20261104T151717Z.json
Live miner	◌ PAUSED (miner): no live miner this round	artifacts/dero-rft.md §0.3; harness/round.json tracks.miner

R26 · 2026-11-04 · +22102026.r26

NODE-ONLY ROUND PLUS FORENSICS DATASET V2 BY DEFECT CLASS: SYNC-HOLD DECAY, DUPLICATE-RACE COUNTING, TYPED EXPLORER ERRORS, A SECOND AUDITOR AND THE V2 FRONTEND

WHAT CHANGED

- Miner track `◌ PAUSED (miner)`: no miner code changed; G17, G18, G21 and S4b / S4c report PAUSED while `harness/round.json` says `tracks.miner = paused`.
- P2P (node-net): sync-hold strikes decay and an accepted relayed block that brings us level forgives the peer; when no peer synced, held peers that are ahead are tried (B99); a duplicate miniblock delivered by two peers inside the PoW-verify window is counted `duplicate_race`, not `insert_failed`, and every refusal line carries the error text (B103); the invalid_pow policy is kept and its comment corrected (B102). Counting and logging only; consensus guard PASS.
- RPC (node-rpc): NameToAddress for an unknown name answers `-32032` (B100); the explorer range methods refuse unknown parameter names with `-32602` and reads below the prune point with `-32031` before any store read (B104); GetRandomAddress keeps its decoy window at every height (B43: the simulator failures were stale decoys).
- Crypto and disclosure: a structural transcript-binding test pins the proof transcript identical at the head, Release152 and the genesis-era tag, with no invalid proof built (FQ4); FQ5 says no public-

data test separates a defect-accepted proof from an honest one (B113).

- Monetary forensics: dataset v2 settles the v1 pending-fee blocks from RAW bodies (B96), tags rows by defect class with bound honesty (G31) and class separation (G32), replays the held archive through the HF3 path to our treehash (B123), and has a second independent auditor (audit2; G29 on every version). Go to Monetary forensics.
- Soundness review for this round: dero-soundness-r29.html, with its "What can be known" section; the Round 25 page stays unchanged.
- Tooling: S5 shadow teardown (B98), byte-exact src-arch (B93), network-class area metrics (B105), scrubbed argv in new snapshots (B119).
- Live r26 readings `measured 2026-11-05`: restart 07:48:21, 62 min read-only; 55/55 samples synced, lag 0, 13–14 outgoing / 0 incoming, 0 ERROR / 0 WARN, 0 recovered panics, 0 bans; duplicate_race 767 with insert_failed 0 in 55/55 samples (B103; each refusal line names its collision), −32032 for an unknown name (B100), −32602 for unknown parameter names (B104), the reorg monitor across a restart (starts 2, B60), B18 observed 4; pending an event: B99 decay and forgiveness (0 holds), B47 hold, B48 lane quota, B57, B61; B85 −32031 unit-proven only (never sent live).

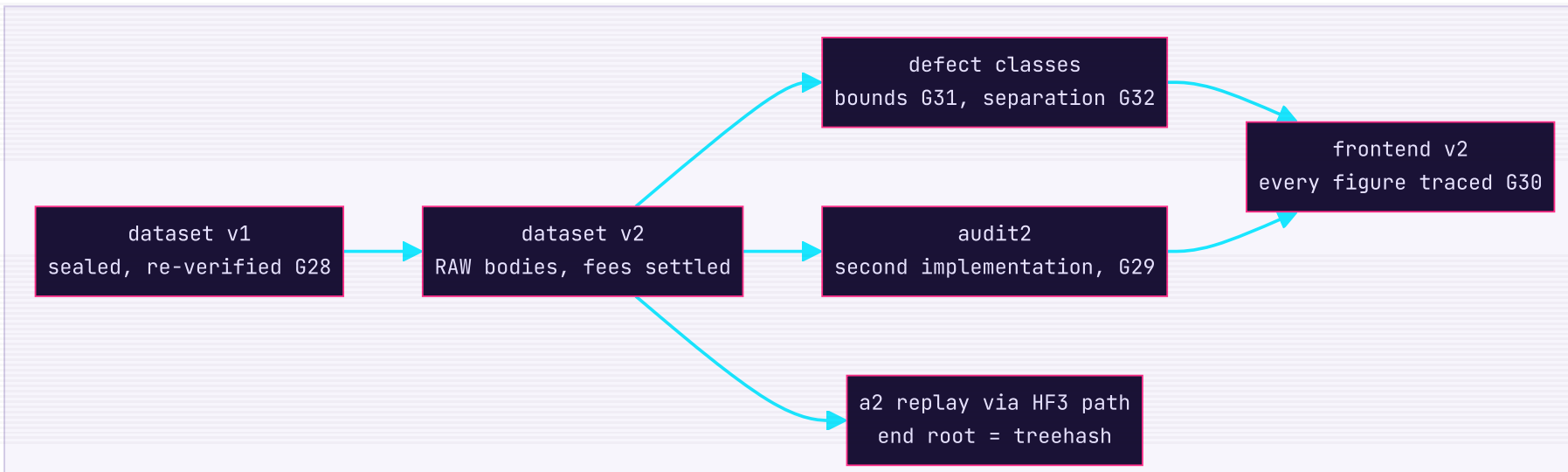
WHY

> The user asked (rft §2.15, §0.8.9) for the forensics to separate the defect classes and to say honestly what each bound is; the miner stays paused. R26 started from the R25 measurement: insert_failed counted duplicate deliveries (B103), a malformed probe read topoheight 0 silently (B104), and sync holds never decayed (B99). Six dimensions ran as builder plus fresh-context verifier; no claim was refuted, and the PARTLY findings went in as corrected claims. The merge found two interface defects (B123, B125), each fixed with a test.

CODE

R26 adds no code excerpt here: the node changes are counters, typed errors and a decay rule, and the forensics are described by their dataset below.

FLOW



EVIDENCE

MEASURED	RESULT	SOURCE
Workflows (R26)	`open loop wf_4edf0851-905: 6 dimensions (harness-tooling, forensics-data, forensics-classes, disclosure-cryptoinfl, node-net, node-rpc), builder + verifier each, 0 REFUTED; integration wf_159d7b8a-386: merge fea0206b (117 files), audit2 and frontend2 c802e4bc`	`artifacts/dero-rft.md §13 R stat fea0206b c802e4bc`
Merge-detected defects	`B125: 4 class files unlisted in the v2 manifest (G28 FAIL) → class_files; B123: the a2 replay skipped the HF3 step → p2p.Add_Complete_Block_HF3, 31,025 blocks to our treehash`	`git show fea0206b; harness/runtime/r26_merge_a`
pipeline_20261104T175125 (full run 1)	`S1-S3k PASS; S3b FAIL: 1327/1329 Go mutations, 2 STALE (B103/B99 rewrote their anchors in p2p/consensus_obs_r25.go) → re-anchored, KAT test_gomut_anchors_r26.py GA1-GA2 now checks every anchor before S3b (d3189e1a)`	`pipeline record pipeline_20`
pipeline_20261104T193301 (full run 2, install)	`S1-S6 PASS: mutations 1329/1329, race 26 packages 0 races, shadow 120/120 blocks 240 treehashes 0 mismatches (control the live r25 build), installed 3.6.0-153.DEROHE.STARGATE+22102026.r26 (backups bin/*.22102026.r25-backup); S7 FAIL: 14 r25 forensics area anchors on the rewritten forensics_intel.json → per-round sealed copy, KAT test_area_seal_r26.py (c643634f)`	`pipeline record pipeline_20`
pipeline_20261104T213409 (project-only, run 3)	`PASS: KATs 35/35, harness mutations 34/34, gates 29/32 (+3 PAUSED, miner), coherence 0 drifts`	`pipeline record pipeline_20`
pipeline_20261104T220624 (full run 4, --no-install)	`guard PASS; format PASS; flaky 32 ×3, 0 flaky; tests PASS; soundness 29 tests; KATs 47 result files PASS; mutations 1329/1329; race 26 packages 0 races; S4b / S4c PAUSED (miner); shadow 120/120 blocks, 240 treehashes, 0 mismatches, 0 regressions (control bin/derod.22102026.r25-backup (the live node's build) against the r26 candidate); install SKIP (--no-install; installed by run 2); S7 gates 29/32 (+3 PAUSED, miner)`	`pipeline record pipeline_20`
Installed derod	`3.6.0-153.DEROHE.STARGATE+22102026.r26, sha256 f6ff3b005872f1cd791310f769e7b67c9dc5d00e340b47c0ebae9fd12bcc5692`	`shasum -a 256 bin/derod aft`

MEASURED	RESULT	SOURCE
Forensic dataset v2	see Monetary forensics (every figure there traces to the dataset, G30)	forensics/r26/manifest.json
pipeline_20261105T003812 (project-only, closing run after the R26 Part D commit)	PASS: KATs 35/35, harness mutations 34/34, gates 29/32 (+3 PAUSED, miner); coherence 0 drifts	pipeline record pipeline_20 harness/loop/GATES.md
Live r26 node: restart and window	3.6.0-153.DEROHE.STARGATE+22102026.r26 running since 2026-11-05 07:48:21 (pid 93240), read to 08:50	harness/runtime/r26_live_wi lstart, GetInfo version)
Live r26 node health	55/55 samples synced, lag 0, 13-14 outgoing / 0 incoming, 0 ERROR / 0 WARN, 0 recovered panics, 0 bans	harness/runtime/r26_live_wi lines); harness/runtime/error_censu
Live r26 features confirmed	duplicate_race 767 with insert_failed 0 in 55/55 samples (B103; each refusal line names its collision), -32032 for an unknown name (B100), -32602 for unknown parameter names (B104), the reorg monitor across a restart (starts 2, B60), B18 observed 4	harness/runtime/r26_live_wi (GetConsensusStats, metrics own calls)
Live r26 features without an event in the window	◌ B99 decay and forgiveness (0 holds), B47 hold, B48 lane quota, B57, B61; B85 -32031 unit-proven only (never sent live)	harness/runtime/r26_live_wi (metrics: p2p_sync_peer_hol 0; rpc_state_unavailable_to
Live r26 emission audit	218 audited, 0 alerts, issued = formula 6,672,750, payees 1,367 / 1,367, mean 4,273 µs/block, 1 reorg	harness/runtime/r26_live_wi (GetEmissionAudit 08:50:33)
B36 24 h emission (the r25 segment)	r25 segment 2026-11-04 06:57:28 → 2026-11-05 07:48:16 re-audited offline: 4,846 blocks 7,855,177..7,860,022 ok, 0 alerts, issued = formula 149,014,500; payee counts not readable	harness/runtime/r26_b36_emi
Live r26 disclosure track	S1–S4 PASS (16:52 UTC); census 15,506 proof scalars, 0 non-canonical	harness/runtime/disclosure_ disclosure_census_20261105T
Live miner	◌ PAUSED (miner): no live miner this round	artifacts/dero-rft.md §0.3; tracks.miner

R27 · 2026-11-05 · +22102026.r27

NODE-ONLY AREA-ENHANCEMENT ROUND: TYPED PARAMETER ERRORS THAT NAME THE FIELD, COUNTED RECOVERS, VISIBLE BANS, A REPORTED CONSENSUS ENV OVERRIDE, EMISSION PAYEE RECORDS, AND A HARNESS WHOSE EVERY TOOL ANSWERS --HELP

WHAT CHANGED

- Miner track ◌ PAUSED (miner): no miner code changed; G17, G18, G21 and S4b / S4c report PAUSED while `harness/round.json` says `tracks.miner = paused`.
- RPC (rpc-metrics): a parameter-decode failure answers `-32602` naming the field and the accepted names (B132; R26 had answered "invalid parameters: [-32602] invalid parameters"); the four untyped panic recovers of the emission and supply audits count `rpc_internal_error`, and `rpc_error_responses_total{code}` counts every error answer (B143).
- P2P and config (p2p-config): a ban reloaded from `ban_list.json` is visible as `p2p_bans_active` and `p2p_bans_loaded_total`, also in GetSyncInfo (B142); a consensus parameter set through the environment is reported by an ERROR line and `config_consensus_env_overrides`, never refused (B141, the user decides). Consensus guard PASS: config/ and blockchain/ unchanged.
- Emission and glue (glue-global, harness-measure, tools-kats): payee counts in the series and archive records (B36 fields), a height-0 archive hang and a dropped series record fixed, the rwc adapters report failed writes (B144), `tools/emission-audit -archive` prints its payees (B139). The three globals amount and offset fixes are held: globals.go is a content-pinned decoder input of the sealed dataset forensics/r25 (B147).
- DVM and blockchain (dvm-blockchain): tests only. KATs for every SC builtin that had no test (dvm coverage 72.2 → 84.4 %) and difficulty vectors (difficulty.go 41.5 → 81.9 %); six consensus-visible DVM behaviours are pinned and documented, never changed (B145).
- Harness and tooling (tools-kats, harness-measure, artifacts-memory): every harness tool answers `--help` without running (B120: 33 → 0), `gen_deroarch.py build --strict` names the miner whose raw file lost anchors (B137), a KAT mutation counts as proved only by its own KAT (B138), the frontend2 KAT runs 852.83 → 244.67 s (B133), `live_window.py` reads the after-restart window (B135).
- Soundness review for this round: dero-soundness-r29.html; the Round 26 page stays unchanged.
- Live r27 readings `measured 2026-11-05`: restart 17:51:48, 60 min read-only; 60/60 samples synced, lag 0, 14 outgoing / 0 incoming, tip block age 11.0 s, 0 ERROR / 0 WARN in 1,199 lines, 0 recovered panics; −32602 names the field and the accepted names in 3/3 malformed probes, rpc_error_responses_total{code="-32602"} and rpc_params_refused_total 0 → 3 (B132, B143); the 18 pre-registered rpc series read 0 before any event (B143); the reloaded ban visible as p2p_bans_active 1, p2p_bans_loaded_total 1, dero_p2p_ban_list_entries 1 and in GetSyncInfo (B142); config_consensus_env_overrides 0 and no override line (B141); series checkpoints carry payees 14 / 296 / 960 (B36); duplicate_race 742 with insert_failed absent in 60/60 samples (B103); reorg monitor starts 3 (B60); pending an event: B99 decay and forgiveness (0 holds), B47

hold, B48 lane quota (0 inbound), B57 (0 rejections), B61; B85 −32031 unit-proven only (never sent live).

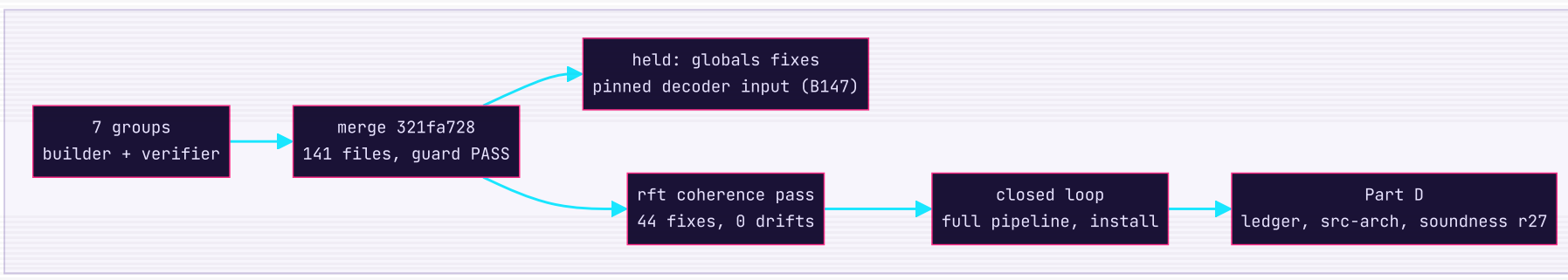
WHY

> R27 (rft §8 Round 27 plan, seven groups) enhanced the areas themselves: the harness, measurements, tools, KATs, artifacts and the memory mirror, p2p, config, rpc, metrics, the glue, globals, dvm and blockchain, with dvm/ and blockchain/ limited to tests and observability; the miner stays paused. R27 started from the R26 measurement: the −32602 message doubled its code and named no field (B132), the project-only wall time had grown 512 → 1,901 s (B133), and area_intel booked recorded audit findings as regressions (B134). Seven groups ran as builder plus fresh-context verifier; three verifiers corrected their patches (tools-kats, p2p-config, rpc-metrics) and no claim was merged unverified. The merge found one interface defect (B147: a fix in a content-pinned file) and held it with its tests.

CODE

R27 adds no code excerpt here: the node changes are typed errors, counters and report-only start-up checks.

FLOW



EVIDENCE

MEASURED	RESULT	SOURCE
Workflow (R27)	wf_eae21b40-28b: 7 groups (harness-measure, tools-kats, artifacts-memory, p2p-config, rpc-metrics, glue-global, dvm-blockchain), builder + verifier each; merge 321fa728 (141 files, +13,555 / -1,876); rft coherence pass 99720ff1 (44 fixes) and 79fe6b78	harness/runtime/r27_merge_2 stat 321fa728
Merge checks	consensus guard PASS (0 consensus-package files vs cee1d80f); go test 13 packages ok; KAT mutations 954/954; go_mutate gm-r27 119/119 CAUGHT + 9 re-targeted 9/9; Go mutations 1,329 + 124 - 5 held = 1,448	harness/runtime/r27_merge_2
Merge-detected defect B147	globals/globals.go is a content-pinned decoder input of the sealed forensics/r25 (MF12 FAIL): the fixes, their 3 tests and 5 mutations held in harness/loop/held/	harness/runtime/r27_merge_2
pipeline_20261105T133652 (full run 1)	S1-S3 PASS (flaky 80 ×3, 0 flaky); S3k FAIL: test_gomut_anchors_r26 GA1, gm-r27-p2pcfg-cfg-version-date anchored on the round tag +22102026.r26 and went STALE at the r27 version bump → round-independent anchor, KAT GA3 (72266d4f)	pipeline record pipeline_20
pipeline_20261105T140432 (full run 2, install)	guard PASS; format PASS; flaky 80 ×3, 0 flaky; tests PASS; soundness 29 tests; KATs 53 result files PASS; mutations 1448/1448; race 27 packages 0 races; S4b / S4c PAUSED (miner); shadow 120/120 blocks, 240 treehashes, 0 mismatches, 0 regressions (control bin/derod (the live node's r26 build) against the r27 candidate); install PASS (atomic rename, backups bin/*.22102026.r26-backup); S7 gates 29/32 (+3 PAUSED, miner), KATs 35/35, harness mutations 34/34	pipeline record pipeline_20
Installed derod	3.6.0-153.DEROHE.STARGATE+22102026.r27, sha256 1c2d2b9123a75bd2d26867e7af4d2407c915429eb62f62f1dc29df5d5442bb47	shasum -a 256 bin/derod aft
pipeline_20261105T163040 (project-only, closing run after the R27 Part D commit)	PASS: gates 29/32 (+3 PAUSED, miner)	pipeline record pipeline_20
B51 GetRandomAddress CPU per call (live, r26 binary)	22.7 / 22.5 ms CPU per call (2 × 200 calls, 100 addresses per call, idle subtracted), client wall p50 17.5 / 17.1 ms; loopback only, under benchlock	artifacts/dero-rft.md §7 B5
Live r27 node: restart and window	3.6.0-153.DEROHE.STARGATE+22102026.r27 running since 2026-11-05 17:51:48 (pid 77276), read 17:53:50 → 18:52:50 (60 samples)	harness/runtime/r27_live_wi sample 121 s after the dero
Live r27 node health	60/60 samples synced, lag 0, 14 outgoing / 0 incoming, tip block age 11.0 s, 0 ERROR / 0 WARN in 1,199 lines, 0 recovered panics	harness/runtime/r27_live_wi lines); harness/runtime/error_censu harness/runtime/r27_live_em block age)
Live r27 features confirmed	-32602 names the field and the accepted names in 3/3 malformed probes, rpc_error_responses_total{code="-32602"} and rpc_params_refused_total 0 → 3 (B132, B143); the 18 pre-registered rpc series read 0 before any event (B143); the reloaded ban visible as p2p_bans_active 1, p2p_bans_loaded_total 1, dero_p2p_ban_list_entries 1 and in GetSyncInfo (B142); config_consensus_env_overrides 0 and no override line (B141); series checkpoints carry payees 14 / 296 / 960 (B36); duplicate_race 742 with insert_failed absent in 60/60 samples (B103); reorg monitor starts 3 (B60)	harness/runtime/r27_live_pr harness/runtime/r27_live_wi harness/runtime/r27_live_em

MEASURED	RESULT	SOURCE
Live r27 features without an event in the window	◌ B99 decay and forgiveness (0 holds), B47 hold, B48 lane quota (0 inbound), B57 (0 rejections), B61; B85 -32031 unit-proven only (never sent live)	harness/runtime/r27_live_wi (metrics: p2p_sync_peer_hol blockchain_block_rejected_t
Live r27 emission audit	230 audited, 0 alerts, issued = formula 7,011,000, payees 1,508 / 1,508, mean 3,774 µs/block, 2 reorgs	harness/runtime/r27_live_em (GetEmissionAudit 18:57:17)
Live r27 disclosure track	S1–S4 PASS (02:57 UTC); census 15,422 proof scalars, 0 non-canonical	harness/runtime/disclosure_ disclosure_census_20261106T
Live miner	◌ PAUSED (miner): no live miner this round	artifacts/dero-rft.md §0.3; tracks.miner

R28 · 2026-11-06 · +22102026.r28

THE CLOSURE ROUND: EVERY OPEN ITEM OF R1–R27 AUDITED (156 UNIQUE: 40 CLOSED BEFORE, 82 CLOSED IN R28, 34 OPEN WITH A BLOCKER), AND NINE GROUPS THAT ENHANCED THE HARNESS, TOOLS, KATS, ARTIFACTS, MEMORY, P2P, CONFIG, RPC, METRICS, GLUE, GLOBALS, DVM, BLOCKCHAIN AND FORENSICS

WHAT CHANGED

- Miner track `◌ PAUSED (miner)`: no miner code changed; G17, G18 and G21 report PAUSED while `harness/round.json` says `tracks.miner = paused`; S4b ran its offline A/B because `globals/` and `metrics/`, which the miner links, changed (staged vs installed 1.002 H/cpu-s like for like, no live miner); S4c SKIP (server unchanged); the 11 paused miner rows wait for the user (rft §0.3).
- Closure audit (fresh context, `harness/runtime/r28_closure_audit_20261106.txt`): the 134 items of the nine "Open after Round N" lists of `dero-r1-r27.html`, the 4 R23b standing constraints, the 179 rft §7 rows and the 21 rows the merge created, 156 unique; 82 closed in R28 with a record path each, 34 left open with a blocker class and the observable that closes them. Closed in Round 28 · Open after Round 28.
- P2P and config (p2p-config): the pending-event rows B47, B48, B57 and B99 closed by unit tests plus an executable scenario each (`harness/intel/event_scenarios.py`, 22/22); `harness/intel/config_truth.py --go` compares every documented flag and environment name with the code (0 DRIFT); the chunk PoW pool counters `p2p_chunk_pow_*` for L24-21.
- RPC and metrics (rpc-metrics): GetInfo `tx_pool_unmined` (B42), counted malformed HTTP bodies (B150), HELP and TYPE for every /metrics family from an embedded registry (`metric_registry.py`, 304 families, B151), B61 closed (one tip only: topoheight = height), B51 decided with an A/B (keep the decoy slack).
- Glue and globals (glue-global): the held B147 globals fixes landed (ParseAmount exact, FormatMoneyPrecision 128-bit, GetOffset sign), the sealed r25 dataset re-verified through pinned snapshots; `harness/intel/b36_payee_read.py` for the B36 24 h payee read (EQUAL proven live on an interim checkpoint).
- DVM and blockchain (dvm-blockchain, tests only): process_transaction_sc 0.0 → 96.4 % and blockchain/ coverage 55.5 → 68.1 % (B146); the burn-tx and SC install/call KATs; report-only verdicts kept for the consensus-visible rows (B10, B11, B31, B33, B45, B145, B157).
- Forensics (forensics-closure): r26 sealed in the §6 order; dataset v3 `forensics/r28` with the run-C replay MATCH; B109, B123, B124, B126 closed; B76 resolved with its scope; B97, B107, B110 open for R29 with their data blockers.
- Harness, tools, KATs, artifacts and memory (harness-measure, tools-kats, artifacts-memory, backlog-sweep): `backlog_truth.py` checks every §7 row against its records (0 drifts), `roundlists.py` every round list (B149), 89/89 tools with a KAT, masked raw scrapes (B135), memory dead-name and dead-symbol KPIs (B140), the pubscrub home-form rule (B156), and the closed-loop and Part D fixes B171 (the S3b preflight event), B172 (a dead-reference misread), B173 (two mutations after the B147 landing) and B174 (the two §1 rules).
- Soundness review for this round: dero-soundness-r29.html; the Round 27 page stays unchanged.
- Live r28 readings `measured 2026-11-06`: restart 16:10:24, 60 min read-only; 60/60 samples synced, lag 0, 13–14 outgoing / 0 incoming, tip block age 4.3 s, 0 ERROR / 0 WARN in 1,180 lines, 0 recovered panics; GetInfo tx_pool_unmined in 60/60 samples (B42); the chunk PoW counters with slow = checks − pool_seen in 13/13 scrapes and a cache miss of 10.0 % over the window (B46/L24-21); the HTTP-layer −32600 answer counted, 0 → 1 (B150); HELP on 283/283 families, TYPE on 273/283 (B151; the gap is 9 vmrange histograms by design and B176); metrics_dump_posts_total registered at 0 (B153); env overrides 0, the reloaded ban, the 18 pre-registered rpc series, duplicate_race with insert_failed absent and the reorg monitor hold on r28; pending an event: B47, B48 (0 inbound), B57, B99 registered with no event in 3,540 s / 193 blocks; B61 topoheight = height 0/60; B85 −32031 never sent (rule). B36: the r27 segment read offline over 22.3 h (the restart came before 24 h): 4,342 blocks ok, 0 alerts, issued = formula 133,516,500, payees 28,461 = explained (28,459 verified), 43/43 checkpoints equal the live series.

WHY

The user asked (2026-11-05, rft §2.16) to close or resolve every remaining open item of the previous rounds, to enhance the harness, measurements, tools, memory control, KAT suites, artifacts, the memory mirror, p2p, config, rpc, metrics, the proof glue, globals, dvm and blockchain, and to keep dero-rft.md coherent; the miner stays paused. R28 built a closure inventory at its start (140 entries), split it into nine groups with a builder and a fresh-context verifier each, merged only the verified patches and closures, then had a fresh-context auditor re-derive the inventory from the ledger and rft §7 (161 entries, 156 unique) and a coherence verifier check the rft against its records. Verifiers refuted three proposals (a duplicate L24-21 row and the CLOSED verdicts for B63, B64 and B67, which stay implemented and pending an event) and corrected several claims; the merge found and fixed its interface defects (B165: a double landing, three conflicts and an unregistered metric family; B167: the r28 re-pin) and left tooling rows for R29 (B166, B168, B170).

CODE

R28 adds no code excerpt here: the node changes are counters, typed fields and test-only KATs, and the closures are records.

FLOW



EVIDENCE

MEASURED	RESULT	SOURCE
Workflow (R28)	wf_63f53449-9e7: 9 groups (harness-measure, tools-kats, artifacts-memory, p2p-config, rpc-metrics, glue-global, dvm-blockchain, forensics-closure, backlog-sweep), builder + verifier each; merge 10039edf (9 verified patches, 197 file entries)	harness/runtime/r28_merge_2
Merge checks	consensus guard PASS (0 consensus-package files vs 5a32c5b2; 7 allowed vs bf274e5); go test of the touched packages ok; -race p2p, rpc, metrics, cmd/derod/rpc 0 DATA RACE; KATs 65 files; Go mutations unioned 1,448 → 1,580	harness/runtime/r28_merge_2
Closure audit (fresh context)	156 unique items: closed before R28 40, closed in R28 82, open 34 (blocker class and closing observable each); 10 cited go tests re-run PASS; backlog_truth 0 drifts	harness/runtime/r28_closure
rft coherence pass	bb1c9565 + 86374336: G25 2,569 statements 0 drifts, backlog_truth 179 rows 0 drifts, config_truth PASS	git show --stat bb1c9565 86
pipeline_20261106T045640 (full run 1)	S1-S3k PASS (flaky 67 ×3, 0 flaky; soundness 29 tests); S3b crashed before any mutation: heavy_preflight passed stage= to event(stage, …) (TypeError; the r27b stage had never run in a full pipeline) → for_stage, KAT RH9 with the real event signature (B171, f8829666)	pipeline record pipeline_20
pipeline_20261106T080005 (full run 2, install)	guard PASS; format PASS; flaky 67 ×3, 0 flaky; tests PASS; soundness 29 tests; KATs 65 result files PASS; mutations 1579/1579; race 27 packages 0 races; S4b PASS (ran: globals/metrics changed; offline --bench, staged vs installed 1.002 H/cpu-s); S4c SKIP (server unchanged); shadow 120/120 blocks, 240 treehashes, 0 mismatches, 0 regressions (control bin/derod (the live node's r27 build) against the r28 candidate); install PASS (atomic rename, backups bin/*.22102026.r27-backup); S7 gates 29/32 (+3 PAUSED, miner), KATs 35/35, harness mutations 34/34	pipeline record pipeline_20
Installed derod	3.6.0-153.DEROHE.STARGATE+22102026.r28, sha256 2dad9f246bbf038b5c079704f98d5d2c9ec9c4f31a9fe51127b8d132c7bcbee7	shasum -a 256 bin/derod aft
pipeline_20261106T111617 (project-only, closing run after the R28 Part D commit)	PASS: gates 29/32 (+3 PAUSED, miner)	pipeline record pipeline_20
B36 payee read (the r27 segment, offline, 22.3 h)	the r27 segment read offline over 22.3 h (the restart came before 24 h): 4,342 blocks ok, 0 alerts, issued = formula 133,516,500, payees 28,461 = explained (28,459 verified), 43/43 checkpoints equal the live series	harness/runtime/r28_b36_emi
Live r28 node: restart and window	3.6.0-153.DEROHE.STARGATE+22102026.r28 running since 2026-11-06 16:10:24 (pid 80281), read 16:11:53 → 17:10:53 (60 samples)	harness/runtime/r28_live_wi sample 89 s after the derod
Live r28 node health	60/60 samples synced, lag 0, 13-14 outgoing / 0 incoming, tip block age 4.3 s, 0 ERROR / 0 WARN in 1,180 lines, 0 recovered panics	harness/runtime/r28_live_wi harness/runtime/error_censu harness/runtime/r28_live_em block age)
Live r28 features confirmed	GetInfo tx_pool_unmined in 60/60 samples (B42); the chunk PoW counters with slow = checks - pool_seen in 13/13 scrapes and a cache miss of 10.0 % over the window (B46/L24-21); the HTTP-layer -32600 answer counted, 0 → 1 (B150); HELP on 283/283 families, TYPE on 273/283 (B151; the gap is 9 vmrange histograms by design and B176); metrics_dump_posts_total registered at 0 (B153); env overrides 0, the reloaded ban, the 18 pre-registered rpc series, duplicate_race with insert_failed absent and the reorg monitor hold on r28	harness/runtime/r28_live_pr harness/runtime/r28_live_wi
Live r28 features without an event in the window	◌ B47, B48 (0 inbound), B57, B99 registered with no event in 3,540 s / 193 blocks; B61 topoheight = height 0/60; B85 -32031 never sent (rule)	harness/runtime/r28_live_wi (pending-event lines); harness/runtime/event_scena

MEASURED	RESULT	SOURCE
Live r28 emission audit	`221 audited, 0 alerts, issued = formula 6,734,250, payees 1,417 / 1,417, mean 4,787 µs/block, 2 reorgs`	harness/runtime/r28_live_em (GetEmissionAudit 17:12)
Live r28 disclosure track	`S1–S4 PASS (00:21 UTC); census 14,840 proof scalars, 0 non-canonical`	harness/runtime/disclosure_ disclosure_census_20261107T
Live miner	◌ PAUSED (miner): no live miner this round	artifacts/dero-rft.md §0.3; tracks.miner

R28d · 2026-11-07 · +22102026.r28d

THE COMMUNITY-DEV MERGE SUB-ROUND: THE VERIFIED DEROFDN COMMUNITY-DEV ITEMS MERGED BY ROWS (NEW OR MEASURED-BETTER ONLY), EACH WITH ITS OWN TEST HERE, AND THE FOREIGN BUILD MEASURED ONCE ON A CLONE

WHAT CHANGED

- Miner track ◌ PAUSED (miner): G17, G18 and G21 report PAUSED; the user's miner runs are reward-only (rft §0.3).
- Chain sync (B177): `chain_block_list` in `p2p/rpc_chain_request.go` stops at the first unreadable topo index instead of sending a zero block id; live: 6 chain requests served, 0 refused.
- Wallet RPC (B178): `scinvoke` takes an optional `fees`; 0 keeps the automatic fee; the consensus-guarded `rpc/` package is unchanged.
- Proof review (B179, B190): the binding census over 128 scalar × base sites and the Bulletproofs assumption table (14 DISCHARGED, 2 VIOLATED = disclosure C1, 2 UNCLEAR), as tests with anchors; no construction.
- Forensics (B180): dataset v4 `forensics/r28d`, the ring-size × payload-format criterion in FQ2 (60 of 75), audit 26/26; G28 re-verifies four versions.
- Tests (B181, B182, B188): the nonce mutations at ring 2 and 16, the registration-check pin, the inputs manifest (hashes only).
- Node run (B189): the community-dev build on an APFS clone followed the same chain below 8,000,000 (120/120 block hashes equal with r28d).
- Live r28d readings `measured 2026-11-07`: restart 17:47:55, 62/62 synced, 0 ERROR/WARNING, emission 0 alerts; the r28 B36 24 h payee read CONFIRMED before the restart.

WHY

> The user asked (2026-11-06, rft §2.17) to evaluate the DEROFDN community-dev branch and merge everything relevant into Release151 and dero-rft.md with no dependency on the evaluation clone, without degrading or duplicating the existing work. The evaluation (CD1–CD3) classified every item with a reproduced test; the merge took only NEW or measured-BETTER rows, report-only rows stay rows, and the consensus-visible fork row at 8,000,000 stays the user's decision (B184).

CODE

Two node-side changes and test-only additions; the fork row is not adopted.

FLOW



EVIDENCE

MEASURED	RESULT	SOURCE
Merge	`M-1..M-5 + B188 + the B182 pin; 3 non-test Go files changed, 0 files deleted, 0 tests removed; consensus guard PASS`	harness/runtime/r28d_merge_20261106.txt
pipeline_20261106T222539 (full run 1)	`S3k FAIL: KAT HS1, roundlists did not require the parent ledger page of a sub-round (fixed e821de90)`	pipeline record pipeline_20261106T222539
pipeline_20261106T233228 (full run 2, install)	`S1–S6 PASS: mutations 1,588/1,588, race 27 packages 0, shadow 120 blocks 0 mismatches, install by atomic rename (backups bin/*.22102026.r28-backup); S7 G22 FAIL on two new cryptography test files (fixed 6719490d)`	pipeline record pipeline_20261106T233228
pipeline_20261107T025211 (project-only)	`PASS; gates 29/32 + 3 PAUSED (miner)`	pipeline record pipeline_20261107T025211
B36 payee read (the r28 segment, 24 h)	`EQUAL at 7,871,000, span 87,436 s, payees 31,320 = explained 31,320, 0 alerts: CONFIRMED`	harness/runtime/r28_b36_payees24h_20261107.txt

MEASURED	RESULT	SOURCE
D3 node run (B189)	`community-dev build on an APFS clone 16:35:24 → 17:47:44: version 3, formula supply equal, 120/120 hashes equal with r28d, 1 peer-side ERROR`	harness/runtime/r28d_d3_noderun_20261107.txt
Live r28d window	`17:51:03 → 18:52:03: 62/62 synced, lag 0, tip age max 52.3 s, 0 ERROR/WARNING in 2,228 lines, chain requests served 6 / refused 0, emission 0 alerts, event scenarios 22/22`	harness/runtime/r28d_live_window_20261107.txt; harness/runtime/r28d_live_probes_20261107.txt
Live miner	`◌ PAUSED (miner): no miner measurement this round`	artifacts/dero-rft.md §0.3; harness/round.json tracks.miner

R29 · 2026-11-08 · +22102026.r29

THE DECISIONS ROUND: THE USER'S 2026-11-08 DECISIONS IMPLEMENTED WITH A TEST EACH, THE COMMUNITY-DEV RESIDUE CLOSED BY MEASUREMENT, A REGISTRATION CENSUS OVER THE WHOLE CHAIN, INBOUND LIVE, AND LONG JOBS WHOSE LIVENESS IS MEASURED AS WORK

WHAT CHANGED

- Miner track `◌ PAUSED (miner)`: no miner code changed; G17, G18 and G21 report PAUSED; S4b ran its offline A/B (staged vs installed 1.001 H/cpu-s like for like, no live miner); the user's miner runs are reward-only (rft §0.3).
- Decisions (the user, 2026-11-08, rft §1): the pool ranks by milli-fee per byte (B19, guard construct `mempool-milli-fee-rank`); derod refuses to start on a consensus Settings env override (B141, `cmd/derod/config_env_r29.go`); the SC recover line demoted to V(2) (B157, logging-only); the rpc/ money copy 128-bit (B144, guard construct `rpc-money-copy-128bit`); B35, B148, B169 closed by decision; miner stays PAUSED.
- Live observability: a jhttp bridge 500 becomes a counted JSON-RPC body (B175); TYPE apart from HELP (B176); the 8 lazy counter families pre-registered at start (B152).
- Wallet glue: walletapi `FormatMoneyPrecision` calls the globals implementation (B144; fail-before 23,631 of 60,027 pairs wrong).
- Community-dev residue: B190 re-graded against the papers the user downloaded (16 DISCHARGED, 3 VIOLATED = C1, 2 UNCLEAR; no construction); B183 CD-04 cost +0.50/+0.75 ms, stays REJECTED; B182 registrations 0..7,398,264 complete from a read-only multi-source fetch: 542,692 of 2,141,434 have 2 leading zero bytes, none fewer.
- Forensics: run D, a fourth node replay to treehash `ca11cc21` (roots byte-identical); B107 and the B97 residual closed; B110 SC payouts of run C exact; B111 (A) answered on a stopped-node APFS clone at the r29 restart: at 7,498,540: 2,141,704 balance keys, 50,284 SC meta, root ca11cc21… EQUAL to the recorded root; at the tip 7,875,451: 2,141,955 keys; registrations through 7,498,539 = 2,141,703: 1 key unattributed (open).
- Consensus and peers: D20 capture skew fixed (B194); D16 grade kept (B195); with inbound live, one identity on 2 slots (B196, report-only); the HF4 fork watch unchanged (B184).
- Harness and tools: liveness is work, not a heartbeat (B198: `harness/kit/hangkit.py`, `harness/kit/coproc.py`, gate G33, stallwatch; every hang fixed became a test that hangs the old code); B166, B168, B170, B155, B192, B197 fixed; the merge found B199 and B200.
- Soundness review for this round: dero-soundness-r29.html; the Round 28d page stays unchanged.
- Live r29 readings `measured 2026-11-08`: 3.6.0-153.DEROHE.STARGATE+22102026.r29 running since 2026-11-08 15:33:00 PST (the standard command, no new flags), read 15:33:40 → 16:33:40 (61 samples, the first 39 s after the start); 61/61 samples synced, lag 0, tip age mean 12.5 s (max 45.3), 13–15 outgoing, incoming 4 → 27 (max 28), 0 ERROR / 0 WARN, 0 restarts, RPC read errors 0; 192 blocks (7,875,456..7,875,648), 0 reorgs, 0 finality violations, block_rejected_total 0; pending an event: B47, B57, B99 no event in 3,600 s / 192 blocks; B48: inbound EVENT +28 but the proven-lane counters stayed 0 (the lane is consulted only when a netgroup's inbound limiter refuses); B61 topoheight = height 0/61; B85 −32031 0. Open after Round 29.

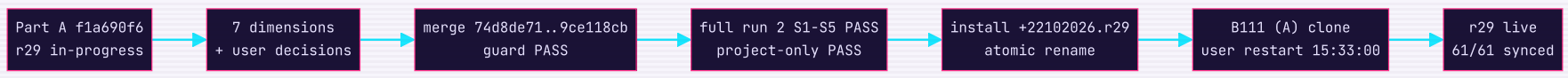
WHY

> The user asked (2026-11-07, rft §1) to finish the community-dev residue (B190 A15/A18, the B182 ranges, B183 CD-04, B192 budget) next to the open §7 work, then (2026-11-08) decided the standing list: B1 (a), B19 (a), B35 (b), B111 (A), B141 (a), B148 (a), B157 (a), B169 (b), miner (b), B182 option (ii) with a fast resumable fetcher, and B144 option (a); and asked that any stuck task be fixed with its fix built into the harness. R29 ran seven dimensions on light clones, serially (resource_intel REDUCE), merged them in seven merge commits, and kept every consensus-visible change inside a reviewed guard construct (S1 PASS: 9 consensus-package files).

CODE

R29 adds no code excerpt here: the node changes are a pool-order construct, a fail-closed start check, counted error bodies, pre-registered counters and a log level, each with its test and mutations.

FLOW



EVIDENCE

MEASURED	RESULT	SOURCE
Merge	seven merge commits 74d8de71..9ce118cb, liveness stack 010edd04; Go mutations 1,619 entries (31 new); consensus guard S1 PASS (9 consensus-package files vs bf274e5, constructs mempool-milli-fee-rank, rpc-money-copy-128bit, zero-signer-recover-return, logging-only)	artifacts/dero-rft.md §12 R29; harness/runtime/r29_rows/
pipeline_20261108T051138 (full run 1)	S3k FAIL (a stale soundness known section, then 3 KAT-sweep finds)	artifacts/dero-rft.md §12 R29
pipeline_20261108T062853 (full run 2, --no-install)	guard PASS; format PASS; flaky 18 ×3, 0 flaky; tests PASS; soundness 29 tests; KATs PASS; mutations 1619/1619; race 27 packages 0 races; S4b PASS (offline, staged vs installed 1.001 H/cpu-s like for like, no live miner); S4c SKIP (server unchanged); shadow 120/120 blocks, 240 treehashes, 0 mismatches, 0 regressions; S6 SKIP (--no-install); S7 FAIL s7_replay (the r28d pins moved; re-pinned 28f9a2ce)	pipeline record pipeline_20261108T062853
pipeline_20261108T123859 (closing project-only run)	PASS: KATs 35/35, harness mutations 34/34, gates 30/33 (+3 PAUSED, miner), 0 failing; 117 steps, wall 2,904 s of 3,500 s	pipeline record pipeline_20261108T123859
Installed derod	3.6.0-153.DEROHE.STARGATE+22102026.r29 by atomic rename (manual S6 (atomic rename, 2026-11-08 ~13:30 PST)), sha256 6595baf82c29febb…; backups bin/*.22102026.r28d-backup	artifacts/dero-rft.md §12 R29; harness/runtime/r29_live_probes_20261108.txt (bin/derod sha256)
B111 (A) Q-CS1 on a stopped-node clone	at 7,498,540: 2,141,704 balance keys, 50,284 SC meta, root ca11cc21… EQUAL to the recorded root; at the tip 7,875,451: 2,141,955 keys; registrations through 7,498,539 = 2,141,703: 1 key unattributed (open)	harness/runtime/r29_b111_qcs1.json
Live r29 node: restart and window	3.6.0-153.DEROHE.STARGATE+22102026.r29 running since 2026-11-08 15:33:00 PST (the standard command, no new flags), read 15:33:40 → 16:33:40 (61 samples, the first 39 s after the start)	harness/runtime/r29_live_window_20261108.txt; harness/runtime/r29_live_probes_20261108.txt
Live r29 node health	61/61 samples synced, lag 0, tip age mean 12.5 s (max 45.3), 13–15 outgoing, incoming 4 → 27 (max 28), 0 ERROR / 0 WARN, 0 restarts, RPC read errors 0	harness/runtime/r29_live_window_20261108.txt; harness/runtime/r29_live_probes_20261108.txt (error census)
Live r29 chain events	192 blocks (7,875,456..7,875,648), 0 reorgs, 0 finality violations, block_rejected_total 0	harness/runtime/r29_live_window_20261108.txt
Live r29 features confirmed	version = registry in 61/61 samples; inbound live after the user's UDP 11011 forward (B1 (a)): incoming 4 → 27; metric registry 304 families, 0 problems; event scenarios 22/22	harness/runtime/r29_live_probes_20261108.txt; harness/runtime/event_scenarios_20261108T163750.json
Live r29 features without an event in the window	◌ B47, B57, B99 no event in 3,600 s / 192 blocks; B48: inbound EVENT +28 but the proven-lane counters stayed 0 (the lane is consulted only when a netgroup's inbound limiter refuses); B61 topoheight = height 0/61; B85 -32031 0	harness/runtime/r29_live_window_20261108.txt (pending-event lines); harness/runtime/r29_live_probes_20261108.txt (B48)
Live r29 emission audit	219 audited, 0 alerts, issued = formula 6,703,500	harness/runtime/r29_live_window_20261108.txt (GetEmissionAudit at the window end)
Live r29 observatories	consensus 20 detectors: ANOMALY 3 (D15, D16, D19), OBSERVED 1 (D13), NORMAL 16, G26 PASS; peers 40 (13 out / 27 in): OBSERVED 2, ANOMALY 4 (D2, D6, D9, D11 = B196), G27 PASS	harness/runtime/r29_live_probes_20261108.txt (consensus_intel, peer_intel --live)
Forensics on the measured build (G28)	PASS, 4 versions re-verified (r25, r26, r28, r28d); r28d: 3 full replays byte-exact, 11/11 derived files equal	harness/runtime/r29_live_probes_20261108.txt (verify_dataset.py forensics --gate)

MEASURED	RESULT	SOURCE
Live miner	◌ PAUSED (miner): no live miner this round	artifacts/dero-rft.md §0.3; harness/round.json tracks.miner

MONETARY FORENSICS: DATASET V2 OF THE HF3 WINDOW, BY DEFECT CLASS

Dataset `forensics/r28d` version 4: 38 files, manifest SHA-256 `28f9a2ce17fd3112d949604ec02dbbdd8bd4991c3ad91a2c032524c8ae2cf444` , built at commit `e431c3e9a13a` ; it supersedes r28 and reads 8 files of the sealed dataset v1 by SHA-256. Replay (gate G28): 3 full runs, byte-exact yes, report `89aa2553e5ad469f` , 11 of 11 derived files equal. Independent audit (gate G29): 27 challenges, 27 hold, 0 fail, 0 unresolved. Every figure here is a span naming its file and SHA-256 (gate G30); every class figure also carries its class, bound, coverage and scope (gates G31 and G32).

> **Still open now (rft §0.8.9 rule twelve):** DC2 (proof soundness) is present in the current consensus code at the round head: its three C1 anchors re-read OK; any aggregate over genesis .. now is unbounded

THE VERDICT BOARD, BY DEFECT CLASS

Each class is graded on its own. No figure sums across classes, and a DC1 amount is never a supply figure.

DC1-OBS · DUPLICATION LADDER, AMOUNT PUBLIC

≥ 0

TXIDs of the table whose amount public data determines: coverage 0 of 75; bound `lower-bound` . Sum of their amounts: none determinable: DC1-obs is empty under the recorded criteria (`lower-bound` ; never a supply figure).

- criterion P1 (a public plaintext value field): 0 of 75
- criterion S-R22 (a shared randomness commitment, held bodies only): 0 of 75 within held bodies

DC1-HIDDEN · DUPLICATION LADDER, AMOUNT NOT PUBLIC

75

of the 75 detected DC1 TXIDs are DC1-pending: neither observable nor evidenced as hidden (DC1-hidden by evidence: 0). Rule: DC1-hidden only beyond an evidenced FQ3 boundary (none known), so the rest is DC1-pending. No amount bound for DC1-hidden can be stated until the FQ3 boundary exists (not inside the window: the cross-tx signature is absent from all 1,796 held bodies (1,769 of 1,769 window TXIDs), so no height in the window separates signature / no signature; the upstream code never changed (code_side)).

- per-tx grade: NOT-DECIDABLE-FROM-PUBLIC-DATA (per tx, derived/grades.json); 2 at E1, 73 at E2
- two bodies under one TXID: pending: one body was fetched per TXID; a block tx list carries only the TXID, which excludes the proof (transaction/transaction.go:509), so a second body is not observable from RAW blocks

DC2 · PROOF SOUNDNESS

unbounded

value created under the defect, scope genesis .. now: bound `unbounded` (lower bound 0, no upper bound; graded once: NOT-DECIDABLE-FROM-PUBLIC-DATA); audit grade NOT-DECIDABLE-FROM-PUBLIC-DATA.

- present in the code path at the round head: yes; at Release153: yes; at the genesis-era tag: yes
- disclosure track C1: CONFIRMED, state upstream

DC3 · PUBLIC-VALUE IDENTITIES

2,536,167,750

window emission (atomic), bound `exact` over 82,477 of 82,477 blocks; class grade: PROVEN-NO-INFLATION-IN-SCOPE (scope: window 7,415,466..7,497,942 (82,477 blocks); public emission only).

- fees in `derived/classes.json` : 855,173, burns 230,470,000 (`exact` , over 82,477 of 82,477 blocks: the file still reads dataset v1, base r25)
- the audit's own sums of the same fields over dataset v2: fees 855,173, burns 230,470,000 (exact over 82,477 of 82,477 heights); challenge D2-19 HOLDS: the class file is corrected at its source (B126), not on this page
- fee verdict V2-FEES: NOT-DECIDABLE-FROM-PUBLIC-DATA (a fee is paid out of a hidden balance)

DC0 · A FINDING OUTSIDE EVERY CLASS

0

TXIDs in two or more window blocks outside the HF3 table: bound `exact` over 82,477 of 82,477 window blocks.

DEFECT CLASSES, COVERAGE AND BOUNDS

Every aggregate of `derived/classes.json`. The bar is the coverage; a hatched bar is an unknown denominator or numerator (pending). Bound badges: exact, lower-bound, unbounded.

FIGURE AND SCOPE	CLASS	VALUE	BOUND	COVERAGE (NUMERATOR OF DENOMINATOR, ANCHORS)
HF3 table entries (the DC1 TXIDs consensus code names) scope: window 7,415,466..7,497,942 (the 75 HF3 entries min..max)	**DC1**	75	exact	75 of 75 derived/hf3.json rows / blockchain/hardfork_fixes.go:27 (derived/hf3.json table.entries)
DC1 bodies held RAW (checks i-iii) scope: window 7,415,466..7,497,942 (the 75 HF3 entries min..max)	**DC1**	75	exact	75 of 75 derived/hf3.json counts.bodies_raw / derived/hf3.json rows
"duplicates" entries carried by two RAW blocks scope: window 7,415,466..7,497,942 (the 75 HF3 entries min..max)	**DC1**	67	exact	67 of 67 derived/hf3.json rows[].carriers / derived/hf3.json table.groups.duplicates
"false parity" entries carried by one RAW block scope: window 7,415,466..7,497,942 (the 75 HF3 entries min..max)	**DC1**	8	exact	8 of 8 derived/hf3.json rows[].carriers / derived/hf3.json table.groups
DC1 TXIDs with two different bodies held scope: window 7,415,466..7,497,942 (the 75 HF3 entries min..max)	**DC1**	pending: one body was fetched per TXID; a block tx list carries only the TXID, which excludes the proof (transaction/transaction.go:509), so a second body is not observable from RAW blocks	unbounded	pending of 75 derived/tx/*.json duplicates / derived/hf3.json rows
DC1 TXIDs whose amount public data determines (DC1-obs) scope: window 7,415,466..7,497,942 (the 75 HF3 entries min..max)	**DC1**	≥ 0	lower-bound	0 of 75 derived/fq.json FQ2 / derived/hf3.json rows
sum of DC1-obs amounts scope: window 7,415,466..7,497,942 (the 75 HF3 entries min..max)	**DC1**	none determinable: DC1-obs is empty under the recorded criteria	lower-bound	0 of 75 derived/fq.json FQ2 / derived/hf3.json rows
TXIDs in two or more window blocks outside the HF3 table scope: window 7,415,466..7,497,942 (the 75 HF3 entries min..max)	**DC0**	0	exact	82,477 of 82,477 derived/window.jsonl.gz rows[].tx_hashes / derived/window.jsonl.gz rows
value created under the proof-soundness defect scope: genesis .. now	**DC2**	unbounded	unbounded	0 of pending no public-data test separates the cases (fq.json FQ5) / pending: every proof since genesis
window emission (formula coinbase) = CalcSupply difference scope: window 7,415,466..7,497,942 (the 75 HF3 entries min..max)	**DC3**	2,536,167,750	exact	82,477 of 82,477 derived/emission.json per_block.supply_identity_ok / derived/emission.json per_block.blocks
window fees, every block scope: window 7,415,466..7,497,942 (the 75 HF3 entries min..max)	**DC3**	855,173	exact	82,477 of 82,477 derived/fees.json v2 (v1 known blocks + v1 pending blocks settled) / derived/emission.json fees (v1 window blocks)
window public burns, every block scope: window 7,415,466..7,497,942 (the 75 HF3 entries min..max)	**DC3**	230,470,000	exact	82,477 of 82,477 derived/fees.json v2 (v1 known blocks + v1 pending blocks settled) / derived/emission.json fees (v1 window blocks)

> **A fixed path is not a restored supply.** A fix status says that a code path no longer admits new cases; it changes no historical grade and restores no value.

> **An unknown is pending, never zero.** An undecidable quantity shows as unbounded or pending, and every aggregate names its class, scope, coverage and bound.

> **A historical replay is not a current-state proof.** A replay that ends at our own state root shows that our node's state is consistent with the blocks it accepted; it says nothing about hidden amounts.

STILL OPEN NOW? THE CODE PATH OF EACH CLASS

For each class, the code lines that admit it, re-read at the round head and at the upstream tags (4 revisions; `derived/presence.json`). "Present" means the path is in that code; a closure is a later line that stops new cases (it restores nothing).

- **HEAD** `5a32c5b21d6e` : the round head
- **Release152** `1d3c4b7bc8b6` : the Release152 tag (HF3 release)
- **Release153** `c4e6f6b40db9` : the newest upstream tag in this repository
- **DeroHE** `e25b1a7f5f1d` : the earliest tag in this repository (genesis era, 2020-12-19)

DC1

- the HF3 exemption table (75 TXIDs)
- a table TXID enters the verification cache before any check
- closure: from HF3 the cache key is the hash of the whole tx (height-bound skip)
- closure (node-local): relay/RPC admission refuses the 75 TXIDs by ID

REVISION	PATH PRESENT	ANCHORS PRESENT	CLOSURES PRESENT	WHERE
HEAD	yes	4 of 4	2 of 2	blockchain/hardfork_fixes.go:27; blockchain/transaction_verify.go:232; blockchain/transaction_verify.go:435; glue/txprecheck/window.go:140
Release152	yes	3 of 4	1 of 2	blockchain/hardfork_fixes.go:27; blockchain/transaction_verify.go:232; blockchain/transaction_verify.go:435
Release153	yes	3 of 4	1 of 2	blockchain/hardfork_fixes.go:27; blockchain/transaction_verify.go:232; blockchain/transaction_verify.go:435
DeroHE	no	0 of 4	0 of 2	none

DC2

- inner-product generator derived from the checked challenge (C1 anchor 1)
- the blinding term on the same base (C1 anchor 2)
- the final inner-product equality (C1 anchor 3)

REVISION	PATH PRESENT	ANCHORS PRESENT	CLOSURES PRESENT	WHERE
HEAD	yes	3 of 3	0 of 0	cryptography/crypto/proof_verify.go:430; cryptography/crypto/proof_verify.go:452; cryptography/crypto/proof_innerproduct.go:307
Release152	yes	3 of 3	0 of 0	cryptography/crypto/proof_verify.go:430; cryptography/crypto/proof_verify.go:452; cryptography/crypto/proof_innerproduct.go:307
Release153	yes	3 of 3	0 of 0	cryptography/crypto/proof_verify.go:430; cryptography/crypto/proof_verify.go:452; cryptography/crypto/proof_innerproduct.go:307
DeroHE	yes	3 of 3	0 of 0	crypto/proof_verify.go:410; crypto/proof_verify.go:432; crypto/proof_innerproduct.go:307

DC3

- the emission formula
- the homomorphic balance update (public deltas are what DC3 can check)

REVISION	PATH PRESENT	ANCHORS PRESENT	CLOSURES PRESENT	WHERE
HEAD	yes	2 of 2	0 of 0	blockchain/supply.go:5; blockchain/transaction_execute.go:241
Release152	yes	2 of 2	0 of 0	blockchain/supply.go:5; blockchain/transaction_execute.go:241
Release153	yes	2 of 2	0 of 0	blockchain/supply.go:5; blockchain/transaction_execute.go:241
DeroHE	no	0 of 2	0 of 0	none

The transcript pin (FQ4): answered. Detectability from public data (FQ5): no public-data test separates a proof accepted under the defect from an honest one: consensus reads one boolean per payload, the proof shape is fixed by the ring size, and scalar canonicality is already censused (FX-N3); a shape or size detector is infeasible (FX-N6).

WHAT THE CURRENT STATE CAN AND CANNOT PROVE

Q-CS1 ARE ALL SPENDABLE BALANCES PUBLICLY ENUMERABLE?

every registered account and SC tree entry is enumerable from the state tree (a key count is public); the balances they hold are ciphertexts, so enumeration does not make them summable (Q-CS2)

status	answered (scope: the held replaywall archive at topo 7467515, not our anchor 7,498,540)
scope	topo 7,467,515, state version 18,960: APFS clone of the replaywall archive copy (read through a clone, never in place)
balance-tree keys	2,141,530
SC tree keys	1,319,156 (173,338 of compressed-point length: structural only: a 33-byte key has the length of a compressed point (candidate holder-balance entry); not decoded); SC meta keys 50,276
root check	census root `d3345594b8bb324b` equal to the recorded archive root: yes
earlier count	2,141,950 (EXT to this dataset (an earlier round's measurement, not re-derivable here); different scope: R22 counted the live store at a later version; the difference is not a finding)

Q-CS2 IS THE AGGREGATE OF BALANCES BOUND BY A PUBLICLY CHECKABLE CONSERVATION RELATION?

no: every balance is a ciphertext under its holder's own key and changes only by ciphertext addition; opening any balance needs that holder's secret key, and no aggregate key or aggregate opening

exists in the code; consensus checks one proof boolean per payload and never compares any aggregate with the formula supply, which RPC reports as CalcSupply (answered (INTERPRETATION of SRC; audit refutes by default); 9 of 9 anchors re-read).

- `cryptography/crypto/algebra_elgamal.go:34` : an account balance is an ElGamal pair (Left, Right)
- `blockchain/transaction_execute.go:74` : a new balance is an encryption of zero under the account's own public key
- `blockchain/transaction_execute.go:239` : each payload adds a per-account ciphertext delta built from the statement (C[i], D)
- `blockchain/transaction_execute.go:241` : the stored balance is updated by ciphertext addition only
- `cryptography/crypto/algebra_elgamal.go:69` : ciphertext addition is component-wise point addition
- `walletapi/daemon_communication.go:573` : opening a balance uses the holder's secret key
- `cmd/derod/rpc/rpc_dero_getinfo.go:83` : the reported supply is the emission formula, not a sum of balances
- `blockchain/transaction_verify.go:454` : consensus accepts a payload on one proof boolean
- `cmd/derod/rpc/monetary_integrity.go:109` : the node's own integrity report states that encrypted value is invisible to it

Q-CS3 CAN A SUPPLY STATEMENT BE PROVEN FROM THE CURRENT STATE ALONE?

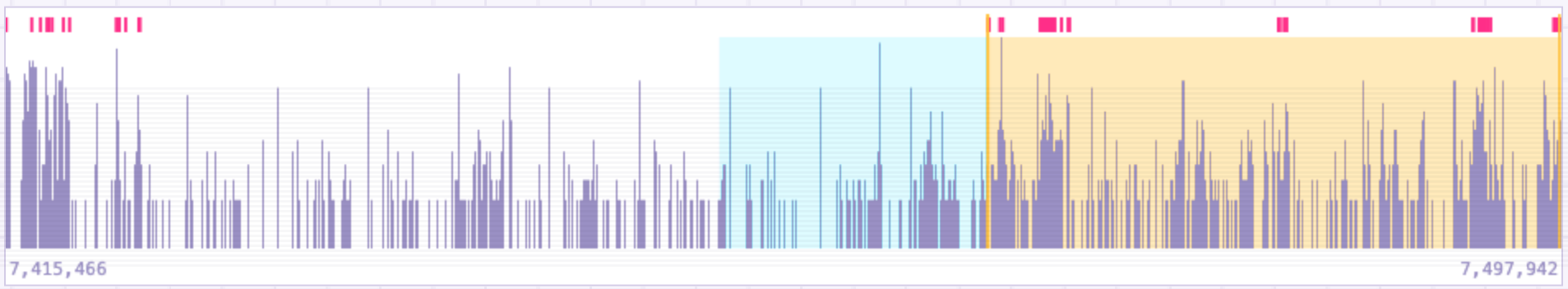
no (answered (INTERPRETATION of SRC; audit refutes by default)). The missing-invariant certificate:

missing invariant	a publicly checkable opening of the aggregate of all balances (or a consensus-checked relation between it and CalcSupply)
shown absent by	blockchain/transaction_execute.go:74; blockchain/transaction_execute.go:241; walletapi/daemon_communication.go:573; cmd/derod/rpc/rpc_dero_getinfo.go:83; blockchain/transaction_verify.go:454
consequence	a complete public accounting of the supply of this ledger cannot be derived from public data; every aggregate over hidden balances is unbounded while DC2 is open
joins	disclosure track FX-N7 (summing the balance tree: infeasible)

Designs that would restore a verifiable baseline (FQ6, fork required, design only): FX-F5 Opt-in migration to a state whose supply is the exact sum of migrated balances (FQ6) (FORK-REQUIRED, design); FX-F6 A new genesis with sound proofs and a public aggregate rule (FQ6) (FORK-REQUIRED, design).

WINDOW EXPLORER

All 82,477 blocks from 7,415,466 to 7,497,942, decoded in your browser from data embedded in this page: the rows of the sealed `forensics/r25/derived/window.jsonl.gz` (SHA-256 `f5f79b1385 6ec50c`) with the fees of the 1,575 blocks dataset v1 left pending settled from `derived/blocks_fees.jsonl.gz` (still pending: 0). Bars: transactions per pixel column (square-root scale). Magenta ticks: the 142 blocks that carry a table TXID. Amber lines: replay walls at 7,467,516, 7,497,836 and the node-replay stop at 7,467,546. Cyan band: heights whose post-state root our replaywall archive holds. Amber band: the 30,427 heights with pre and post roots from the node replay of this dataset.



txs per column · HF3 carrier · replay wall / node-replay roots · archive root held

Show every block ← → page 1 of 3,300 · 82,477 blocks height go

HEIGHT	BLOCK ID	TIME (UTC)	MINIBLOCKS	TXS	FEES	DEPTH
7,415,466	a8c4e55b7ceec11f…	2026-08-02 04:47	10	1	80	A HF3
7,415,467	9e7526a850753362…	2026-08-02 04:47	10	2	480	A
7,415,468	f5f820d44171b2ff…	2026-08-02 04:48	10	0	0	A
7,415,469	a8400feb1ac5c531…	2026-08-02 04:48	10	0	0	A
7,415,470	a019c9aa02bf3f2d…	2026-08-02 04:48	10	1	80	A HF3
7,415,471	1706c89c954da239…	2026-08-02 04:48	10	0	0	A
7,415,472	423d60d63be2c6a0…	2026-08-02 04:49	10	0	0	A
7,415,473	92735f1714ff7971…	2026-08-02 04:49	10	1	40	B
7,415,474	f508ceb9a49ef99e…	2026-08-02 04:49	10	0	0	B
7,415,475	e677bdf7a88d4e23…	2026-08-02 04:49	10	0	0	B
7,415,476	3835f9a0d726af77…	2026-08-02 04:49	10	0	0	B
7,415,477	e11db45a702289b6…	2026-08-02 04:50	10	0	0	B
7,415,478	16d83534cf5ba04c…	2026-08-02 04:50	10	1	120	B
7,415,479	9df08553ec3047c7…	2026-08-02 04:50	10	0	0	B
7,415,480	7f064efd2e149c51…	2026-08-02 04:51	10	0	0	B
7,415,481	07c7fcf665c438ea…	2026-08-02 04:51	10	0	0	B

HEIGHT	BLOCK ID	TIME (UTC)	MINIBLOCKS	TXS	FEES	DEPTH
7,415,482	0b63c63200d00275…	2026-08-02 04:51	10	0	0	B
7,415,483	761221d481fcd162…	2026-08-02 04:51	10	0	0	B
7,415,484	d896f886b224818f…	2026-08-02 04:52	10	0	0	B
7,415,485	fc0b37f613a0d61c…	2026-08-02 04:52	10	1	0	B
7,415,486	a6e7884b91bed4c0…	2026-08-02 04:53	10	0	0	B
7,415,487	f1ffe537b43f25ac…	2026-08-02 04:53	10	0	0	B
7,415,488	f2ba439ad4dfe9df…	2026-08-02 04:53	10	0	0	B
7,415,489	277ed5abc8e91196…	2026-08-02 04:54	10	0	0	B
7,415,490	fd6737de9b64bdae…	2026-08-02 04:54	10	0	0	B

Select a block.

THE HF3 EXCEPTION TABLE (DC1), TXID BY TXID

The 75 entries of the table are the DC1 TXIDs consensus code names: 67 under “duplicates” and 8 under “no duplicates, but false parity”; Tx_Parity true for 17. The 142 carrier blocks: 67 TXIDs sit in two blocks, 8 in one; dataset v2 finds 67 TXIDs in two or more RAW blocks, all in the table (0 outside it). These rows are the sealed dataset v1 (re-verified byte for byte; `forensics/r25/derived/hf3.json`).

Exception path under the pinned code: proof verification skipped (blockchain/transaction_verify.go:232-233); nonce bypassed on 74 carriers; the parity override changes execution on 73; today the relay and RPC refuse these bodies (glue/txprecheck/window.go:136-145). Proof under today’s verifier where the ring state is held: pass 1, fail 1, not decidable 73 (outcomes only, rft §0.8.1 rule six).

Filter all

TXID	TABLE HEIGHT	LINE	GROUP	TX_PARITY	CARRIERS	NONCE BYPASSED	PARITY CHANGES	PROOF TODAY	EVIDENCE	FEE	BC
0e75ae6662c08627470a7ea97d2115830f2b0e392efdba3c58ace068adc99aa6	7,415,466	95	duplicates	yes	7,415,466, 7,415,470	1	1	not decidable: ring state unavailable	E2	80	a4 aa
42f2876c9052281208556e7f6e0f51ef7edd84f1f99fba48acc84b52321f30d	7,416,808	94	duplicates	yes	7,416,808, 7,416,813	1	1	not decidable: ring state unavailable	E2	80	30 fc
a16a9d9395518fc4ad2a1f1ac3a878ca35d9d819e0dd3086e495f3c2f4a089db1	7,417,290	93	duplicates	yes	7,417,290, 7,417,295	1	1	not decidable: ring state unavailable	E2	80	7(69
3478d883230b579a88607d5f5d1a52aadd4355b38ba794352d54a829f45f5656	7,417,631	92	duplicates	no	7,417,631, 7,417,635	1	1	not decidable: ring state unavailable	E2	80	7a 44

TXID	TABLE HEIGHT	LINE	GROUP	TX_PARITY	CARRIERS	NONCE BYPASSED	PARITY CHANGES	PROOF TODAY	EVIDENCE	FEE	BC
f00cc25c28947e52c48b450d259b5a06daf9bce580bb5503e9780878d236d4f90	7,417,751	91	duplicates	no	7,417,751, 7,417,757	1	1	not decidable: ring state unavailable	E2	80	6c fd
60bfbd73a60060d6502b13871f55a47ebfb732bcb8f5685602688f7eae36fa3e	7,417,912	104	no duplicates, but false parity	yes	7,417,912	1	1	not decidable: ring state unavailable	E2	80	a8 be
912dbaf89800086dd56271 93c4ae3264da9bba216763 7f078380f5b0f06a24520	7,418,504	90	duplicates	no	7,418,504, 7,418,510	1	1	not decidable: ring state unavailable	E2	80	b9 00
e0d83b72e7ab11213f78ca799459 8d2b988799d36786e94a2018c765133 39ba8	7,418,813	89	duplicates	no	7,418,813, 7,418,817	1	1	not decidable: ring state unavailable	E2	80	e8 85
e2fdb507438dc80667e73d60a176588 11a1181234e9bbddcf81ec05de1c7a51d	7,421,309	88	duplicates	no	7,421,309, 7,421,312	1	1	not decidable: ring state unavailable	E2	80	dc 00
e53ea2a813bdccbff19ef24010e7e5bef0bf2408423eb71a86dbca3cbca22530	7,421,373	87	duplicates	no	7,421,373, 7,421,377	1	1	not decidable: ring state unavailable	E2	80	94 5c
d9d7e7de41ca9c81693eca4e3ea56ae02b6680887c35fc1b1b2da719673 35404	7,421,416	86	duplicates	no	7,421,416, 7,421,417	1	1	not decidable: ring state unavailable	E2	80	ec 2b

TXID	TABLE HEIGHT	LINE	GROUP	TX_PARITY	CARRIERS	NONCE BYPASSED	PARITY CHANGES	PROOF TODAY	EVIDENCE	FEE	BC
d81f2d3911e388f88b2e427d3f6cc8ddc272c69aa80fe84e15ce8654620713 28	7,421,470	85	duplicates	no	7,421,470, 7,421,476	1	1	not decidable: ring state unavailable	E2	80	1(c9
b9c2b724a684eb1b278bb3f185a807e46b3ef22a87c5cb8ed694a32df1661df0	7,421,790	84	duplicates	yes	7,421,790, 7,421,793	1	1	not decidable: ring state unavailable	E2	80	0: be
6843437e70503ed702d5adad3d29e7aa0ed8427dd98eda0e42b77d24e7567bc8	7,422,487	83	duplicates	yes	7,422,487, 7,422,490	1	1	not decidable: ring state unavailable	E2	80	c2 84
86a15dad0a5ca6b8b317398a56113e95850a1628868fb3284963e30e046aac5d	7,422,548	82	duplicates	no	7,422,548, 7,422,553	1	1	not decidable: ring state unavailable	E2	80	c4 14
414f79b8ff3c862eaa5727c659154221888 1c751ba4fe052c9aaeeec0cfedfcd	7,467,498	102	no duplicates, but false parity	yes	7,467,498	1	0	pass	E1	121	7(0f
0045a6d4c51a8578b6c97f2b3764156535835 05f21032eb6a808cd1643a300a7	7,467,516	103	no duplicates, but false parity	yes	7,467,516	1	0	fail	E1	121	da c0
3659ae2be646c4384bcc4adc94bdf9cfc23b366faa128a88f9ea5522e3d08d61	7,467,538	81	duplicates	yes	7,467,538, 7,467,544	1	1	not decidable: ring state unavailable	E2	121	c9 a5

TXID	TABLE HEIGHT	LINE	GROUP	TX_PARITY	CARRIERS	NONCE BYPASSED	PARITY CHANGES	PROOF TODAY	EVIDENCE	FEE	BC
404180f833143669d3b347f4d8ceb1067b7a1ed057efcad7abcd7ad5397451338	7,467,569	80	duplicates	no	7,467,569, 7,467,572	1	1	not decidable: ring state unavailable	E2	121	dc b8
0511f2b2a8ee9408b216cb1db6129937119fb76e99a1dcd1544f49462da58a38	7,467,585	101	no duplicates, but false parity	no	7,467,585	1	1	not decidable: ring state unavailable	E2	121	9(7d
4ad3ef9af337b1e92d895f9342ad9a9235d601291cb42fb6b726fbda35933fe7	7,468,150	79	duplicates	no	7,468,150, 7,468,152	1	1	not decidable: ring state unavailable	E2	121	4(e6
f389f4b63e9256d0524455889ca06eee694bac0eefda658add5d4dbd37b29d60c	7,468,300	78	duplicates	yes	7,468,300, 7,468,303	1	1	not decidable: ring state unavailable	E2	121	2t da
9fba85d0ed98249e9b36ffbbaf15f6ae9bf9be061f3200b2f572c3e2f34b39ea	7,470,282	77	duplicates	no	7,470,282, 7,470,283	1	1	not decidable: ring state unavailable	E2	121	b8 3f
ea445f18926af580fe8bf95466f3f7b9eea3eea0295a791323ee1d0ab4eddb0c	7,470,357	76	duplicates	yes	7,470,357, 7,470,361	1	1	not decidable: ring state unavailable	E2	121	0t e0
0a0671cf132cc2c353df8d28631581c762acefac200123c84dd039c34e7cd336	7,470,386	75	duplicates	no	7,470,386, 7,470,387	1	1	not decidable: ring state unavailable	E2	121	f5 db

TXID	TABLE HEIGHT	LINE	GROUP	TX_PARITY	CARRIERS	NONCE BYPASSED	PARITY CHANGES	PROOF TODAY	EVIDENCE	FEE	BC
8692d178675b0ab019f765ab8b8f5b739c0668f361519312 23c4e3b63883f89a	7,470,403	74	duplicates	no	7,470,403, 7,470,409	1	1	not decidable: ring state unavailable	E2	121	c 65
1877ddc33990d521dbd1702f8872441c642641cbb528042 432da8d22b722fdbe	7,470,452	73	duplicates	no	7,470,452, 7,470,455	1	1	not decidable: ring state unavailable	E2	121	3 97
6d882984bad971d4becb9ac15ecd8cb3e3e0ee783d6d6c1d1ce95df126257 32f	7,470,465	100	no duplicates, but false parity	yes	7,470,465	1	1	not decidable: ring state unavailable	E2	121	0 9e
92dce2c525287 32ef76ba717f582b784a50bca61bc83942683 8cc641d28f545a	7,470,477	72	duplicates	yes	7,470,477, 7,470,480	1	1	not decidable: ring state unavailable	E2	121	1 67
62fa2cc6f10da7275d54f0448bd7028a1d1d7a92a99e204c3f4e2c08571 032f7	7,470,527	71	duplicates	no	7,470,527, 7,470,529	1	1	not decidable: ring state unavailable	E2	121	1 2b
556d78cfc579c2897be7d875e08a702860b867e63650ad2d07c393e50e1e484c	7,470,582	70	duplicates	no	7,470,582, 7,470,587	1	1	not decidable: ring state unavailable	E2	121	2 1a
00c6c9a78c4b40a03dc08e3637ee4e34103b39cfe2c6efaa8038295 09f0506d6	7,470,648	69	duplicates	no	7,470,648, 7,470,651	1	1	not decidable: ring state unavailable	E2	121	2 56

TXID	TABLE HEIGHT	LINE	GROUP	TX_PARITY	CARRIERS	NONCE BYPASSED	PARITY CHANGES	PROOF TODAY	EVIDENCE	FEE	BC
a70950ff17747e3b347d30b3db5631a5f97cd967a937eac2dcc11763c768549 0	7,470,706	68	duplicates	yes	7,470,706, 7,470,711	1	1	not decidable: ring state unavailable	E2	121	1l 92
60e4547d48b6245550 81cb8f7c97f858323 4c3a112bb586c868ad5ff00eddaa4	7,470,802	67	duplicates	no	7,470,802, 7,470,805	1	1	not decidable: ring state unavailable	E2	121	6: 35
2f39de57b82f02787fc15537a29da2e4cdd8c350d2d2bc7f96b0d49ac57fcbaa	7,470,879	66	duplicates	no	7,470,879, 7,470,883	1	1	not decidable: ring state unavailable	E2	121	c(ce
1e8c9c7764ea63e39a9b5fa590135 78bf3a7cdf00edfa059cf5f1da6dc08b0d9	7,470,896	65	duplicates	yes	7,470,896, 7,470,897	1	1	not decidable: ring state unavailable	E2	121	5(3a
6e59f6b5645 3219ab9722a47870c4b2d76b29dd da068c069dcf59ba14e3e98aa	7,470,910	64	duplicates	no	7,470,910, 7,470,915	1	1	not decidable: ring state unavailable	E2	121	3: 47
6034f404619a7fa7bad529eb6758a6ba30184b100a5f8a8f921c99bebe309514	7,470,924	63	duplicates	yes	7,470,924, 7,470,925	1	1	not decidable: ring state unavailable	E2	121	b9 01
09d0260ea572349c4869b58cb596383 48e1ed4bc57c511619d36fd1fd484211f	7,470,943	99	no duplicates, but false parity	no	7,470,943	1	1	not decidable: ring state unavailable	E2	121	75 a6

TXID	TABLE HEIGHT	LINE	GROUP	TX_PARITY	CARRIERS	NONCE BYPASSED	PARITY CHANGES	PROOF TODAY	EVIDENCE	FEE	BC
9b75eb3bc2c2f10cb84977e4648e5a9f4bebcfb73075dd27a1fa1a14d0f6dfb0	7,470,983	62	duplicates	no	7,470,983, 7,470,986	1	1	not decidable: ring state unavailable	E2	121	65e3
b6cf0ef15e9af69d7b2f040ba2e0a75f41a414201069187fdec994584298e8f7	7,471,011	61	duplicates	no	7,471,011, 7,471,016	1	1	not decidable: ring state unavailable	E2	121	37c
2e2b49f284f9d206fdb6dcf4a9ac9259426c61036bdf52036f1baf24b21269ac	7,471,053	60	duplicates	yes	7,471,053, 7,471,055	1	1	not decidable: ring state unavailable	E2	121	34ab
879f99d24bde3eaadd0ee348d3eef5e61b13bc9ab7c554ef6fc1a679b7de6	7,471,414	59	duplicates	no	7,471,414, 7,471,415	1	1	not decidable: ring state unavailable	E2	121	60be
e0a6f89071b89ee8f479845399e78c17d31f3dbc8e1c4370e22ec5a4fcd214bc	7,471,775	58	duplicates	no	7,471,775, 7,471,779	1	1	not decidable: ring state unavailable	E2	121	1f6
37df76020e1823aade3b130fe53571b93d6c5c34e35d1d522c7c5cd449e843770	7,471,805	98	no duplicates, but false parity	no	7,471,805	1	1	not decidable: ring state unavailable	E2	121	d37c
364538c7073f9c5e3b99eeec33a9bfb830a5d3d0cefb6c44d63b22b90c82a655	7,471,828	57	duplicates	no	7,471,828, 7,471,832	1	1	not decidable: ring state unavailable	E2	121	4080

TXID	TABLE HEIGHT	LINE	GROUP	TX_PARITY	CARRIERS	NONCE BYPASSED	PARITY CHANGES	PROOF TODAY	EVIDENCE	FEE	BC
e42eb301e2f0863de346ada3a531cdf70ca224a90dae2e861a21407a0c76bfc5	7,482,913	56	duplicates	no	7,482,913, 7,482,919	1	1	not decidable: ring state unavailable	E2	121	a860
9becfdff8e750fcd741f240ae701ca134fd54b1c3d1db9dbf9f02545a794046	7,482,959	55	duplicates	no	7,482,959, 7,482,960	1	1	not decidable: ring state unavailable	E2	121	6ad0
6aa20567ff23705aa91a4c607e8387d147b309f8eeb9f8198149c21c439c3692	7,482,968	54	duplicates	no	7,482,968, 7,482,973	1	1	not decidable: ring state unavailable	E2	121	flfc
3ccf648db6d26295d68ec1c88cdc736a10caa35ad83b856cb9bb304f893c5043	7,483,214	53	duplicates	no	7,483,214, 7,483,220	1	1	not decidable: ring state unavailable	E2	121	0a0f
d856184d636c3ef8545979c902df15d6a1bf41b4a4b81d81762f519a6acfb689	7,483,355	52	duplicates	no	7,483,355, 7,483,358	1	1	not decidable: ring state unavailable	E2	121	42ba
bba77f85c72d2b896e1fca1e39327a3a57eca4d722326a524db2042df04766ee	7,493,238	51	duplicates	no	7,493,238, 7,493,239	1	1	not decidable: ring state unavailable	E2	121	78aa
b43be6528061e4eb1ca6e7bcfcd419e42839bbeec5cdcbf7e664be2fd501dad2	7,493,246	50	duplicates	no	7,493,246, 7,493,249	1	1	not decidable: ring state unavailable	E2	121	5111

TXID	TABLE HEIGHT	LINE	GROUP	TX_PARITY	CARRIERS	NONCE BYPASSED	PARITY CHANGES	PROOF TODAY	EVIDENCE	FEE	BC
fd79549ad416cbec243d9ce54014b26f888f1cfd0cfe1aed5ddе978a11dcaeab	7,493,549	49	duplicates	no	7,493,549, 7,493,551	1	1	not decidable: ring state unavailable	E2	121	2c 2f
aa0faf67a2856ee30ed4e275c1079d1572b1bcb03d4a9ca682bd825e11d81a43	7,493,552	97	no duplicates, but false parity	no	7,493,552	0	1	not decidable: ring state unavailable	E2	121	ba 83
9eba39af652ee2a0d4835d964cffcd0be4ac6f1031338e87c8095390 2e4ce0026	7,493,588	48	duplicates	no	7,493,588, 7,493,591	1	1	not decidable: ring state unavailable	E2	121	3 7a
1eb43df8ab1abd8934c74cc0d08b5d9fef02604ca6500a7cf947d263b882cafe	7,493,600	47	duplicates	no	7,493,600, 7,493,606	1	1	not decidable: ring state unavailable	E2	121	f bc
74b791784269 90bcd071cd26829900027e78b6d36e77b57e050a0bba4398ed1f	7,493,663	46	duplicates	no	7,493,663, 7,493,664	1	1	not decidable: ring state unavailable	E2	121	c f1
ba3834ef15b7d714836e9944c46c8795066b328ff678c5cc44c7cdf0a3fa5ebd	7,493,682	45	duplicates	no	7,493,682, 7,493,683	1	1	not decidable: ring state unavailable	E2	121	c 1e
ed09fcce52484862a01fcb0e03d6799ba16d7304e4b0af1d60cde4ce399c1207	7,493,752	44	duplicates	no	7,493,752, 7,493,756	1	1	not decidable: ring state unavailable	E2	121	7 6b

TXID	TABLE HEIGHT	LINE	GROUP	TX_PARITY	CARRIERS	NONCE BYPASSED	PARITY CHANGES	PROOF TODAY	EVIDENCE	FEE	BC
17cd5710eee2cb27cdb1a68dfe63e955260ea8dc3e59e234a20bed14cb409970	7,493,772	43	duplicates	no	7,493,772, 7,493,778	1	1	not decidable: ring state unavailable	E2	121	1a80
e5676c4e2cc98429c890a30bdc1db231935dd52c91d17aa077a24a87a2dc35f9	7,493,786	42	duplicates	no	7,493,786, 7,493,789	1	1	not decidable: ring state unavailable	E2	121	3e39
2110829 55f1fda468ba0cd604e14dc5d84f4eb096f3c349f4ad05757c979dce6	7,493,898	41	duplicates	no	7,493,898, 7,493,903	1	1	not decidable: ring state unavailable	E2	121	ee08
85d52fe6f310f4af1d11611ef3577d477b76af3c0d050e2f4493e16822f68a18	7,494,011	40	duplicates	no	7,494,011, 7,494,017	1	1	not decidable: ring state unavailable	E2	121	9a85
5a14d851628864 8133b8cea98afa1b9a2f783a3ea345b191e57d388b64ca5ecf	7,494,167	39	duplicates	no	7,494,167, 7,494,172	1	1	not decidable: ring state unavailable	E2	121	abde
7158814 8dade214248f46fb912ac933fc60f2a0ba7d69b3b7317067c366e8312	7,497,519	38	duplicates	no	7,497,519, 7,497,521	1	1	not decidable: ring state unavailable	E2	121	9279
337677e2cad7c2633099ab65b5f5ce5a099b9c76a5cc7846c587d189f55ad95d	7,497,568	37	duplicates	no	7,497,568, 7,497,569	1	1	not decidable: ring state unavailable	E2	121	3c7e

TXID	TABLE HEIGHT	LINE	GROUP	TX_PARITY	CARRIERS	NONCE BYPASSED	PARITY CHANGES	PROOF TODAY	EVIDENCE	FEE	BC
411012a7711794d2c641acbd4e1cd14dfd5e20d6085ab4748af28e7d6a3a5b99	7,497,675	36	duplicates	no	7,497,675, 7,497,679	1	1	not decidable: ring state unavailable	E2	121	2c de
ab6cca967482928 25e81885 50f3fadcd43734d4a1e73514c03702adb456f8926	7,497,706	35	duplicates	no	7,497,706, 7,497,710	1	1	not decidable: ring state unavailable	E2	121	af 8e
a86a316b2d9f33e5e855292 5e7cd5049990 33b38c4e2aba6aa3a25f7a2eba003	7,497,722	34	duplicates	no	7,497,722, 7,497,728	1	1	not decidable: ring state unavailable	E2	121	c7 c8
4d56a1a150063b6bd810a33fc669df8e2698475f0b3c17d96cab738b25ba8393	7,497,836	33	duplicates	no	7,497,836, 7,497,838	1	1	not decidable: ring state unavailable	E2	121	d3 87
a9b0c1d11295853 4b5f3eec186832cc e8598627 8afe1729 937469fff18c1684a	7,497,847	32	duplicates	no	7,497,847, 7,497,852	1	1	not decidable: ring state unavailable	E2	121	c7 ba
94f8ced9b3cd409 93173c53ea0bffc9 9d1db3a8d7a8cfea 2cbed7a7fba52468 4	7,497,892	31	duplicates	no	7,497,892, 7,497,897	1	1	not decidable: ring state unavailable	E2	121	9(38
2aabb168b2b7210 bc2d1620 ddbbd038 de4ae458 47cc77a2 7be4389a ea388a5e9	7,497,921	30	duplicates	no	7,497,921, 7,497,926	1	1	not decidable: ring state unavailable	E2	121	6f 3c

TXID	TABLE HEIGHT	LINE	GROUP	TX_PARITY	CARRIERS	NONCE BYPASSED	PARITY CHANGES	PROOF TODAY	EVIDENCE	FEE	BC
ab1 da12 1afe 1785 2ee2 4f19 b836 e602 d338 9b56 9048 8382 23ac d754 247d 7ff9 1	7,497,942	29	duplicates	no	7,497,942, 7,497,948	1	1	not decidable: ring state unavailable	E2	121	48 df

INFLATION ACCOUNTING, WITH BOUNDS

DC3: DEMONSTRABLE FROM PUBLIC FIELDS

QUANTITY	VALUE	EVIDENCE	BOUND
Window emission = CalcSupply after the window − CalcSupply at its start (atomic)	2,536,167,750 = 1,674,208,886,850 − 1,671,672,719,100	DERIVED (v1)	`exact`
Coinbase = base reward + fees; supply identity	82,477 and 82,477 of 82,477 blocks	DERIVED (v1)	`exact`
Window fees, every tx-carrying block (atomic)	855,173 over 1,726 blocks, pending 0	DERIVED (v2, audit D2-08 HOLDS)	class file: `exact`; audit: exact
Window public burns (atomic)	230,470,000	DERIVED (v2)	class file: `exact`; audit: exact
Gap to our anchor: fees, burns (atomic)	1,567, 0 over 21 blocks	DERIVED (v2)	audit: exact
Blocks v1 left pending, settled by v2	1,575 of 1,575 (fees 838,373, burns 230,470,000); v1 known rows that disagree: 0	DERIVED	—
SC transactions; their payouts	199 in 192 blocks; payouts pending: SC payouts are produced by SC execution (blockchain/transaction_execute.go), which a decoder does not run and the node replay does not export	DERIVED / pending	pending

Rule, verbatim: fee = transaction.Fees(), burn = sum of payload BurnValue (the node decoder, as v1 derive.py); public fields only; SC payouts stay pending with the reason. Scope: tx-carrying blocks 7,415,466..7,498,540 (window 7,415,466..7,497,942 and the gap to our anchor). Emission grade (v1): **PROVEN-NO-INFLATION-IN-SCOPE**, scope the formula coinbase of the 82,477 window blocks: each has one COINBASE miner tx with value 0, the base reward is the formula (node and Python agree on every row), and the sum equals CalcSupply(end+1) - CalcSupply(start). It says nothing about hidden (encrypted) amounts, and nothing about fees, burns or SC payouts of blocks whose bodies are pending.

DC1 AND DC2: NOT DECIDABLE FROM PUBLIC DATA, AND WHY

- DC1 amounts: DC1-obs 0 of 75; the rest is DC1-pending (75). The external figure of the disclosure: the external 23,886,517.3 figure stays EXT (disclosure track C6); no DERIVED input reproduces it.
- DC2: value created under the proof-soundness defect is unbounded over genesis .. now (NOT-DECIDABLE-FROM-PUBLIC-DATA); unbounded (lower bound 0, no upper bound): an accepted proof under the defect and an honest one pass the same verifier (FQ5); the structural verifier (D2-12) shows encodings, not soundness; a second soundness verifier is NO-GO (rft §9)
- Hidden balances as a whole: Q-CS3 answers no, so no figure on this page is a statement about the supply of the ledger.

REPLAY AND REPRODUCIBILITY

One command re-derives every derived/ file of dataset v2 from raw/ (and the pinned v1 files) into a scratch directory and compares every SHA-256: `sh forensics/r28d/replay/replay.sh` (gate G28).

recorded runs	3 runs bound to this manifest, 3 full PASS, byte-exact yes, report `89aa2553e5ad469f`, last 2026-11-08 09:19:37-08:00
re-derived	11 of 11 equal; tx bodies in the store 1,796; wall-one link blocks passing check i: pending
tools pinned	15 (SHA-256 in the manifest)

THE NODE REPLAY OF THE WINDOW'S END

Dataset v2 carries two node replay runs from our archive state (root `d3345594b8bb324b`): byte-identical yes, 31 topos 7,467,515–7,467,545, then the node's own checks rejected topo 7,467,546 (Invalid TX: tx `ca63e2c4d5190290`, in the table no; Verify_Transaction_NonCoinbase_CheckNonce_Tips (blockchain/transaction_verify.go:72-200, error at :189)). End state, as recorded: pending: the run reached topo 7467545, not 7,498,540.

Overturned by the audit (D2-14 HOLDS): The merge-stage run through the sync path's entry (`harness/runtime/r26_merge_a2_hf3_replay_20261104.json`) added 31,025 blocks (918 txs) from 7,467,516 to 7,498,540, completed, last root `ca11cc2106c86d9d`, equal to our own treehash at the anchor (`ca11cc2106c86d9d`): yes; 3 of 3 walls accepted.

> This is a statement about our node's state, not about hidden amounts: a historical replay is not a current-state proof. State roots held before the archive: bound: the replay from the archive top state reaches our own treehash at the anchor.

INDEPENDENT AUDIT

A second implementation (`harness/forensics/audit2.py`, audit2-1; shares code with the builder: no) re-ran every check from bytes and refuted by default. Bound to data `2d3a1aa7a99960a3` (29 files): 27 challenges, 27 HOLD, 0 FAIL, 0 UNRESOLVED; overturns over all runs 0 (0 distinct); 0 downgraded.

Overturns on record (audit-1, kept under `audit/history/`, report `bdaf2bfc1e40921d`): 0 (none), unresolved none.

Overturns on record (audit-2, kept under `audit/history/`, report `71ccb13bbd27391f`): 0 (none), unresolved none.

Not attempted (NO-GO): a second, independent SOUNDNESS verifier of the range proof / inner-product argument, instead the structural / public-value verifier D2-12 (pass/fail per named check only).

VERDICT	QUESTION	AUDIT GRADE	OPEN OR FAILED
V2-FEES	did the public fees and burns of the window + gap create value beyond the emission formula?	**NOT-DECIDABLE-FROM-PUBLIC-DATA**	none
V2-DUP-BODY	is a TXID held with two different bodies (DC1 two-bodies; DC0 outside the table)?	**NOT-DECIDABLE-FROM-PUBLIC-DATA**	none
V2-END-STATE	does a replay of 7,467,516..7,498,540 from the archive seed end at our own treehash?	**NOT-DECIDABLE-FROM-PUBLIC-DATA**	none
V-STATE-ROOTS	what were the pre/post balance-tree roots of the window's blocks (A10c)?	**NOT-DECIDABLE-FROM-PUBLIC-DATA**	none
V-DC2	did the proof soundness defect (DC2) create value over genesis .. now?	**NOT-DECIDABLE-FROM-PUBLIC-DATA**	none

RESEARCH QUESTIONS, STATUS

QUESTION	WHAT IT ASKS	STATUS
FQ1	DC1 census: two bodies under one TXID, parity, exception path; any duplicate outside the table (DC0)	answered (partial: two-bodies pending)
FQ2	DC1 amount observability: is a DC1 tx amount determined by public data?	answered for the recorded criteria, window-wide (every window TXID held)
FQ3	the randomness-reuse boundary: which wallet versions reused payload randomness, from which height the signature stops	answered for the window: code side and on-chain census (no boundary inside the window)
FQ4	DC2 presence in the current code (head, Release152, genesis-era tag) + structural transcript KAT	answered
FQ5	DC2 detectability from public data	answered (INTERPRETATION, re-tested every round)
FQ6	what would restore a verifiable baseline (design only, FORK-REQUIRED)	pending: FX-F5 / FX-F6 rows belong to the disclosure track (B114)
Q-CS1–Q-CS3	the current-state questions	answered (scope: the held replaywall archive at topo 7467515, not our anchor 7,498,540); answered (INTERPRETATION of SRC; audit refutes by default); answered (INTERPRETATION of SRC; audit refutes by default)

ID	CLAIM	RESULT	EVIDENCE
D2-01	the ledger entry of r28d names the manifest the dataset now has	HOLDS	RAW: ledger 106631411c20a8e0, manifest 106631411c20a8e0, sealed False
D2-02	every dataset file re-hashes to the manifest, none missing or unlisted, class files listed	HOLDS	RAW: listed 35, on disk 35, mismatch 0, missing 0, unlisted 0, class files listed True
D2-03	the v1 inputs v2 reads are pinned and unchanged; v1 is sealed and unchanged	HOLDS	RAW: 8 inputs, changed 0; v1 sealed True, manifest equal True
D2-04	the RAW chain 7,412,665..7,498,540 is one tips chain from our anchor, and the 2,799 wall-1 link blocks are its blocks 7,412,665..7,415,463	HOLDS	RAW: walk 85,876 blocks 7,498,540->7,412,665, contiguous True, multi-tip 0; wall-1 rows equal 2,799/2,799

ID	CLAIM	RESULT	EVIDENCE
D2-05	the 1,796 RAW tx bodies are every body the window + gap + wall-1 window need, each passing check (iii), 0 missing	HOLDS	RAW: needed (own) 1,796 (window+gap 1,794, wall-1 window 2), verified 1,796, missing 0, extra 0, txid mismatch 0
D2-06	the 1,796 derived tx rows (type, fee, burns, ring sizes, roothash, reference, SC data, body hash) equal an independent decode	HOLDS	DERIVED: equal 1,796/1,796, differ 0
D2-07	the 1,747 per-block fee/burn/SC rows equal own sums	HOLDS	DERIVED: equal 1,747/1,747, differ 0
D2-08	fees.json totals: window fees 855,173, burns 230,470,000, v1's 1,575 pending-fee blocks all settled, v1 known rows unchanged	HOLDS	DERIVED: own {'gap_burn_sum': 0, 'gap_fee_sum': 1567, 'gap_txblocks': 21, 'sc_tx_blocks': 192, 'sc_txs': 199, 'settled': 1575, 'settled_burn_sum': 230470000, 'settled_fee_sum': 838373, 'v1_known_rows_disagreeing': 0, 'window_burn_sum': 230470000, 'window_fee_sum': 855173, 'window_txblocks': 1726}; claimed equal True
D2-09	SC: 199 SC txs in 192 blocks; their payouts stay pending (no decoder runs SC execution)	HOLDS	DERIVED: own sc_txs 199, sc_tx_blocks 192; claimed 199/192; payouts 'pending: SC payouts are produc...'
D2-10	summary.json counts equal the audit's counts	HOLDS	DERIVED: claimed bodies 1796, txblocks 1747; own 1796, 1747
D2-11	duplicate-body census (B97): of 67 TXIDs carried by more than one RAW block, none is held with two different bodies, and the rest is unobservable, not zero (dc1-two-bodies stays unbounded)	HOLDS	RAW: multi-block 67 (in table 67, outside 0); 2+ sources 67: equal 67, different 0; single source 0; no copy 0; RAW: all TXIDs with 2+ independent copies: 68 (equal 68, different 0); tx lists repeating a TXID 0; INTERPRETATION: a block commits to the TXID only; the TXID is Keccak-256 of the core statement, which excludes the proofs (audit decoder: core_len < body length for every proof-carrying body), so the RAW blocks cannot show which body a carrier held; bodies are compared across independent holders instead
D2-12	structural verifier (B97, second implementation): every proof-carrying body of 1,560 is well-formed: wire layout per ring size, every point and scalar canonically encoded, ring shape	HOLDS	RAW: bodies 1,796 {'NORMAL': 1361, 'REGISTRATION': 236, 'SC': 199}; points 132,504, scalars 41,410; failed checks none; decode errors 0
D2-13	node replay runs are byte-identical per entry point, start from the archive seed, and follow the RAW chain	HOLDS	DERIVED: runs ['A', 'B', 'C'] identical True; seed root == v1 archive root True; seed blid == walk True
D2-14	run(s) ['C'] through the HF3 entry (p2p.Add_Complete_Block_HF3) complete 7,467,516..7498540 and end at our own treehash; the pre-fix stop of runs A/B at 7,467,546 is a tool artifact	HOLDS	DERIVED: own replay: completed, 31,025 blocks, root ca11cc21 == recorded treehash True; walls accepted {'wall2': True, 'wall3': True, 'wall4': True}; own roots 7a4a620913da4ea1 == dataset run roots ['7a4a620913da4ea1']; RAW: HF3 carriers in 7,467,516..7,467,546: 3, Parity differs with/without the HF3 step on 1
D2-15	roothash witnesses: RAW Statement.Roothash equals the held archive root at its reference topo ({'archive_differ': 0, 'archive_equal': 200, 'differ': 0, 'equal': 690})	HOLDS	RAW: own archive witnesses equal 200, differ 0; DERIVED: own replay roots as witnesses: equal 690, differ 0, outside 870
D2-16	A10c (carried from v1): the archive roots 7,453,290..7,467,515 are the canonical roots of those topos	HOLDS	DERIVED: dataset claim: bound: the replay from the archive top state reaches our own treehash at the anchor; dataset run roots == own replay roots True; own end-state match True; own replay witnesses differ 0
D2-17	G31 bound honesty: the 12 figures each carry class, scope, coverage, bound_type, and no forbidden phrase	HOLDS	DERIVED: problems none; forbidden phrases none
D2-18	FQ1 / DC0: the census figures (window distinct TXIDs, multi-block TXIDs, all in the table, carriers, parity) equal an independent count; DC0 = 0	HOLDS	RAW: own {'carriers_total': 142, 'dc0_window_duplicates': 0, 'duplicates_two_carriers': 67, 'false_parity_one_carrier': 8, 'of_which_in_table': 66, 'table_txids_outside_window_carrier': 1, 'tx_parity_true': 17, 'window_distinct_txids': 1769, 'window_tx_carrying_blocks': 1726, 'window_txids_in_two_or_more_blocks': 66}; DERIVED: claimed {'carriers_total': 142, 'dc0_window_duplicates': 0, 'duplicates_two_carriers': '67 of 67', 'false_parity_one_carrier': '8 of 8', 'of_which_in_table': 66, 'tx_parity_true': 17, 'window_distinct_txids': 1769, 'window_tx_carrying_blocks': 1726, 'window_txids_in_two_or_more_blocks': 66}
D2-19	classes.json DC3 figures reflect dataset v2 (fees and public burns of the window)	HOLDS	DERIVED: classes base r25: fees {'bound_type': 'exact', 'value': 855173} over 82477 blocks, burns {'bound_type': 'exact', 'value': 230470000}; v2 own: fees 855,173, burns 230,470,000 over 82477/82477
D2-20	G32 records half: every v2 derived/raw record carries its defect class	HOLDS	DERIVED: v2 records 39,163: tagged 39,163, pending 0; classes.json records {'derived/blocks_fees.jsonl.gz': {'DC3': 1747}, 'derived/roots_window.jsonl.gz': {'DC3': 31025}, 'derived/txs.jsonl.gz': {'DC1': 75, 'DC3': 1721}, 'raw/tx_bodies.jsonl.gz': {'DC1': 75, 'DC3': 1721}, 'raw/wall1_blocks.jsonl.gz': {'DC3': 2799}}
D2-21	presence.json: every anchor re-reads as recorded at its rev (36 anchors); DC2 present at the head is the headline	HOLDS	SRC: re-read equal 36/36; differ none; revs not in clone none; DC2 present at head True
D2-22	invariant.json: its Q-CS2/Q-CS3 anchors exist at the head, and the Q-CS1 census root is the held archive root	HOLDS	SRC: anchors ok 9/9; Q-CS1 root == v1 archive root at 7467515: True
D2-23	the v2 dataset text holds no exploit content and no private information (no address, path, IP, host)	HOLDS	DERIVED: files 30, patterns 15, private terms 34, hits 0
D2-24	(a2) TreeSection from ONE source stays BLOCKED and pending, with evidence, never graded	HOLDS	DERIVED: class pending; verdict 'BLOCKED from this host: every source tha...'
D2-26	dc1_structure.json: the ring size / payload format counts of the 75 DC1 txs equal an independent count (own HF3 parse, own walk, own decoder)	HOLDS	DERIVED: own {'criterion': 60, 'criterion:single_inclusion': 7, 'criterion:two_inclusions': 53, 'ring16+rpctype1': 15, 'ring2+rpctype0': 60, 'single_inclusion:ring16+rpctype1': 1, 'single_inclusion:ring2+rpctype0': 7, 'two_inclusions:ring16+rpctype1': 14, 'two_inclusions:ring2+rpctype0': 53}; decoded 75/75, two carriers 67; claimed equal True
D2-27	fq.json: the FQ2 correction adds only criterion S-RT and a dated note to r28's fq.json; S-RT agrees with the own count; DC1-obs unchanged	HOLDS	DERIVED: rest equal True; S-RT '60 of 75 carry it (53 of 67 two-inclusion, 7 of 8 single)', agrees True; dc1_obs 0 (base 0); corrections 1
D2-28	walls.json (33 rows + replay_stops) and inclusions.json (67 rows + the file) each carry the defect class its rule gives	HOLDS	DERIVED: walls equal 33/33 (unwalked 0, differ none); replay_stops equal True; inclusions equal 67/67 (own multi-block 67, rows == own True, own classes {'DC1': 67}); file class equal True; pending 0

EVIDENCE CLASSES AND DEFECT CLASSES

CLASS	WHAT IT IS	EXAMPLE IN THIS DATASET
RAW	bytes from our own chain or a source after checks i–iii, with SHA-256 and provenance	the 1,796 tx bodies (missing 0)
DERIVED	a deterministic computation from RAW by a pinned tool, re-derived by replay	1,796 decoded tx rows; the window fees 855,173
EXT	a claim with its source, never evidence	the Foundation's external figure (disclosure track C6); the earlier key count 2,141,950
INTERPRETATION	a labelled reading that names its DERIVED or SRC inputs	Q-CS2: answered (INTERPRETATION of SRC; audit refutes by default)
DC0–DC3	the defect class of a finding (rft §0.8.9); one class per finding, never summed across	12 class figures: exact 8, lower-bound 2, unbounded 2

ROUND OVER ROUND

ROUND	WHAT EXISTED	WHAT IT ADDED
R24	the disclosure track: 11 claims re-tested each round, 12 new facts	claims tested against the tree and the live node
R25 dataset v1	manifest `cc80176e3dde0c67`, sealed yes; 82,477 heights RAW, 75 table bodies RAW; 5 new facts	replay, an independent audit, content pins, the first frontend
R26 dataset v2	manifest `2fd3061cf7affc91`, sealed yes; 1,796 tx bodies RAW, 1,575 pending-fee blocks settled, 5 new facts	defect classes with bounds, presence and the missing-invariant certificate; the node replay; a second audit (27 challenges); this frontend by class
R27 (planned)	dataset v2 re-verified byte for byte	class tags on every record (B126), the class file on dataset v2, a rebuilt node replay (B123)

Sources of every figure: `audit/audit_report.json`, `audit/challenges.json`, `derived/a2.json`, `derived/classes.json`, `derived/fees.json`, `derived/fq.json`, `derived/inclusions.json`, `derived/invariant.json`, `derived/presence.json`, `derived/state_roots.json`, `derived/summary.json`, `derived/walls.json`, `forensics/ledger.json`, `forensics/r25/derived/emission.json`, `forensics/r25/derived/hf3.json`, `forensics/r25/derived/window.jsonl.gz`, `forensics/r25/raw/anchors.json`, `forensics/r25/raw/missing.json`, `harness/loop/disclosure_track.json`, `harness/runtime/forensics_replay.json`, `harness/runtime/r26_merge_a2_hf3_replay_20261104.json`, `manifest.json`.

SUBSYSTEM INDEX

Changes grouped by subsystem. Round links jump to the section; paths are repository-relative.

MINER AND GPU

ROUND	CHANGE	FILES
R1	TCP_NODELAY dialer, reconnect back-off, submitShare	`cmd/dero-miner/miner.go, gpu_worker.go`
R14	GPU wins re-hashed on CPU, getwork fd leak, atomic job_counter, --cpu-qos	`cmd/dero-miner/miner.go, thread_darwin.go`
R15	est_earned with finals, session counters, nw_from_job, 15 s window	`cmd/dero-miner/stats.go, miner.go`
R16	paired SHA-256, GPU stale accounting, bench log path	`cmd/dero-miner/astropair/, pairhash.go, gpu_worker.go`
R17	unlocked mining threads, paid vs accepted	`cmd/dero-miner/miner.go, outcome.go`
R18	--gpu-mode=split, GPU auto-respawn	`cmd/dero-miner/gsplit/engine.go, gpu_worker.go, gpu-miner-poc/gpu_split_server.swift`
R19	split audit-strike policy	`cmd/dero-miner/gsplit/engine.go`
R20	GPU kernels, log rotation, exit reasons, getwork read deadline	`gpu-miner-poc/gpu_split_server.swift, cmd/dero-miner/minerlog.go`
R21	stale-drop on final flip, cgo yield, --gpu-max busy union, run ledger	`cmd/dero-miner/gpu_split.go, gsplit/engine.go`
R22	--gpu-max duty bucket, exact PoW-target compare, stale-batch SHA skip, opt-in auto duty tuner, r22 stage-1 table kernels and per-kernel spans, submit token bucket	`cmd/dero-miner/gsplit/duty.go, powtarget.go, dutytune.go, gpu-miner-poc/gpu_split_server.swift`
R23	run-ledger Hashes/Height restored, contention fields, noise-gated duty tuner, job-latency metrics	`cmd/dero-miner/runledger.go, contention.go, dutytune.go, job_latency.go`
R23b	--supervised console exit, --phase-duty (off by default) and per-phase metrics, --cpu-max charged on CPU share	`cmd/dero-miner/miner.go, phase_duty.go, cpushare.go, metrics.go`
R23c	phase-exact expected shares and luck (the 5/(9D) model over-stated expectation ×1.034–1.072) and final_hash_fraction in REWARD STATS; the split comment names DERO_GSPLIT=8192,3,2 as the quiet-box A/B; CPU path, handoff and GPU flow measured with no engine change (tests and 6 mutations only)	`cmd/dero-miner/outcome.go, miner.go, expect_phase_r23c_test.go, gpu_split.go (comment), gsplit/handoff_r23c_test.go, gsplit/overhead_test.go`

SUFFIX SORT

ROUND	CHANGE	FILES
R14	descriptor sort behind SuffixSort hook, three guards	`cmd/dero-miner/dsa/, fastsort.go, astrobwt/astrobwtv3/gpu_bridge.go, pow.go`

ROUND	CHANGE	FILES
R15	bounds-check reslices, Verify 1/1024, PGO	cmd/dero-miner/dsa/dsa.go, default.pgo
R16	native C sort via cgo, tail room fix	cmd/dero-miner/dsac/, dsa/dsa.go
R19	NEON fast paths	cmd/dero-miner/dsac/descriptor.c
R20	16-lane NEON singleton scatter	cmd/dero-miner/dsac/descriptor.c
R21	zero-run tie-break keys, opt-in GPU sort probe	cmd/dero-miner/dsac/, gsplit/engine.go
R22	phase profile (merge 43 %), six variants refuted, DSA_PROF builds on arm64	cmd/dero-miner/dsac/descriptor.c
R23	no change: the box, not the sort, set the R23 ceiling (§39.1)	none
R23b	clang PGO of the C descriptor sort refuted (×0.99–1.01); per-core sort speed sets the ~100k ceiling	none
R23c	NEON pair singleton write (DSA_PAIR) and branch-free group split (DSA_GSCAN): thread CPU per sort ×1.035–1.053, bit-exact on 120,000 fresh texts; fused radix pass 2 refuted (×1.137 slower)	cmd/dero-miner/dsac/descriptor.c, dsac/r23c_fresh_test.go

P2P AND SYNC

ROUND	CHANGE	FILES
R1	--lowlatency KCP, latency-ordered miniblock broadcast	p2p/controller.go, p2p/connection_pool.go, cmd/derod/main.go
R4	fill_common and LocatePruneTopo bootstrap panics	p2p/common.go, p2p/rpc_handshake.go
R5	grandfather table, v2-remnant drop and ban	blockchain/hardfork_fixes.go, p2p/common.go
R6	changeset nil block, 5-minute peer persistence	p2p/rpc_changeset.go, p2p/controller.go
R11	GetObject failure classes, P5 tx-fetch budget	p2p/object_errors.go, p2p/rpc_notifications.go
R13	bounded frames, deadlines, misbehaviour scoring	p2p/call.go, p2p/penalty.go, p2p/rpc_cbor_codec.go
R15	unknown_tip chunk guard, escalating v2 ban	p2p/chunk_server.go, p2p/common.go
R17	one KCP packet per message, first-relay scoring, 24 h whitelist	p2p/rpc_cbor_codec.go, p2p/first_relay.go, p2p/peer_pool.go
R18	pipelined block fetch, bounded prewarm, KCP window 256	p2p/sync_pipeline.go
R19	pipelined TreeSection, tx admission dedupe	p2p/ (sync, bootstrap)
R20	tx-ask hold, compact block reconstruction	p2p/tx_ask.go
R21	grey quarantine, known-tx inventory, frame byte budget	p2p/grey_quarantine.go
R22	stuck-peer eviction (F4), --p2p-nat=probe, known-tx sweep fix, per-peer INV bucket, short GetObject replies counted, txcanon relay admission	p2p/stuck_peer.go, p2p/nat/probe.go, p2p/tx_known.go, p2p/inv_budget.go, p2p/tx_flow.go, glue/txcanon/
R23	fast stuck-peer re-eviction, txprecheck before state reads with relaying-peer scoring, txcanon observes accepted-block txs only (CR23-2)	p2p/stuck_peer.go, p2p/tx_flow.go, p2p/object_errors.go, glue/txprecheck/, glue/txcanon/
R23b	HF3 table Parity on every block-add path (CR23b-1), relay admission window and HF3 TXID refusal, registration canonical-form relay policy (CR23b-2), lazy sync height gate	p2p/hf3_parity.go, chain_sync.go, compact_block.go, tx_flow.go, sync_height_gate.go, glue/txcanon/txcanon.go
R23c	sync PoW prewarm overlapped with the previous block's topo fsync (×1.255–1.331 blocks/s, loopback shadow); ban-list lookup index; pruned, bounded penalty table; inbound /24 and /48 aggregate bans	p2p/sync_prewarm_overlap.go, chain_sync.go, bans.go, penalty.go, controller.go, counter_registry.go
R24	compact-block txs observed after the block is accepted (B18, SP24-1); prewarm counters count each synced block once and book overlap time separately (B34, SP24-2 / CP24-2); cross-response prewarm overlap through a non-blocking peek of the next pipelined response (SP24-3); single-flight difficulty in the sync prewarm (CP24-1: CPU per synced block ×0.947 verifier, ×0.916 builder); the chain locator no longer spins on a topo read error (SP24-5)	p2p/compact_block.go, chain_sync.go, sync_pipeline.go, sync_prewarm_overlap.go, chunk_server.go, counter_registry.go
R25	peer offences readable: per-peer score, ban count and reasons in GetPeerInfo (B63), per-reason offence counter made visible (B64), Chain requests counted (B65), ban sources in a sidecar (B66), INV clips per connection (B67), unserved objects and short answers counted (B68), stuck holds listed (B69); a peer that answers a sync with nothing is held from sync, escalating (B47); observed peer addresses removed from Go sources (B70)	p2p/peer_observe.go, peer_info.go, penalty.go, bans.go, chain_sync.go, inv_budget.go, stuck_peer.go
R26	sync-hold strikes decay and an accepted relayed block that brings us level forgives the peer; held peers ahead are tried when no peer synced (B99); a duplicate miniblock from two peers inside the PoW-verify window counts as duplicate_race, not insert_failed, and refusal lines carry the error text (B103); the two-invalid_pow policy kept and its comment corrected (B102)	p2p/penalty.go, p2p/peer_observe.go, p2p/rpc_notifications.go, p2p/consensus_obs_r25.go, p2p/counter_registry.go, p2p/connection_pool.go
R27	a ban reloaded from ban_list.json is visible: gauge p2p_bans_active, counter p2p_bans_loaded_total and the same two fields in GetSyncInfo (B142); counting only, consensus guard PASS	p2p/bans.go, p2p/peer_observe.go, p2p/counter_registry.go, p2p/connection_pool.go, cmd/derod/rpc/rpc_dero_getsyncinfo.go
R28	the chunk PoW verification pool is counted (p2p_chunk_pow_* families, L24-21; the cache-miss rate is read live on r28); the pending-event rows B47, B48, B57 and B99 closed by unit tests plus executable scenarios (event_scenarios.py 22/22); derod --help states the random p2p port default (B1)	p2p/chunk_server.go, p2p/r28_chunk_pow.go, p2p/r28_p2p_config_test.go, p2p/r28_p2p_coverage_test.go, harness/intel/event_scenarios.py

RPC AND WEBSOCKETS

ROUND	CHANGE	FILES
R12	DERO.GetSyncInfo, target_height, no stack traces in errors	cmd/derod/rpc/rpc_dero_getsyncinfo.go, rpc_errors.go

ROUND	CHANGE	FILES
R13	DERO.GetMiningOutcome	cmd/derod/rpc/mining_outcome.go
R15	GetBlockTemplate 9·D on finals, hashrate estimates	cmd/derod/rpc/rpc_dero_getblocktemplate.go, websocket_getwork_server.go
R17	template refresh_due	cmd/derod/rpc/websocket_getwork_server.go
R19	Subscribe, /health, range and pool queries, /metrics	cmd/derod/rpc/ (RPC_METHODS.md)
R20	Host/Origin guard, GetTxStatus, getwork presence	cmd/derod/rpc/
R21	foreign-origin ws budgets, WatchTx, TxPool pushes, job frame once	cmd/derod/rpc/
R22	opt-in --rpc-tx-cache, canonical GetRandomAddress keys, txcanon on SendRawTransaction	cmd/derod/rpc/tx_result_cache.go, rpc_dero_getrandomaddress.go, rpc_dero_sendrawtransaction.go
R23	one getwork dispatcher, per-miner job numbering newest-last, finality fields, GetFeeEstimate, GetEmissionAudit	cmd/derod/rpc/job_dispatch.go, getwork_dispatch.go, rpc_dero_gettxstatus.go, rpc_dero_getfeeestimate.go, emission_audit.go
R23b	settled-tx GetTxStatus cache, Reorg event and finality monitor, GetMonetaryIntegrity, getwork miner-absence metric and WARNING	cmd/derod/rpc/tx_status_final_cache.go, reorg_events.go, monetary_integrity.go, getwork_absence.go
R23c	ordered, lossless Block/MiniBlock/TxPool pushes (per-client queues, one push worker); at most 256 txids or SC keys per remote call, refusals scored; GetMonetaryIntegrity summary without list copies, with issued, burns and payees; getwork flip-delay counters	cmd/derod/rpc/websocket_server.go, ws_hardening.go, ws_tx_events.go, param_bounds.go, rpc_dero_gettransactions.go, rpc_dero_getsc.go, monetary_integrity.go, getwork_flip.go, websocket_getwork_server.go
R24	JSON-RPC without the exact Content-Type gets a JSON-RPC −32600 body instead of an empty 415/405 (B38, RPCWS24-1); remote bodies carry at most 1,024 txids + SC keys and pay ceil(n/8) item tokens on HTTP and websockets, with the two bypasses the verifier found closed (B29, RPCWS24-2); GetMonetaryIntegrity copies the auditor result instead of a hand-made field list (CI24-1); GetRandomAddress decoys corrected for tree depth (CI24-3); L9 and slow-flip getwork counters (GWO24-3)	cmd/derod/rpc/r24_rpcws.go, param_bounds.go, ws_hardening.go, rpc_abuse.go, emission_audit.go, r24_decoy_sampler.go, rpc_dero_getrandomaddress.go, getwork_flip.go
R25	typed errors where handlers panicked or answered wrongly: a state read below the prune point gets −32031 before any read (B85); every input-reachable panic(of the 31 in cmd/derod/rpc returns a typed error (B86); GetBlockHeaderByHash no longer returns an all-zero header with no error for an unknown hash (B80); GetTransaction gains per-tx txs_status; GetInfo total_supply_basis (BD6); the wallet RPC parses Content-Type like derod (B40); new read-only DERO.GetConsensusStats	cmd/derod/rpc/r25_rpcrobust.go, rpc_dero_getencryptedbalance.go, rpc_dero_getblockheaderbyhash.go, rpc_dero_gettransactions.go, rpc_dero_getinfo.go, consensus_stats_r25.go, walletapi/rpcserver/
R26	NameToAddress for an unknown name answers the typed −32032 (B100); GetBlockHeadersRange, GetMiniBlocks and GetDifficultyHistory refuse unknown parameter names with −32602 and reads below prunedheight with −32031 before any store read, and GetDifficultyHistory no longer panics on a pruned-shape record (B104); GetRandomAddress applies its 5-topo decoy window at every height, clamped at the prune floor (B43 root cause: stale decoys in the simulator)	cmd/derod/rpc/rpc_dero_nametoaddress.go, rpc_dero_explorer.go, rpc_dero_getrandomaddress.go
R27	a parameter-decode failure answers −32602 naming the field and the accepted names, with data {field, accepted}, on all 53 method-table entries (B132, typed_new); the four untyped panic recovers of GetEmissionAudit, GetMonetaryIntegrity and GetSupplyAudit count rpc_internal_error, typed refusal counters are pre-registered and rpc_error_responses_total{code} counts every error answer (B143); a ban-list gauge (B142)	cmd/derod/rpc/r27_rpcmetrics.go, emission_audit.go, monetary_integrity.go, rpc_dero_getsupplyaudit.go, node_metrics.go, websocket_server.go, RPC_METHODS.md
R28	GetInfo tx_pool_unmined (B42); every JSON-RPC error body written outside jrpc2 is counted in rpc_error_responses_total{code} at its writer (B150); the /metrics page carries HELP and TYPE for every family from an embedded registry, 304 families (B151); B61 closed by the one-tip rule	cmd/derod/rpc/rpc_dero_getinfo.go, r28_rpcmetrics.go, r24_rpcws.go, rpc_abuse.go, param_bounds.go, metrics/registry_r28.go, metrics/metric_registry.json, harness/intel/metric_registry.py
R29	a jhttp bridge 500 becomes a counted JSON-RPC error body (B175); the 8 lazily created counter families are pre-registered at start (B152); TYPE is tracked apart from HELP (B176); the rpc/ display copy of FormatMoneyPrecision is 128-bit via the reviewed guard construct rpc-money-copy-128bit (B144)	cmd/derod/rpc/r29_bridge_errors.go, cmd/derod/rpc/r29_lazy_preregister.go, p2p/r29_lazy_preregister.go, metrics/registry_r28.go, rpc/

WALLET

ROUND	CHANGE	FILES
R6b	wallet CLI built; Release142→152 fixes catalogued	cmd/dero-wallet-cli, walletapi/
R18	wallet RPC Origin/Host guard, /install_sc limit	walletapi/rpcserver/origin_guard.go
R19	atomic saves, --password on restore, sync_loop exit, paged transfers	walletapi/
R20	parallel history discovery, bounded daemon calls, HTTP timeouts	walletapi/
R21	wakes on Height push, fileverify fix	walletapi/, cmd/dero-wallet-cli
R22	ring-order nonce-reuse fix (C3), ErrRingOrderExhausted, signature input checks (C5), EncryptWithKey (C6)	walletapi/transaction_build.go, wallet_transfer.go, wallet.go, cipher.go
R23	C4 receiver payment-proof check, burn-aware (CR23-1); integrated-address port size check (CR23-5); zero seed refused; reconnect fixes, Connected=false on drop	walletapi/payment_proof_verify.go, wallet_transfer.go, wallet.go, daemon_connectivity.go
R23b	no wallet change; history resync after a reorg (RPCWS23B-E) found, not fixed	none
R23c	history resync after a reorg in one call, keeping the valid entries (RPCWS23B-E fixed); the receiver-proof payload leak (CI-9) found, report-only	walletapi/history_sync_plan.go, daemon_communication.go, r23c_history_sync_test.go

ROUND	CHANGE	FILES
R24	V2 receiver proofs are withheld when two transfers to the wallet repeat one ephemeral key E, so a proof can no longer disclose another payload's key (B32 / CI-9, CI24-2; wallet-local, no format change)	walletapi/r24_v2proof.go, wallet.go, wallet_memory.go

SECURITY

ROUND	CHANGE	FILES
R13	remote crash found in review, scoring and bans	p2p/penalty.go, p2p/call.go
R18	outbound /16 diversity, gossip validation, HTTP limits	p2p/diversity.go, cmd/derod/rpc/
R19	UDP reflection deadline, IPv6 /64 bans, slowloris timeouts, XSWD loopback	p2p/, cmd/derod/rpc/, walletapi/xswd
R20	DNS rebinding and cross-site websocket fix, TLS 1.2 floor	cmd/derod/rpc/, p2p/
R21	getwork blob-length panic (remote crash), ALLOWED_ORIGINS=null bypass	cmd/dero-miner/gpu_split.go, cmd/derod/rpc/
R22	TLS 1.2 AEAD/curve floor, per-peer INV bucket, miner submit bucket and LastError gate, BUILD_STAMPS hex, prune Lstat	p2p/tls_policy.go, p2p/inv_budget.go, cmd/dero-miner/, cmd/derod/prune_preflight.go
R23	remote RPC bucket, scoring and escalating bans; automated-client bans (IPv6 /64); getwork per-IP cap, bad-share bans, one-slot gate; getwork TLS floor; random TLS serial	cmd/derod/rpc/rpc_abuse.go, getwork_guard.go, getwork_tls.go, p2p/controller.go
R23b	rpc.serverInfo closed to remote and cross-site callers, global anomaly surge budget, SendRawTransaction header peek and admission window, unknown tx type refused from the header, random getwork TLS serial	cmd/derod/rpc/rpc_abuse.go, ws_hardening.go, rpc_dero_sendrawtransaction.go, websocket_getwork_server.go, glue/txprecheck/window.go
R23c	trusted remote keys score ×1 during a surge (RPC-1); /24 and /48 aggregate scoring for untrusted keys (RPC-2); per-key ban checked before the global budget (SEC23b-2); ban and penalty lookups no longer scan	cmd/derod/rpc/rpc_abuse.go, p2p/bans.go, p2p/penalty.go
R24	pre-warm escape closed: trust is revoked by probing, trusted keys' unknown names score into their /24 or /48, probes never earn trust (B28, SEC24-1); aggregate-only refusals are rate-limited, not scored (SEC24-2); the penalty table bound is hard, by insert-time eviction (SEC24-3); a returning-peer accept lane for addresses with a verified handshake (B30 node-local part, SEC24-4)	cmd/derod/rpc/rpc_abuse.go, p2p/penalty.go, p2p/proven_lane.go, p2p/controller.go, p2p/rpc_handshake.go
R25	proven-lane quota per netgroup and a quarter burst for fresh remote RPC keys (B48); rebroadcast skips txs a recent block carries (B42); the pool cap holds under concurrency and the summary follows the mined set (B44); relay and RPC admission refuse non-reduced proof scalars (BD5 / FX-N4)	p2p/proven_lane.go, cmd/derod/rpc/rpc_abuse.go, p2p/tx_flow.go, glue/txpolicy/txpolicy.go, glue/txcanon/txcanon.go

STORAGE AND OPS

ROUND	CHANGE	FILES
R3	salvage, archive, fastsync playbook	operational (no code)
R6	peer and ban list persistence	p2p/controller.go
R12	atomic peers.json and ban_list.json	p2p/persist.go
R20	derod log rotation flags, darwin metrics, datadir disk metric	cmd/derod/main.go, metrics
R21	--prune-history preflight, root binary untracked	cmd/derod/main.go
R22	storage growth measured read-only; storefs.go non-atomic writes reported (consensus package)	blockchain/storefs.go (report only), harness/intel/datadir_intel.py
R23	per-block emission auditor and offline tool; storefs.go non-atomic writes still reported only	glue/emission/, tools/emission-audit/, cmd/derod/rpc/emission_audit.go
R23b	emission auditor reorg-aware flows and burns, archive mode, HF3 forensic report, crash-safe series and issued_emission; launchd plist generator	glue/emission/archive.go, hf3.go, series.go, tools/emission-audit/modes.go, tools/launchd/
R23c	emission auditor: per-account check, coinbase-shape ALERT, deep-rewind segment restart (window 4,096, IssuedEmission reset), ×0.64 CPU; read-only tools/storecheck; storefs atomic writes still report-only	glue/emission/emission.go, cmd/derod/rpc/emission_audit.go, tools/emission-audit/main.go, tools/storecheck/
R24	tools/launchd-derod: a derod launchd plist generator (KeepAlive, ExitTimeOut 120, stdin /dev/null; B35, not installed); tools/nodehealth (node health without a miner, always sends Content-Type); tools/getworkprobe (a getwork client that never submits)	tools/launchd-derod/, tools/nodehealth/, tools/getworkprobe/, cmd/derod/main.go
R25	tools/launchd-derod -preflight dry run that refuses to install while a derod runs on the same store (B35); getworkprobe --metrics; benchlock --qos (opt-in); the sync-CPU herd fix cuts CPU per synced block ×0.923 but does not raise blocks/s (B46 refuted as a lever)	tools/launchd-derod/, tools/getworkprobe/, harness/pipeline/benchlock.py

HARNESS

ROUND	CHANGE	FILES
R2	reward_audit.py, mining_dashboard.sh	reward_audit.py, mining_dashboard.sh
R7	corpus, miners, page generator, G1–G12, DK01–DK20	harness/
R8	EXT root, DK21, DK22	harness/external/, harness/kat/verify_deroarch.py
R10	console triage, DK23–DK25	harness/miners/mine_triage.py
R11	consensus guard, fix_build_verify pipeline	harness/consensus_guard.py, harness/pipeline/fix_build_verify.py
R12	log_intel, datadir_intel, go_mutate, G14, G15	harness/intel/
R13	coverage miner, G16 ratchet	harness/miners/mine_coverage.py
R18	H per CPU-second bench gate	harness/pipeline/fix_build_verify.py

ROUND	CHANGE	FILES
R20	S3a, S3k, metrics_intel, G18	harness/pipeline/, harness/intel/
R21	like-for-like S4b, regime cards, run ledger	harness/pipeline/fix_build_verify.py
R22	G19 binding gate, miner_runs and chain join, go_mutate INVALID, atomic snapshots, S4c, race_split (IK38)	harness/intel/miner_runs.py, go_mutate.py, harness/pipeline/fix_build_verify.py
R23	contention census and G20, S4c self-A/B, payout ledger and G21, S3 no-race path, localdiff per-func split, HF3 soundness tests	harness/intel/contention.py, miner_runs.py, harness/run_gates.py, harness/pipeline/fix_build_verify.py, harness/miners/mine_localdiff.py
R23b	G22 soundness gate and S3s, idle ledger, G17 binding next to regime, G20 full-run record, S3a re-runs race-skipping tests, localdiff own-copy citations, DK18 unlabelled seeds; with_clients cleanup (bfe3e54)	harness/soundness.py, run_gates.py, intel/miner_runs.py, pipeline/fix_build_verify.py, miners/mine_localdiff.py, kat/verify_deroarch.py, cmd/derod/rpc/getwork_session_test.go
R23c	round registry harness/round.json (KAT RR1–RR4), in-repo benchlock, payout census (KAT IK23cP1–2), mine_evolution + DK35, ledger R1 → R23 comparison and enhancement curve; install backup tag keeps its letter suffix (4f0dda2); evolution_git.tsv line 1 no longer carries HEAD (8a086cd)	harness/round.json, pipeline/benchlock.py, pipeline/fix_build_verify.py, intel/payout_census.py, miners/mine_evolution.py, kat/test_round_registry_r23c.py, kat/test_intel_r23c_payouts.py, kat/verify_deroarch.py
R24	the miner track pause switch: G17, G18, G21 and S4b/S4c report PAUSED (miner) only while harness/round.json says paused (H24-1, KAT TR1–TR7); known test failures listed with evidence and S3 re-runs a listed flake ×3 but never excuses a first-run DATA RACE (B21, H24-3); area intelligence (harness/areas.json, area_intel.py, gate G23); the Foundation-disclosure track (disclosure_track.json / .py, gate G24)	harness/kit/roundtracks.py, run_gates.py, pipeline/fix_build_verify.py, loop/KNOWN_TEST_FAILURES.json, intel/go_mutate.py, areas.json, intel/area_intel.py, intel/disclosure_track.py, loop/disclosure_track.json, kat/test_round_tracks_r24.py, kat/test_area_intel_r24.py, kat/test_disclosure_track_r24.py
R25	standing derod error census (error_census.py) and a prune guard for every probe (kit/pruneguard.py); area tooling and tooling_intel.py (B87), a rebuilt scrubbed memory mirror (B88), live readings booked to the binary's round (B89), evolution ORDER early warning (B90), meta review (§6 step 19a); resume checkpoint round_checkpoint.py (§0.10); install backup tags read without the next literal's letter; the shadow control is the live node's build after an install	harness/intel/error_census.py, kit/pruneguard.py, intel/tooling_intel.py, intel/round_checkpoint.py, loop/meta_review.py, pipeline/fix_build_verify.py, kat/test_install_tag_r25.py
R26	S5 always tears its shadow down (B98); the paused-miner clean-clone project run and log_intel across DST (B101); src-arch byte-exact on a double run and the B93b publish scrub (B93); network-class area metrics (B105); privkit.scrub_argv for new snapshots (B119); argument parsers for tooling_intel and mine_triage (B117); from the closed loop: every Go-mutation anchor checked before S3b (B129) and volatile area KPI sources sealed per round (B130)	harness/pipeline/fix_build_verify.py, harness/intel/log_intel.py, harness/gen_deroarch.py, harness/intel/area_intel.py, harness/kit/privkit.py, harness/kat/test_gomut_anchors_r26.py, harness/kat/test_area_seal_r26.py
R27	every harness tool answers --help without running (B120: 33 → 0); gen_deroarch build --strict names the miner whose raw file lost anchors (B137); a KAT mutation is proved only by its own KAT (B138); area_intel metric class record (B134); the frontend2 KAT memoises its dataset readers (B133); live_window.py for the after-restart read (B135); the memory mirror checks the scrubbed bytes	harness/intel/*.py, harness/gen_deroarch.py, harness/kit/katmut.py, harness/intel/area_intel.py, harness/intel/live_window.py, tools/memory-mirror/sync_mirror.py, harness/kat/test_tools_kats_r27.py, test_harness_measure_r27.py, test_artifacts_memory_r27.py
R28	backlog_truth.py (every §7 row against its records), roundlists.py (B149), a KAT for every tool (89/89), masked raw scrapes (B135), memory dead-name and dead-symbol KPIs (B140), the closure audit; closed-loop fixes B171 (the S3b preflight event keyword) and B172 (a dead-reference misread)	harness/intel/backlog_truth.py, harness/kit/roundlists.py, harness/intel/tooling_intel.py, harness/intel/live_window.py, harness/pipeline/fix_build_verify.py, harness/kat/test_*_r28.py

TX POOL AND ADMISSION

ROUND	CHANGE	FILES
R24	node-local admission policy shared by RPC and relay: identical bytes verified once (flight), proof-invalid bytes remembered, a pool cap with fee-rate eviction that protects the txs a mining template carries; mined txs no longer count as pending in GetTxStatus / GetFeeEstimate; new pool fields (TXP-2 … TXP-7). B19 integer fee ranking VERIFIED, its fix not shipped (consensus guard S1)	glue/txpolicy/txpolicy.go, glue/txcanon/txcanon.go, cmd/derod/rpc/recent_mined.go, tx_pool_status.go, template_txs.go, rpc_dero_getfeeestimate.go, rpc_dero_gettxstatus.go, rpc_dero_sendrawtransaction.go

CONSENSUS OBSERVABILITY

ROUND	CHANGE	FILES
R25	the CO24 node-side gaps closed as logging, counters and RPC fields only: miniblock refusals by reason (B55), block fees in GetBlockHeadersRange (B56), block rejections by reason with the sending peer (B57), alt_blocks_count assigned (B58), first-seen times by RPC (B59), a persisted reorg monitor (B60), header miners[] at the block's topoheight (B61); detector D19 (B62)	p2p/consensus_obs_r25.go, cmd/derod/rpc/consensus_stats_r25.go, cmd/derod/rpc/blockheader.go, reorg_events.go, harness/intel/consensus_intel.py

CRYPTO AND SUPPLY

ROUND	CHANGE	FILES
R25	report-and-pin: which non-reduced proof-scalar encodings Verify accepts is pinned by an honest-proof test (BD9 closed); exactly one final miniblock per block from height 399,000, so B22 is refuted as a live under-payment; B33 re-anchored; the HF3 table covers replay walls 2 and 3, not wall 1 (B81)	cryptography/crypto/r25_scalar_reencode_test.go, blockchain/r25_final_count_test.go, harness/loop/disclosure_track.json

ROUND	CHANGE	FILES
R26	FQ4: a structural transcript-binding test pins every challenge, its preimage, the u_x base and the inner-product call identical at the head, Release152 and the genesis-era tag (no invalid proof is built, rft §9); FQ5: no public-data test separates a defect-accepted proof from an honest one, so DC2 is not decidable per tx (B113)	cryptography/crypto/r26_transcript_binding_test.go, harness/loop/disclosure_track.json

MONETARY FORENSICS

ROUND	CHANGE	FILES
R25	dataset v1 of the HF3 window (every height RAW, every table body RAW), content-pinned replay (G28), an independent second-implementation audit with its overturn kept on record (G29), and the ledger forensics sections (G30); see Monetary forensics on this page	harness/forensics/, tools/forensics/, forensics/r25/, artifacts/src/forensics_section.py, harness/forensics/facts.py
R26	dataset v2 (forensics/r26): every window tx body RAW, the v1 pending-fee blocks settled (B96); defect classes, bounds and presence (G31, G32; B107–B114); the a2 node replay of the held archive through the HF3 path ends at our treehash (B123); a second independent auditor, audit2 (G29 on every version; B97 PARTLY, B127, B128); the v2 frontend by defect class; see Monetary forensics on this page	harness/forensics/v2.py, classes.py, bounds.py, audit2.py, facts.py, tools/forensics/a2/, forensics/r26/, artifacts/src/forensics_section.py
R28	r26 sealed in the §6 order; dataset v3 forensics/r28 with the run-C replay MATCH and audit-3 24/24; B109, B123, B124, B126 closed; B76 resolved with its scope	forensics/r28/, harness/forensics/v3.py, harness/forensics/pinsnap.py
R29	run D, a fourth node replay to treehash ca11cc21 (roots byte-identical); B107 walls and inclusions tagged; B97 67/67 TXIDs from 2+ sources; B110 SC payouts of run C exact; B111 Q-CS1 at 7,498,540 on a stopped-node clone; B182 registrations 0..7,398,264 complete (542,692 with 2 leading zero bytes)	harness/forensics/, harness/runtime/r29_forensics_20261108.txt, r29_b111_qcs1.json, r29_b182_deep_census.json

CONFIG

ROUND	CHANGE	FILES
R27	a consensus parameter that is environment-settable (B141) is reported, not refused: one ERROR line per Settings field that differs from its default and the metric config_consensus_env_overrides; config/ and blockchain/ unchanged, their defaults and constants pinned by a test	cmd/derod/config_env_r27.go, cmd/derod/main.go, config/r27_consensus_params_test.go
R28	config_truth.py --go: every documented flag and environment name against the code (10 AGREE, 0 DRIFT, 0 MISSING); B141 stays report-only (refusing to start is the user's decision)	harness/intel/config_truth.py
R29	derod refuses to start on a consensus Settings environment override, naming the variable and not its value (B141, the user's decision (a))	cmd/derod/config_env_r29.go, cmd/derod/main.go

EMISSION AND GLUE

ROUND	CHANGE	FILES
R27	the series and archive records carry payee counts (B36 fields); the archive no longer hangs on a height-0 report and ReadSeriesTail keeps the first record of a window that starts on a line boundary; the glue/rwc adapters report a failed write and do not re-send on a short write (B144); tools/emission-audit -archive prints its payees (B139). The globals amount and offset fixes are held for the next dataset version (B147)	glue/emission/archive.go, series.go, emission.go, glue/rwc/rwc.go, rwc_nhyoor.go, tools/emission-audit/modes.go, main.go, harness/loop/held/r27_globals_amount_offset.patch
R28	the held globals fixes landed (B147: ParseAmount exact and bounded, FormatMoneyPrecision 128-bit, GetOffset sign; the sealed r25 re-verified through pinned snapshots); the B36 24 h payee read is one command (b36_payee_read.py); B144 partly fixed (the rpc copy waits for a guard decision)	globals/globals.go, globals/r28_globals_test.go, harness/loop/held/landed/, harness/forensics/pinsnap.py, harness/intel/b36_payee_read.py
R29	the Bulletproofs/Zether assumption table re-graded against the user-held papers: 16 DISCHARGED, 3 VIOLATED (= C1), 2 UNCLEAR, as anchored tests, no construction (B190)	cryptography/crypto/r29_assumption_anchors_test.go, harness/external/r29_inputs_manifest.json

DVM AND BLOCKCHAIN (TESTS ONLY)

ROUND	CHANGE	FILES
R27	KATs for every SC builtin that was at 0 % (dvm coverage 72.2 → 84.4 %), the hash-gas census, the difficulty vectors (difficulty.go 41.5 → 81.9 %); six consensus-visible DVM behaviours pinned and documented, never changed (B145); the first blockchain/ -coverpkg record (B146)	dvm/r27_builtins_test.go, dvm/r27_hash_gas_census_test.go, blockchain/r27_difficulty_kat_test.go, blockchain/testdata/r27_difficulty_vectors.json
R28	process_transaction_sc 0.0 → 96.4 % and blockchain/ coverage 55.5 → 68.1 % (B146); burn-tx and SC install/call KATs (B33); report-only verdicts with guard readings for B10, B11, B31, B45, B145, B157	blockchain/r28_sc_flow_test.go, dvm/r28_dvm_kat_test.go

MEMPOOL AND BLOCKCHAIN

ROUND	CHANGE	FILES
R29	the pool ranks by milli-fee per byte (pool order only; guard construct mempool-milli-fee-rank, B19); the SC recover log line demoted to V(2) (logging-only, B157)	blockchain/mempool/mempool.go, blockchain/transaction_execute.go

WALLET API

ROUND	CHANGE	FILES
R29	walletapi FormatMoneyPrecision calls the globals implementation (B144; fail-before 23,631 of 60,027 pairs wrong)	walletapi/

HARNESS AND TOOLS

ROUND	CHANGE	FILES
R29	liveness is work, not a heartbeat (B198: hangkit, coproc, gate G33, stallwatch, bounded sh()); roundlists.py the one round-list reader (B166); go_mutate STALE for a missing test (B168); booked ownership pins (B170); the G29 lint memo (B155); STEP_BUDGETS re-derived (B192); open_round() (B197); the merge finds B199 and B200	harness/kit/hangkit.py, harness/kit/coproc.py, harness/kit/roundlists.py, harness/intel/go_mutate.py, harness/intel/area_intel.py, harness/pipeline/

CEILINGS AND OPEN ITEMS

WHY 150 KH/S IS OUT OF REACH ON THIS MACHINE

In split mode the GPU does stage 1 and the final SHA-256; the CPU sorts. After R21 the two bind together: the GPU is busy ~95 % of the time while staged texts wait on the CPU sort ~74 % of the time. Speeding up one side buys at most the other side's headroom, and the live rate is whatever CPU the miner gets. R23 closed the loop on a quiet box: the same r22 miner made 92.5k H/s on 17.31 of 18 budget cores where the law predicts 91,382, so ~90–100k is the box ceiling and the remaining payout levers are uptime and the paid fraction (ARCHITECTURE §39.1). R23b recomputed it from the per-core sort rate: about 100k H/s, and 150k needs ×1.44 per-core sort speed, which nothing found provides (R23b workflow HR23b-6).

QUANTITY	VALUE	SOURCE
Descriptor sort DSA_PAIR + DSA_GSCAN (R23c)	thread CPU per sort ×1.035–1.053; live +1–3 % CPU-bound, 0–1 % quiet (estimate)	R23c workflow sortfuse; verify-sortfuse
Geometry 8192,3,2 (R23c proxy)	GPU busy per hash ×0.72–0.82; +2.9–3.7 % on a quiet box (estimate); CPU capacity 93.15–93.9k	R23c workflow gpuflow; verify-gpuflow
Live P-core split rate (R23c)	5,963 H per P-core-second; E-cores 0.38× P carry 16–17 % of the CPU-seconds	R23c workflow cpupath; verify-cpupath
Ceiling from the per-core sort rate (R23b)	~100k; 101.8–104.2k on all 20 cores with nothing else running; 150k needs ×1.44 per-core sort	R23b workflow HR23b-6
Live sort rate (R23b)	~5,650 H/cpu-s, 0.95 of the P-core rate: no engine overhead left to recover	R23b workflow HR23b-1, HR23b-6
Agent job at nice 15 next to the miner (R23b)	~2.6 cores of a 6-thread job (~15k H/s, estimate); utility QoS ×0.12	R23b workflow HR23b-2
GPU geometry 8192 (R23b proxy)	staged capacity ×1.34–1.74; live ≈ 0 while CPU-bound	R23b workflow HR23b-4
Clang PGO of the C sort (R23b)	REFUTED · ×0.99–1.01	R23b workflow HR23b-5
Quiet box, r22 miner, G20 census quiet (R23)	92.5k at 17.31/18 cores; law 91,382	RUN_REPORT Round 23; ARCHITECTURE §39.1
What 150k needs on this box (R23)	~28.9 cores against 20 (estimate)	ARCHITECTURE §39.1
Contended sample 01:53 (R23)	62.5k at 11.32 cores vs 94.7k at the full budget: -32k estimate	ARCHITECTURE §39.1
Saturated feeder signals	feeder_ring_full 0.99, feeder_cpu_backlog 0.9 in every regime: not regime signals	ARCHITECTURE §39.1
GPU-only staged ceiling at the live geometry 4096,4,2	~83k (staged / busy ≈ 78.5k / 0.946)	RUN_REPORT Round 22; ARCHITECTURE §37.2
Box total, GPU-capped plus spare CPU hashing on its own	~90–100k, depending on CPU duty	RUN_REPORT Round 22
What 150k needs	~1.8× the GPU and ~1.47× the sort	RUN_REPORT Round 22
Contention law, live (other sessions loading the box)	H/s = 4,817 × miner cores + 8,000 (r = 0.991)	ARCHITECTURE §38.1
Split CPU side / full CPU pair hash	5,792 / 3,205 H/cpu-s	ARCHITECTURE §37.2
CPU budget at `--cpu-max=90`, split finishing only	18 × 5,792 = 104k	ARCHITECTURE §37.2
r22 stage-1 kernels	GPU time ×0.742, ≈ 0 end to end until a geometry change	ARCHITECTURE §38.5
Six sort variants and five kernel levers (R22)	REFUTED · refuted with measurements	RUN_REPORT Round 22
Moving the sort to the GPU (op 3)	REFUTED · ~9 % net loss live; ~+2 % at best	RUN_REPORT Round 21
Earlier full-hash ceiling (before split mode)	~43k H/s	ARCHITECTURE §34.1
Before R14: "M1 hardware ceiling" ~7,680 H/s	REFUTED · it was the ceiling of SA-IS	ARCHITECTURE §13.1, §30.5
K-way ILP interleave of SA-IS (pre-R1 study)	REFUTED · ≤1.04×, net negative	ARCHITECTURE §14

OPEN AFTER ROUND 21

- A live A/B of `--cpu-max=100` (estimated +4.5–8k H/s, workflow cpusort-5; not measured). `still open` R22 shipped the opt-in `--cpu-max=auto` tuner; the live A/B moved to the list below. R23: HR23-1 "+17 % at 90 %" refuted (90 vs 100 within noise).
- `--min-peers=20` : only ~15–17 peers are reachable. `still open` R22: 15 outgoing peers are the whole reachable set.
- Forward UDP 11011 for inbound peers. Inbound has been 0 since Round 2. `still open` R22: the router answers no NAT mapping protocol (`--p2p-nat=probe`). R23: still 0 inbound (user action).
- F4: stuck-peer eviction. `closed in R22: shipped (p2p/stuck_peer.go)`
- F6: mutex profile. `closed in R22: done, no lock limits throughput`

OPEN AFTER ROUND 22

- A live `--cpu-max=auto` / `--gpu-max=auto` A/B on a quiet box (other sessions' jobs took ~14k H/s at 06:44–06:58). `still open` R23 shipped the noise-gated tuner and the G20 census that refuses a contended A/B.
- Geometry A/B `DERO_GSPLIT=8192,5,3` : ABBA, abort if restarts or retries rise or stale drops pass 3 %. `still open`
- S4c end to end on a changed GPU server: run it once with `--gpu-bench` before relying on it. `path shipped in R23, not yet run` `--gpu-bench` on an unchanged server is now a self-A/B (bit-exact, ratio in [0.93, 1/0.93]); `--gpu-bench-only` runs S4c alone. The live GPU was busy (ARCHITECTURE §39.2).
- Forward UDP 11011. `still open`
- C1 and C7 upstream fixes (decoder changes need coordinated activation). `still open`
- `blockchain/storefs.go` writes block and tx files non-atomically: a consensus package, reported only. `still open`
- Deferred by the R22 RPC dimension (RPCWS22-D): getwork N-miner dispatch, fee-estimation and finality RPCs, walletapi gaps.
  `closed in R23: shipped (getwork dispatcher, finality fields, GetFeeEstimate, walletapi reconnect fixes)`

OPEN AFTER ROUND 23

- Forward UDP 11011 (user action): inbound is still 0. `still open` R23b: the 14 outgoing peers are every usable known peer (17 known, 2 stuck, 1 on the v2 chain); only inbound reachability can add more.
- `--cpu-max` / `--gpu-max=auto` and `DERO_GSPLIT=8192,5,3` A/Bs on a quiet box. `still open` R23b measured proxies only (GPU staged capacity ×1.34–1.74 at 8192, ≈ 0 live while CPU-bound) and shipped the CPU-share cap; the box was contended by other agents' jobs.
- A 24 h live read of the payout-path counter `getwork_stale_height_jobs_total` / blocks: the dispatcher gain (+0.3–2 %) is an estimate until then.
  `read in R23b: 1.48 stale-height jobs per block (FOR-4), harmless` Dead time ≤ ~5–10 ms per block (estimate); the dispatcher gain stays an estimate.
- An emission-auditor derod shadow run.
  `R23b: archive mode audited 19,183 more blocks, all clean` A loopback sync A/B with the auditor on and off showed no measurable cost (SP23b-3).
- C1/C7, DVM hash gas and the SC unregistered-destination burn: upstream, coordinated activation. `still open`
- `blockchain/storefs.go` non-atomic writes (consensus package). It is also on the soundness review's open list for the double-spend concern. `still open`
- Miner uptime is the largest payout lever (≈6.3 DERO forgone 19:55–01:29): run it detached (tmux/launchd). `shipped in R23b: --supervised + tools/launchd` Measured before it: uptime 0.312 over 72 h, every run ended by a typed console exit (PAY-1).
- Range-proof soundness (C1 of the disclosure audit): the fix changes which proofs are valid, so it needs a coordinated fork. No release carries one (soundness review). `still open` R23b: no node-local, non-consensus mitigation exists (SND-5).
- The 23,886,517.3 DERO figure needs the Foundation's own forensics: no node can check it (the table holds no amounts; the amounts are encrypted). `still open` R23b HF3 forensic report: 1 of the 75 bodies exists in any local store (SND-2).

OPEN AFTER ROUND 23B

- Range-proof soundness (C1): still in the code; it needs a coordinated upstream fork, and no node-local fix exists. `still open` R23c: the per-account auditor proves only that published deltas reached the right accounts (CI-11).
- Value created inside encrypted amounts, and the 23,886,517.3 DERO figure, cannot be checked from a node. `still open`
- The global anomaly surge can be triggered on purpose, which makes honest remote clients easier to ban (the live RPC binds loopback only).
  `closed in R23c: trusted keys score ×1 during a surge (RPC-1)` An established wallet at 12

invalid calls a minute was banned after its 11th call on r23b and is never banned on r23c; a fresh key with the same pattern is still banned.

- No /24 aggregate key for remote-RPC scoring. `closed in R23c: /24 and /48 aggregate for untrusted keys (RPC-2)` 16 rotating keys got 4,800 of 4,800 probes through in 30 min before, 572 after. Keys pre-warmed to trusted still escape it (open after Round 23c).
- `blockchain/storefs.go` non-atomic writes are not addressed (consensus package). `still open` R23c shipped the read-only `tools/storecheck` detector instead (0 problems on 20,000 clone blocks).
- RPCWS23B-E: walletapi history resync after a reorg needs one sync call per reorged entry group and can throw away all history after a shallow reorg; found, not fixed. `closed in R23c: one call, valid entries kept (RPCWS23C-1)` A 2-group reorg needed 3 calls and now needs 1; an entry above the daemon tip no longer stalls the sync.
- Optional fixes not applied in the merge: SEC23b-2 order (apply the global factor only after the per-key ban check); CR23b-4 (check the error from `rand.Read` in `globals/random.go`); RPCWS23B-C (snapshot without copying the report lists). `SEC23b-2 closed in R23c (RPC-3)` a banned key drew 580 global tokens before, 0 after. `REFUTED · CR23b-4 in R23c` go1.26.6 `crypto/rand.Read` never returns an error and calls fatal() on failure, so the check would be dead code. `RPCWS23B-C closed in R23c (RPCWS23C-5)` about ×15 cheaper, 256 B instead of 9,984 B per call.
- Quiet-box A/Bs: `--phase-duty` (−1.7 % to +8.9 %, estimate), the CPU-share `--cpu-max` (+4–12 % while contended, estimate) and `DERO_GSPLIT=8192,5,3` (+5–8 % on a quiet box, estimate). Run the G20 census first. `still open` R23c: the geometry to A/B is `8192,3,2` (+2.9–3.7 %, estimate); 8192,5,3 is no better than 8192,4,2 and holds batches ×2.3 longer.
- Forward UDP 11011 (user action). Orphans are 2.7 % of regular shares; halving delivery latency is worth about +1.2 % DERO (estimate, PAY-4). `still open` R23c census: 3 of the 5 lost finals reached our chain first and lost the lower-BLID tie-break, so latency can recover at most about 1.9 % of found shares.
- A 24 h read of the idle ledger once the miner runs under launchd (ARCHITECTURE §40.2). `still open` R23c: uptime 0.2667 of 86.46 h with the miner not under launchd; 34 % of the idle hours overlapped derod downtime (PAY23c-1).
- Public copy: the public copy's history was rewritten so no commit carries peer IPs or external-implementation names (sanitised commits bc09e72, 581e21d and 5eb7251 for r21', r22 and r23). `done`

STANDING CONSTRAINTS

- No consensus-visible change is made locally. The SC zero-signer revert (R7, R11) and the range-proof defect (R8) stay as upstream behaviour; so do the C1, C2, C4, C7 and C8 decoder and format findings (R22).
- Do not bump the Go module directive past 1.23 without fixing `random_code_gen.go`: go 1.24 randseednop breaks its reproducibility (ARCHITECTURE §34.6).
- No Metal benchmarks on the live box while the miner's GPU is running (R18 incident).
- `--gpu-max` is a real cap from R22: restart the miner with `--gpu-max=96`, not 90 (ARCHITECTURE §38.6).

OPEN AFTER ROUND 23C

R23c extends Round 23 across the miner and the node. This is the list after measurement: the r23c miner was read live on 2026-11-02 01:08–01:28 (harness/runtime/r23c_live_window_20261102.txt), which closed the live A/B and the live TOTAL / staged / GPU-busy items and read the flip counters; items not yet measured (the 8192,3,2 ABBA, the --phase-duty A/B, the 24 h emission read, uptime under launchd) stay `◌ pending measurement`; the chips on the "Open after Round 23b" list above show what R23c closed. Together they seed Round 24.

- Live A/B of r23c against r23b in H per CPU-second over equal-load windows (G20 census first). The sort win is an ESTIMATE: +1–3 % CPU-bound, 0–1 % on a quiet box (verify-sortfuse); S4b was REFUSED on a contended box. Baseline r23b 09:27–09:37: 90.5k H/s at 17.53 cores, 5,164 H/cpu-s, GPU busy 93.1–95.3 %. MEASURED 2026-11-02 01:08–01:28 (n=81, load avg 50–57, semi-quiet): 5,435 H/cpu-s at 17.37 cores = +5.2 % against that baseline and +3.2 % against same-day r23b 15:14 (5,265); above the 0–1 % quiet-box estimate, so part may be load or regime variance (PLAUSIBLE magnitude). Source: harness/runtime/r23c_live_window_20261102.txt. `measured: sort win CONFIRMED`
- Live TOTAL and staged H/s, GPU busy of the 96 % cap, CPU backlog and the binding label of the r23c miner. Whether a faster sort feeds the GPU is regime-dependent: quiet box ≤ +1.6 % staged; contended windows had GPU busy 73.3–85.7 % with the sort binding (verify-handoff). MEASURED 2026-11-02 01:08–01:28: TOTAL 94,273 H/s mean (median 96,297; +4.2 % against 90.5k), staged 87,904, GPU busy 95.6 % of the 96 % cap with the duty bucket sleeping 15.2 s/min, CPU backlog 36.9 s/min: the GPU co-binds (regime law confirmed on a semi-quiet

box); 0 split restarts, 0 audit faults, 0 GPU/CPU mismatches. Sources: harness/runtime/r23c_live_window_20261102.txt, r23c_live_extract_20261102.txt.

`measured: GPU co-binds`

- Quiet-box ABBA of DERO_GSPLIT=8192,3,2 against 4096,4,2 (miner ≥ 17 cores, GPU busy ≥ 92 %, ≥ 30 min per arm; abort on any restart, retry or mismatch, or stale above 2.5 % of staged). +2.9–3.7 % ESTIMATE (CPU capacity 93.15–93.9k); stage 1 at 8192 costs ×1.197 at tries above 30, so the size of the gain needs the live run. The 2026-11-02 reading (GPU co-binds at 95.6 % of 96) makes this the next H/s lever; user-run.

  `◌ pending measurement (user-run)` `◌ R24: PAUSED (miner, §0.3)`
- E-core sort speed: the QOS_BACKGROUND run (×0.960, 6/6) probably never ran on the E-cores; a pinned-core bench is needed. `open` `◌ R24: PAUSED (miner)`
- TOTAL and staged H/s count stale_sha_skipped texts whose SHA never runs (engine.go:1093, miner.go:968): 0.30–0.42 % over-count, which also moves expected DERO and luck. Fix with a separate counter; keep hashes_total monotonic. `open` `◌ R24: PAUSED (miner)`
- P/E CPU split in the miner (proc_pid_rusage v6 → per-P-core and per-E-core seconds): E-frac swings 0.10–0.17 and moves H/cpu-s by up to 8 %. Restate the live law with the verifier's decomposition (residuals −1.8 to +2.7 % over 12 windows), not the one that counts own-hash IPC twice. `open` `◌ R24: PAUSED (miner)`
- blockchain/storefs.go atomic writes and the topo F_FULLFSYNC per block (7.1 ms builder, 4.93 ms verifier): guarded consensus package, report-only. tools/storecheck found 0 problems on 20,000 clone blocks. `report-only` `R24: still report-only (CP24-5, guarded blockchain/)`
- Spoofed-source datagrams drain the pre-TLS accept budget (about 10 datagrams/s for all unknown inbound, 1/s per /16): the fix is a return-routability cookie, a p2p wire change. No effect today (0 inbound). `report-only`

  `R24: mitigated node-locally: returning-peer lane (SEC24-4); the cookie stays report-only`
- Remote RPC item bound is per call: a batch of 256 calls × 256 txids (65,536 txids, about 0.6 s CPU) is still admitted. A per-body aggregate cap or item-weighted cost is open. `open`

  `closed in R24: 1,024 items per remote body, item-weighted cost, two bypasses closed (RPCWS24-2; base batch 5.8–6.0 s child CPU, RPCWS24-3)`
- Aggregate RPC scoring can be escaped by pre-warming: 16 keys that make 70 cheap calls over 10.5 min become trusted and pass 4,800/4,800 probes; heavy new honest clients in one /24 lose up to 13.3 % of calls for about 15 min. Only when RPC binds a non-loopback address.

  `open`

  `closed in R24: probing revokes trust; probes never earn it (SEC24-1: 4,800 → 573 probes)`
- Penalty-table bound is soft: with every ban still running the table held 68,537 records against the 65,536 bound. `open`

  `closed in R24: hard bound by insert-time eviction (SEC24-3: peak 85,536 → 65,536)`
- CI-9: a receiver payment proof discloses Keccak(sk·E), the V2 payload key, for an E the proof does not bind, so a crafted proof can decrypt another V2 transfer's payload. Payload confidentiality only; wallet-side. `report-only`

  `mitigated in R24 (wallet-side): repeated-E proofs withheld (CI24-2)`
- BURN_TX can never verify: Value is not serialised and verification refuses 0 (deflation only; a fix is a consensus change). `report-only` `R24: still report-only (consensus)`
- An SC payout that panics part-way still audits as sc_unexplained (consensus quirk); in SC-call blocks a Left-only move between accounts audits ok_sc (right_only). `report-only`

  `R24: still report-only (consensus)`
- Sync prewarm metrics: p2p_sync_pow_prewarmed_total counts overlapped blocks twice and p2p_sync_prewarm_seconds_total includes overlapped time. Live catch-up rate under r23c is unmeasured (the ×1.255–1.331 is a loopback shadow). `open`

  `closed in R24: each block counted once, overlap time on its own counter (SP24-2 / CP24-2: 17.48 → 10.00 PoW per block, shadow)`
- Miner uptime: 0.2667 of 86.46 h (≈112 DERO forgone) to 0.2735 of 87.26 h (≈102 DERO), ESTIMATE; the live miner was not under launchd or --supervised. derod has no supervisor: 21.65 of the 63.40 idle hours (34 %, ≈35 DERO) overlapped derod downtime. The measured r23c miner (2026-11-02) still ran from a terminal, not launchd, after typed console exits on 11-01 at 16:13 and 22:39: uptime under launchd is unmeasured and stays the biggest payout lever. `open (user action)`

  `R24: derod launchd generator shipped, not installed (GWO24-5); miner uptime PAUSED`
- Live read after the restart: getwork_dead_normal_jobs_total / getwork_final_flips_total and getwork_final_flip_delay_seconds (simulated L9 mean 10.36 ms, about 0.1 %); the template cache after a peer's miniblock (111 of 1,794 jobs ≥ ~1 ms old) is in guarded blockchain/miner_block.go. MEASURED 2026-11-02 01:29 over 7,163 s of r23c derod: 16 dead normal jobs, 128 final flips, flip delay count 14, sum 0.738 s (mean 52.7 ms, buckets 6.0–113.6 ms): about 5× the simulated mean on a small sample. Source: harness/runtime/r23c_live_extract_20261102.txt. `measured (above the simulation)`

  `REFUTED · R24: the 5× is a definition mismatch, not a dispatcher defect (GWO24-1); L9 counter added; live read PAUSED (miner): no miner on r24`
- 24 h emission read after install: 0 alerts and payees_explained = payees outside SC-payout blocks. PARTIAL 2026-11-02 01:28: status OK, 489 blocks audited (segment 7,844,298–

7,844,779, about 2.4 h), 0 alerts, emission_consistent true, payees 3,331/3,331 explained, 3,525 µs/block mean (max 27,276); the 24 h read stays pending.

◌ partial: 489 blocks, 0 alerts; 24 h pending

◌ R24: partial, 0 alerts; 24 h read pending

- Quiet-box A/B of --phase-duty=balanced: upper bound +8.6–8.8 % (ESTIMATE, assumes H/s proportional to CPU duty). ◌ pending measurement ◌ R24: PAUSED (miner)
- GPU-split tail after a job invalidation: 19 of 19 stale-flagged regular finds were orphaned (1.06 % of found shares); harness metrics_intel and miner_intel still use 5/(9D) (KATs IK11, IK13). open ◌ R24: PAUSED (miner)
- Forward UDP 11011 (user action): inbound 0. Latency can recover at most about 1.9 % of found shares; 3 of the 5 lost finals reached our chain first and lost the lower-BLID tie-break (PAY23c-8 corrected). open (user action) R24: still open (user action)
- Range-proof soundness C1: needs a coordinated fork; no node-local fix exists. open (upstream) R24: still open (upstream); disclosure track re-tested

OPEN AFTER ROUND 24

R24 is a node-only round (the miner track is paused, rft §0.3). This is the list after measurement: the user restarted derod onto r24 on 2026-11-02 at 15:31:03 and it was read at 15:41:24 and 16:32:01 (harness/runtime/r24_live_first_20261102.txt, r24_live_hour_20261102.txt); readings that need an event the hour did not have (a tx-carrying block, a sync, 24 h) stay ◌ pending measurement. Each item names its backlog id (B..., BD...) or finding id; the second chip on the "Open after Round 23c" list above shows what R24 did to each of those items. Sources: harness/loop/runs/r24_dimension_results_20261102.json, harness/runtime/area_proposals.json, harness/loop/disclosure_track.json.

- Restart derod onto r24 and read it: GetMonetaryIntegrity must agree with GetEmissionAudit (disclosure check S3 FAILS on the running r23c node: audit issued_matches_formula true, payees 14,550; integrity false, 0), the compact-block observation counter, the L9 / slow-flip counters and the prewarm counters (BD7, B18, B34, B39). Source: harness/runtime/disclosure_live_20261102T175638Z.txt. MEASURED 2026-11-02 on the restarted r24 node: S1–S4 PASS twice, S3 18 shared fields equal (BD7 closed; harness/runtime/disclosure_live_20261103T003206Z.txt); the compact-block and prewarm counters stay 0 because the hour had no tx-carrying block (0 txs in 217 blocks) and no sync, so B18 and B34 stay pending; the flip counters are PAUSED (no miner).

measured: S3 PASS (BD7 closed)

R25 measured 2026-11-04: B18 observed live (block_tx_roundtrip_observed_total 4 on the 4 tx blocks); B34 one synced block, 10 prewarmed PoW (n = 1)

- 24 h single-segment emission read once derod has stayed up 24 h: 0 alerts, payees_explained = payees (B36). r24 after 1 h: 219 audited, 0 alerts, payees 1,447 / 1,447 (harness/runtime/r24_live_hour_20261102.txt); the 24 h read is due from 2026-11-03T15:31. ◌ pending measurement

◌ R25: the r25 restart (2026-11-04 06:57:28) reset the segment again: the 24 h read can start 2026-11-05 06:57; 63 min read 0 alerts (rft §13 R25)

- 24 h live read of getwork_final_flip_l9_seconds and getwork_final_flip_slow_total; tools/getworkprobe can drive the dispatcher without a miner. The r23c "5×" flip delay was a conditional-vs-unconditional comparison, not a defect (GWO24-1, z = 1.60) (B39). On r24 no miner connected: 0 flips after 1 h. ◌ PAUSED (miner)

◌ R25: 75 min read on the r24 binary, 0 slow windows (PO25-2); 24 h read pending

- Template cache in guarded blockchain/miner_block.go: rebuild it when the pool's normal-miniblock count changes, about 0.04 % of hashing time (ESTIMATE, lower bound; GWO24-4). report-only (upstream) R25: still report-only, re-anchored (PO25-3)
- B19: the mempool ranks txs by integer fee/size, so every mainnet tx ranks 0 (58/58 non-registration txs in 20,000 blocks). The exact tie-break needs a reviewed guard construct or an upstream change; the optional patch makes the consensus guard FAIL (TXP-1) (B19). verified; fix not shipped (S1) ◌ R25: needs the user's guard decision (SP25-7)
- GetInfo.tx_pool_size / mempool_count still count mined txs, and p2p rebroadcast can re-announce a mined tx (TXP-2): upstream API meaning. report-only

R25: rebroadcast half closed (SP25-5); the GetInfo half stays report-only

- Pool cap is soft under concurrency (45,000 B against a 30,000 B cap with 6 admissions in flight); after a block clears the pool the cached summary can refuse a tx for up to 250 ms; heap per pooled tx 3.9–21.4 KB (TXP-5, ESTIMATE 0.3–0.6 GB at the default cap). open

closed in R25: the cap holds under concurrency and the summary follows the mined set (SP25-4); heap per tx stays ESTIMATE

- Intermittent proof failure of a wallet-built tx in the simulator (2 of ~10 rounds), probably the walletapi flake family (TXP-10, B21). open (unexplained)

R25: reproduced independent of the admission policy; root cause pending (SP25-8)

- Batch / multi-scalar proof verification for TPS is in guarded cryptography/ (ESTIMATE ~250 tx/s on 20 cores; TXP-8). report-only R25: still report-only

- B30: the return-routability cookie stays a p2p wire change; the r24 lane protects only peers with a verified handshake, and its 4,096-entry set can be filled by an operator with real handshakes (a per-netgroup quota is suggested) (B30). `mitigated; cookie report-only` `R25: netgroup quota on the proven lane (SP25-2); the cookie stays report-only`
- Heavy fresh clients in one /24 are rate-limited at 11.1 % until trusted at 10 min (DERO_RPC_REMOTE_RATE scales it); the adaptive floor (~55 unknown names/min per /24) is the same for fresh and pre-warmed keys; the proven set is in memory only (SEC24 open items). `open` `R25: a fresh remote key holds a quarter of the burst until trusted (SP25-3)`
- Read the new security metrics after the restart: rpc_abuse_trust_revoked_total, rpc_abuse_aggregate_short_total, p2p_penalty_evictions_total, p2p_inbound_proven_lane_admits_total / _refusals_total; live effect stays nil while RPC binds loopback and inbound is 0 (B1). Read on r24 after 1 h: all 0 (harness/runtime/r24_live_hour_20261102.txt). `◌ read: 0 (no remote traffic)` `◌ R25: read after the r25 restart`
- A fresh remote websocket key can still spend its 2,000-token burst on 58 full calls (~14.8k txids, ~0.26 s CPU) once per ~10 s refill (ESTIMATE; RPCWS24 residual) (B29). `open (residual)` `R25: quarter burst for fresh keys (SP25-3)`
- The wallet RPC server uses the same jhttp bridge: empty 415 and exact Content-Type match, not changed in r24 (RPCWS24-7) (B38). `open` `closed in R25: the wallet RPC parses Content-Type like derod (B40)`
- -race report in TestR19MethodsOnSimulatorChain: the test swaps the global chain while the mining-outcome resolver reads it (RPCWS24-5) (B21). `open (test)` `closed in R25: one chain-swap lock (B41)`
- B21: upstream walletapi Test_Payload_TX, Test_Creation_TX_morecheck and walletapi/xswd TestXSWDServer are red on base too; Test_R22_MultiPayload_Distinct_Blinders flakes 1/8 (the tx stays in the pool after the 2 mined blocks, PLAUSIBLE); go mutations that target a known-flaky test could count a flake as CAUGHT (B21). `partial (classified, tests open)` `R25: re-verified, still upstream-red (HT25-17)`
- CI24-3 decoy sampler: R24_DECOY_SLACK=2 shipped (modelled MAP success 7.0 %, +9.6 ms CPU per call) against 1 (10.3 %, about +4 ms, ESTIMATE); the GetRandomAddress remote weight is now 8. `open (choice)` `R25: live CPU per call 25.1–40.9 ms on the r24 node (CD25-1); the weight decision is open`
- CI24-2: an E binding third parties can check needs a proof-format change; a copycat entry's Data is still decrypted with the reused keystream; withholding is griefable for one tx fee (B32). `mitigated wallet-side; format upstream` `R25: a new residual for partial-history wallets (CD25-5)`
- Sync blocks/s gain of the prewarm herd fix is not established on a contended box (order-balanced ×1.034 builder, ×1.008 verifier, run position dominates): re-run on a quiet box with one run per bench-lock hold (CP24-3) (B34). `◌ pending measurement` `REFUTED · R25: sync blocks/s is set by the one-at-a-time block add, not the prewarm (PO25-1: CPU ×0.923)`
- Peers that advertise blocks they cannot serve are re-synced every 4 s without end (77 such lines in the last 150k live log lines): score repeated GetObject ServerErrors (syncp2p open item). `open` `closed in R25: sync hold, escalating (B47); live after the restart`
- Parallel chunk PoW verification at tip (chunk_server.go:161) is unmeasured; measure its cache-miss rate first (CP24-6). `open` `R25: still unmeasured`
- storefs atomic writes, the topo F_FULLFSYNC (~7 ms per block of Add) and the storefs directory scan per uncached Load_Block_* call: guarded blockchain/ (CP24-5) (B11, B31). `report-only` `R25: still report-only`
- derod launchd plist (tools/launchd-derod) was not run under real launchd: the user installs it after stopping the terminal derod (GWO24-5) (B35). `◌ pending (user action)` `R25: -preflight dry run shipped (PO25-4); the install is the user's step`
- Area proposals feed (harness/runtime/area_proposals.json, run 20261102T123330): 22 proposals, 0 high: 2 medium history-drift (AP-disclosure-history-drift-9c579b, AP-harness-history-drift-5a5e4f: gates@r24 recorded 1/1 and 4/4, the records now compute 2/2 and 5/5), 1 medium tests-unregistered (AP-mempool-tests-unregistered-b3af0c: the opt-in simulator tests TestR24EvictionSparesTemplateTxs, TestR24TxpoolLoad), 18 low backlog-unanchored (B18, B19, B22, B28–B36: rows with no tree anchor) and 1 low kat-unassigned (AP-harness-kat-unassigned-29adeb: test_evolution_order_r24.py). `open (triage into §7)` `R25: triaged; history drift pinned (B52, HT25-1); the new feed is in Open after Round 25`
- The disclosure area is OK with go_mutate 0/0 over cryptography/: its coverage is met by G22 alone (verify-areaintel residual). `open` `R25: unchanged (G22 covers it)`
- FX-N4: relay/mempool policy refusing proof scalars ≥ Order (txcanon): feasible, policy only, encoding hygiene, not a soundness mitigation (BD5). `feasible-not-implemented` `closed in R25: implemented (SP25-6)`
- FX-N9: GetInfo states that total_supply is CalcSupply(height), not a coin count (BD6). `feasible-not-implemented` `closed in R25: GetInfo total_supply_basis (BD6)`

- FX-F1 … FX-F4: the fork-required designs (an independent inner-product generator, binding the blinding scalar into the challenges, canonical decoding, activation mechanics): each needs an external review and a coordinated fork; none can ship from this tree (BD1–BD4). `design (upstream)` `R25: the designs are unchanged (upstream)`
- BD9: whether a non-reduced proof scalar verifies like its reduced value needs an executable test with a full statement (defensive, no proof construction); census of 100,000 blocks: 0 of 15,866 scalars ≥ Order (BD9). `◌ pending (verify)` `closed in R25: pinned by an honest-proof test (CD25-2)`
- Range-proof soundness C1 still needs a coordinated upstream fork; no node-local fix exists (B8). `open (upstream)` `R25: still open (upstream); the forensics dataset grades the table txs NOT-DECIDABLE`
- Forward UDP 11011 (user action): inbound 0 (B1). `open (user action)` `R25: still open (user action)`
- Miner backlog (8192,3,2 ABBA, --phase-duty A/B, E-core bench, stale_sha_skipped counter, P/E split, GPU-split tail, uptime under launchd): untouched while the miner track is paused; resume by setting tracks.miner to active (B3, B4, B25, B26, B27, B37, B5). `◌ PAUSED (miner)` `◌ R25: PAUSED (miner)`
- miner_runs' idle ledger books the user's deliberate pause as forgone payout (H24-5): fix when the miner track resumes. `◌ PAUSED (miner)` `◌ R25: PAUSED (miner)`
- Harness: fix_build_verify --project-only has no --root option (a clone without bin/ cannot run it); the out-of-repo export tool also skips a leading-zero IPv4 token (H24 open items). `open` `closed in R25: --root re-executes the named repo's pipeline (B49, HT25-5)`
- area_intel books a live counter of 0 as a measured value even when the counted event never happened (r24 compact_observed 0 over 217 blocks with 0 txs): a count metric needs an event denominator before a 0 is a reading (self-detected at the R24 measurement) (B54). `open` `closed in R25: a count needs its event before a 0 is a reading (B54, HT25-3)`
- CO24-N1 (B63): Per-peer offence score and last reasons are not exposed over RPC (consensus observatory, area peers). `open (R25, node-side)` `R25: implemented node-side; live after the restart`
- CO24-N2 (B64): No per-reason offence counter (invalid_pow, malformed_request, undecodable, invalid_tx, timeout) (consensus observatory, area peers). `open (R25, node-side)` `R25: implemented node-side; live after the restart`
- CO24-N3 (B65): Inbound Chain requests served are not counted (consensus observatory, area peers). `open (R25, node-side)` `R25: implemented node-side; live after the restart`
- CO24-N4 (B66): Ban list entries carry no reason or origin (consensus observatory, area peers). `open (R25, node-side)` `R25: implemented node-side; live after the restart`
- CO24-N5 (B67): Per-peer INV budget clips are node-wide only (consensus observatory, area peers). `open (R25, node-side)` `R25: implemented node-side; live after the restart`
- CO24-N6 (B68): Per-peer object-serve failures are only in the log (consensus observatory, area peers). `open (R25, node-side)` `R25: implemented node-side; live after the restart`
- CO24-N7 (B69): Stuck-peer holds are not listed (consensus observatory, area peers). `open (R25, node-side)` `R25: implemented node-side; live after the restart`

OPEN AFTER ROUND 25

R25 is node-only (the miner track is paused, rft §0.3) plus the monetary-forensics engine and the tooling track. This is the list after measurement: the user restarted derod onto r25 on 2026-11-04 at 06:57:28 and it was read until 08:00:18 (harness/runtime/r25_live_window_20261104.txt); rows whose event did not occur in the window stay pending. Each row names its rft §7 id; the forensics items come from the dataset's audit and stage reports.

- Measured 2026-11-04 (restart 06:57:28, 63 min, rft §13 R25): GetConsensusStats, the B55/B56/B58/B59/B60 counters and fields, counter-mode peer observatory, ban_reasons.json, typed not-found (B80), txs_status, total_supply_basis and B18 confirmed live; 0 ERROR/WARN, so G15 is clear. Still waiting for an event: the B47 sync hold, the B48 lane quota, B57 rejected-block attribution, B60 persistence across a restart, B61; B85 −32031 is unit-proven and never sent live on purpose. `◌ pending (no event in the window)`
- B103: insert_failed refusals (834 in 63 min) count concurrent duplicate miniblock deliveries (IsCollision before the PoW check, collision again under the lock): PLAUSIBLE by code path; split the reason and log the error text (p2p, non-consensus). `open (R26)` `R26: counted as duplicate_race, error text logged (B103); live split after the restart`
- B104: a probe with wrong GetBlockHeadersRange param names silently read topoheight 0 (−32098, no log line, self-inflicted); probes must take param names from the handler json tags; the node ignores unknown params and does not type a range below the prune point. `open (R26)` `closed in R26: unknown params -32602, below-prune ranges -32031 (B104)`
- B105: area_intel books network-behaviour KPIs (integrator-only final share, a peer 82,442 blocks behind) as node regressions; 5 high proposals closed as network observations. `open (R26)` `closed in R26: network-class metrics propose observations (B105)`

- Forensics (a2): a state tree at topo 7,415,465 from an archival source, then our own replay of the window that must end at our treehash ca11cc21... at 7,498,540; this is what would bind the archive roots (audit A10c UNRESOLVED, V-STATE-ROOTS pending) (B95, B76).
  `open (R26)`
  `◌ R26: blocked, same-IP session drop (B95, user decision); the held archive replays to our treehash (B123)`
- Forensics: blocks whose fees, burns and SC payouts are pending because their bodies are not in raw/; the replaywall archive's bodies settle part of them under check (iii) (B96). `open (R26)`
  `closed in R26 for fees and burns: 1,575 of 1,575 settled in dataset v2 (B96); SC payouts pending`
- Forensics: the proof results are single-implementation (the node verifier); a second proof verifier, and the duplicate-body census (two bodies under one TXID), are open (B97).
  `open (R26+)`
  `R26: census and a second structural verifier done; a second soundness verifier NO-GO (B97)`
- Forensics: a genesis replay is expected to stop at replay wall 1 (7,412,675, not in the HF3 table); not run (B81 corrected in memory). `◌ pending (not run)`
- Harness: a failed S5 leaves its shadow clone behind (19 GB, removed by hand after pipeline_20261103T165745) (B98). `open` `closed in R26: S5 tears its shadow down (B98)`
- Peers and security residuals from the verifiers: B47 sync-hold strikes never decay and are not cleared by notified blocks (all peers ahead held means no sync); Offenders() holds penalty_mu for 22.6–23.4 ms per GetPeerInfo on a full table (loopback only); two invalid_pow offences never ban despite the comment (policy pinned) (B99, B102). `open`
  `R26: strikes decay, forgiveness, held-peer fallback (B99); policy kept (B102)`
- RPC residuals: NameToAddress for an unknown name answers untyped −32098; GetEncryptedBalance with no topoheight reads topoheight 0, the B85 path on any pruned node; whether GetRandomAddress's topoheight−5 can fall below the prune point after a fresh fastsync is pending (B100). `open` `closed in R26: -32032 for an unknown name (B100)`
- Harness residuals: log_intel sorts by the local-time string across the DST change (latent); a clean clone cannot run --project-only because miner_intel exits 2 without a miner log while the miner is paused (B101). `open` `closed in R26: DST-safe sort, paused-miner skip (B101)`
- Scrub and determinism: pubscrub misses 3 of 5 planted secret kinds (a sentence-final IPv4 leaks with exit 0; fix the IPv4 lookahead in the scrub and export tools, outside the repo); gen_deroarch writes the build time, so its double run is never byte-exact (B93). `open`
  `closed in R26: byte-exact src-arch, B93b scrub (B93); residual B122`
- B32 residual (CD25-5): a wallet with TrackRecentBlocks > 0 keeps partial history, so the r24 V2 receiver-proof guard cannot see older entries; decide between withholding and documenting (wallet-local). `open (decision)` `R26: decided, accepted residual, pinned (B32)`
- B51: the live CPU per GetRandomAddress on the r24 node is 25.1–40.9 ms/call (verifier 1.0–2.3× the 18 ms the remote weight 8 was calibrated on); RPC binds loopback only; decide whether to raise the weight (read on the r24 binary). `open (decision)`
- B19: the exact fee-rate ranking cannot be reached without guarded consensus code; the user decides (a reviewed allowed construct, or upstream). `◌ open (user decision)`
- B43: the intermittent wallet-tx proof failure in the simulator reproduces independent of the admission policy; root cause pending (decode the 2 captured failing txs). `open`
  `closed in R26: root cause stale decoys, fixed (B43)`
- B36 and B39 (live): the 24 h single-segment emission read can start no earlier than 2026-11-05 06:57 local (the r25 restart reset the segment; 08:00: 227 audited, 0 alerts, payees 1,451/1,451); the 24 h flip-counter read needs a miner (PAUSED). `◌ pending measurement`
- B35: install derod under launchd (the user's step: -preflight must say READY with the terminal derod stopped); uptime under launchd unread. `◌ open (user action)`
- B92: the only skill, dero-miner-tuning, carries node rounds while the miner is paused (skill_fresh needs the R25 section, added this round); a node-operations skill is still a decision. `open (decision)`
  `R26: decided, split into a node-operations skill; the skill file is pending (B92)`
- Report-only, guarded consensus packages: B45 (template-cache dead windows), B31/B11, B30 (return-routability cookie, a p2p wire change), the one-at-a-time block add that sets sync blocks/s (B46). `report-only`
- Miner backlog: untouched while the miner track is paused (B2–B7, B25–B27, B37, B50).
  `◌ PAUSED (miner)`
- Range-proof soundness C1: needs a coordinated fork; no node-local fix exists; the forensics dataset grades every table tx NOT-DECIDABLE-FROM-PUBLIC-DATA for that reason (B8).
  `open (upstream)`
- Area proposals feed (harness/runtime/area_proposals.json, run 20261108T165653): 109 proposals (high 0, low 99, medium 10); high and medium: AP-consensus-network-observation-658583 top1_final_share: 0.289 (r28) -> 0.381 (r29), worse than 10% noise (lower is better) [netw; AP-consensus-network-observation-f26bbb integrator_only_final_share: 0.3253 (r28) -> 0.421 (r29), worse than 10% noise (lower is b; AP-consensus-network-observation-fcb1b5 detectors_anomaly: 2 (r28) -> 3 (r29), worse than 0% noise (lower is better) [network KPI ; AP-consensus-observatory-observed-999cd9 D13-

v2-remnant OBSERVED: Peers still on the pre-HF3 v2 remnant chain (7 connected below v3; AP-peers-network-observation-09ac26 anomaly: 2 (r28) -> 4 (r29), worse than 0% noise (lower is better) [network KPI (metric-re; AP-peers-network-observation-47e833 lag_max: 0 (r28) -> 2874039 (r29), worse than 0% noise (lower is better) [network KPI (met; AP-peers-network-observation-8976de v2_seen_7d: 1 (r28) -> 3 (r29), worse than 0% noise (lower is better) [network KPI (metric; AP-peers-network-observation-b9adb1 lag_max = 2874039 trips > 25 [network KPI (live-alert): an observation of other nodes, not; AP-peers-observatory-observed-c262e3 D3 OBSERVED: Stuck peers evicted (log) (2 finding(s)); AP-peers-observatory-observed-f4d4fe D1 OBSERVED: Peers on the v2 remnant chain (3 finding(s)) (triage into §7, B52). `open (triage)`

OPEN AFTER ROUND 26

R26 is node-only (the miner track is paused, rft §0.3) plus forensics dataset v2 by defect class. This is the list after implementation: +22102026.r26 is installed and the running derod is still r25 until the user restarts it, so every live row is pending. Each row names its rft §7 id; the second chip on the “Open after Round 25” list above shows what R26 did to each of those items.

- Measured 2026-11-05 (restart 07:48:21, 62 min, rft §13 R26): the B103 duplicate_race split, −32032 (B100), −32602 (B104) and the reorg monitor across a restart (B60) confirmed live; 0 ERROR/WARN, so G15 is clear. Still waiting for an event: B99 strike decay and forgiveness (0 holds), B47, B48, B57, B61. `◌ pending (no event in the window)`
- Forensics (a2) state at topo 7,415,465: decided 2026-11-05, not pursued: the HF3-path archive replay already ends at our treehash (B123); wall 1 has no held archive; reopen only if one appears (B95). `decided (not pursued)`
- B132–B134 (R26 measurement): the −32602 message doubles its code and names no field; the project-only wall time 512 → 1,901 s; area_intel books recorded forensics audit findings as regressions. `open (R27)`

  `R27: B132 fixed (the field is named), B133 partly fixed (frontend2 KAT 852.83 → 244.67 s), B134 fixed (record class)`
- Forensics: SC payouts of the window stay pending; FQ2/FQ3 window-wide randomness census re-run on the 1,796 RAW v2 bodies; the on-chain boundary height (B110). `open`
- Forensics: the duplicate-body census compares 0 of 67 multi-block TXIDs because each is held from one independent holder; a second holder is needed; a second soundness verifier stays NO-GO (B97). `open`
- Forensics: v2 records carry class pending, so G32's record half is vacuous (B126); the balance-tree census at 7,498,540 needs a stopped-node clone (B111). `open`
- Tooling: 33 harness tools without an argument parser (B120); the PRIVATE history keeps raw argv in snap_*.json and the LAN address in shadow_*.json (B121, decided 2026-11-05: no rewrite; a sync-time check refuses those paths, 109/109 excluded); pubscrub machine-model and tool-path pass (B131, fixed 2026-11-05: control FN 6 → 0); box-model residual on the migration report (B122). `open` `R27: B120 fixed (--help census 33 → 0); B122 still open`
- Carried: B19 guard decision (user), B51 weight decision and live read, B35 launchd install (user), B36 24 h emission (read 2026-11-05: 4,846 blocks, 0 alerts; payee counts not readable), report-only B11/B30/B31/B45/B46, C1 upstream (B8), the paused miner rows. `◌ carried`

  `R27: B51 live CPU per call read on r26 (22.7 / 22.5 ms); the slack decision stays open`

OPEN AFTER ROUND 27

R27 is node-only (the miner track is paused, rft §0.3), an area-enhancement round over seven groups. This is the list after implementation: +22102026.r27 is installed and the running derod is still r26 until the user restarts it, so every live row is pending. Each row names its rft §7 id; the second chip on the “Open after Round 26” list above shows what R27 did to each of those items.

- Measured 2026-11-05 (restart 17:51:48, 60 min, rft §13 R27): the −32602 field names (B132), the pre-registered rpc series and rpc_error_responses_total (B143), the reloaded ban as p2p_bans_active / p2p_bans_loaded_total / dero_p2p_ban_list_entries (B142) and config_consensus_env_overrides 0 (B141) confirmed live, payee counts in the r27 series checkpoints (B36); 0 ERROR/WARN. Self-detected B149: r27 was missing from the area registry rounds, added at measurement. `measured 2026-11-05`
- Held for the next dataset version: the globals ParseAmount, FormatMoneyPrecision and GetOffset fixes (B144, B147; harness/loop/held/r27_globals_amount_offset.patch); the 64-bit FormatMoneyPrecision copies in the wallet stay unchanged. `held`

  `R28: B147 landed at the merge (harness/loop/held/landed/); B144 partly fixed (the rpc copy needs a guard decision, the walletapi copy is R29 work)`
- DVM behaviours pinned and documented, never changed (consensus-visible, report-only, B145); blockchain/ coverage 54.0 → 55.4–55.5 % with process_transaction_sc at 0.0 %

(B146). open

R28: B146 closed (process_transaction_sc 96.4 %); B145 report-only (pinned, never changed)

- Privacy residual: the raw /metrics scrapes carry peer labels and stay in the scratchpad (B135 partly fixed); the emission-audit -archive payee line is not under test (B139 residual); memory dead references (B140). open

  R28: B135 and B139 closed; B140 closed at Part D (memory dead references 6 → 0)
- B133: the project-only wall time on r27 is 1,377 s (closing run pipeline_20261105T163040; 1,901 s on r26), still above R25's 512 s. partly fixed

  R28: B133 closed (measured and attributed)
- Carried: B99 decay and forgiveness, B47, B48, B57, B61 (no event yet); B19 guard decision (user), B51 slack decision (live CPU per call read in R27 on r26: 22.7 / 22.5 ms), B35 launchd install (user), report-only B11/B30/B31/B45/B46, C1 upstream (B8); forensics B97, B110, B111, B126; B122; the paused miner rows. ◌ carried

  R28: B47, B48, B57, B99, B61, B51, B126 closed; B8, B11, B30, B31, B45 report-only, mitigated or upstream; B46 closed, its open half is L24-21; B19, B35, B111 user decisions; B97, B110 open; the miner rows paused

CLOSED IN ROUND 28

R28 is the closure round: a fresh-context audit read every open item of Rounds 1 to 27 (the “Open after Round 21” to “Open after Round 27” lists above and rft §7) and the rows the R28 merge created: 156 unique items, 40 closed before R28, 82 closed in R28, 34 still open, each open one with its blocker and closing observable. Part D closed two of the open ones with records (B140, B156) and the closed loop and Part D found and fixed four new ones (B171, B172, B173, B174), so this list has 88 items and the next one holds the 32 still open. An item is closed only with a record path; CLOSED here also covers REPORT-ONLY, REFUTED, DECIDED and UPSTREAM verdicts that need no further action from this project.

ITEM	WHAT	R28 VERDICT	RECORD
B8	Range-proof soundness C1 needs a coordinated upstream fork.	REPORT-ONLY	artifacts/dero-rft.md §7 B8; harness/runtime/r28_closure_audit_20261106.txt
B9	The 23,886,517.3 DERO figure, and value inside encrypted amounts, are not node-checkable (1 of the 75 HF3 bodies is in any local store, SND-2).	CLOSED	artifacts/dero-rft.md §7 B9; harness/runtime/r28_closure_audit_20261106.txt
B10	C7, DVM unmetered hash gas (sha3/keccak of under 1 MiB costs a flat 25k gas), SC unregistered-destination burn: consensus pricing / coordinated activation.	REPORT-ONLY	artifacts/dero-rft.md §7 B10; harness/runtime/r28_closure_audit_20261106.txt
B11	blockchain/storefs.go non-atomic writes (consensus package): report only.	REPORT-ONLY	artifacts/dero-rft.md §7 B11; harness/runtime/r28_closure_audit_20261106.txt
B30	Spoofed-source datagrams drain the pre-TLS accept budget (~10/s all unknown inbound, 1/s per /16)	MITIGATED	artifacts/dero-rft.md §7 B30; harness/runtime/r28_closure_audit_20261106.txt
B31	Topo F_FULLFSYNC per block (7.1 ms builder, 4.93 ms verifier), guarded blockchain/storetopo.go:99	REPORT-ONLY	artifacts/dero-rft.md §7 B31; harness/runtime/r28_closure_audit_20261106.txt
B33	BURN_TX never verifies (Value not serialised; 0 refused)	REPORT-ONLY	artifacts/dero-rft.md §7 B33; harness/runtime/r28_closure_audit_20261106.txt
B45	GWO24-4: dead getwork windows come from the 100 ms template cache (blockchain/miner_block.go:397)	REPORT-ONLY	artifacts/dero-rft.md §7 B45; harness/runtime/r28_closure_audit_20261106.txt
B145	DVM behaviour pinned by R27 KATs (consensus-visible; documented, never changed): EXISTS/LOAD after DELETE of the same key in ONE invocation fails the call ('invalid, ...)	REPORT-ONLY	artifacts/dero-rft.md §7 B145; harness/runtime/r28_closure_audit_20261106.txt
BD1	FX-F1 (C1/C2): an independent inner-product generator (own hash-to-point label, scaled by the challenge) instead of u_x = H^o	REPORT-ONLY	artifacts/dero-rft.md §7 BD1; harness/runtime/r28_closure_audit_20261106.txt
BD2	FX-F2 (C1): bind the blinding scalar (with t-hat, s_tau) into the preimage of the inner-product challenges, compare final points canonically.	REPORT-ONLY	artifacts/dero-rft.md §7 BD2; harness/runtime/r28_closure_audit_20261106.txt

ITEM	WHAT	R28 VERDICT	RECORD
BD3	FX-F3 (C3): canonical decode of proof scalars (< Order) and points (R22 x >= P, flag bytes).	REPORT-ONLY	artifacts/dero-rft.md §7 BD3; harness/runtime/r28_closure_audit_20261106.txt
BD4	FX-F4: activation mechanics for any proof change (no proof-version field, tx Version must be 1; height-keyed like the HF3 cache key).	REPORT-ONLY	artifacts/dero-rft.md §7 BD4; harness/runtime/r28_closure_audit_20261106.txt
B50	H24-5: the miner_runs idle ledger books the user's deliberate pause as forgone payout (75.1 h idle, uptime 0.285, 127.98 DERO ESTIMATE)	FIXED	artifacts/dero-rft.md §7 B50; harness/runtime/r28_closure_audit_20261106.txt
B39	Flip delay live mean 52.7 ms (n=14) against the simulated 10.36 ms	REFUTED	artifacts/dero-rft.md §7 B39; harness/runtime/r28_closure_audit_20261106.txt
B141	Config: a consensus parameter is environment-settable.	MITIGATED	artifacts/dero-rft.md §7 B141; harness/runtime/r28_closure_audit_20261106.txt
B47	A peer that advertises blocks it cannot serve is re-synced every 4 s with no end (77 such lines in the last 150k derod log lines)	CLOSED	artifacts/dero-rft.md §7 B47; harness/runtime/r28_closure_audit_20261106.txt
B48	Security residuals: the proven-peer lane (4,096 entries) can be filled by an operator with 4,096 real handshakes (per-netgroup quota suggested)	CLOSED	artifacts/dero-rft.md §7 B48; harness/runtime/r28_closure_audit_20261106.txt
B99	Peers: B47 sync-hold strikes never decay and are not cleared by notified blocks (all peers ahead held = no sync)	CLOSED	artifacts/dero-rft.md §7 B99; harness/runtime/r28_closure_audit_20261106.txt
B57	CO24-C3: Name the sending peer (pseudonymous id) on block rejection lines and add blockchain_block_rejected_total{reason}: a rejected block cannot be attributed today.	CLOSED	artifacts/dero-rft.md §7 B57; harness/runtime/r28_closure_audit_20261106.txt
B61	CO24-C7: Resolve header miners[] against the block's topoheight state, not Topo_store.Read(height) (correct only while topoheight == height).	CLOSED	artifacts/dero-rft.md §7 B61; harness/runtime/r28_closure_audit_20261106.txt
B43	TXP-10: intermittent proof failure of a wallet-built tx in the simulator (2 of ~10 rounds)	CLOSED	artifacts/dero-rft.md §7 B43; harness/runtime/r28_closure_audit_20261106.txt
B44	Pool cap residuals: soft under concurrency (45,000 B against a 30,000 B cap with 6 admissions in flight, verifier)	FIXED	artifacts/dero-rft.md §7 B44; harness/runtime/r28_closure_audit_20261106.txt
B85	Self-caused by the R25 discovery probe, a node bug (§6 step 1a class): GetEncryptedBalance at a topoheight below the prune point panics on purpose (panic(err) after ...)	FIXED	artifacts/dero-rft.md §7 B85; harness/runtime/r28_closure_audit_20261106.txt
B55	CO24-C1: Count and log miniblock refusals by reason in NotifyMiniBlock (stale height, unregistered miner, unexpandable tip, final propagated): today they return an error ...	CLOSED	artifacts/dero-rft.md §7 B55; harness/runtime/r28_closure_audit_20261106.txt
B56	CO24-C2: Expose a block's fees in BlockHeader_Print / GetBlockHeadersRange: fees per block are unmeasurable without decoding every tx (observatory fees_total = pending).	CLOSED	artifacts/dero-rft.md §7 B56; harness/runtime/r28_closure_audit_20261106.txt
B58	CO24-C4: Assign GetInfo alt_blocks_count (declared, never set): it reads 0 while the log shows 99 side-block lines in the window.	CLOSED	artifacts/dero-rft.md §7 B58; harness/runtime/r28_closure_audit_20261106.txt
B59	CO24-C5: Record each block's (and miniblock's) first-seen time and expose it by RPC: arrival lag (timestamp skew against this clock) is measurable only from the DEBUG ...	CLOSED	artifacts/dero-rft.md §7 B59; harness/runtime/r28_closure_audit_20261106.txt
B60	CO24-C6: Persist the reorg monitor across restarts and log each reorganisation at INFO (counts reset at every start).	CLOSED	artifacts/dero-rft.md §7 B60; harness/runtime/r28_closure_audit_20261106.txt

ITEM	WHAT	R28 VERDICT	RECORD
B65	CO24-N3: Inbound Chain requests served are not counted: chain-request amplification cannot be measured: no counter or log of served Chain requests and their sizes	CLOSED	artifacts/dero-rft.md §7 B65; harness/runtime/r28_closure_audit_20261106.txt
B66	CO24-N4: Ban list entries carry no reason or origin: ban_list.json maps address -> expiry only	CLOSED	artifacts/dero-rft.md §7 B66; harness/runtime/r28_closure_audit_20261106.txt
B68	CO24-N6: Per-peer object-serve failures are only in the log: peer_intel reads 'could not serve' and 'answered N blocks' from derod.log	CLOSED	artifacts/dero-rft.md §7 B68; harness/runtime/r28_closure_audit_20261106.txt
B69	CO24-N7: Stuck-peer holds are not listed: p2p_stuck_peer_held is a count	CLOSED	artifacts/dero-rft.md §7 B69; harness/runtime/r28_closure_audit_20261106.txt
BD6	FX-N9: GetInfo states that total_supply is CalcSupply(height) (additive field or doc), not a coin count.	CLOSED	artifacts/dero-rft.md §7 BD6; harness/runtime/r28_closure_audit_20261106.txt
B108	Bound honesty gate G31 + KAT test_forensics_bounds_r26.py (lower-bound labelled total, DC2 as 0, cross-class sum, fix status rewriting a historical grade, forbidden ...)	CLOSED	artifacts/dero-rft.md §7 B108; harness/runtime/r28_closure_audit_20261106.txt
B112	Q-CS2/Q-CS3: the missing-invariant certificate from source (balance encoding, homomorphic update, what an aggregate opening needs), refute-by-default audit	ANSWERED	artifacts/dero-rft.md §7 B112; harness/runtime/r28_closure_audit_20261106.txt
B113	FQ4/FQ5: DC2 presence at head / Release152 / genesis-era tag and the structural transcript-binding KAT (no accepted-invalid proof, §9)	ANSWERED	artifacts/dero-rft.md §7 B113; harness/runtime/r28_closure_audit_20261106.txt
B114	Pages: soundness "What can be known" section, ledger #fx-classes / #fx-presence / #fx-invariant and the verdict board by class, src-arch section 10 "Defect classes" line	CLOSED	artifacts/dero-rft.md §7 B114; harness/runtime/r28_closure_audit_20261106.txt
B92	Skills: the only skill, dero-miner-tuning (1,024 lines), carries node rounds while the miner is PAUSED.	CLOSED	artifacts/dero-rft.md §7 B92; harness/runtime/r28_closure_audit_20261106.txt
B95	Forensics (a2): a state tree at topo 7,415,465 from an archival source via TreeSection, then our own replay of 7,415,466..7,498,540 that must end at our treehash ...	CLOSED-OBSOLETE	artifacts/dero-rft.md §7 B95; harness/runtime/r28_closure_audit_20261106.txt
B32	CI-9: a receiver payment proof discloses Keccak(sk·E), the V2 payload key, for an unbound E: a crafted proof decrypts another V2 payload (payload privacy only).	CLOSED	artifacts/dero-rft.md §7 B32; harness/runtime/r28_closure_audit_20261106.txt
B21	Upstream walletapi tests are red on pristine Release153 too (Test_Payload_TX, Test_Creation_TX_morecheck).	CLOSED	artifacts/dero-rft.md §7 B21; harness/runtime/r28_closure_audit_20261106.txt
B42	TXP-2 residuals: GetInfo tx_pool_size / mempool_count still count mined txs until HouseKeeping (upstream API meaning)	FIXED	artifacts/dero-rft.md §7 B42; harness/runtime/r28_closure_audit_20261106.txt
B51	CI24-3 decoy sampling: R24_DECOY_SLACK=2 shipped (modelled MAP success 7.0 %, +9.6 ms CPU per call) against 1 (10.3 %, ~+4 ms, ESTIMATE)	CLOSED	artifacts/dero-rft.md §7 B51; harness/runtime/r28_closure_audit_20261106.txt
B53	Coherence drifts found by harness/intel/coherence.py at R24 (G25): see §12 R24 entry	CLOSED	artifacts/dero-rft.md §7 B53; harness/runtime/r28_closure_audit_20261106.txt
B109	Class separation and presence gate G32 + KAT test_forensics_classes_r26.py	CLOSED	artifacts/dero-rft.md §7 B109; harness/runtime/r28_closure_audit_20261106.txt
B124	Forensics tooling (FD-5, self-detected by forensics-data): writing v2 blobs into v1's tier B store broke v1's full replay, because harness/forensics/replay.py ...	CLOSED	artifacts/dero-rft.md §7 B124; harness/runtime/r28_closure_audit_20261106.txt

ITEM	WHAT	R28 VERDICT	RECORD
B126	Merge-detected interface defect: classes.py tag_dataset and G32's record check read the v1 record layout (derived/tx/*.json, derived/block/*.json, window.jsonl.gz)	CLOSED	artifacts/dero-rft.md §7 B126; harness/runtime/r28_closure_audit_20261106.txt
B122	Publish scrub residual: pubscrub reports box-model 1 on artifacts/release151-migration-report.html with both the previous and the R26 scrubber (pre-existing; the other ...)	CLOSED	artifacts/dero-rft.md §7 B122; harness/runtime/r28_closure_audit_20261106.txt
B133	Tooling regression flagged by area_intel (R26 measurement, proposal high): tooling.project_wall_s 512 s (r25) → 1,901 s (r26), the project-only pipeline wall time	CLOSED	artifacts/dero-rft.md §7 B133; harness/runtime/r28_closure_audit_20261106.txt
B135	Privacy: derod /metrics exports p2p_first_relay_total{peer="<IPv4>"}	CLOSED	artifacts/dero-rft.md §7 B135; harness/runtime/r28_closure_audit_20261106.txt
B139	Tools (R27 Part A sweep): tools/emission-audit had 0 tests and its -archive mode printed no payees	CLOSED	artifacts/dero-rft.md §7 B139; harness/runtime/r28_closure_audit_20261106.txt
B146	blockchain/ coverage record (first -coverpkg figure): the union over the 9 test packages that reach blockchain/ is 54.0 % (1,369/2,533) at cee1d80f and 55.4–55.5 % ...	CLOSED	artifacts/dero-rft.md §7 B146; harness/runtime/r28_closure_audit_20261106.txt
B147	Merge-detected (R27 merge, harness/runtime/r27_merge_20261105.txt): (1) glue-global changed globals/globals.go (ParseAmount exact, FormatMoneyPrecision 128-bit, ...)	CLOSED	artifacts/dero-rft.md §7 B147; harness/runtime/r28_closure_audit_20261106.txt
B148	Resource load of the project's own workflow (2026-11-05; proposed by r27b as B135, renumbered at integration).	MITIGATED	artifacts/dero-rft.md §7 B148; harness/runtime/r28_closure_audit_20261106.txt
B149	Self-detected (R27 measurement, 2026-11-05): R27 Part D set harness/round.json r27 but did not add r27 to harness/areas.json rounds (R26 Part D had added r26, 935c8fe7), ...	CLOSED	artifacts/dero-rft.md §7 B149; harness/runtime/r28_closure_audit_20261106.txt
L24-9	Batch / multi-scalar proof verification for TPS (TXP-8, ESTIMATE ~250 tx/s on 20 cores)	REPORT-ONLY	harness/runtime/r28_glue_global_20261106.txt §consensus guard (identity hook S1 FAIL, reverted); harness/runtime/r28_closure_audit_20261106.txt
L24-25	disclosure area go_mutate 0/0 over cryptography/: coverage met by G22 alone	CLOSED	cryptography/crypto/r28_verify_guards_test.go; harness/intel/go_mutations.json; harness/runtime/r28_closure_audit_20261106.txt
L25-8	Genesis replay expected to stop at replay wall 1 (7,412,675, not in the HF3 table)	CLOSED	forensics/r28/derived/walls.json wall1 rows; rft §0.8 "Replay walls vs the table"; harness/runtime/r28_closure_audit_20261106.txt
L25-24	Area proposals feed: triage every high/medium proposal into §7 or close it	CLOSED	harness/areas.json tooling.checkpoint_age better none + alert_if; KAT test_harness_measure_r28.py HS3; harness/runtime/r28_closure_audit_20261106.txt
L24-24	Area proposals feed of run 20261102T123330	CLOSED	artifacts/dero-rft.md §7 B52; harness/runtime/r28_closure_audit_20261106.txt
N28-01	HT11 mutation is round-anchored (the literal ORDER list): it goes stale at every round open and had to be re-anchored (R29) at R28 Part A (B149 class)	CLOSED	harness/kat/test_tools_kats_r28.py TK18; harness/runtime/r28_closure_audit_20261106.txt
N28-02	area history drift: kats@r27 recorded '25/25' but the records now compute '26/26'	CLOSED	harness/areas.json rounds pinned_why; KAT test_harness_measure_r28.py HS4; harness/runtime/r28_closure_audit_20261106.txt

ITEM	WHAT	R28 VERDICT	RECORD
N28-03	getworkops: no registered metric measured since r24 although the getwork counters are on /metrics at 0 (miner paused)	CLOSED	harness/intel/area_intel.py staleness/stale_exempt; KAT HS2; harness/runtime/r28_closure_audit_20261106.txt
N28-04	17 of 78 harness tools have no KAT (kat_coverage_pct 78.2)	CLOSED	harness/runtime/tooling_20261106T012838.json tools_with_kat 78; harness/runtime/r28_closure_audit_20261106.txt
N28-05	round_checkpoint finds derod with pgrep -f (a pattern match, §5 asks for pid-based waits; can match a foreign process)	CLOSED	harness/kat/test_tools_kats_r28.py TK19; harness/runtime/r28_closure_audit_20261106.txt
N28-06	memory-mirror box-proc check is case-sensitive and misses a lower-case process name (verifier advisory)	CLOSED	memory-mirror/MANIFEST.sha256; KAT test_artifacts_memory_r28.py AM8; harness/runtime/r28_closure_audit_20261106.txt
N28-07	wasm wallet over the nhooyr adapter with a > 2 MiB reply untested	CLOSED	glue/rwc/r28_wasm_outer_test.go TestR28WasmWalletTransportLargeReply; harness/runtime/r28_closure_audit_20261106.txt
N28-08	pins.py --help pinned until the next dataset version (content-pinned tool)	CLOSED	harness/runtime/r28_closure_audit_20261106.txt (N28-08: tools_without_help [] and tools_without_help_pinned [] on the tree; pins.py --help rc 0); harness/runtime/r28_closure_audit_20261106.txt
N28-09	generator determinism read is 'pending' in the default tooling_intel run (needs --determinism)	CLOSED	harness/runtime/tooling_20261106T012838.json determinism 3/3; harness/runtime/r28_closure_audit_20261106.txt
N28-11	--last 28 is the default from R28 Part A (B149) but renders no R28 content until Part D (0 "Round 28" strings)	CLOSED	harness/kat/test_artifacts_memory_r28.py AM13; harness/runtime/r28_closure_audit_20261106.txt
N28-12	r26 dataset still unsealed: seal order requires B126 tagging first (§6 step 3a), so R28 Part A could not seal it	CLOSED	forensics/ledger.json r26 sealed manifest 2fd3061c; harness/runtime/r28_closure_audit_20261106.txt
B150	R28 rpc-metrics: rpc_error_responses_total{code} missed every JSON-RPC error body written outside jrpc2: 405/415 −32600, the abuse table's 429 −32029/−32030, the …	FIXED	artifacts/dero-rft.md §7 B150; harness/runtime/r28_closure_audit_20261106.txt
B151	R28 rpc-metrics: /metrics on r27 carried HELP for 2 of the 278 families on the page (298 source-declared) and 33 TYPE lines (30 KCP_*, 3 dero_*), so readers depended on …	FIXED	artifacts/dero-rft.md §7 B151; harness/runtime/r28_closure_audit_20261106.txt
B153	R28 rpc-metrics: Dump_metrics_data_directly (METRICS_SERVER push) panicked on a non-string name (unchecked assertion), did not query-escape its job/instance labels and …	FIXED	artifacts/dero-rft.md §7 B153; harness/runtime/r28_closure_audit_20261106.txt
B154	R28 rpc-metrics, report-only: rpc.Argument.UnmarshalJSON answers 'unknown data type unknown data type' (it prints the type through DataType.String, so the letter is lost)	REPORT-ONLY	artifacts/dero-rft.md §7 B154; harness/runtime/r28_closure_audit_20261106.txt
B157	SC burn conservation by path (R28 dvm-blockchain DB28-1/DB28-2): the burn is REFUNDED on a failing call, a missing SCID or entrypoint, empty or unparseable code, an …	REPORT-ONLY	artifacts/dero-rft.md §7 B157; harness/runtime/r28_closure_audit_20261106.txt
B162	Self-detected (R28 tools-kats; alias N28-T1): corpus/tags.tsv is git-ignored.	FIXED	artifacts/dero-rft.md §7 B162; harness/runtime/r28_closure_audit_20261106.txt

ITEM	WHAT	R28 VERDICT	RECORD
B163	Self-detected (R28 tools-kats; alias N28-T2): mine_coverage spent 110 of 125 s in 201 recursive greps (cProfile)	FIXED	artifacts/dero-rft.md §7 B163; harness/runtime/r28_closure_audit_20261106.txt
B164	Self-detected (R28 tools-kats; alias N28-T3): mine_evolution counted the planned round R29 (no commit yet) as an item 'a miner could not anchor' (evolution.json ...)	FIXED	artifacts/dero-rft.md §7 B164; harness/runtime/r28_closure_audit_20261106.txt
B165	Merge-detected (R28 merge): (1) forensics-closure landed the r27 held globals patch while glue-global held its superset r28 landing patch with the switch off: the merged ...	FIXED	artifacts/dero-rft.md §7 B165; harness/runtime/r28_closure_audit_20261106.txt
B167	Merge hazard (R28 forensics-closure FC28-8, V-2, V-3): dataset v3 forensics/r28 content-pins every non-test decoder input, so the merged non-test changes to ...	FIXED	artifacts/dero-rft.md §7 B167; harness/runtime/r28_closure_audit_20261106.txt
B140	Memory dead references (the R27 tooling_intel memory_dead_refs KPI)	CLOSED (Part D)	harness/runtime/tooling_20261106T110928.json (memory_dead_refs 0, memory_dead_names 0, mirror_drift 0, every rule-2 target met); rft §7 B140
B156	Publish copies kept '~/' home forms	CLOSED (Part D)	harness/runtime/tooling_20261106T050055.json (scrub_page_home_forms {} and residual {} on 15/15 pages, control 12 plants FN 0 FP 0); rft §7 B156
B171	heavy_preflight passed stage= to event(stage, ...): the full pipeline stopped at S3b	FIXED (R28 closed loop)	harness/loop/runs/pipeline_20261106T045640.jsonl; harness/kat/test_resource_hygiene_r27.py RH9 (38/38); rft §7 B171
B172	tooling_intel REF_RX read 'bin/derod.round5-backup' as a dead 'bin/derod.round5'	FIXED (R28 Part D)	harness/kat/test_artifacts_memory_r27.py AM4 (23/23); rft §7 B172
B173	Two globals.ParseAmount Go mutations went wrong at the merge's B147 landing (one named a test without the input, one became an equivalent mutant)	FIXED (R28 closed loop)	harness/runtime/r28_closedloop_20261106.txt; harness/intel/go_mutations.json; rft §7 B173
B174	coherence.py and backlog_truth H4 disagreed on which round's prompt §1 holds after an implementation	FIXED (R28 Part D)	harness/kat/test_backlog_truth_r28.py BT8 (24/24); rft §7 B174

OPEN AFTER ROUND 28

R28 is node-only (the miner track is paused, rft §0.3). This is the list after implementation: +22102026.r28 is installed and the running derod is still r27 until the user restarts it, and the B36 24 h read of the r27 segment comes first (from 2026-11-06 17:51:52). Every row is an item the closure audit left open, with its blocker and the observable that closes it; the chip names the blocker class.

- B1: Forward UDP 11011 (user action). Inbound has been 0 since Round 2. Blocker: user step: forward UDP (not TCP) 11011 on the router, or choose (b)/(c) of the decision request. Closes when: GetInfo incoming > 0 (GetSyncInfo inbound_since_start > 0) on 127.0.0.1:10102. `user-decision`

  `R29: DONE by the user (decision (a)): UDP 11011 forwarded; incoming 4 → 27 in the R29 window`
- B19: Mempool FeesPerByte is integer fee/size, so almost every tx ranks 0. Blocker: user decision (a) reviewed guard ALLOWED construct, (b) hook outside blockchain/, (c) upstream. Closes when: with (a)/(b): TestR28FeeRankIsIntegerPerByte inverted (a 2,098-byte tx with a higher fee ranks above 0) and guard S1 PASS. `user-decision`

  `R29: FIXED (decision (a)): the pool ranks by milli-fee per byte (guard construct mempool-milli-fee-rank)`
- B35: derod has no supervisor: 21.65 of 63.40 idle miner hours (34 %, ≈35 DERO, ESTIMATE) overlapped derod downtime. Blocker: user step: stop the terminal derod (pid 77276) and install the tools/launchd-derod plist, or accept unsupervised downtime (option b). Closes when: tools/launchd-derod -preflight READY and launchctl shows the derod job with derod running under it. `user-decision`

  `R29: CLOSED by the user's decision (b): keep the terminal derod`
- B36: the 24 h emission read with payees. R28 measurement: the user restarted before the r27 segment reached 24 h, so it was read offline over 22.3 h (4,342 blocks ok, 0 alerts, issued = formula 133,516,500, payees 28,461 = explained). Blocker: the live 24 h EQUAL read needs a segment of 86,400 s; the r28 segment reaches it at 2026-11-07 16:10:27. Closes when: python3 -I harness/intel/b36_payee_read.py reports CONFIRMED (EQUAL, span >= 86,400 s,

0 alerts) before any restart; the flip counters stay PAUSED (miner). `partial (22.3 h offline)` `R28d: CONFIRMED (24 h payee read on the r28 segment, 2026-11-07)`

- B63: CO24-N1: Per-peer offence score and last reasons are not exposed over RPC: GetPeerInfo (p2p/peer_info.go PeerInfo) has no penalty score field. Blocker: no scored offence has happened on the live node (fields present on 14/14 peers, all 0). Closes when: GetPeerInfo penalty_score / penalty_events > 0 or an offenders entry on a live read. `◌ pending-event`
- B64: CO24-N2: No per-reason offence counter (invalid_pow, malformed_request, undecodable, invalid_tx, timeout): /metrics counts bans only. Blocker: no counted offence on the live node (7 reasons pre-registered at 0). Closes when: p2p_penalty_events_total{reason} > 0 on live /metrics. `◌ pending-event`
- B67: CO24-N5: Per-peer INV budget clips are node-wide only: a flood is visible as one counter, not attributable to a connection. Blocker: no per-peer INV clip on the live node (sum 0 over 14 peers). Closes when: PeerInfo inv_txids_clipped > 0 for a peer on a live read. `◌ pending-event`
- BD5: FX-N4: relay/mempool policy refusing proof scalars >= Order (txcanon, like the R23b registration check). Blocker: no non-canonical proof scalar seen on mainnet (0 of 15,422 in 100,000 blocks). Closes when: a txcanon admission refusal counted live (scalar >= Order) or the census finding one. `◌ pending-event`
- B76: State-root chain of the window: roots are not consensus-committed. Blocker: topos 7,415,466..7,453,289 have no held state (B95 decided not pursued). Closes when: a held archival state at or below topo 7,415,466 whose replay ends at treehash ca11cc21 at 7,498,540. `data-unavailable` `R29: unchanged (data-unavailable)`
- B97: Forensics: proof results are single-implementation (the node verifier, audit A11). Blocker: a second independent holder of the 67 multi-block TXID bodies is needed (new acquisition). Closes when: audit2 D2-11 with_2plus_sources > 0 in forensics/r<N>/audit/audit_report.json. `data-unavailable` `R29: residual CLOSED: 67/67 multi-block TXIDs from 2+ sources`
- B107: Forensics classes (§0.8.9): tag every v2 row with DC1 / DC2 / DC3 / DC0. Blocker: derived/walls.json 33 rows + replay_stops and derived/inclusions.json 67 rows still read class pending. Closes when: G32 / D2-20 read walls.json and inclusions.json with 0 rows of class pending. `r29-work` `R29: CLOSED: walls and inclusions tagged (G32 0 pending)`
- B110: FQ2/FQ3: DC1 amount observability and the randomness-reuse boundary. Blocker: SC payouts: 110 of 199 SC txs need an a2 SC-payout exporter (R29 work), 89 have no held state. Closes when: forensics/r<N>/derived/fq.json SC payouts graded for the 110 run-C txs. `data-unavailable` `R29: run C SC payouts exact; 89 txs below 7,467,516 stay data-unavailable`
- B111: Q-CS1: balance-tree enumeration at our anchor on a clone (never the live store; needs a stopped-node clone or a held archive) — accounts, ... Blocker: Option A needs the user to stop derod for an APFS clone of the datadir (never done by an agent, rft §5). Closes when: classstats census at 7,498,540 on that clone, or the user keeps ANSWERED at held-archive scope (option B). `user-decision` `R29: ANSWERED at 7,498,540 on a stopped-node clone (decision (A)); 1 key unattributed`
- B144: glue-global (R27): globals.ParseAmount rounded digits past the 5th decimal silently (0.000009 DERO → 1 atomic unit, 0.000005 → 0) and lost the last ... Blocker: rpc/wallet_rpc.go copy: rpc/ is in consensus_guard CONSENSUS (S1 FAIL), needs a user decision on a guard ALLOWED entry. Closes when: harness/loop/held/r28_rpc_money_copy.patch applied with S1 PASS and TestR28WalletFormatMoney 20,000/20,000. `user-decision` `R29: FIXED, both copies (decision (a))`
- L24-21: Parallel chunk PoW verification at tip (p2p/chunk_server.go:161): cache-miss rate measured live on r28 (rft §13 R28): 10 checks per block, slow = checks − pool_seen in 13/13 scrapes, cache miss 10.0 % over the window (185 / 1,850 checks), 5.7 ms verify wall per block. `measured`
- B175: a malformed JSON-RPC body is answered by the upstream bridge with HTTP 500 and plain text, and rpc_error_responses_total{code="-32700"} stays 0 (R28 measurement). Closes when: the parse-error path is counted and answered as a JSON-RPC body, with a failing-before test. `r29-work` `R29: FIXED (counted JSON-RPC body)`
- B176: dero_tip_age_seconds writes its own HELP and gets no TYPE line (TYPE 273/283 on the live r28 page; the other 9 are vmrange histograms by design). Closes when: every registered non-vmrange family has a TYPE line on a live scrape. `r29-work` `R29: FIXED (TYPE tracked apart from HELP)`
- B152: R28 rpc-metrics: 20 source-declared /metrics families are created lazily and were absent from the live r27 page (absent ≠ 0): ... Blocker: 20 lazily created families absent from /metrics: pre-register at 0 (B143 pattern) then metric_registry.py build on a fresh scrape. Closes when: the 20 families present at 0 on /metrics of the next installed binary. `r29-work` `R29: FIXED for the counters (pre-registered at start); the histograms report-only`
- B155: Tooling (R28 harness-measure, proposed for the forensics owner): G29's exploit/privacy lint (harness/forensics/audit_check.py scan_dataset) re-scans ... Blocker: G29 lint memo touches a content-pinned verifier (forensics KATs, B147 guard). Closes when: run_gates G29 time below 64.1 s with the forensics KATs green. `r29-work` `R29: FIXED (G29 lint memo)`

- B166: Self-detected (R28 merge): B149 is implemented twice: harness-measure's kit/roundlists.py (KPI round_lists_missing, KAT HS1/HS12) and tools-kats' ... Blocker: B149 has two implementations (roundlists.py and tooling_intel.round_list_gaps). Closes when: round_list_gaps calls roundlists.py; duplicate KAT checks retired with mutations re-anchored. `r29-work` `R29: CLOSED (roundlists.py the one reader)`
- B168: Proposal (R28 glue-global GG28-7): go_mutate counts a mutation whose test name does not exist as MISSED (go test -run with no match passes, '[no ...). Blocker: go_mutate counts a mutation with a missing test name as MISSED. Closes when: go_mutate reports STALE for a missing test (go test -list first). `r29-work` `R29: FIXED (STALE for a missing test)`
- B169 (alias N28-10): Decision request (R28 artifacts-memory N28-10): harness/round.json's r26 github_sync evidence names the user's public handle (1 hit at 8fb2916f and ...). Blocker: user decision (a) keep the handle in the private registry or (b) reword the r26 evidence to the R15 output. Closes when: harness/round.json r26 github_sync evidence reworded (b), or the decision (a) booked in the registry. `user-decision` `R29: CLOSED by the user's decision (b)`
- B170: Self-detected (R28 merge): a registry edit re-attributes booked history: glue-global added globals/ to cryptosupply code_paths (GG28-8) and ... Blocker: area_intel recomputes 11 booked values after registry ownership edits. Closes when: each booked round pins its code_paths / gates (or a dated correct per value). `r29-work` `R29: FIXED (ownership pins)`
- Miner track PAUSED (rft §0.3), 11 rows: B2, B3, B4, B5, B6, B7, B25, B26, B27, B37, B20. Blocker: the user's decision to resume miner enhancements. Closes when: the user asks to resume miner enhancements (rft §0.3 'How to resume'): registry tracks.miner = active and a miner connected (getwork_connects_total > 0; 0 at the R28 start, harness/runtime/r28_start_20261105.txt). `◌ paused-miner` `◌ R29: still PAUSED (the user's decision: miner (b))`

OPEN AFTER ROUND 29

R29 is node-only (the miner track is paused, rft §0.3). This is the list after the measurement: the user restarted derod onto +22102026.r29 at 2026-11-08 15:33:00 PST and it was read live for 61 minutes. Every row is an item still open in rft §7, with the observable that closes it; the chip names the blocker class.

- B111: Q-CS1 answered at 7,498,540 (2,141,704 balance keys, root ca11cc21 EQUAL). Residual: registrations through 7,498,539 total 2,141,703, so 1 key is unattributed. Closes when: the extra key is attributed (a registration or an account created another way) with an anchor. `open residual`
- B110: SC payouts graded for run C (7,467,516..7,497,942: 800 atomic, 110/110 txs). Residual: 89 SC txs below 7,467,516 have no held state. Closes when: a held state below 7,467,516 or the residual graded data-unavailable in a dataset version. `data-unavailable`
- B48: inbound is live (B1 (a), incoming up to 28 in the R29 window), but the proven-lane counters stayed 0: the lane is consulted only when a netgroup's inbound limiter refuses. Closes when: a proven-lane admit, refusal or quota eviction is counted live. `◌ pending-event`
- B47, B57, B99: no event in 3,600 s / 192 blocks of the R29 window (registered, scenarios 22/22). Closes when: each counter moves on a live read. `◌ pending-event`
- B63, B64, B67: per-peer offence score, per-reason offence counters, per-peer INV clips: implemented, no scored offence on the live node yet. Closes when: a nonzero reading on GetPeerInfo or /metrics. `◌ pending-event`
- BD5: txcanon refuses proof scalars >= Order; no non-canonical scalar seen on mainnet. Closes when: a refusal is counted live or the census finds one. `◌ pending-event`
- B76: the state-root chain of the window: topos 7,415,466..7,453,289 have no held state. Closes when: a held archival state at or below 7,415,466 replays to ca11cc21. `data-unavailable`
- B190: A15 (narrowed) and A21 (scope) stay UNCLEAR after the R29 re-grade. Closes when: each is graded against a held input with an anchor (no construction). `r30-work`
- B196: one node identity can hold 2 slots (one in, one out): derod dedupes by address only. REPORT-ONLY (upstream behaviour). Closes when: the user decides on a one-slot-per-identity change. `user-decision`
- B193: the memory part is fixed; skill freshness is checked at Part D. Closes when: tooling_intel reports skill_fresh for the node skill after the R29 measured section. `r30-work`
- B184: the HF4 fork watch (version 4 at 8,000,000 on DEROFDN community-dev only): unchanged at the R29 watch. Standing: re-read every round; adopting a fork row is the user's decision. `◌ standing`
- Miner track PAUSED (rft §0.3), 11 rows: B2, B3, B4, B5, B6, B7, B25, B26, B27, B37, B20. Closes when: the user asks to resume miner enhancements (rft §0.3 'How to resume'). `◌ PAUSED (miner)`

DISCLOSURE TRACK

The Foundation's 2026-10-20 range-proof disclosure, re-tested every round against this tree and the live node (src-arch "The Foundation's disclosure, tested against this tree"). Source: `harness/loop/disclosure_track.json`, gate G24 (`harness/intel/disclosure_track.py`): every anchor re-read, no verdict change without evidence, no exploit content. A blank cell means the claim was not tested that round.

CLAIM	R8	R23	R23B	R24	R25	F
C1 · A soundness defect exists in the transfer-proof verification path	CONFIRMED			CONFIRMED	CONFIRMED	
C2 · Mechanism: a prover-adjustable generator unbound from the transcript	PARTLY-CONFIRMED			PARTLY-CONFIRMED	PARTLY-CONFIRMED	
C3 · The broken component is the sole magnitude gate, and ordinary testing cannot see it	PARTLY-CONFIRMED		PARTLY-CONFIRMED	PARTLY-CONFIRMED	PARTLY-CONFIRMED	
C4 · Inherited from the ConsenSys Anonymous Zether lineage	PARTLY-CONFIRMED			PARTLY-CONFIRMED	PARTLY-CONFIRMED	
C5 · Present from the introduction of the proof path (genesis)	PARTLY-CONFIRMED			PARTLY-CONFIRMED	PARTLY-CONFIRMED	
C6 · HF3 double-spend: 23,886,517.3 DERO created (floor), fourteen bodies undecodable	PARTLY-CONFIRMED		PARTLY-CONFIRMED	PARTLY-CONFIRMED	PARTLY-CONFIRMED	
C7 · 5,000,000 DERO burned in the same era (attribution unproven)	PARTLY-CONFIRMED			PARTLY-CONFIRMED	PARTLY-CONFIRMED	
C8 · Support ends; no rollback	◌ CANNOT-DETERMINE-LOCALLY			◌ CANNOT-DETERMINE-LOCALLY	◌ CANNOT-DETERMINE-LOCALLY	
X-EXEMPT · Own finding: the 75-TXID HF3 table exempts its TXIDs from Proof.Verify	CONFIRMED	PARTLY-CONFIRMED	PARTLY-CONFIRMED	PARTLY-CONFIRMED	PARTLY-CONFIRMED	
X-PARITY · Own finding: the HF3 Parity override reads the zero value off the p2p sync path	CONFIRMED		CONFIRMED	CONFIRMED	CONFIRMED	
X-SUPPLY · Operator takeaway: the published supply is a formula, so it cannot show inflation	CONFIRMED	CONFIRMED		CONFIRMED	CONFIRMED	

New in R26: NF26-1 FQ4: the transfer-proof transcript is the same at the head, at Release152 and at the genesis-era tag `MEASURED`; NF26-2 Behavioural cross-check on honest proofs: every one-field change is refused, and by which check `MEASURED`; NF26-3 FQ5: no public-data test separates a proof accepted under the defect from an honest one (interpretation) `◌ INTERPRETATION`; NF26-4 Live on the r25 node (2026-11-04): S1-S4 PASS; the 24 h emission read is still pending `MEASURED`; NF26-5 Census of the last 100,000 blocks (R26): 0 scalars >= Order `MEASURED`

Fix designs that need a coordinated fork:

- FX-F1 An independent inner-product generator `design` (C1, C2)
- FX-F2 Bind the blinding scalar into the transcript and compare canonically `design` (C1)
- FX-F3 Canonical decoding of proof scalars and points `design` (C3)
- FX-F4 Activation mechanics for a proof-system change `design` (C1, C8)
- FX-F5 Opt-in migration to a state whose supply is the exact sum of migrated balances (FQ6) `design` (X-SUPPLY, C1)
- FX-F6 A new genesis with sound proofs and a public aggregate rule (FQ6) `design` (X-SUPPLY, C1)

Node-local measures (no fork):

- FX-N1 Per-round re-test of every disclosure claim (anchor drift sentinel) `implemented` (C1, C2, C3, C4, C5, C6, C7, C8)
- FX-N2 Live supply and integrity cross-check `implemented` (X-SUPPLY, X-EXEMPT)
- FX-N3 Proof-scalar canonicality census `implemented` (C3)

- FX-N4 Relay/mempool policy: refuse non-reduced proof scalars `feasible-not-implemented` (C3)
- FX-N5 Patch the verifier locally (bind mu or change u_x) `◌ infeasible` (C1)
- FX-N6 A shape or size policy as a detector `◌ infeasible` (C3)
- FX-N7 Sum the balance tree to audit the supply `◌ infeasible` (X-SUPPLY)
- FX-N8 Refuse the 75 HF3 TXIDs at relay and RPC by ID `implemented-earlier` (X-EXEMPT)
- FX-N9 Say in GetInfo that total_supply is formula-derived `feasible-not-implemented` (X-SUPPLY)
- FX-N10 Structural transcript-binding KAT (a presence sentinel for class DC2) `implemented` (C1, C2, C5)

AREAS: EVERY PART OF THE STACK, SCORED EACH ROUND

One card per area of `harness/areas.json`, computed by `harness/intel/area_intel.py` from the pipeline and go_mutate records, the gate ledger, the derod logs and a read-only `/metrics` capture. Each value shows the line it was read from; a value without an anchor is not drawn. A paused area (the miner, dero-rft §0.3) is shown as PAUSED, never as failing. The proposals feed (`harness/runtime/area_proposals.json`) is what the next round triages into dero-rft §7 and §8 (§0.4).

Scorecard run 20261108T165653 (2026-11-08T16:56:53-08:00), round r29: 18 areas, 17 active, 1 paused; states OK 15, WATCH 2, PAUSED 1; 109 proposals; 0 values refused for a missing anchor.

HARNESS ITSELF (PIPELINE, GATES, KATS, REGISTRY, MUTATIONS)

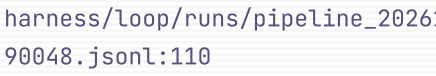
OK owner **harness**

go_mutate **3/3** in `gomut_20261108T062853.json`

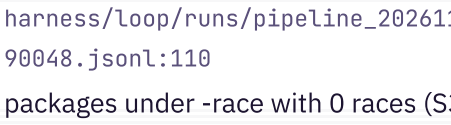
gates **6/6** PASS in `pipeline-20261108T123859`

open backlog **B20, B41, B49, B52, B54, B98, B101, B105** · proposals **0**

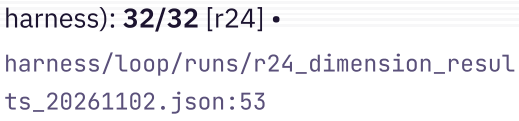
- flaky new tests (S3a): **0** [r29] → from 0 [r28d] `harness/loop/runs/pipeline_20261108T062853.jsonl:8`



- new tests re-run x3 (S3a): **18** [r29] ↑ from 10 [r28d] `harness/loop/runs/pipeline_20261108T062853.jsonl:8`



- go_mutate caught/total over the whole tree (S3b): **1619/1619** [r29] → from 1588/1588 [r28d] `harness/runtime/gomut_20261108T062853.json:2`



- verify_deroarch harness mutations proved (S7): **34/34** [r28d] → from 34/34 [r28] `harness/loop/runs/pipeline_20261107T190048.jsonl:110`
- gates PASS in the closed loop (S7): **29/32 (+3 PAUSED, miner)** [r28d] from 29/32 (+3 PAUSED, miner) [r28] `harness/loop/runs/pipeline_20261107T190048.jsonl:110`
- stand-alone KAT files passing in the closed loop (S7): **35/35** [r28d] → from 35/35 [r28] `harness/loop/runs/pipeline_20261107T190048.jsonl:110`
- packages under -race with 0 races (S3c): **27** [r29] → from 27 [r28d] `harness/loop/runs/pipeline_20261108T062853.jsonl:20`
- r24 tracks KAT mutations proved (verify-harness): **32/32** [r24] • `harness/loop/runs/r24_dimension_results_20261102.json:53`
- gates of the area PASS: **5/5** [r28d] → from 5/5 [r28] `harness/loop/GATES.md:2634`

+3 more in harness/runtime/area_intel.json

SECURITY: DOS, AI AGENTS, INFLATION-ADJACENT ABUSE (RPC + P2P SCORING, BANS, BUDGETS)

OK owner **security**

go_mutate **181/181** in `gomut_20261108T062853.json`

gates **1/1** PASS in `pipeline-20261108T123859`

open backlog **B30** · proposals **0**

- live p2p_inbound_aggregate_refusals_total: **0.0** [r29] → from 0.0 [r28] `harness/runtime/area_prom_20261108T165650.txt:17`



- live p2p_bans_total (since derod start): **2.0** [r29] ↑ from 0.0 [r28] `harness/runtime/area_prom_20261108T165650.txt:16`



- probes that pass for keys pre-warmed to trusted (30 min scenario S1): **573 of 4,800** [r24] ↓ from 4,800/4,800 [r23c] `harness/loop/runs/r24_dimension_results_20261102.json:140`



- r24 security mutations caught (verifier re-run): **15/15** [r24] • `harness/loop/runs/r24_dimension_results_20261102.json:165`
- live rpc_abuse_bans_total (since derod start): **0.0** [r29] → from 0.0 [r28] `harness/runtime/area_prom_20261108T165650.txt:27`
- gates of the area PASS: **1/1** [r28d] → from 1/1 [r28] `harness/loop/GATES.md:2663`
- go_mutate caught/total in the area files: **181/181** [r29] → from 181/181 [r28d] `harness/runtime/gomut_20261108T062853.json:2`
- registered own tests (NEW_TESTS) run in S3, attributed by the file that defines them: **101** [r29] → from 101 [r28d] `harness/loop/runs/pipeline_20261108T062853.jsonl:10`

CRYPTO AND SUPPLY: PROOFS, DECODERS, PAYLOAD PRIVACY, EMISSION AUDITOR, MONETARY INTEGRITY

OK owner **cryptoinfl**

go_mutate **272/272** in `gomut_20261108T062853.json`

gates **2/2** PASS in `pipeline-20261108T123859`

open backlog **B8, B10, B33, B36, B145** · proposals **2**

- modelled MAP attacker success on 16-member rings (ideal 6.25 %): **17.4 %** [r23c] • `harness/loop/runs/r24_dimension_results_20261102.json:190`
- emission auditor alerts on the live node: **0.0** [r29] → from 0.0 [r28] `harness/runtime/area_prom_20261108T165650.txt:6`



- live dero_emission_audit_consistent (1 = audit agrees with the formula): **1.0** [r29] → from 1.0 [r28] `harness/runtime/area_prom_20261108T165650.txt:7`



- GetMonetaryIntegrity totals equal GetEmissionAudit (CI24-1): **no: 0, False, 0/0/0 while the audit said True** [r23c] • `harness/loop/runs/r24_dimension_results_20261102.json:232`
- r24 cryptoinfl mutations caught: **21/21** [r24] • `harness/loop/runs/r24_dimension_results_20261102.json:170`
- live block_tx_roundtrip_broken_total: **0.0** [r29] → from 0.0 [r28] `harness/runtime/area_prom_20261108T165650.txt:3`



- gates of the area PASS: **2/2** [r28d] → from 2/2 [r28] `harness/loop/GATES.md:2652`
- KAT files of the area passing: **3/3** [r29] → from 2/2 [r28d] `harness/loop/runs/pipeline_20261108T062853.jsonl:14`
- go_mutate caught/total in the area files: **272/272** [r29] → from 267/267 [r28d] `harness/runtime/gomut_20261108T062853.json:2`

+1 more in harness/runtime/area_intel.json

GETWORK AND OPS: DISPATCHER, FINAL FLIPS, DEROD SUPERVISION, HEALTH TOOLS

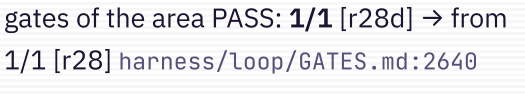
OK owner **getworkops**

go_mutate **134/134** in `gomut_20261108T062853.json`

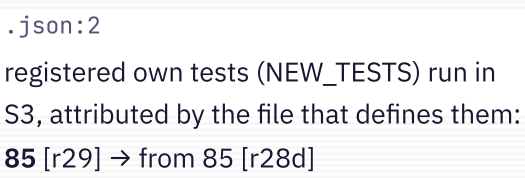
gates **1/1** PASS in `pipeline-20261108T123859`

open backlog **B5, B6, B7, B36, B45** · proposals **1**

- dead normal jobs per final flip: **0.18 (9 / 50)** [r29] → from 0.143 (34 / 238) [r23c] `harness/runtime/area_prom_20261108T165650.txt:9`



- final-flip delay, conditional mean (flips that had a dead window): **52.5556 (0.473 / 9)** [r29] → from 61.2 ms (n=30) [r23c] `harness/runtime/area_prom_20261108T165650.txt:11`



- getwork_final_flip_l9_seconds mean (r24 metric comparable with the model L9): **13.0204 (0.638 / 49)** [r29] → from 14.2 ms (shadow, n=130) [r24] `harness/runtime/area_prom_20261108T165650.txt:13`

- r24 getworkops mutations caught (verifier): **20/20** [r24] • `harness/loop/runs/r24_dimension_results_20261102.json:323`
- gates of the area PASS: **1/1** [r28d] → from 1/1 [r28] `harness/loop/GATES.md:2640`
- go_mutate caught/total in the area files: **134/134** [r29] → from 134/134 [r28d] `harness/runtime/gomut_20261108T062853.json:2`
- registered own tests (NEW_TESTS) run in S3, attributed by the file that defines them: **85** [r29] → from 85 [r28d] `harness/loop/runs/pipeline_20261108T062853.jsonl:10`

MEMPOOL, ADMISSION AND TPS

OK owner **txpool**

go_mutate **92/92** in `gomut_20261108T062853.json`

gates **none registered**

open backlog **B19, B42, B44** · proposals **0**

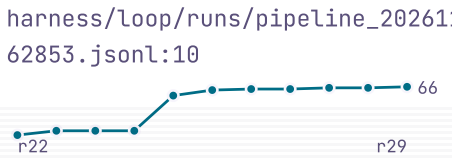
- policy-on / policy-off CPU per admission, median pair ratio (below the cap): **x0.9969 (192 pairs)** [r24] • `harness/loop/runs/r24_dimension_results_20261102.json:508`
- live mempool_count: **0.0** [r29] → from 0.0 [r28] `harness/runtime/area_prom_20261108T165650.txt:15`



- r24 txpool mutations caught (verifier fixed patch): **31/31** [r24] • `harness/loop/runs/r24_dimension_results_20261102.json:547`
- share of mainnet txs that rank 0 under integer FeesPerByte (B19): **58/58 = 100 %** [r24] • `harness/loop/runs/r24_dimension_results_20261102.json:478`
- serial admission CPU per tx (proof-verification bound): **75.9 ms** [r24] • `harness/loop/runs/r24_dimension_results_20261102.json:520`
- go_mutate caught/total in the area files: **92/92** [r29] → from 89/89 [r28d] `harness/runtime/gomut_20261108T062853.json:2`



- registered own tests (NEW_TESTS) run in S3, attributed by the file that defines them: **66** [r29] ↑ from 65 [r28d] `harness/loop/runs/pipeline_20261108T062853.jsonl:10`



CHAIN PERFORMANCE: SYNC PREWARM, POW/DIFFICULTY COST, BLOCK ADD

OK owner **chainperf**

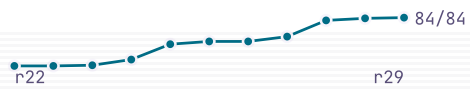
go_mutate **84/84** in `gomut_20261108T062853.json`

gates **none registered**

open backlog **B11, B31** · proposals **1**

- CPU per synced block, candidate/base geomean: **x0.916 (10 pairs)** [r24] • `harness/loop/runs/r24_dimension_results_20261102.json:602`
- PoW prewarms per synced block (10 = no double count): **10.0 (20 / 2)** [r29] → from 10.0 (10 / 1) [r25] `harness/runtime/area_prom_20261108T165650.txt:21`

- r24 chainperf mutations caught (verifier): **11/11** [r24] • `harness/loop/runs/r24_dimension_results_20261102.json:652`
- go_mutate caught/total in the area files: **84/84** [r29] → from 83/83 [r28d] `harness/runtime/gomut_20261108T062853.json:2`



- registered own tests (NEW_TESTS) run in S3, attributed by the file that defines them: **31** [r29] ↑ from 30 [r28d] `harness/loop/runs/pipeline_20261108T062853.jsonl:10`

RPC AND WEBSOCKETS (JSON-RPC BRIDGE, BATCHES, PUSHES, WALLET API TRANSPORT)

OK owner **rpcws**

go_mutate **272/272** in `gomut_20261108T062853.json`

gates **none registered**

open backlog **B40, B80, B85, B86** · proposals **0**

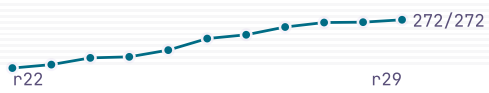
- CPU of a remote 256x256 GetTransaction batch (B29): **5.31 s (admitted)** [r23c] • `harness/loop/runs/r24_dimension_results_20261102.json:383`
- loopback handler cost, r24 / base (verify patch): **x0.987 (verify patch)** [r24] • `harness/loop/runs/r24_dimension_results_20261102.json:460`
- live rpc_recovered_panics_total: **0.0** [r29] → from 0.0 [r28] `harness/runtime/area_prom_20261108T165650.txt:28`



- live rpc_requests_total (since derod start): **5092 (sum of 12 labelled series in the capture)** [r29] • `harness/runtime/area_prom_20261108T165650.txt:29`
- live rpc_ws_push_skipped_total: **0.0** [r29] → from 0.0 [r28] `harness/runtime/area_prom_20261108T165650.txt:41`



- go_mutate caught/total in the area files: **272/272** [r29] → from 263/263 [r28d] `harness/runtime/gomut_20261108T062853.json:2`



- registered own tests (NEW_TESTS) run in S3, attributed by the file that defines them: **196** [r29] ↑ from 191 [r28d] `harness/loop/runs/pipeline_20261108T062853.jsonl:10`

SYNC AND P2P: PEERS, COMPACT BLOCKS, CATCH-UP, LOCATOR, INBOUND

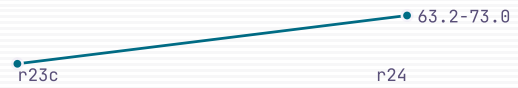
OK owner **syncp2p**

go_mutate **290/290** in `gomut_20261108T062853.json`

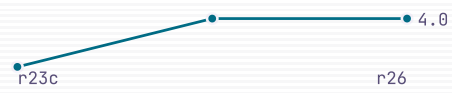
gates **2/2** PASS in `pipeline-20261108T123859`

open backlog **B1, B2, B30** · proposals **0**

- live-source loopback catch-up, blocks/s (shadow): **63.2-73.0** [r24] ↑ from 54.5-65.6 [r23c] `harness/loop/runs/r24_dimension_results_20261102.json:763`



- compact-block txs observed by txcanon (B18; 0 on r23c); r25 (B54): a 0 is a reading only once blockchain_tx_total > 0: **4.0** [r26] → from 4.0 [r25] `harness/runtime/area_prom_20261105T085719.txt:4`



- live incoming peers (B1: 0 until UDP 11011 is forwarded): **27.0** [r29] ↑ from 0.0 [r28] `harness/runtime/area_prom_20261108T165650.txt:18`



- live outgoing peers: **13.0** [r29] → from 14.0 [r28] `harness/runtime/area_prom_20261108T165650.txt:19`
- r24 syncp2p mutations caught (builder + verifier): **17/17** [r24] • `harness/loop/runs/r24_dimension_results_20261102.json:687`
- live dero_sync_lag_blocks: **0.0** [r29] → from 0.0 [r28] `harness/runtime/area_prom_20261108T165650.txt:8`
- gates of the area PASS: **2/2** [r28d] → from 2/2 [r28] `harness/loop/GATES.md:2640`
- KAT files of the area passing: **2/2** [r29] → from 2/2 [r28d] `harness/loop/runs/pipeline_20261108T062853.jsonl:14`
- go_mutate caught/total in the area files: **290/290** [r29] → from 286/286 [r28d] `harness/runtime/gomut_20261108T062853.json:2`

+1 more in harness/runtime/area_intel.json

WALLET (WALLETAPI, WALLET CLI, PAYLOAD PRIVACY)

OK owner **cryptoinfl**

go_mutate **65/65** in `gomut_20261108T062853.json`

gates **none registered**

open backlog **none** · proposals **0**

- upstream walletapi tests red on pristine Release153 too (B21): **1** [r29] → from 1 [r28d] `harness/runtime/gomut_20261108T062853.json:19872`



- go_mutate caught/total in the area files: **65/65** [r29] → from 63/63 [r28d] `harness/runtime/gomut_20261108T062853.json:2`



- registered own tests (NEW_TESTS) run in S3, attributed by the file that defines them: **76** [r29] ↑ from 74 [r28d] `harness/loop/runs/pipeline_20261108T062853.jsonl:10`



STORAGE AND BLOCK MANAGEMENT (STOREFS, TOPO STORE, DATADIR)

OK owner **syncp2p**

go_mutate **8/8** in `gomut_20261108T062853.json`

gates **1/1** PASS in `pipeline-20261108T123859`

open backlog **B11, B31** · proposals **1**

- datadir_intel checks that FAIL: **0** [r29] • `harness/runtime/datadir_20261108T165428.json:19`
- gates of the area PASS: **1/1** [r28d] → from 1/1 [r28] `harness/loop/GATES.md:2644`

- go_mutate caught/total in the area files: **8/8** [r29] → from 8/8 [r28d] `harness/runtime/gomut_20261108T062853.json:2`

- registered own tests (NEW_TESTS) run in S3, attributed by the file that defines them: **0** [r29] → from 0 [r28d] `harness/loop/runs/pipeline_20261108T062853.jsonl:10`

OBSERVABILITY AND INTEL (LOGS, /METRICS, INTEL TOOLS, CONSOLE TRIAGE)

OK owner **harness**

go_mutate **67/67** in `gomut_20261108T062853.json`

gates **2/2** PASS in `pipeline-20261108T123859`

open backlog **none** · proposals **0**

- ERROR lines at tip in the current derod run (log_intel): **0** [r29] → from 0 [r28] `harness/runtime/logintel_20261108T165416.json:60`



- live recovered_panics_total: **0 (sum of 5 labelled series in the capture)** [r29] → from 0 (sum of 5 labelled series in the capture) [r28] `harness/runtime/area_prom_20261108T165650.txt:22`



- gates of the area PASS: **2/2** [r28d] → from 2/2 [r28] `harness/loop/GATES.md:2640`



- KAT files of the area passing: **1/1** [r29] → from 1/1 [r28d] `harness/loop/runs/pipeline_20261108T062853.jsonl:14`
- go_mutate caught/total in the area files: **67/67** [r29] → from 65/65 [r28d] `harness/runtime/gomut_20261108T062853.json:2`
- registered own tests (NEW_TESTS) run in S3, attributed by the file that defines them: **33** [r29] ↑ from 31 [r28d] `harness/loop/runs/pipeline_20261108T062853.jsonl:10`

DOCS, PRIVACY AND PUBLISHING (ARTIFACTS, GENERATORS, PUBSCRUB, GITHUB COPY)

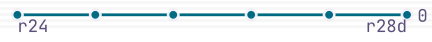
OK owner **docs/artifacts**

go_mutate **0/0** in `gomut_20261108T062853.json`

gates **10/10** PASS in `pipeline-20261108T123859`

open backlog **none** · proposals **0**

- cross-artifact statements that disagree with their canonical source (S7 step harness/intel/coherence.py --record; G25): **0** [r28d] → from 0 [r28] `harness/loop/runs/pipeline_20261107T190048.jsonl:109`



- src-arch rows refused at build (no anchor, no row): **0** [r28d] → from 0 [r28] `harness/loop/runs/pipeline_20261107T190048.jsonl:103`



- gates of the area PASS: **10/10** [r28d] → from 10/10 [r28] `harness/loop/GATES.md:2632`



- KAT files of the area passing: **2/2** [r29] → from 3/3 [r28d] `harness/loop/runs/pipeline_20261108T062853.jsonl:14`
- go_mutate caught/total in the area files: **0/0** [r29] from 0/0 [r28d] `harness/runtime/gomut_20261108T062853.json:2`
- registered own tests (NEW_TESTS) run in S3, attributed by the file that defines them: **0** [r29] → from 0 [r28d] `harness/loop/runs/pipeline_20261108T062853.jsonl:10`

THE FOUNDATION'S DISCLOSURE (SECTION 10 OF DERO-SRC-ARCH): RANGE-PROOF SOUNDNESS, INFLATION VISIBILITY

OK owner **disclosure**

go_mutate **23/23** in `gomut_20261108T062853.json`

gates **2/2** PASS in `pipeline-20261108T123859`

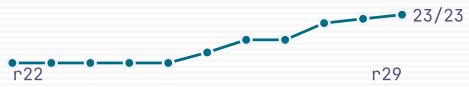
open backlog **B8** · proposals **0**

- audit rows with verdict CONFIRMED (forensics.json): **7** [r29] • `harness/model/raw/forensics.json:12`
- audit rows that survived review (forensics.json): **17** [r29] • `harness/model/raw/forensics.json:12`
- gates of the area PASS: **2/2** [r28d] → from 2/2 [r28] `harness/loop/GATES.md:2652`

- KAT files of the area passing: **1/1** [r29] → from 1/1 [r28d] `harness/loop/runs/pipeline_20261108T062853.jsonl:14`



- go_mutate caught/total in the area files: **23/23** [r29] → from 21/21 [r28d] `harness/runtime/gomut_20261108T062853.json:2`



- registered own tests (NEW_TESTS) run in S3, attributed by the file that defines them: **29** [r29] ↑ from 27 [r28d] `harness/loop/runs/pipeline_20261108T062853.jsonl:10`

MINER (DERO-MINER CPU SORT, GPU SPLIT, PAYOUTS PER HASH)

◌ PAUSED owner **cpusort+gpu+payouts(miner half)**

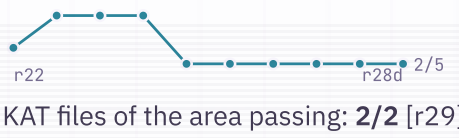
registry status paused `artifacts/dero-rft.md:146`

go_mutate **249/249** in `gomut_20261108T062853.json`

gates **2/5** PASS in `pipeline-20261108T123859` · G17 PAUSED, G18 PAUSED, G21 PAUSED

open backlog **B3, B4, B25, B26, B27, B37, B50** · proposals **0** · held while paused **4**

- live H per CPU-second: **5,435** [r23c] • `harness/round.json:37`
- live TOTAL H/s window mean: **94,273 H/s** [r23c] • `harness/round.json:37`
- gates of the area PASS: **2/5** [r28d] → from 2/5 [r28] `harness/loop/GATES.md:2647`



- KAT files of the area passing: **2/2** [r29] → from 2/2 [r28d] `harness/loop/runs/pipeline_20261108T062853.jsonl:14`



- go_mutate caught/total in the area files: **249/249** [r29] → from 249/249 [r28d] `harness/runtime/gomut_20261108T062853.json:2`



- registered own tests (NEW_TESTS) run in S3, attributed by the file that defines them: **208** [r29] → from 208 [r28d] `harness/loop/runs/pipeline_20261108T062853.jsonl:10`

CHAIN CONSENSUS OBSERVATORY (PARTICIPANTS, BLOCK TIMING, DIFFICULTY RULE, REORGS, CONSENSUS LOG CENSUS)

WATCH owner **consensus**

go_mutate **42/42** in `gomut_20261108T062853.json`

gates **1/1** PASS in `pipeline-20261108T123859`

open backlog **B72** · proposals **7**

- window MEAN block time (consecutive-height timestamp gaps), consensus_intel: **18.612** [r29] → from 18.563 [r28] `harness/runtime/consensus_capture_20261109T005529Z.txt:18`



- window p95 block time: **39.239** [r29] → from 37.575 [r28] `harness/runtime/consensus_capture_20261109T005529Z.txt:20`



- observatory detectors graded ANOMALY (stated statistical test crossed): **3** [r29] ↑ from 2 [r28] `regression` `harness/runtime/consensus_capture_20261109T005529Z.txt:101`



- observatory detectors graded OBSERVED (a consensus rule violated on the chain): **1** [r29] → from 1 [r28] `harness/runtime/consensus_capture_20261109T005529Z.txt:100`
- stored difficulties that differ from the consensus rule recomputed: **0** [r29] → from 0 [r28] `harness/runtime/consensus_capture_20261109T005529Z.txt:35`
- finality violations (GetMonetaryIntegrity reorg monitor): **0** [r29] → from 0 [r28] `harness/runtime/consensus_capture_20261109T005529Z.txt:82`
- share of blocks assembled by participants that mine nothing themselves: **0.421** [r29] ↑ from 0.3253 [r28] `regression` `harness/runtime/consensus_capture_20261109T005529Z.txt:65`
- Nakamoto coefficient of finals (fewest integrators assembling above 1/2 of blocks): **3** [r29] → from 3 [r28] `harness/runtime/consensus_capture_20261109T005529Z.txt:60`
- Nakamoto coefficient of regular miniblocks (fewest miners above 1/2): **4** [r29] → from 4 [r28] `harness/runtime/consensus_capture_20261109T005529Z.txt:59`

+6 more in harness/runtime/area_intel.json

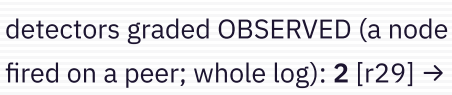
CONSENSUS OBSERVATORY: THE NETWORK PEERS THIS NODE TALKS TO (HEALTH, ACTIVITY, WRONGDOING; PSEUDONYMISED)

WATCH owner **peers**

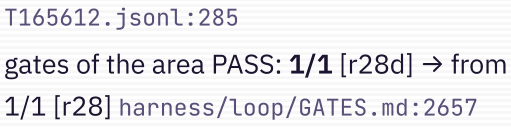
go_mutate **0/0** in `gomut_20261108T062853.json`

gates **1/1** PASS in `pipeline-20261108T123859`

open backlog **B63, B64, B67, B70, B71** · proposals **10**

- detectors graded ANOMALY (stated test crossed): **4** [r29] ↑ from 2 [r28]
  regression
  `harness/runtime/peer_capture_20261108T165612.jsonl:285`
  

- share of p2p_first_relay_total delivered by the top peer (topology dependence): **0.5162** [r29] → from 0.6176 [r28]
  `harness/runtime/peer_capture_20261108T165612.jsonl:285`
  

- largest lag of an active peer behind our tip: **2874039** [r29] ↑ from 0 [r28]
  regression
  `harness/runtime/peer_capture_20261108T165612.jsonl:285`
  

- distinct IPv4 /16 (IPv6 /32) among the active peers, computed privately: **40** [r29] ↑ from 14 [r28]
  `harness/runtime/peer_capture_20261108T165612.jsonl:285`
- detectors graded OBSERVED (a node rule fired on a peer; whole log): **2** [r29] → from 2 [r28]
  `harness/runtime/peer_capture_20261108T165612.jsonl:285`
- active peers at the capture (GetPeerInfo): **40** [r29] ↑ from 14 [r28]
  `harness/runtime/peer_capture_20261108T165612.jsonl:285`
- distinct peers dropped for the v2 remnant chain within 7 days: **3** [r29] ↑ from 1 [r28]
  regression
  `harness/runtime/peer_capture_20261108T165612.jsonl:285`
- gates of the area PASS: **1/1** [r28d] → from 1/1 [r28] `harness/loop/GATES.md:2657`
- KAT files of the area passing: **4/4** [r29] → from 4/4 [r28d]
  `harness/loop/runs/pipeline_20261108T062853.jsonl:14`

+2 more in harness/runtime/area_intel.json

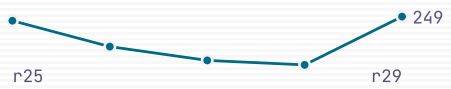
TOOLING & ASSETS (NON-CORE): HARNESS TOOLS, KAT SUITES, GENERATORS, SCRUB/SYNC, SKILLS, MEMORY CONTROL AND MIRROR, SCRIPTS (DERO-RFT §0.9)

OK owner **tooling**

go_mutate **2/2** in `gomut_20261108T062853.json`

gates **none registered**

open backlog **B81, B89, B93, B104, B116, B117, B119** · proposals **0**

- seconds since the round's resume checkpoint (harness/intel/round_checkpoint.py --loop 600, dero-rft §0.10; target <= 1200 while a round is open; r28: a 10-min loop's age is a sawtooth 0..600 s, so no trend is a regression: the alert is > 1200 s): **249** [r29] ↑ from 107 [r28]
  `harness/runtime/tooling_20261108T165620.json:72`
  

- R24/R25 tooling defect classes without a KAT (target 0): **0** [r29] → from 0 [r28]
  `harness/runtime/tooling_20261108T165620.json:123`
  

- generated pages whose double run is not byte-exact (target 0; --determinism): ◌ pending `tooling_intel.json has no scalar kpis.determinism_mismatches`
- flaky tests in the newest full run (S3a); target 0: **0** [r29] → from 0 [r28]
  `harness/runtime/tooling_20261108T165620.json:29`
  

- gate ids used by two different gates (target 0): **0** [r29] → from 0 [r28]
  `harness/runtime/tooling_20261108T165620.json:119`
- KAT mutation proofs caught / total over every KAT step of the newest S7 (INVALID never counts): **100.0** [r29] → from 100.0 [r28]
  `harness/runtime/tooling_20261108T165620.json:19`
- MEMORY.md size (index hygiene, §0.9 rule 4): **8756** [r29] → from 8875 [r28]
  `harness/runtime/tooling_20261108T165620.json:34`
- longest MEMORY.md line (target <= 300): **205** [r29] ↓ from 265 [r28]
  `harness/runtime/tooling_20261108T165620.json:35`
- MEMORY.md lines over 300 chars (target 0): **0** [r29] → from 0 [r28]
  `harness/runtime/tooling_20261108T165620.json:36`

+11 more in harness/runtime/area_intel.json

MONETARY FORENSICS ENGINE: THE HF3-WINDOW DATASET, REPLAY, INDEPENDENT AUDIT, LEDGER FRONTEND (DERO-RFT §0.8)

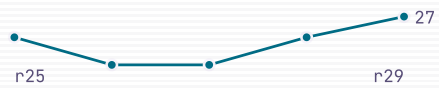
OK owner **forensics**

go_mutate **21/21** in `gomut_20261108T062853.json`

gates **5/5** PASS in `pipeline-20261108T123859`

open backlog **B76, B97** · proposals **0**

- audit challenges that FAIL in the current audit run (overturned, not yet corrected): **0** [r29] → from 0 [r28]
  `harness/runtime/forensics_intel.json:11550`
  

- independent audit challenges that HOLD (audit/audit_report.json): **27** [r29] ↑ from 24 [r28]
  `harness/runtime/forensics_intel.json:11551`
  

- overturns over all audit runs of the newest version (history kept): **0** [r29] → from 0 [r28]
  `harness/runtime/forensics_intel.json:11552`
  

- audit challenges UNRESOLVED (each downgrades its verdict): **0** [r29] → from 0 [r28]
  `harness/runtime/forensics_intel.json:11553`
- verdicts NOT-DECIDABLE-FROM-PUBLIC-DATA (the HF3 TXIDs): **5** [r29] → from 5 [r28]
  `harness/runtime/forensics_intel.json:11559`
- verdicts pending: **0** [r29] → from 0 [r28]
  `harness/runtime/forensics_intel.json:11560`
- verdicts PROVEN-NO-INFLATION-IN-SCOPE: **0** [r29] → from 0 [r28]
  `harness/runtime/forensics_intel.json:11561`
- verdicts PROVEN-INFLATION: **0** [r29] → from 0 [r28]
  `harness/runtime/forensics_intel.json:11562`
- verdicts SPECULATION (EXT only): **0** [r29] → from 0 [r28]
  `harness/runtime/forensics_intel.json:11563`

+9 more in harness/runtime/area_intel.json

CONSENSUS OBSERVATORY

Who makes the chain and the peers this node hears it from, read-only from the running node every pipeline run (rft §0.7). Grades: **OBSERVED** = a rule line of this tree was violated (cited), **ANOMALY** = a stated test crossed its threshold (not an accusation), **NORMAL**, **NO-DATA**/pending. Participants and peers are salted pseudonyms; no address leaves the private capture. Every measurement with its evidence line: src-arch “Consensus observatory — chain” and “Consensus observatory — network peers”.

Chain (consensus_capture_20261109T005529Z.txt, captured 2026-11-09T00:56:08+00:00, binary r29): OBSERVED 1 · ANOMALY 3 · NORMAL 16 · NO-DATA 0.

CHAIN KPI	VALUE
blocks in the window	3000
mean block time (s)	18.612
p95 block time (s)	39.239
participants	41
Nakamoto coefficient, regular miniblocks	4
Nakamoto coefficient, finals	3
top-1 share of finals	0.381
difficulty-rule mismatches	0
finality violations	0

- OBSERVED D13-v2-remnant: Peers still on the pre-HF3 v2 remnant chain (7 connected below v3, 9 drop lines in the window (57 in the whole log, 3 seen in 7 d))
- ANOMALY D15-concentration-final: No participant holds more than 1/3 of the finals (block integration) (0.381)
- ANOMALY D16-final-vs-regular: Each integrator wins finals in proportion to its regular miniblock share (14)
- ANOMALY D19-collective-final: Integrator-only participants together assemble no more than 1/3 of the blocks (0.421)

Peers (peer_capture_20261108T165612.jsonl, captured 2026-11-08T16:56:12-08:00, 3.6.0-153.DEROHE.STARGATE+22102026.r29): OBSERVED 2 · ANOMALY 4 of 15 detectors.

PEER KPI	VALUE
active peers	40
outgoing	13
incoming	27
max lag (blocks)	2874039
median latency (ms)	178.6
distinct /16s	40
top first-relay share	51.6 %
v2-remnant peers seen in 7 d	3

- OBSERVED D1: Peers on the v2 remnant chain (3 finding(s))
- ANOMALY D2: Lagging or over-advertising peers (live) (8 finding(s))
- OBSERVED D3: Stuck peers evicted (log) (2 finding(s))
- ANOMALY D6: Clock skew of peers (1 finding(s))
- ANOMALY D9: First-relay concentration (topology dependence) (1 finding(s))
- ANOMALY D11: One node identity on several addresses (1 finding(s))

Node-side gaps the observatories could not read (R25, node code, so a derod restart):

- CO24-N1 → rft §7 B63: Per-peer offence score and last reasons are not exposed over RPC (area peers)
- CO24-N2 → rft §7 B64: No per-reason offence counter (invalid_pow, malformed_request, undecodable, invalid_tx, timeout) (area peers)
- CO24-N3 → rft §7 B65: Inbound Chain requests served are not counted (area peers)
- CO24-N4 → rft §7 B66: Ban list entries carry no reason or origin (area peers)
- CO24-N5 → rft §7 B67: Per-peer INV budget clips are node-wide only (area peers)
- CO24-N6 → rft §7 B68: Per-peer object-serve failures are only in the log (area peers)
- CO24-N7 → rft §7 B69: Stuck-peer holds are not listed (area peers)

Sources: RUN_REPORT.md, ARCHITECTURE.md and git history of the project repository; Round 20 to 23b workflow outputs for figures marked as workflow findings (R23b: the merge report and the per-dimension findings, cited by id); the soundness review page dero-soundness-r23.html; live read-only RPC reads where a row says so. R23c: the three workflow journals and the builder and verifier reports, cited by dimension and finding id, the pipeline records named in its rows, and read-only reads of the miner log. R25: the node and forensics workflow journals (harness/loop/runs/r25_dimension_results_20261103.json), the merge report, the pipeline records named in its rows, and the forensic dataset forensics/r25/ (every figure of the forensics sections names its file and SHA-256). R27: the merge record harness/runtime/r27_merge_20261105.txt and the pipeline records named in its rows. R28: the merge record harness/runtime/r28_merge_20261106.txt, the closure audit harness/runtime/r28_closure_audit_20261106.txt and the pipeline records named in its rows. Code excerpts are copied from `git show <commit>:<path>` ; the comment lines of the R16 excerpt were trimmed.